RZ/T2, RZ/N2

Quick Start Guide: EtherNet/IP OpENer Sample Program

Introduction

This document describes the setup procedure of the sample program for RZ/T2 and RZ/N2 port of EtherNet/IP™ "OpENer".

Note : In this document, the Cortex®-R52 is referred to as CR52 and the Cortex®-A55 is referred to as CA55.

Target Device

RZ/T series: RZ/T2M, RZ/T2ME, RZ/T2H
RZ/N series: RZ/N2L, RZ/N2H

Contents

1. Overview

This document describes the setup procedure of the sample program for RZ/T2 and RZ/N2 port of EtherNet/IP “OpENer” and explains the procedure for connecting the CODESYS software programmable logic controller (PLC).

In this sample program, these open-source software are used.

- EtherNet/IP OpENer
- FreeRTOS
- lwIP

For demonstration, the application of this sample program has an Exclusive Connection and an Input Only Connection. This document also explains how to connect to the CODESYS software prepared in this package.

1.1 Abbreviations / Definitions

Table 1-1 Abbreviations/Definitions

Index	Abbreviations/Definitions	Description
1	IP	Internet Protocol
2	TCP	Transmission Control Protocol
3	USB	Universal Serial Bus
4	PC	Personal Computer
5	SW	Switch
6	EWARM	Embedded Workbench® for ARM
7	lwIP	lightweight IP
8	CIP	Common Industrial Protocol
9	OSS	Open-Source Software
10	QoS	Quality of Service
11	DLR	Device Level Ring
12	LLDP	Link Layer Discovery Protocol
13	FSP	Flexible Software Package
14	ACD	Address Conflict Detection

1.2 Reference

Technical information is available via Renesas.

Table 1-2 Technical Inputs for RZ/T2M

Index	Technical Inputs
1	r20ut4939egxxxx-rskplus-rzt2m-v1-um.pdf
2	r01uh0916ejxxxx-rzt2m.pdf
3	r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started

Table 1-3 Technical Inputs for RZ/T2ME

Index	Technical Inputs
1	r20ut4939egxxxx-rskplus-rzt2m-v1-um.pdf
2	r01uh1062ejxxxx-rzt2me.pdf
3	r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started

Table 1-4 Technical Inputs for RZ/T2H

Index	Technical Inputs
1	r20ut5405ejxxxx-rzt2hevb.pdf
2	r01uh1039ejxxxx-rzt2h_n2h.pdf
3	r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started

Table 1-5 Technical Inputs for RZ/N2L

Index	Technical Inputs
1	r20ut4984egxxxx-rskplus-rzn2l-v1-um.pdf
2	r01uh0955ejxxxx-rzn2l.pdf
3	r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started

Table 1-6 Technical Inputs for RZ/N2H

Index	Technical Inputs
1	r20ut5522ejxxxx-rzn2hevb.pdf
2	r01uh1039ejxxxx-rzt2h_n2h.pdf
3	r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started

1.3 Limitations and Restrictions / Known Issues

Table 1-7 Limitations and Restrictions / Known Issues

<table>
<tr><th>Device</th><th>Category</th><th>Detail</th></tr>
<tr><td>RZ/T2M
RZ/T2ME
RZ/T2H
RZ/N2H</td><td>EtherNet/IP OpENer</td><td>Implemented the process for Reset Service (Type 0, 1). Since this process involves CPU reset and disconnects the debugger for RAM-execution projects, it is disabled by default with the macro switch.
If you'd like to enable it, please change the macro value. "SAMPLE_EXECUTE_SYSTEM_RESET_PROCESS" in opener_port_main.c.</td></tr>
<tr><td>RZ/T2M
RZ/T2ME</td><td>EWARM 9.60.1 with RZT FSP SC v.2.1.0</td><td>Section allocations in the linker script have changed, and the default icf file may cause build errors. As a workaround, try to build using the modified icf file included in this package.
Please refer to the Note in the last of the section 6.2.2.3 and 6.3.2.3.</td></tr>
<tr><td rowspan="2">RZ/T2H</td><td rowspan="2">EWARM 9.60.2 with RZT FSP SC v.2.2.0</td><td>Some files in the CA55 project with flash boot mode were modified to address the issue of the RZT FSP v2.2.0.(Known Issues No.39 in the FSP quick start guide r01an6434ej0113-rzt2-rzn2-fsp-getting-started.pdf)
Therefore, please note the following.
- The system.c (\opener_single_ca55_cpu0\rzt\fsp\src\bsp\cmsis\Device\RENESAS\Source\)
- The fsp_xspi1_boot.icf file (\opener_single_ca55_cpu0\script)</td></tr>
<tr><td>If you change the boot mode, these files will be overwritten at the time of code generation, so please save it in advance and restore it before building.</td></tr>
<tr><td>RZ/N2L</td><td>EWARM 9.60.2 with RZN FSP SC v.2.1.0</td><td>Section allocations in the linker script have changed, and the default icf file may cause build errors. As a workaround, try to build using the modified icf file included in this package.</td></tr>
<tr><td rowspan="2">RZ/N2H</td><td rowspan="2">EWARM 9.60.2 with RZN FSP SC v.2.1.0</td><td>Some files in the CA55 project with flash boot mode were modified to address the issue of the RZN FSP v2.1.0. (Known Issues No.39 in the FSP quick start guide r01an6434ej0113-rzt2-rzn2-fsp-getting-started.pdf)
Therefore, please note the following.
- The system_core.c (\opener_single_ca55_cpu0\rzn\fsp\src\bsp\cmsis\Device\RENESAS\Source\ca\)
- The bsp_cache.c (\opener_single_ca55_cpu0\rzn\fsp\src\bsp\mcu\all\)
- The fsp_xspi1_boot.icf file (\opener_single_ca55_cpu0\script)</td></tr>
<tr><td>If you change the boot mode, these files will be overwritten at the time of code generation, so please save it in advance and restore it before building.</td></tr>
</table>

2. Features

The "OpENer" is an open-source software for I/O communication adapters of EtherNet/IP. It supports multiple I/O and explicit connections and includes objects and services for making EtherNet/IP-compliant products as defined in the ODVA specification.

This package is the RZ/T2 and RZ/N2 port of OpENer and includes OpENer source codes. Regarding the open-source license of OpENer, please see the following file.

```
common\oss\OpENer\license.txt
```

And this sample software has the functionality required by the ODVA Conformance Test Suite CT20 (hereinafter referred to as CT20).

The Class Objects implemented in this sample software are as follows. For details, please see Chapter 8.3.

Table 2-1 CIP Object Classes supported this sample software.

Object Class #	Object Class Name
0x01	Identity
0x02	Massage Router
0x04	Assembly
0x06	Connection Manager
0x47	Device Level Ring
0x48	QoS
0xF5	TCP/IP Interface
0xF6	Ethernet Link
0x109	LLDP Management

Option: CIP Safety feature

For only RZ/N2L, This sample code also includes the implementation of the CIP Safety feature as an option. **Other device sample code will support CIP Safety feature.**

By default, this feature is disabled, and it can be enabled by modifying the project's defined symbols.

Note: Regarding the CIP Safety feature, we have evaluated its connection and operation with Safety PLC and functional safety boards, but verification using official tools has not been conducted yet.

The objects that become active upon enabling are as follows:

Table 2-2 List of CIP Safety Objects Enabled

Object Class #	Object Class Name
0x39	Safety Supervisor
0x3A	Safety Validator
0x49	Safety Analog Input Point
0x64	Safety Analog Output Point

For details on how to enable the CIP Safety feature, services, and operation, please refer to Chapter 9 Appendix C CIP Safety feature.

3. Requirements

This project has been developed and tested on these environments using the following boards and tools.

Table 3-1 RZ/T2M Requirements

Item	Vender	Description
Board	Renesas Electronics	RZ/T2M RSK Board
IDE	IAR Systems	Embedded Workbench® for ARM Version 9.50.1
	Renesas Electronics	e² studio 2024-01.1 Flexible Software Package (FSP) for Renesas RZ/T series v2.0.0 FSP Smart Configurator 2024-01.1 (FSP v2.0.0) Please download from the link below. https://github.com/renesas/rzt-fsp/releases/tag/v2.0.0
Emulator	IAR Systems	I-jet
	SEGGER	J-Link (Software Version: 7.94h)
GCC Compiler	Arm	12.2.Rel1
Evaluation Software	CODESYS GmbH	CODESYS v3.5.15.10 32-bit

Table 3-2 RZ/T2ME Requirements

Item	Vender	Description
Board	Renesas Electronics	RZ/T2ME RSK Board
IDE	IAR Systems	Embedded Workbench® for ARM Version 9.60.1
	Renesas Electronics	e² studio 2024-04 Flexible Software Package (FSP) for Renesas RZ/T series v2.1.0 FSP Smart Configurator 2024-04 (FSP v2.1.0) Please download from the link below. https://github.com/renesas/rzt-fsp/releases/tag/v2.1.0
Emulator	IAR Systems	I-jet
	SEGGER	J-Link (Software Version: 7.96j)
GCC Compiler	Arm	12.2.Rel1
Evaluation Software	CODESYS GmbH	CODESYS v3.5.15.10 32-bit

Table 3-3 RZ/T2H Requirements

Item	Vender	Description
Board	Renesas Electronics	RZ/T2H Evaluation Board
IDE	IAR Systems	Embedded Workbench® for ARM Version 9.60.2 + Patch* *: Please download from the RZ/T2H product page in Renesas WEB. Or click here EWARM Patch File for RZ/T2H and RZ/N2H.
	Renesas Electronics	e² studio 2024-10 Flexible Software Package (FSP) for Renesas RZ/T series v2.2.0 FSP Smart Configurator 2024-10 (FSP v2.2.0) Please download from the link below. https://github.com/renesas/rzt-fsp/releases/tag/v2.2.0
Emulator	IAR Systems	I-jet
	SEGGER	J-Link (Software Version: 7.98c)
GCC Compiler	Arm	GNU ARM Embedded Toolchain (version 12.2.1.arm-12-24 for CR52) GNU ARM A-Profile (AArch64 bare-metal) (version 10.3.1.20210621 for CA55)
Evaluation Software	CODESYS GmbH	CODESYS v3.5.15.10 32-bit

Table 3-4 RZ/N2L Requirements

Item	Vender	Description
Board	Renesas Electronics	RZ/N2L RSK Board
IDE	IAR Systems	Embedded Workbench® for ARM Version 9.60.2
	Renesas Electronics	e² studio 2024-10 Flexible Software Package (FSP) for Renesas RZ/N series v2.1.0 FSP Smart Configurator 2024-10 (FSP v2.1.0) Please download from the link below. https://github.com/renesas/rzn-fsp/releases/tag/v2.1.0
Emulator	IAR Systems	I-jet
	SEGGER	J-Link (Software Version: 7.98c)
GCC Compiler	Arm	13.3.1.arm-13-24
Evaluation Software	CODESYS GmbH	CODESYS v3.5.21.10 64-bit

Table 3-5 RZ/N2H Requirements

Item	Vender	Description
Board	Renesas Electronics	RZ/N2H Evaluation Board
IDE	IAR Systems	Embedded Workbench® for ARM Version 9.60.2 + Patch* *: Please download from the RZ/N2H product page in Renesas WEB. Or click here EWARM Patch File for RZ/T2H and RZ/N2H.
	Renesas Electronics	e² studio 2024-10 Flexible Software Package (FSP) for Renesas RZ/N series v2.1.0 FSP Smart Configurator 2024-10 (FSP v2.1.0) Please download from the link below. https://github.com/renesas/rzn-fsp/releases/tag/v2.1.0
Emulator	IAR Systems	I-jet
	SEGGER	J-Link (Software Version: 7.98c)
GCC Compiler	Arm	GNU ARM Embedded Toolchain (version 12.2.1.arm-12-24 for CR52) GNU ARM A-Profile (AArch64 bare-metal) (version 10.3.1.20210621 for CA55)
Evaluation Software	CODESYS GmbH	CODESYS v3.5.15.10 32-bit

4. Hardware Setup

4.1 RZ/T2M RSK board

4.1.1 Jumper and Switch configuration

This document describes the major hardware properties. Refer to Renesas Stater Kit+ for RZ/T2M user's manual and schematic for more board details.

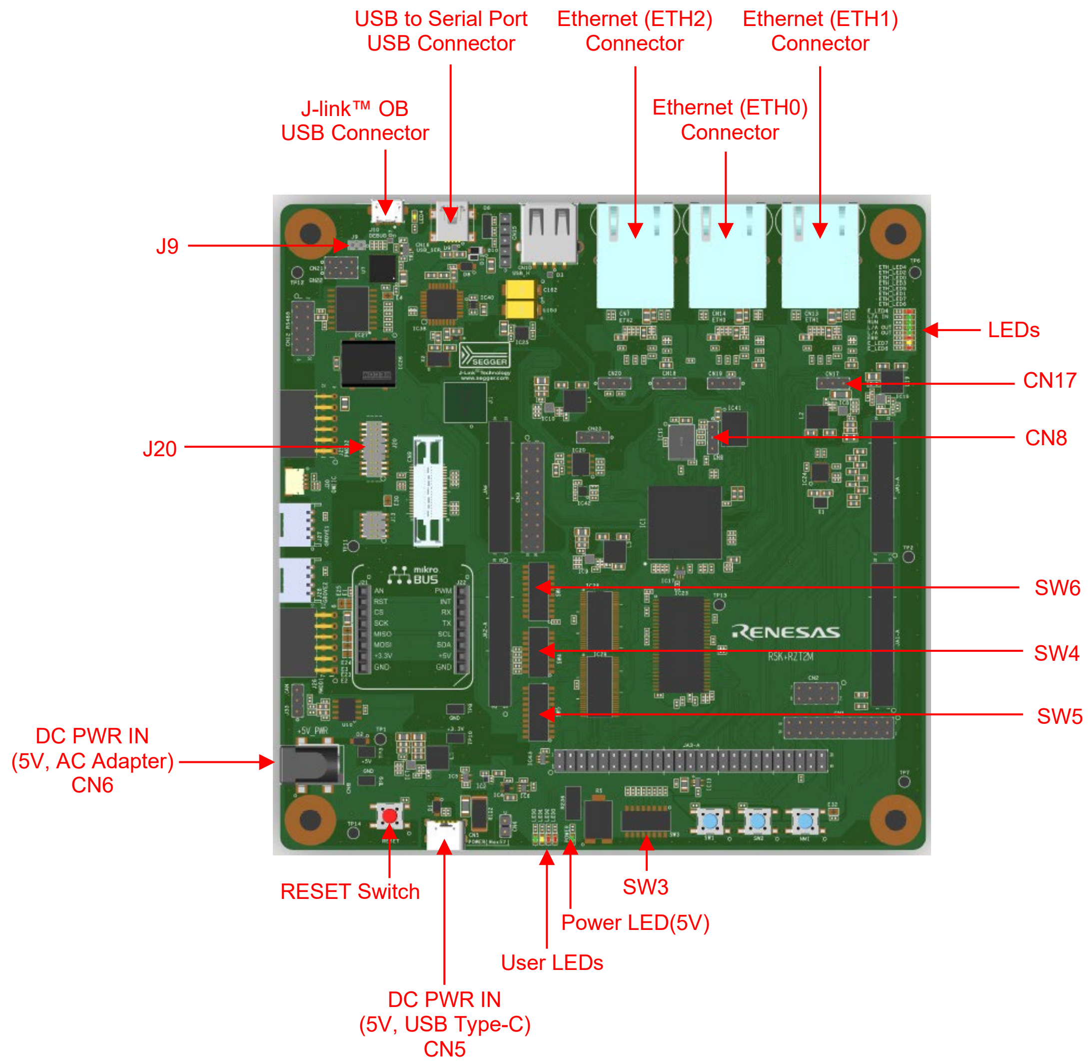


Figure 4-1 RZ/T2M RSK board layout

Table 4-1 Jumper pin settings

Reference	Jumper Position	Description
CN8	Shorted Pin 2-3	Enable QSPI (IC21).
CN17	Shorted Pin 2-3	Connect 1.8V Power rail to VCC1833_2. (When using Ethernet)
J9	Open	Enable the J-Link® OB.

Table 4-2 SW4 Settings

SW4	Setting	Description
SW4-1	ON	16-bit bus boot mode (NOR flash) Sets this SW settings when the program is executed by RAM.
SW4-2	OFF	
SW4-3	ON	
SW4-1	ON	xSPI0 boot mode (x1 boot serial flash) Sets this SW settings when the program is executed by flash writing.
SW4-2	ON	
SW4-3	ON	
SW4-4	ON	JTAG Authentication by Hash is disabled.
SW4-5	OFF	ATCM 1 wait

Table 4-3 SW5 Settings

SW5	Setting	Description
SW5-3	ON	Enable SCI_RTS
SW5-4	OFF	
SW5-5	ON	Enable SCI_RXD
SW5-6	OFF	
SW5-7	OFF	
SW5-8	OFF	Enable SCK3
SW5-9	ON	
SW5-10	OFF	

Table 4-4 SW6 Settings

SW6	Setting	Description
SW6-1	OFF	Enables signals other than the external bus. (CAN, Emulator, I2C, etc.)
SW6-3	ON	Enable TRACE_CTL
SW6-4	OFF	
SW6-5	OFF	Enable SCI_TXD
SW6-6	ON	
SW6-7	OFF	Enable MB_RST
SW6-8	ON	
SW6-9	OFF	Enable CAN_RX_OB
SW6-10	ON	

Other SW settings refer to r20ut4939egxxxx-rskplus-rzt2m-v1-um.pdf.

4.1.2 Setup Board

Setting the board for running sample program is shown below.

1. Connect an emulator to the connector .
   - When you use I-jet or J-Link emulator, connect it to J20 on RZ/T2M RSK board.
   - When you use J-Link On-Board emulator, connect USB micro-B to J10 on RZ/T2M RSK board. (Please disconnect J9 for powering up J-Link OB.)

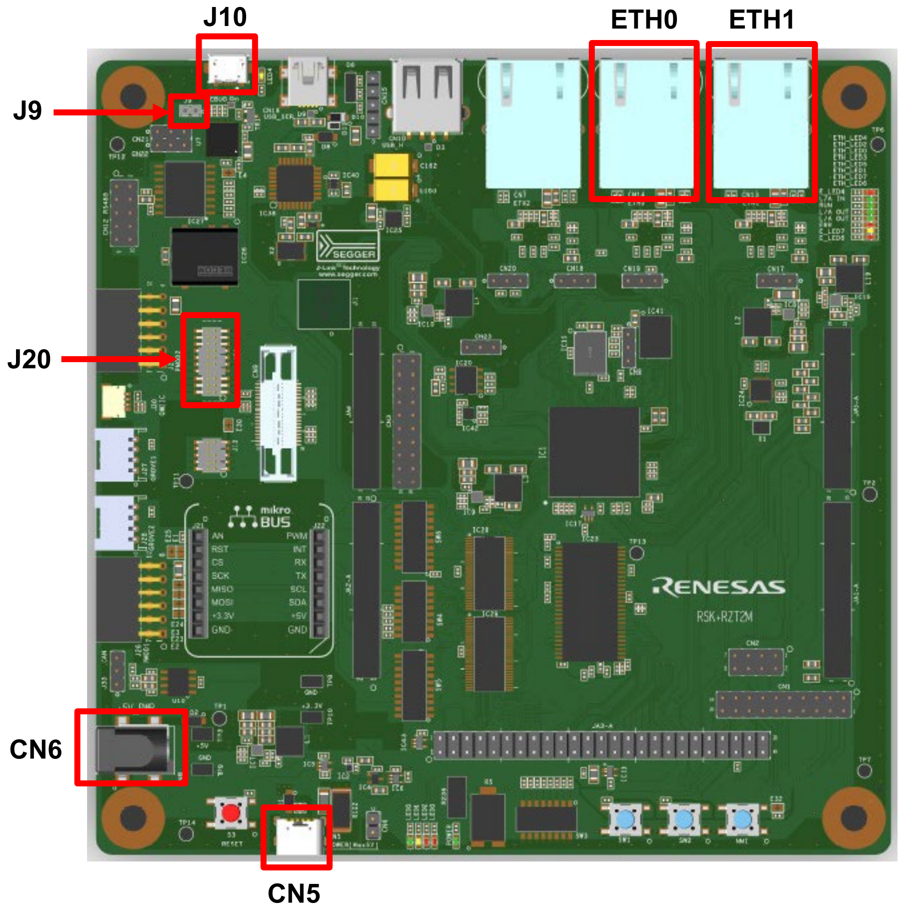


Figure 4-2 Setup RZ/T2M RSK board

2. Power is supplied using USB cable (Type-C) or AC / DC adapter.
   - When using USB cable (Type-C), connect it to the USB connector “CN5” on the RZ/T2M RSK board.
   - When using AC/DC adapter, connect it to the connector “CN6” on the RZ/T2M RSK board.
3. Connect Ethernet Cable to the Ethernet Connector “ETH0” or “ETH1”.

4.2 RZ/T2ME RSK board

4.2.1 Jumper and Switch configuration

For each configuration, see the section 4.1.1 RSK+RZT2M.

4.2.2 Setup Board

For each setting, see the section 4.1.2 RSK+RZ/T2M.

4.3 RZ/T2H Evaluation board

4.3.1 Jumper and Switch configuration

This document describes the major hardware properties. Refer to the Evaluation Board for RZ/T2H user's manual and schematic for more board details.

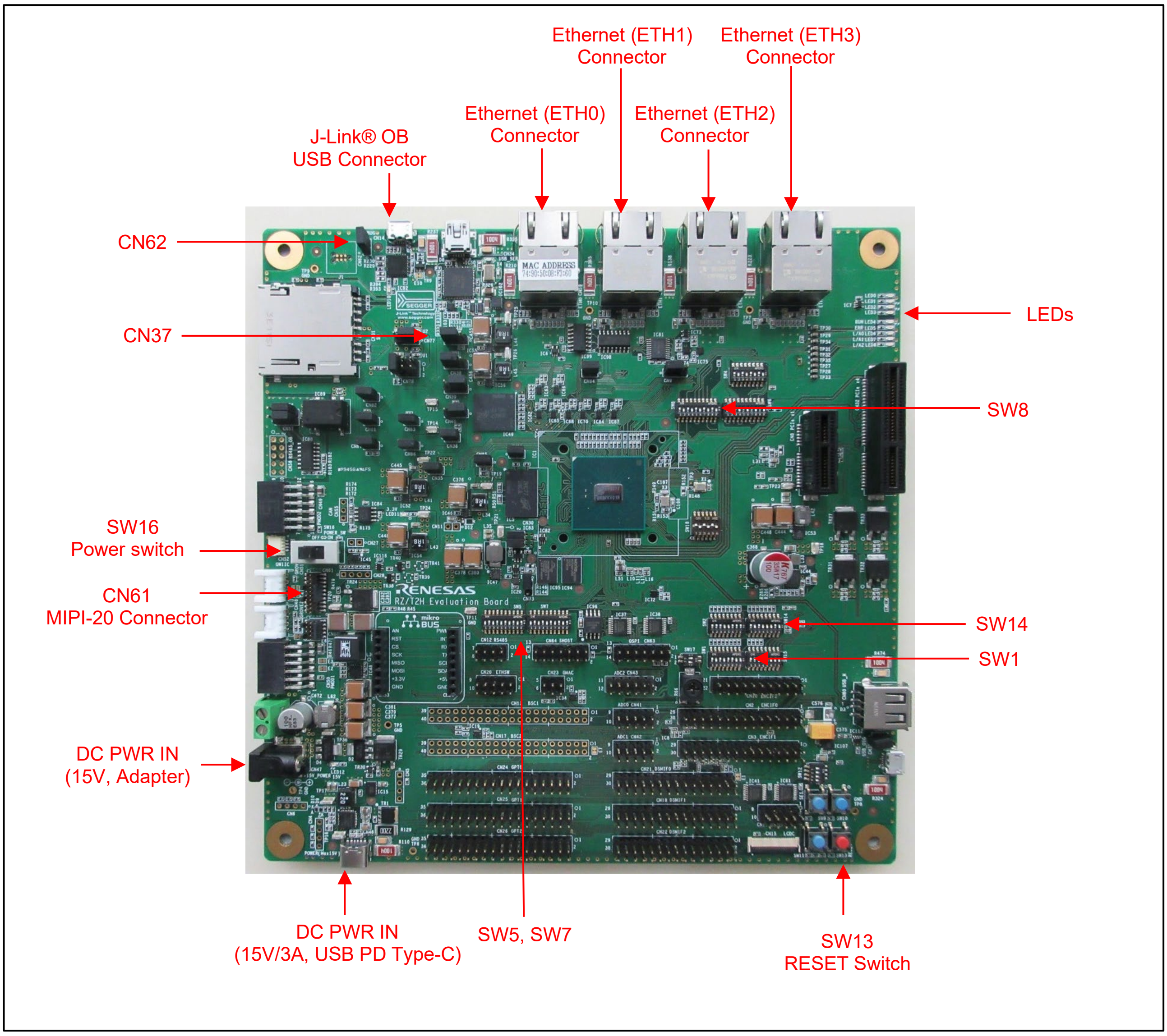


Figure 4-3 RZ/T2H Evaluation Board Layout

Table 4-5 Jumper pin settings

Reference	Jumper Position	Description
CN37	2-3 are short-circuit	3.3-V power is supplied to VCC1833_0. (For Ethernet Port 0)
CN62	Open-circuit	The on-board debugging function J-Link® OB is enabled.

Table 4-6 SW1 settings

SW1	Setting	Description
SW1-1	ON	XTALSEL = 'L' Select oscillator for RZ/T2H clock input.
SW1-3	ON	P35_3,4,5,6 are connected to SW12 and used as user DIPSW inputs.
SW1-6	ON	P01_0,1,2,4,5,6,7 and P02_0,1,2,3 are used as XSPI1 signals.

Table 4-7 SW2 settings

SW1	Setting	Description
SW2-3	OFF	LED3 Enable

Table 4-8 SW5 settings

SW5	Setting	Description
SW5-1	ON	P22_7 is used as ESC_LINKACT0.
SW5-2	OFF	
SW5-3	ON	P23_0 is used as ESC_LINKACT1.
SW5-4	OFF	
SW5-9	ON	P32_2 is used as USER_LED1 control.
SW5-10	OFF	

Table 4-9 SW7 settings

SW7	Setting	Description
SW7-5	ON	P23_5 is used as ESC_LINKACT2.
SW7-6	OFF	

Table 4-10 SW8 settings

SW8	Setting	Description
SW8-1	ON	P18_1 is used as ESC_LED_ERR.
SW8-2	OFF	
SW8-3	ON	P18_0 is used as ESC_LED_RUN.
SW8-4	OFF	
SW8-9	ON	P23_1 is used as USER_LED0.
SW8-10	OFF	

Table 4-11 SW14 settings

SW14	Setting	Description
SW14-1	ON	MD = 2 (L, H, L) xSPI1 boot mode (x1 boot serial flash)
SW14-2	OFF	
SW14-3	ON	
SW14-4	ON	CPU0 ATCM wait cycle = 0 wait cycle
SW14-5	OFF	CPU1 ATCM wait cycle = 1 wait cycle
SW14-6	OFF	Supply voltage of boot peripheral is 3.3 V
SW14-7	ON	JTAG mode = Normal mode

Other SW settings refer to r20ut5316egxxxx-rzt2hevb.pdf.

4.3.2 Setup Board

Setting the board for running sample program is shown below.

1. Connect an emulator to the connector .
   - When you use I-jet or J-Link emulator, connect it to CN61.
   - When you use J-Link On-Board emulator, connect USB micro-B to CN14. (Please disconnect CN62 for powering up J-Link OB.)

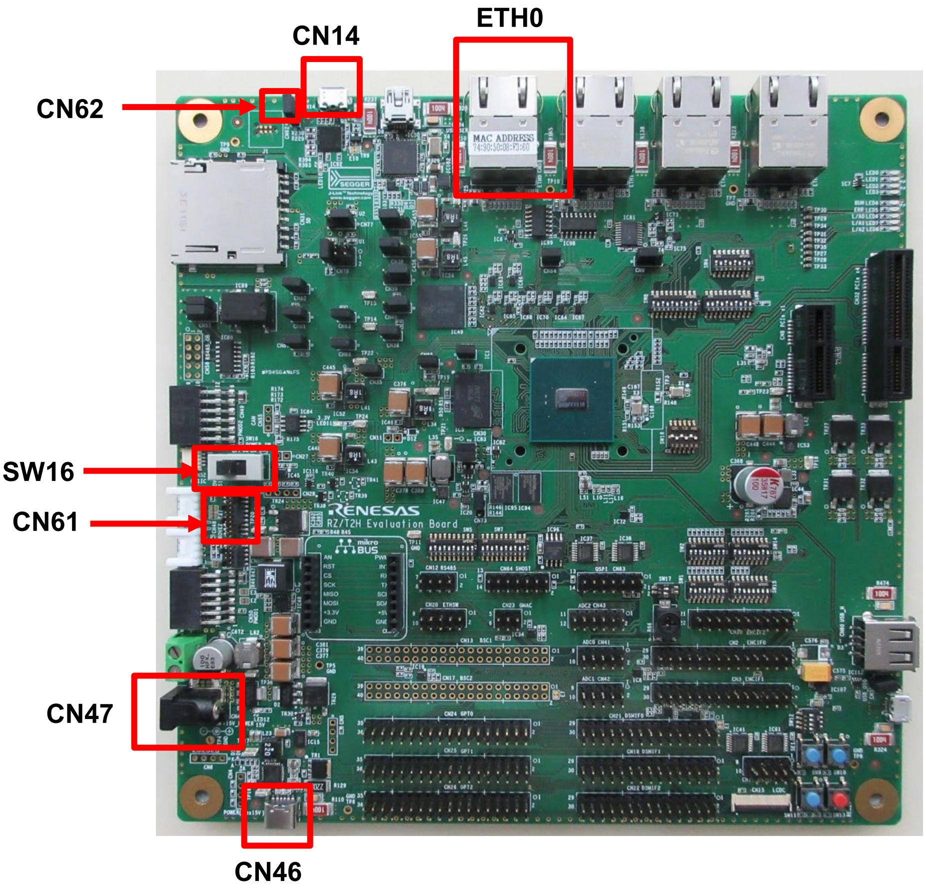


Figure 4-4 Setting RZ/T2H Evaluation Board

2. Power is supplied using a 15V AC/DC adapter or a USB cable (15V/3A USB PD Type-C).
   - When using a 15V AC/DC adapter, connect it to the connector CN47.
   - When using a USB cable (15V/3A USB PD Type-C), connect it to the USB connector CN46.
3. Connect Ethernet Cable to the Ethernet Connector ETH0.
4. Slide the SW16 POWER_SW to ON. (Provide power to the board)

4.4 RZ/N2L RSK board

4.4.1 Jumper and Switch configuration

This document describes the major hardware properties. Refer to Renesas Stater Kit+ for RZ/N2L user's manual and schematic for more board details.

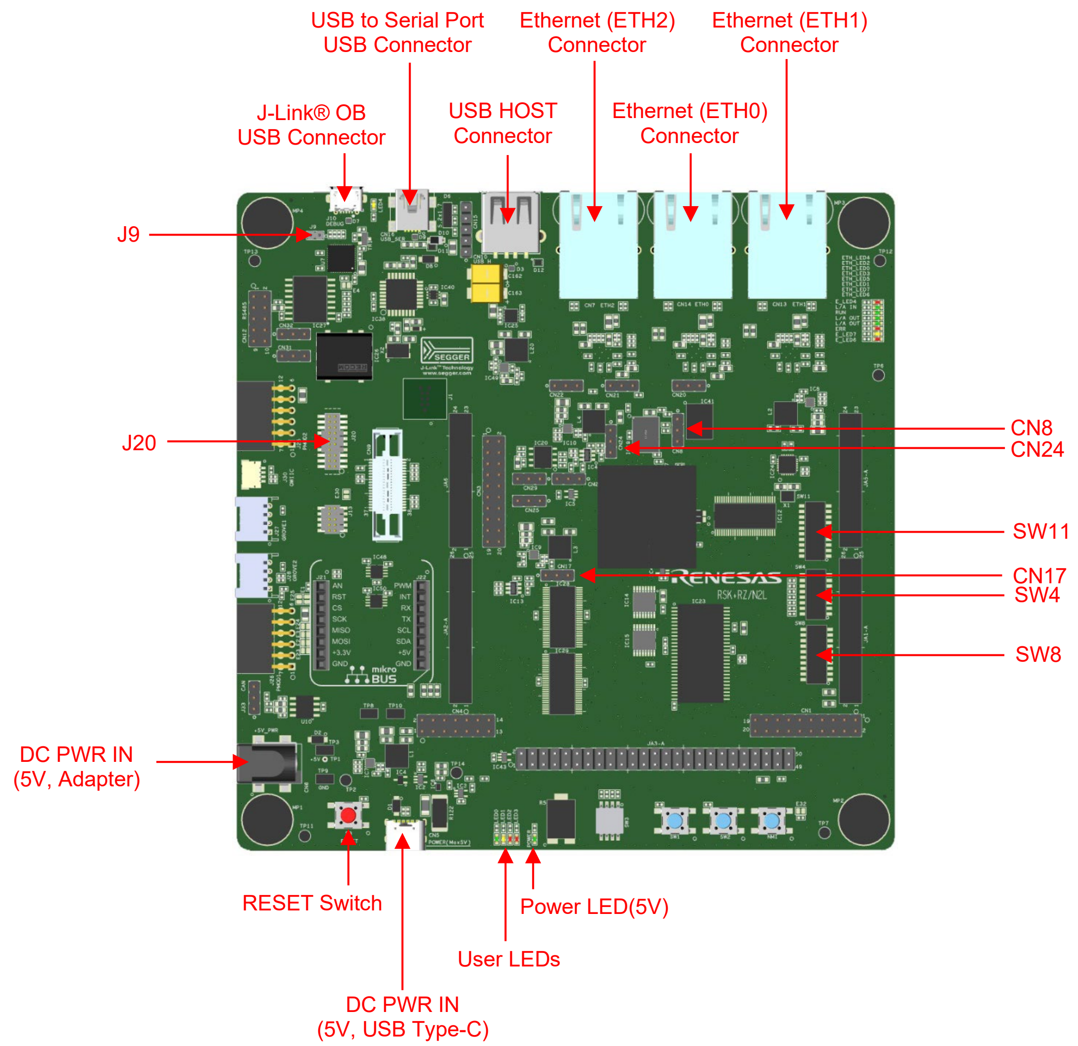


Figure 4-5 RZ/N2L RSK board layout

Table 4-12 Jumper pin settings

Reference	Jumper Position	Description
CN8	Shorted Pin 2-3	Enable QSPI (IC21).
CN17	Shorted Pin 2-3	Connect 1.8V Power rail to VCC1833_2. (When using Ethernet)
CN24	Shorted Pin 2-3	Connect 1.8V Power rail to VCC1833_3. (When using QSPI Flash IC21)
J9	Open	Enable the J-Link® OB.

Table 4-13 SW4 Settings

SW4	Setting	Description
SW4-1	ON	16-bit bus boot mode (NOR flash) Sets this SW settings when the program is executed by RAM.
SW4-2	OFF	
SW4-3	ON	
SW4-1	ON	xSPI0 boot mode (x1 boot serial flash) Sets this SW settings when the program is executed by flash writing.
SW4-2	ON	
SW4-3	ON	
SW4-4	ON	JTAG Authentication by Hash is disabled.
SW4-6	OFF	Enables signals other than the trace signal. (Motor, RS485, etc.)
SW4-7	ON	Enables signals other than the external bus. (CAN, Emulator, I2C, etc.)
SW4-8	OFF	Enable SW3.

Table 4-14 SW8 Settings

SW5	Setting	Description
SW8-1	OFF	Enable the "LED_GREEN" signal.
SW8-2	ON	
SW8-3	OFF	
SW8-4	ON	Enable the "LED5" signal.
SW8-5	OFF	

Table 4-15 SW11 Settings

SW6	Setting	Description
SW11-1	ON	Enable the " LED_RED2" signal.
SW11-2	OFF	
SW11-3	OFF	

Other SW settings refer to r20ut4984egxxxx-rskplus-rzn2l-v1-um.pdf.

4.4.2 Setup Board

Setting the board for running sample program is shown below.

1. Connect an emulator to the connector .
    - When you use I-jet or J-Link emulator, connect it to J20 on RZ/N2L RSK board.
    - When you use J-Link On-Board emulator, connect USB micro-B to J10 on RZ/N2L RSK board. (Please disconnect J9 for powering up J-Link OB.)

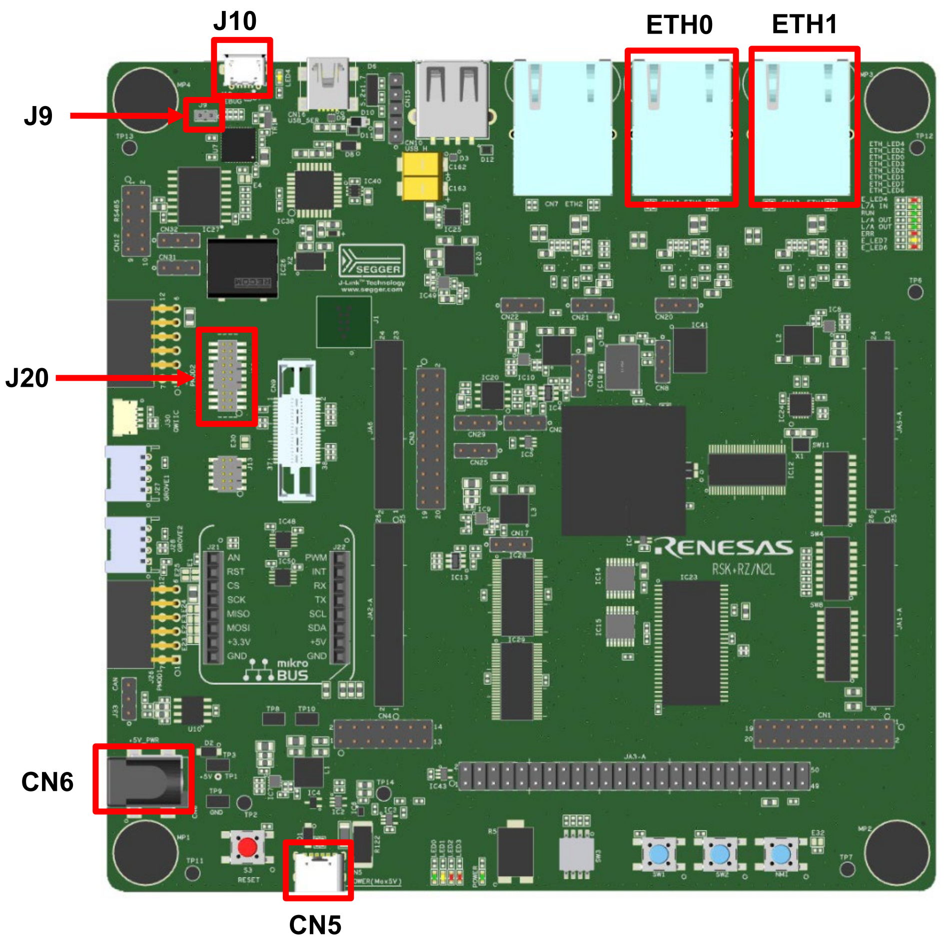


Figure 4-6 Setup RZ/N2L RSK board

2. Power is supplied using USB cable (Type-C) or AC / DC adapter.
   - When using USB cable (Type-C), connect it to the USB connector "CN5" on the RZ/N2L RSK board.
   - When using AC/DC adapter, connect it to the connector "CN6" on the RZ/N2L RSK board.
3. Connect Ethernet Cable to the Ethernet Connector "ETH0" or "ETH1".

4.5 RZ/N2H Evaluation board

4.5.1 Jumper and Switch configuration

This document describes the major hardware properties. Refer to the Evaluation Board for RZ/N2H user's manual and schematic for more board details.

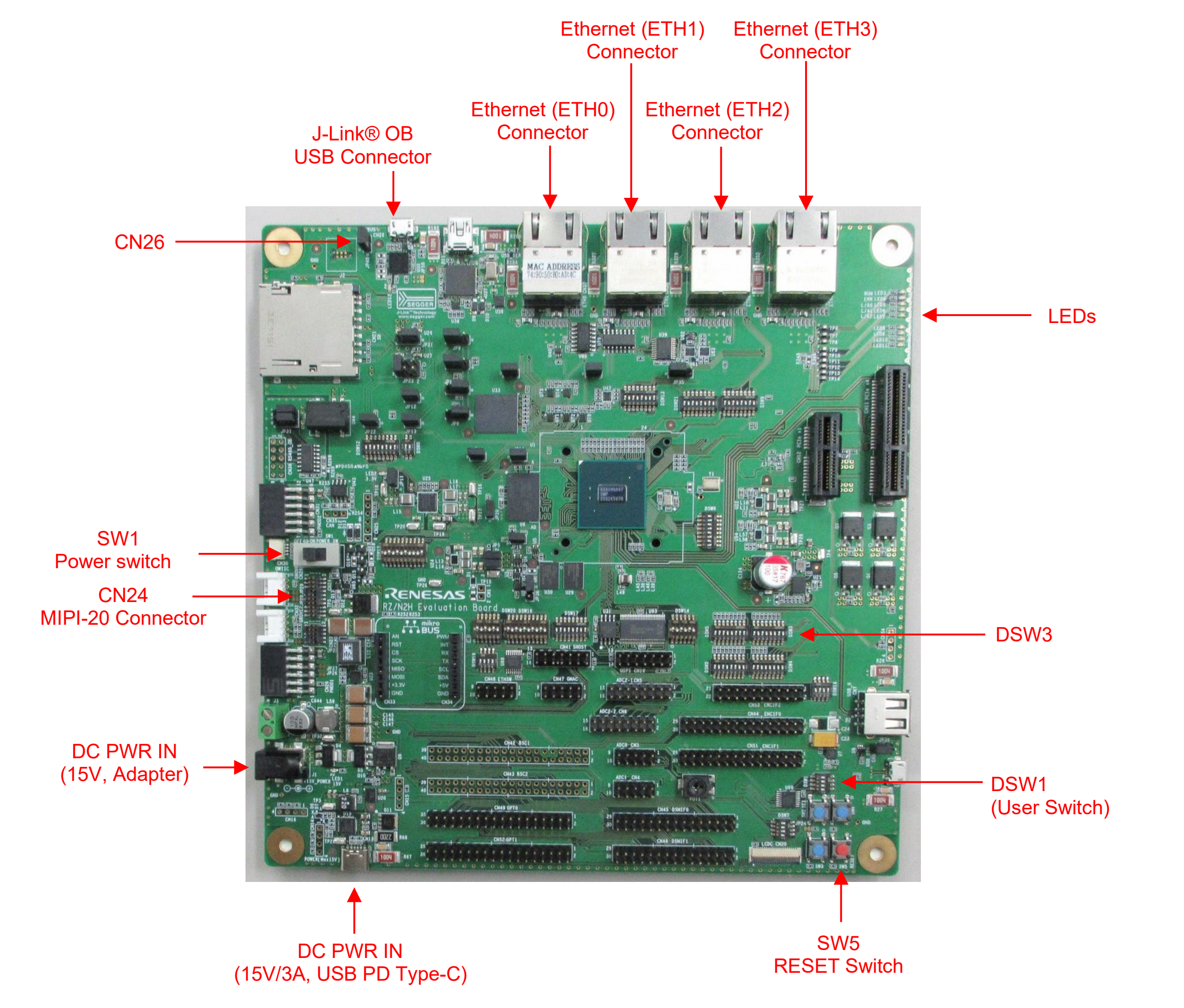


Figure 4-7 RZ/N2H Evaluation Board Layout

Table 4-16 Jumper pin settings

Reference	Jumper Position	Description
CN26	Open-circuit	The on-board debugging function J-Link® OB is enabled.

Table 4-17 DSW3 settings

DSW3	Setting	Description
DSW3-1	ON	xSPI1 boot mode (x1 boot serial flash)
DSW3-2	OFF	
DSW3-3	ON	
DSW3-4	ON	CPU0 ATCM 0 wait
DSW3-5	OFF	CPU1 ATCM 1 wait
DSW3-6	OFF	The power-supply voltage of the boot peripheral is 3.3 V.
DSW3-7	ON	JTAG mode = Normal mode
DSW3-8	Unused	-

Other SW settings refer to r20ut5522egxxxx-rzn2hevb.pdf.

4.5.2 Setup Board

Setting the board for running sample program is shown below.

1. Connect an emulator to the connector.
   - When you use I-jet or J-Link emulator, connect it to CN24.
   - When you use J-Link On-Board emulator, connect USB micro-B to CN26. (Please disconnect JP40 for powering up J-Link OB.)

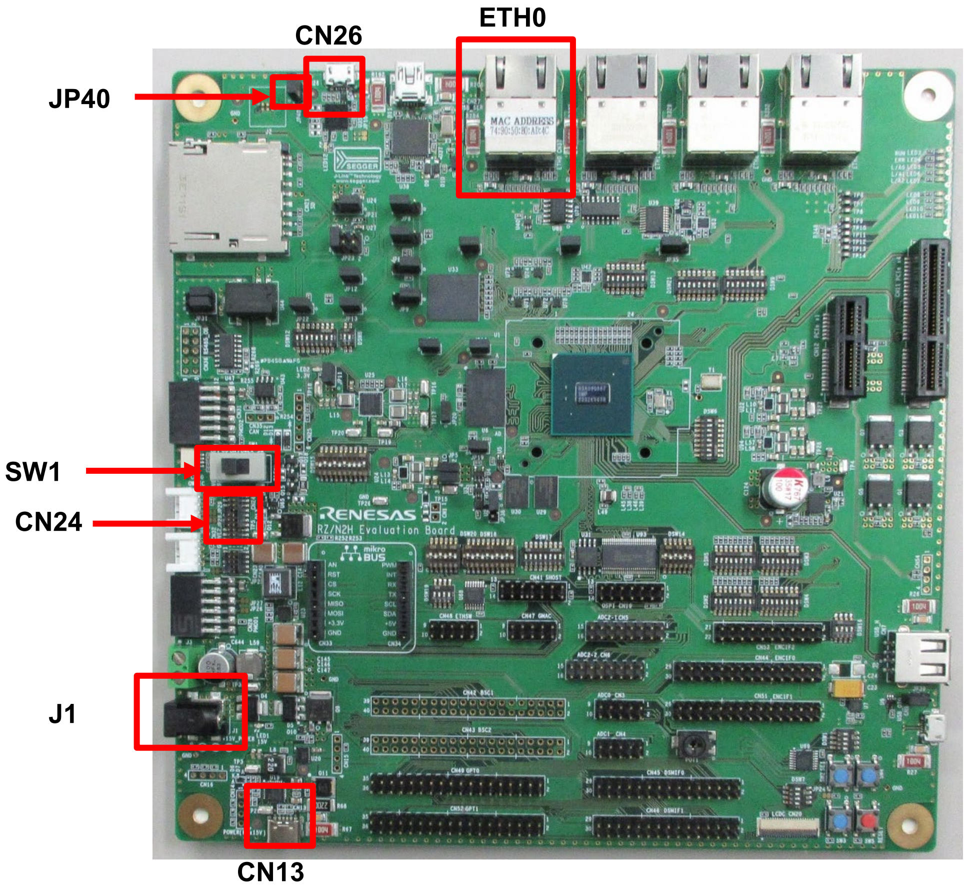


Figure 4-8 Setting RZ/N2H Evaluation Board

2. Power is supplied using a 15V AC/DC adapter or a USB cable (15V/3A USB PD Type-C).
    - When using a 15V AC/DC adapter, connect it to the connector J1.
    - When using a USB cable (15V/3A USB PD Type-C), connect it to the USB connector CN13.
3. Connect Ethernet Cable to the Ethernet Connector ETH0.
4. Slide the SW1 POWER_SW to ON. (Provide power to the board)

5. Setup the Host Device

5.1 Configuration the Host IP Address

Set an IP address that can communicate with the device in the Ethernet adapter settings on the PC side.

This sample software sets the IP address 192.168.1.170 (or 192.168.1.10) and subnet mask 255.255.255.0 for the device by default. Therefore, for example, set as follows on the PC side.

- Example setting on the PC side.
    - ➢ IP address: 192.168.1.100
    - ➢ Subnet mask: 255.255.255.0

5.2 Setup the CODESYS Software

This chapter describes the setup of the CODESYS software.

5.2.1 How to get CODESYS

CODESYS Development system is available from the following web sites.

- CODESYS Store
    - ➢ To create your account and login is required to download the CODESYS tools.
        - ✧ When you create an account as a business customer, you need:
            - VAT Number if you are European VAT registered Customers.
            - Certificate of Registration as Taxpayer (entrepreneur) if you are non-EU customers.
    - ➢ Click the “All versions" tab to get the specified version.

- CODESYS Store North America
    - ➢ To create your account and login is required to download the CODESYS tools.
        - ✧ United States, Canada and Mexico only can be registered in the "Country" form of the "Address Information".
    - ➢ Click the “Versions" tab to get the specified version.

- LINX (Distributer in Japan)
    - ➢ To create your account and login is required to download the CODESYS tools.
        - ✧ Only Japanese companies can create the account.

5.2.2 Startup CODESYS Tools

After installing the CODESYS, please launch the CODESYS tools shown below.

Table 5-1 CODESYS tools

Name	Description	Note
CODESYS V3	IDE	-
CODESYS Gateway V3	Software Gateway	This may be already started by Windows Start Up Process.
CODESYS Control Win V3	Software PLC	This may be already started by Windows Start Up Process.

If the CODESYS is launched properly, the following window is shown.

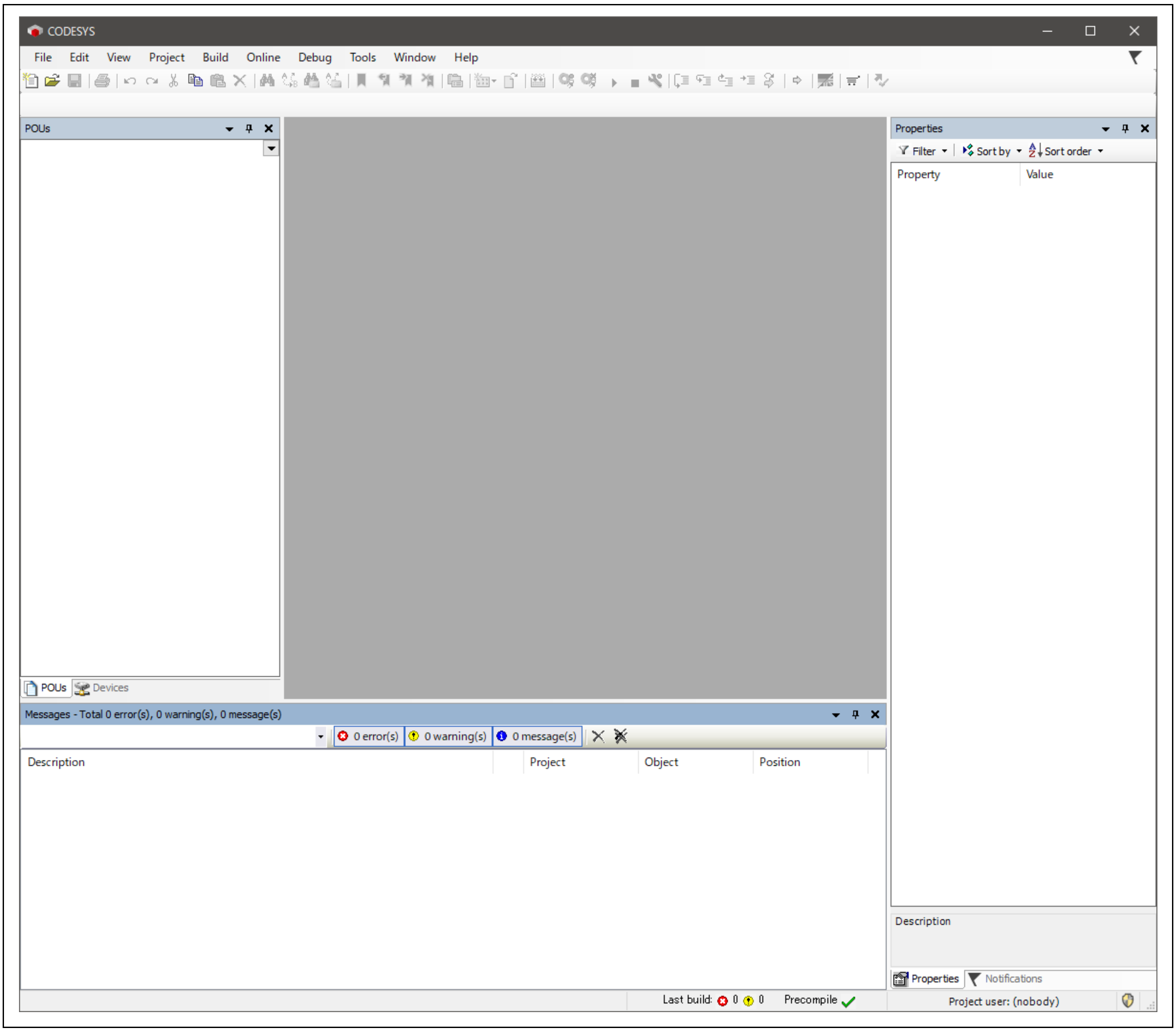


Figure 5-1 CODESYS Initial Window

If the CODESYS Gateway and Control Win SysTray is launched properly, the following icons are shown in notification area of Windows Tool Bar. (The left icon is of the CODESYS Gateway, and the right one is of the CODESYS Control Win SysTray)

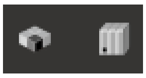


Figure 5-2 CODESYS Icons

Please click each icon and click “Start Gateway” and “Start PLC”.

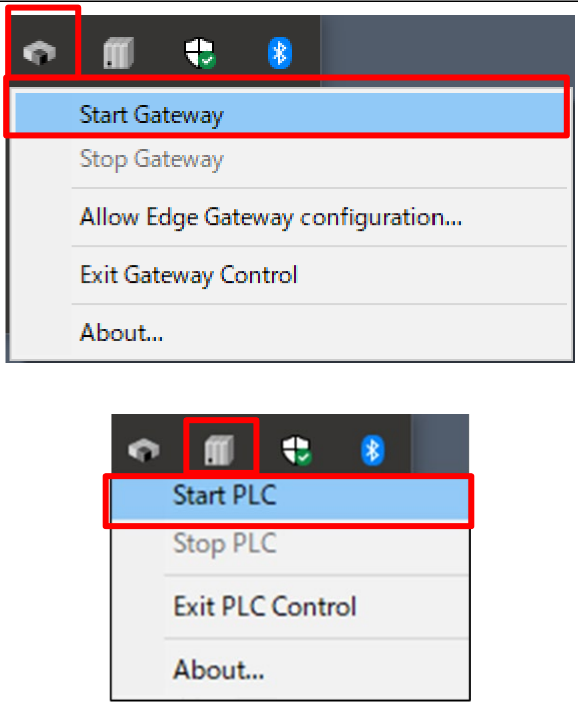


Figure 5-3 Start Gateway and PLC

If the Gateway and PLC are started properly, the icons pigment like the following image.

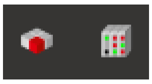


Figure 5-4 Icons successfully started

5.2.3 Install EDS File into CODESYS

In the CODESYS, please open "tools" > "Device Repository" in tool bar.

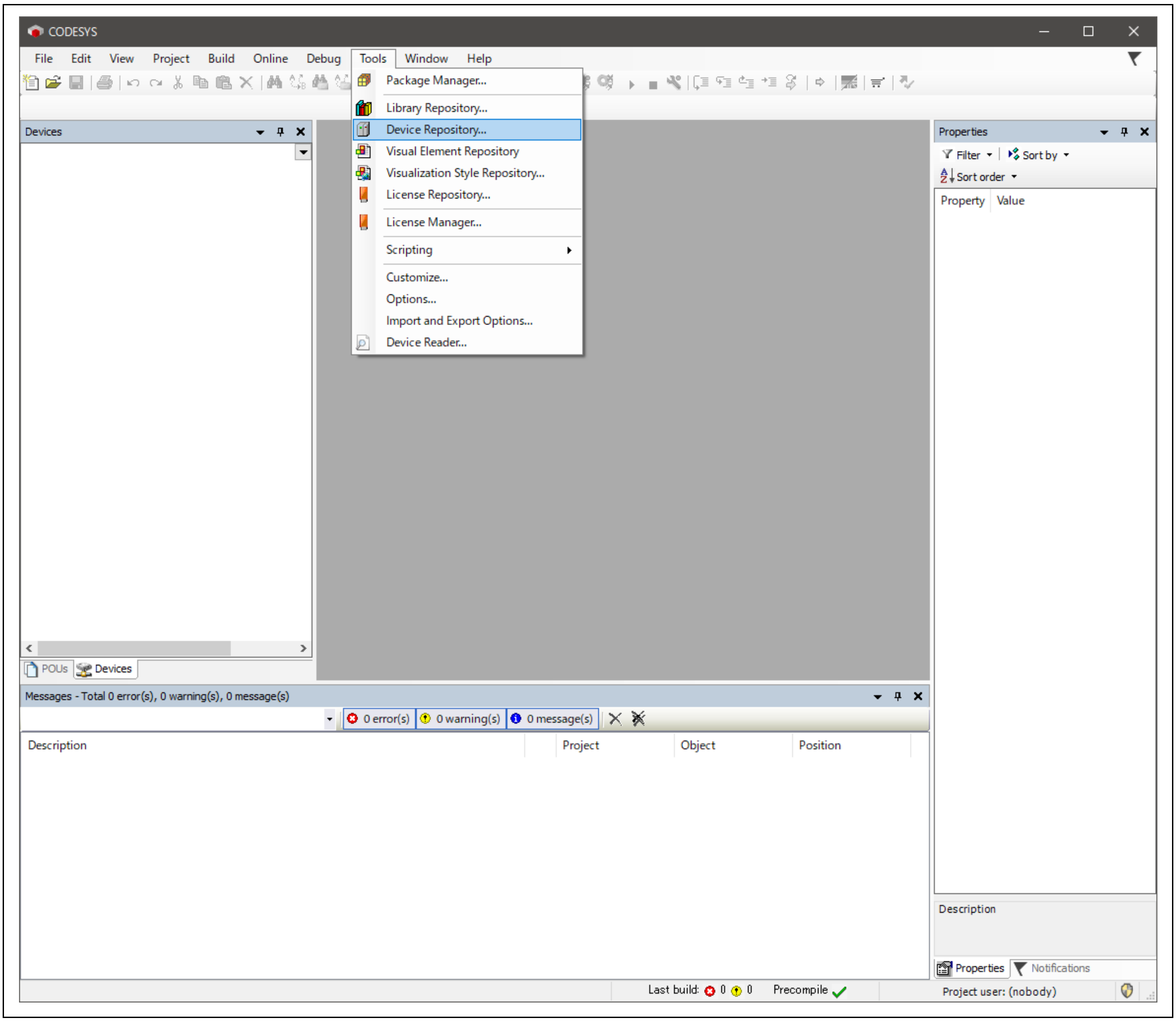


Figure 5-5 Device Repository in CODESYS

Please click the “Install” to open dialog to select EDS file, and select the EDS file “renesas_opener_sample_app.eds” in “scanner” directory.

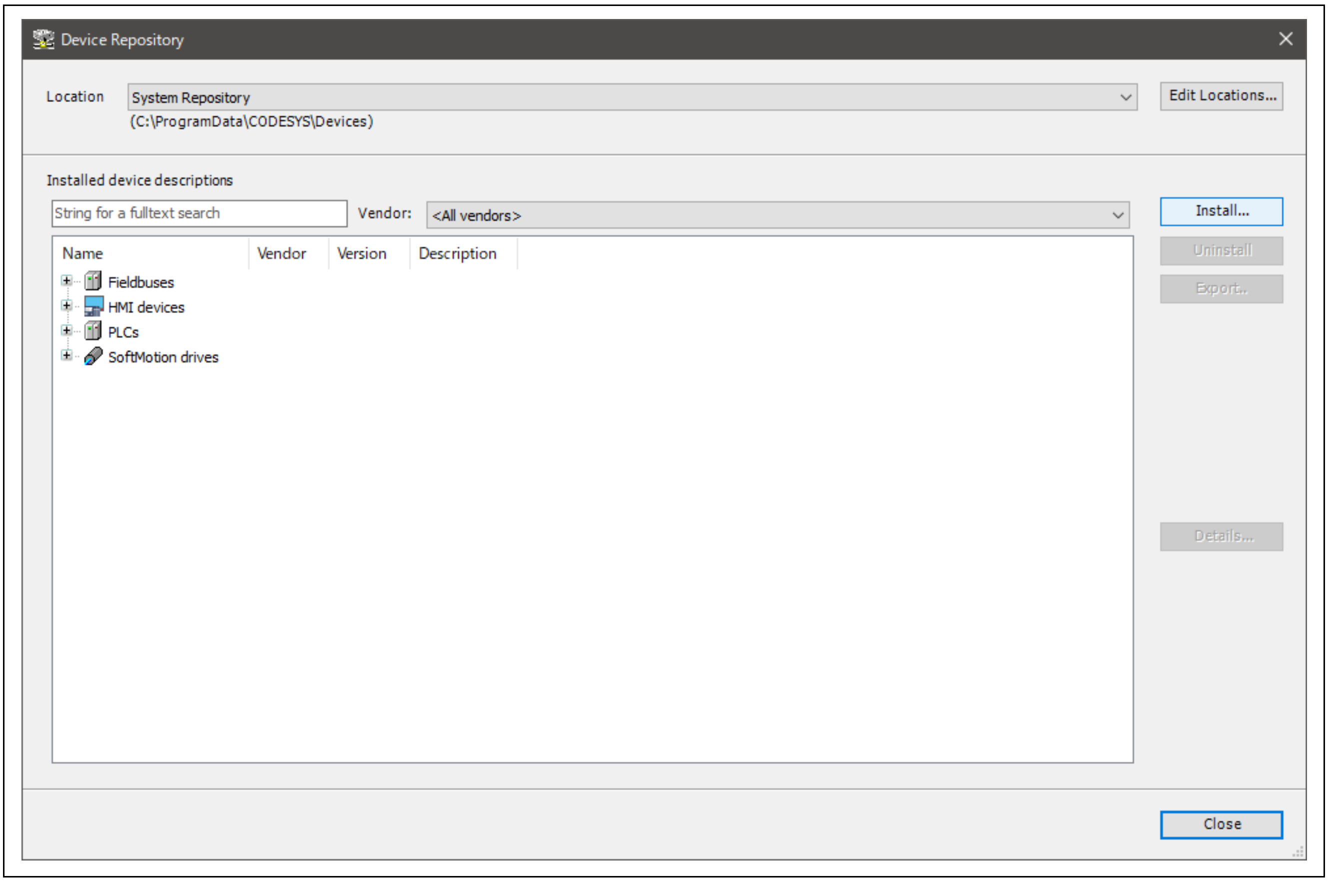


Figure 5-6 Click "Install" in Device Repository Window

If the Renesas OpENer Device is shown in the blue line as EtherNet/IP Remote Adapter, please click “close” to close this window.

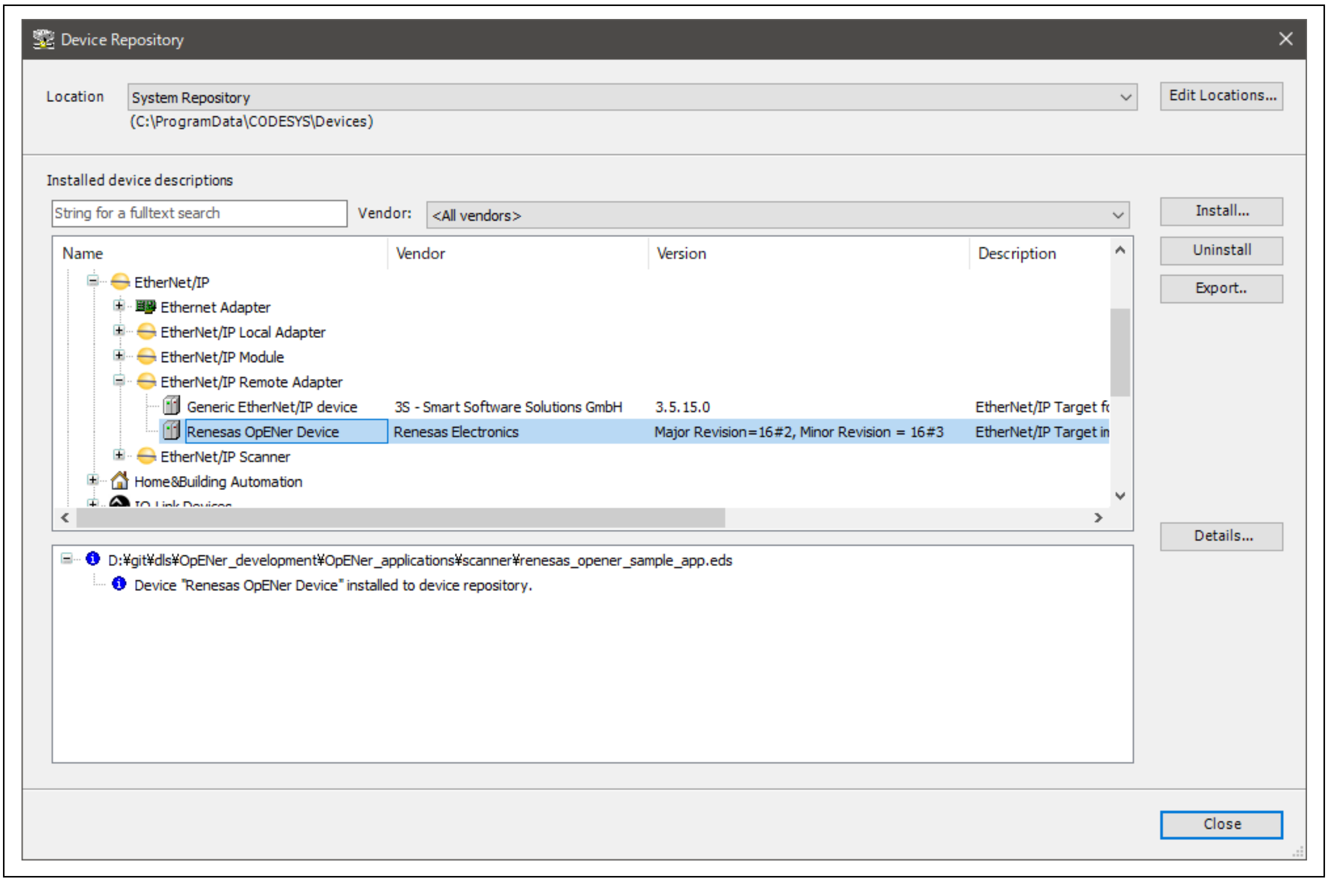


Figure 5-7 EtherNet/IP Remote Adapter

Preparation for using CODESYS is now complete. Continue with the program operation in Chapter 6 and 6.6 and then operate CODESYS again in Chapter 7.

6. Running the Sample Application

This chapter describes how to download and run the program.

Since the procedure varies by device, please refer to the appropriate section according to the device you are using.

- RZ/T2M (Dual-core project): Please refer to the section 6.2.
- RZ/T2ME (Dual-core project): Please refer to the section 6.3.
- RZ/T2H CA55 core (Single-core project): Please refer to the section 6.4.
- RZ/T2H CR52 core (Dual-core project): Please refer to the section 6.5.
- RZ/N2L (Single-core project): Please refer to the section 6.6.
- RZ/N2H CA55 core (Single-core project): Please refer to the section 6.7.
- RZ/N2H CR52 core (Dual-core project): Please refer to the section 6.8.

Before following this chapter, please look at Section 4 for board setup.

6.1 Board IP Address Setting

The IP address is set in src/renesas/application/opener_port_instance.c.

The following addresses are used in the example:

IP address	: 192.168.1.170 (or 192.168.1.10)
Subnet mask	: 255.255.255.0
Default gateway	: 192.168.1.2
Host Name	: OPENER_NETIF0

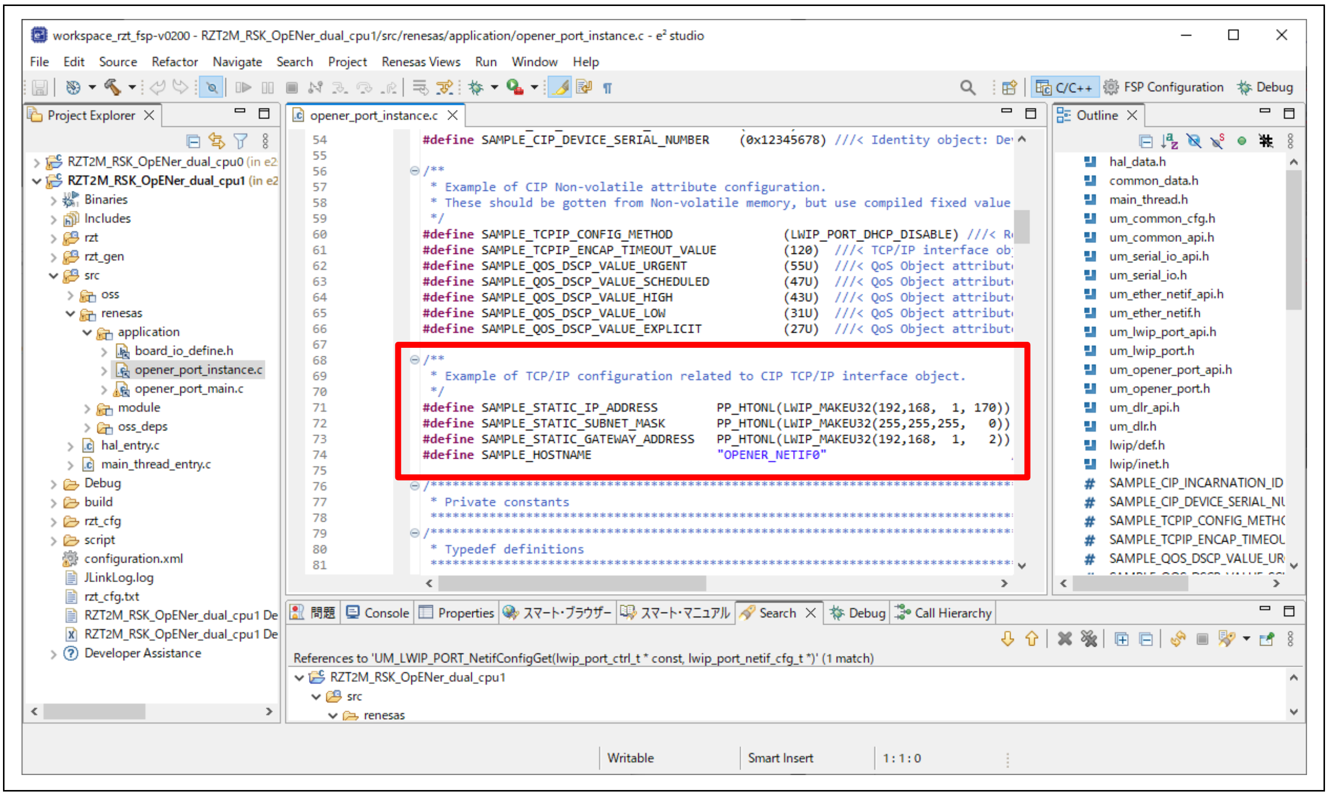


Figure 6-1 Static IP address

6.2 For RZ/T2M (Dual-core project)

The setup differs depending on the IDE.

- When using e² studio, refer to section 6.2.1.
- When using EWARM, refer to section 6.2.2.

6.2.1 Setup sample project for e² studio

The setup differs depending on the boot mode.

- When executing the sample program with RAM, refer to section 6.2.1.2.
- When executing the sample program with flash boot mode, refer to section 6.2.1.3.

6.2.1.1 Startup e² studio

1. Open the e² studio and select a directory as workspace.
2. Click “Import” in File tab.

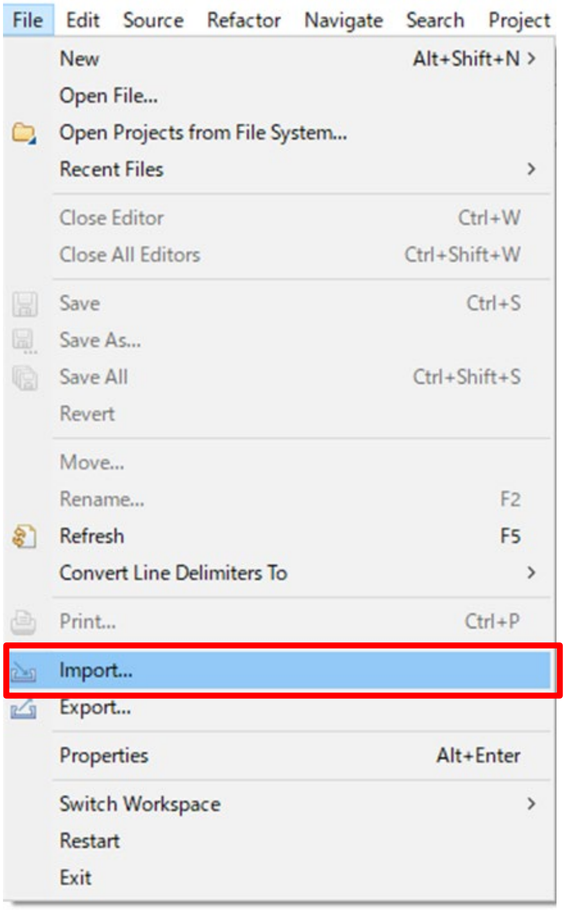


Figure 6-2 e² studio File tab

3. Click “Existing Projects into Workspace” in “General” and Click “Next”.

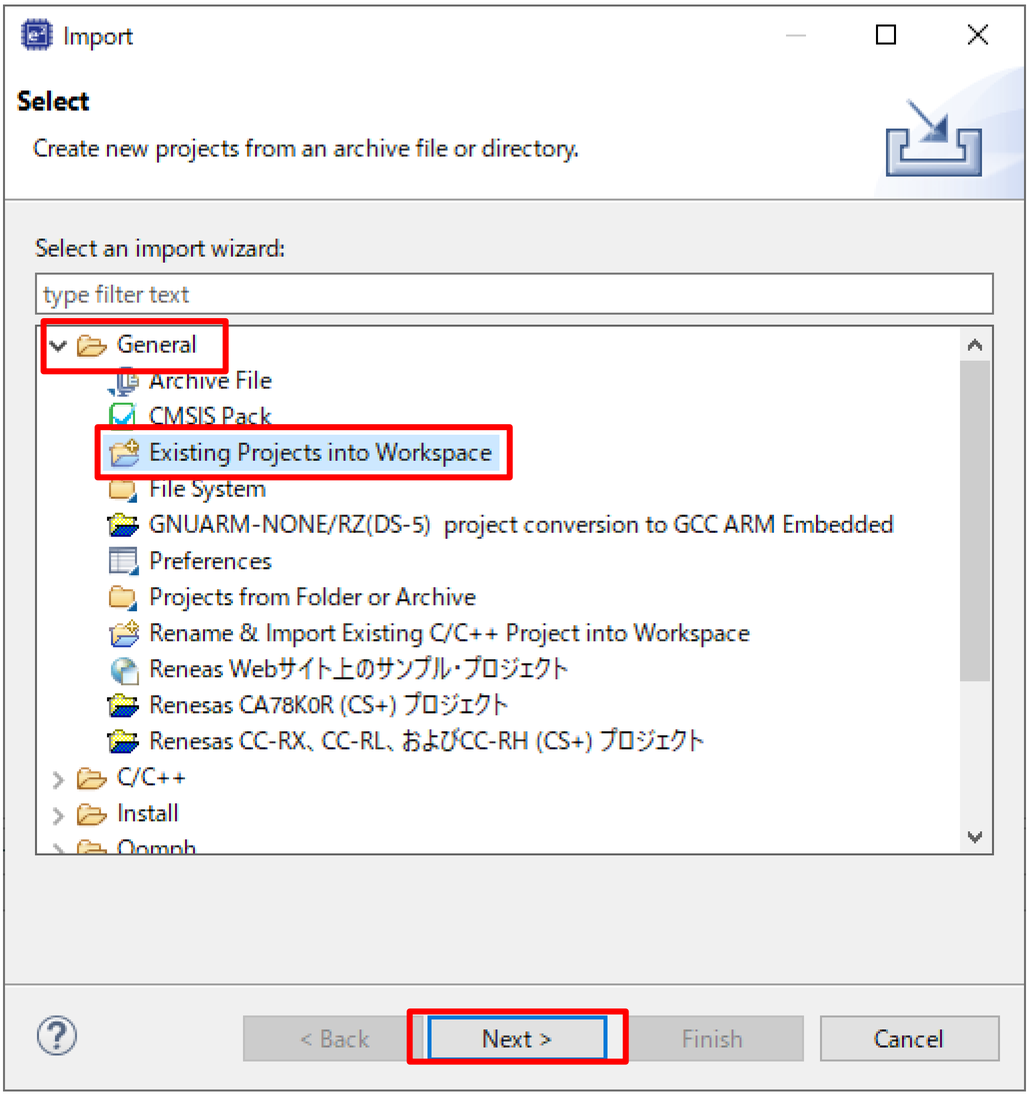


Figure 6-3 Select import type

4. Import project by the following steps.
   A) Set the root directory.
      "(Extracted zip folder path)\project\rzt2m_rsk"
   B) Select 2 projects.
   C) Click "Finish"

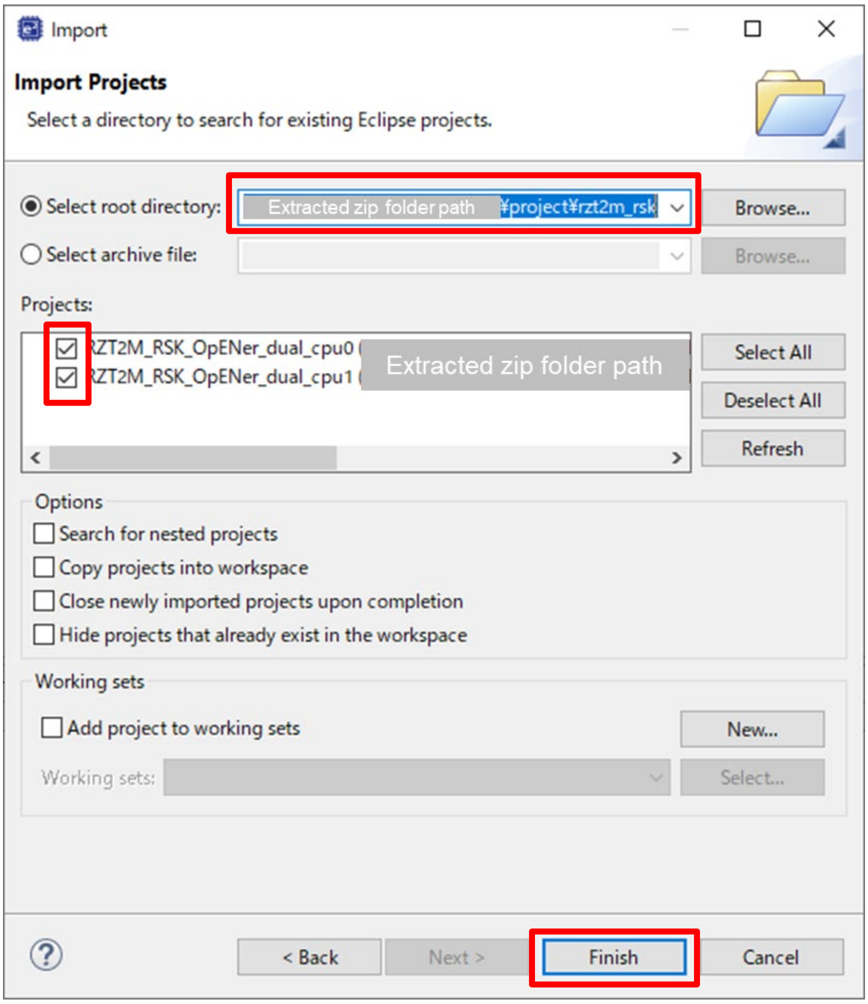


Figure 6-4 Import two projects on e² studio

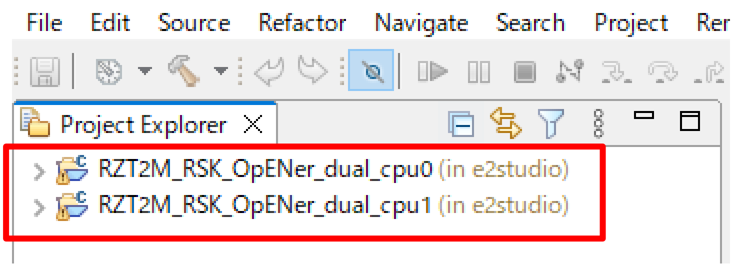


Figure 6-5 Open 2 projects

6.2.1.2 RAM execution

(1) How to generate source code and how to build

A) Click the configuration.xml of the CPU0 project.

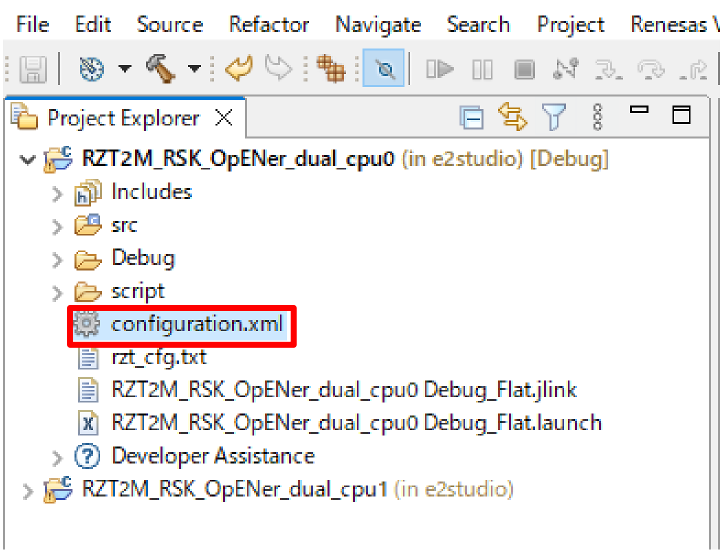


Figure 6-6 Configuration

B) Click “Generate Project Content” button then generate rzt, rzt_gen, rzt_cfg folder.

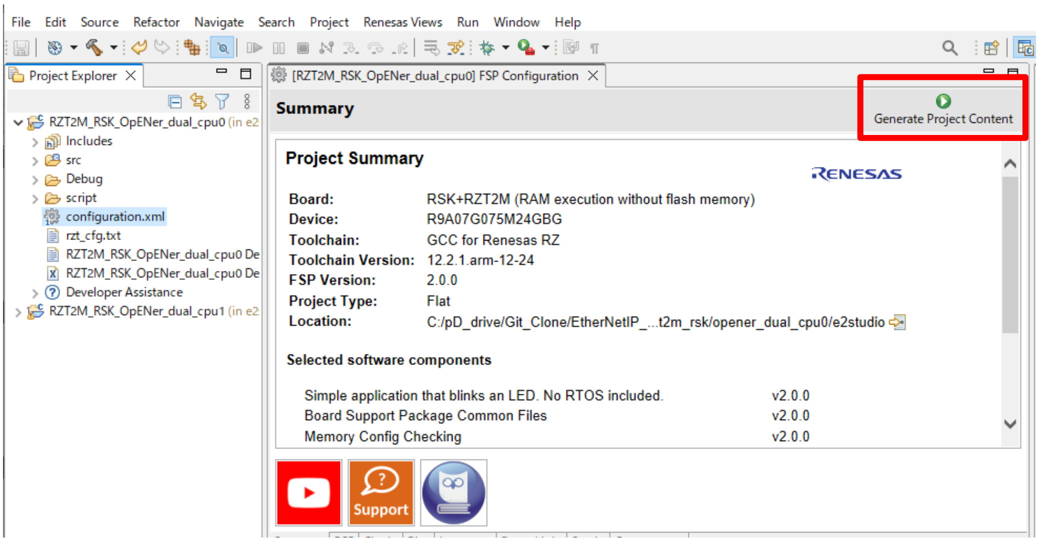


Figure 6-7 Generate Project Content

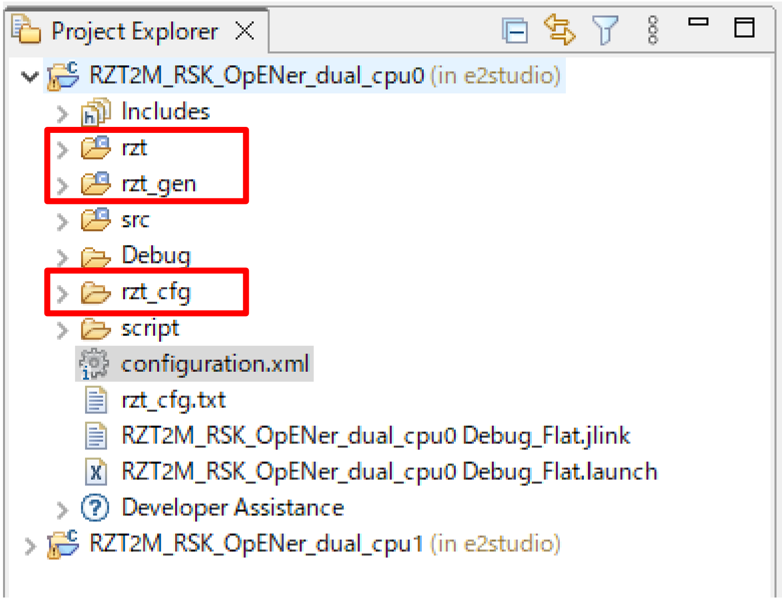


Figure 6-8 Generate project folder

C) Click the Build button in tool bar to build the project and confirm that there is no error message in the build message log.

Figure 6-9 Build button

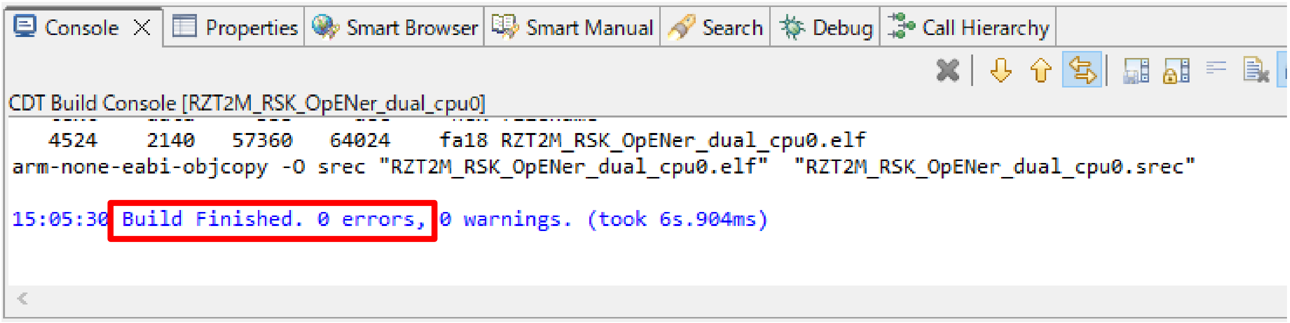


Figure 6-10 Build message

D) For the CPU1 project, generate the folders and build the project, same as the CPU0 project.

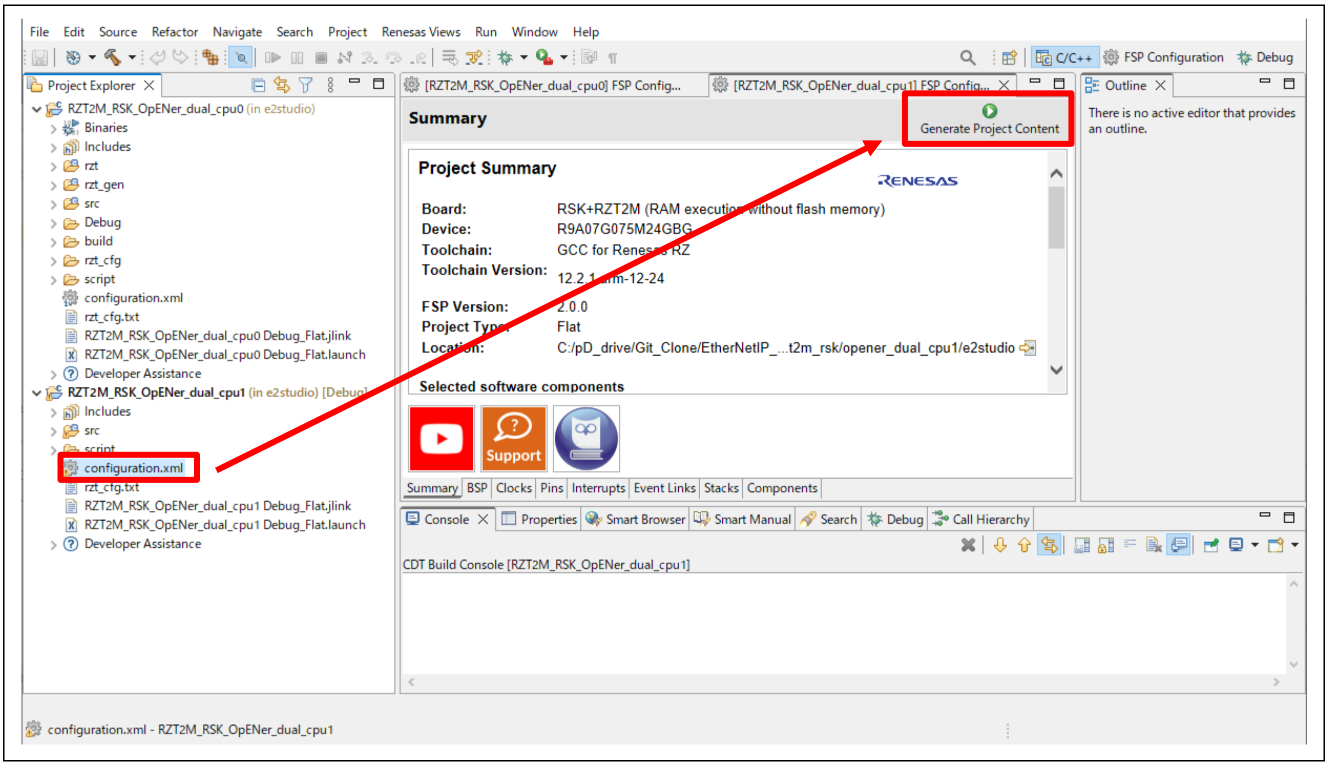


Figure 6-11 Generate folders for the CPU1 project

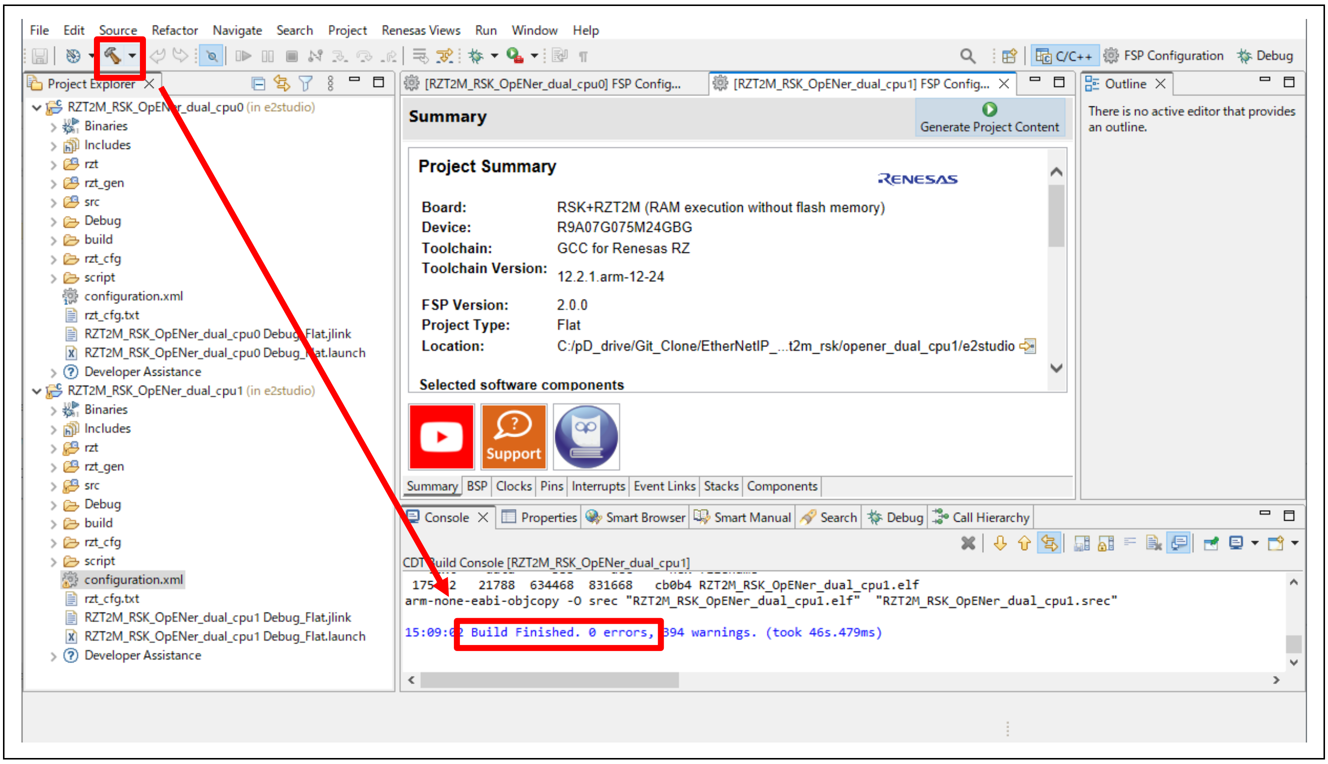


Figure 6-12 Build the CPU1 project

(2) Download application and run debugger

A) Select the CPU0 project and click the Debug button to download the CPU0 program.

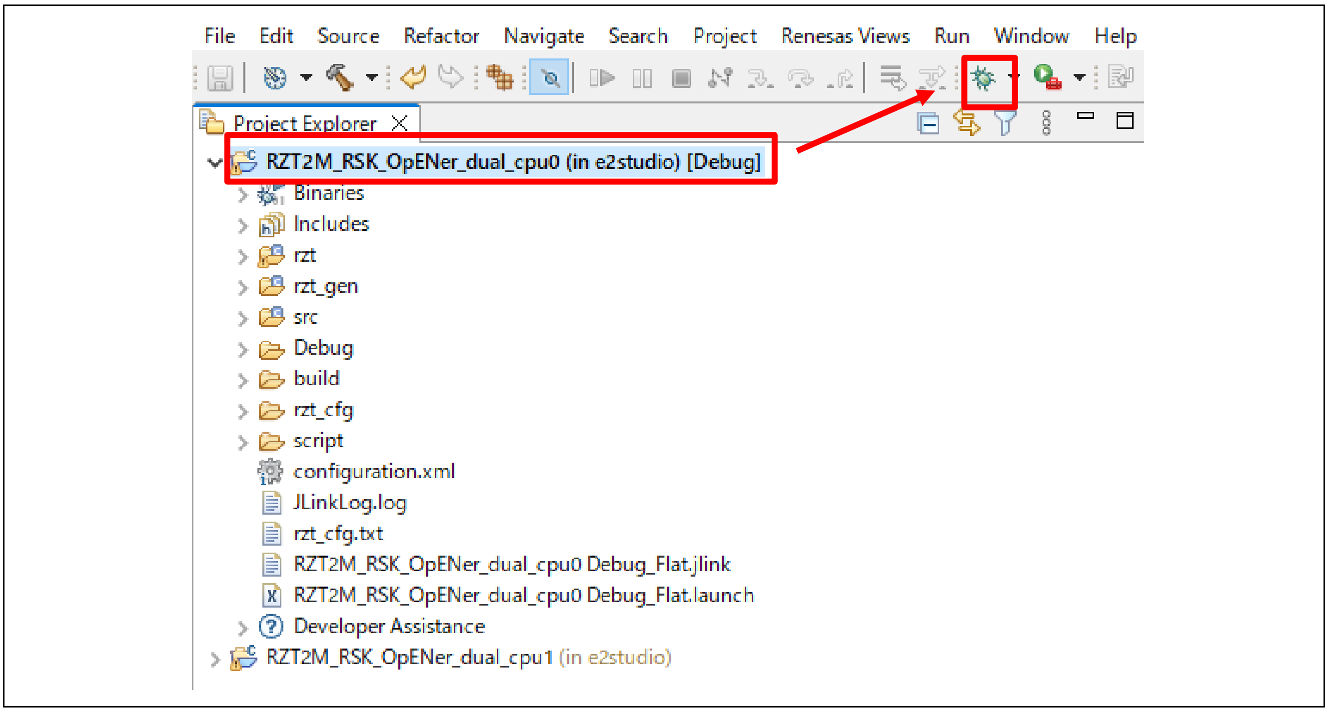


Figure 6-13 Debug button, the CPU0 project

B) Click the “No” button on the Perspective Switch dialog box.

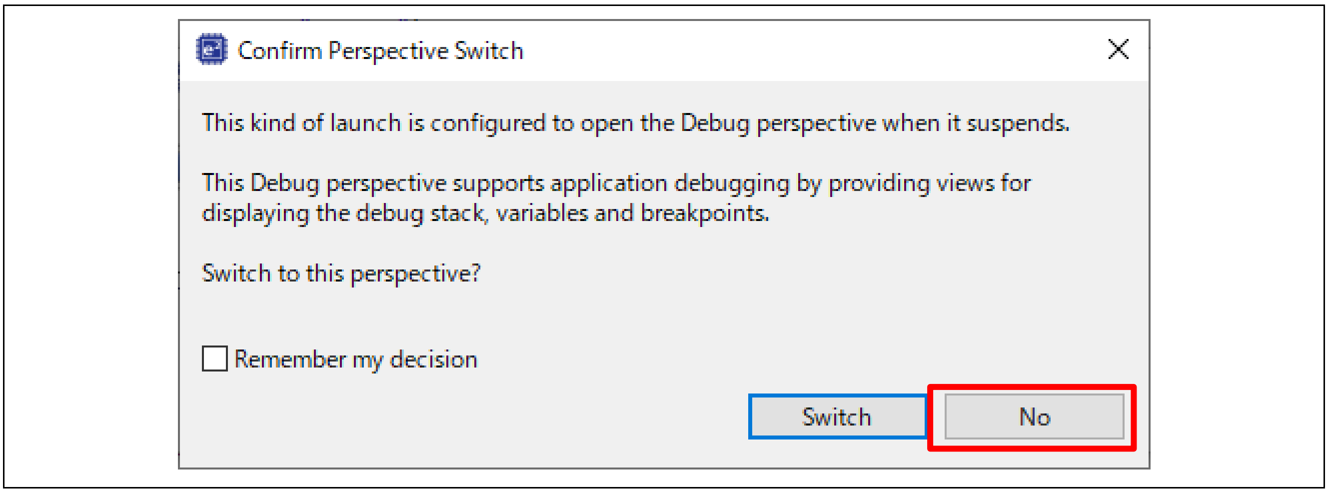


Figure 6-14 Confirm Perspective Switch

C) Select the CPU1 project and click the Debug button to download the CPU1 program.

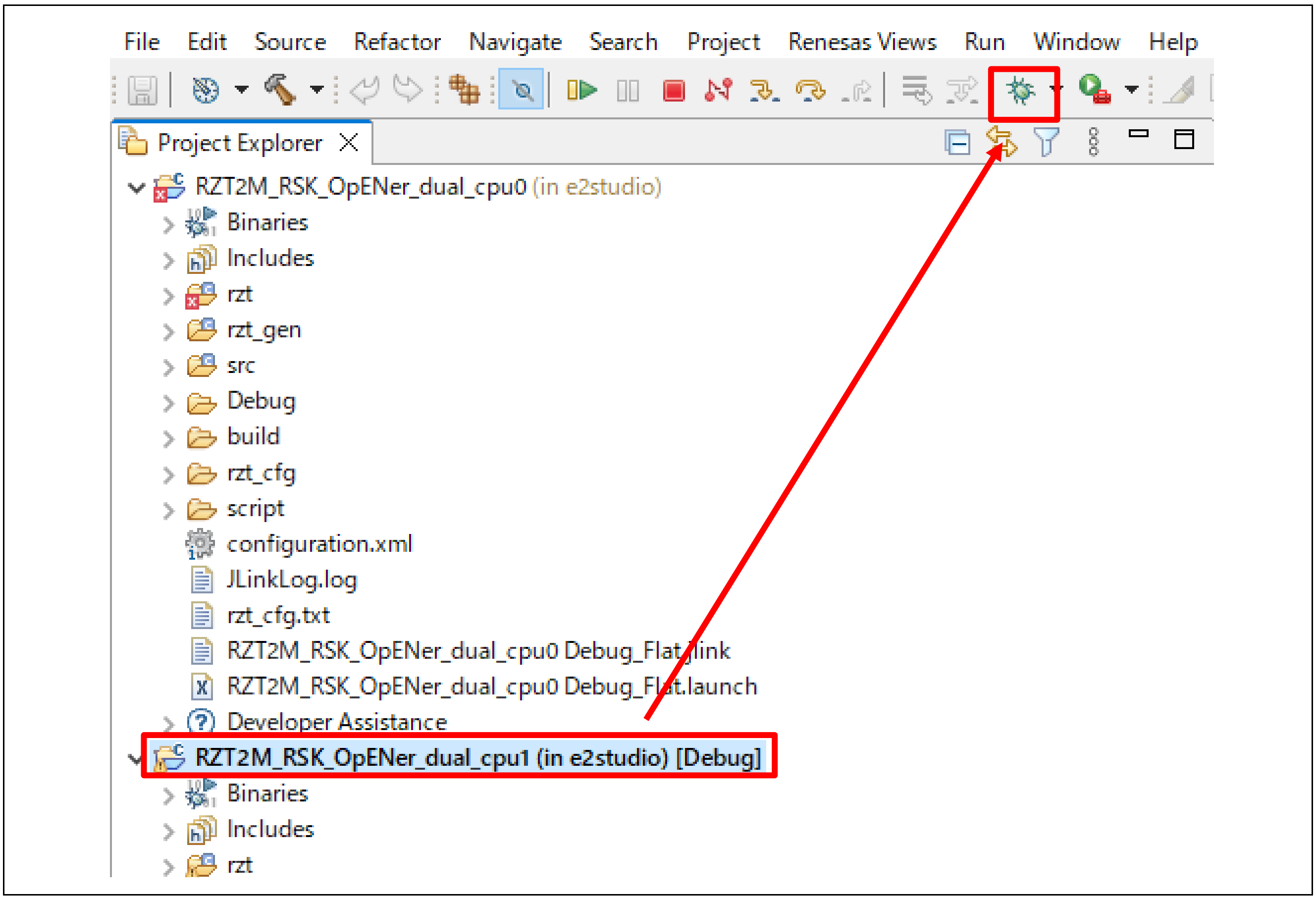


Figure 6-15 Debug button, the CPU1 project

D) Click the “No” button on the Launcher dialog box.

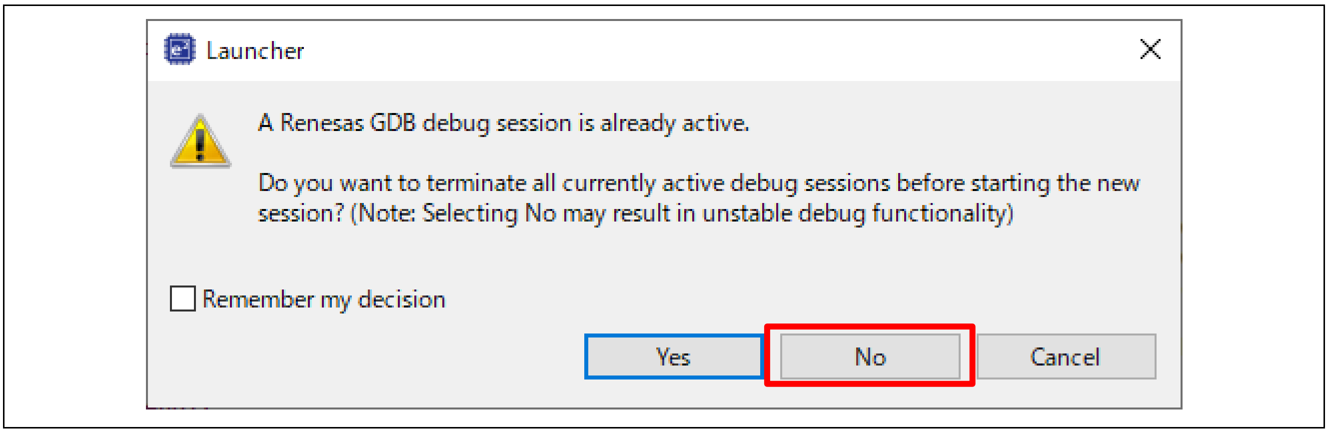


Figure 6-16 Launcher dialog box

E) Click to “Yes” button on the Proceed confirmation dialog box.

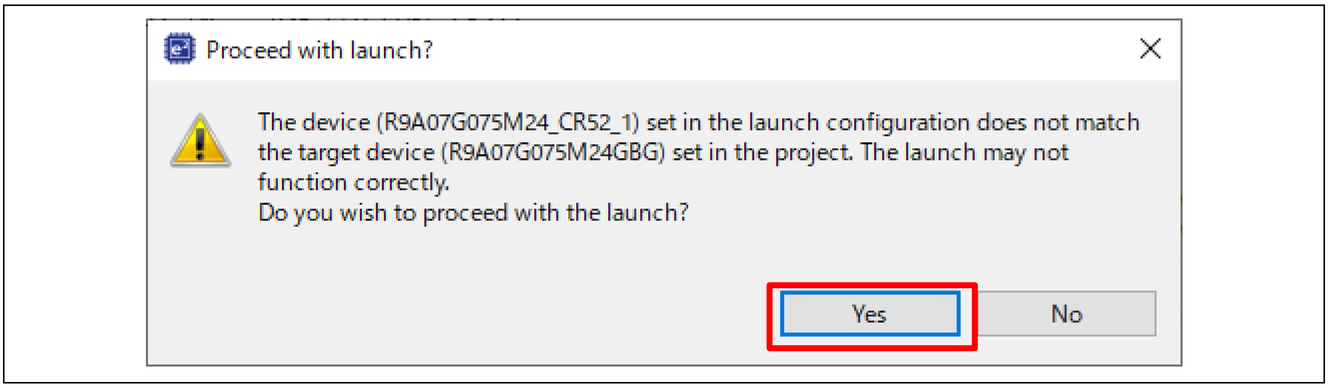


Figure 6-17 Proceed confirmation dialog box

F) Click to “Yes” button on the Perspective Switch dialog box.

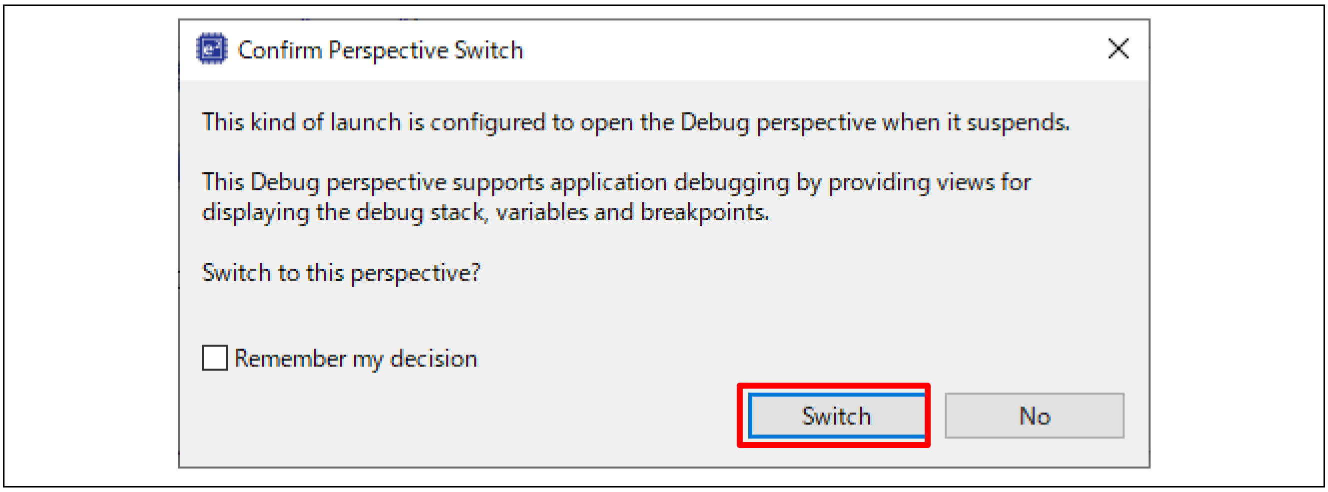


Figure 6-18 Confirm Perspective Switch

G) Run the CPU0 project.

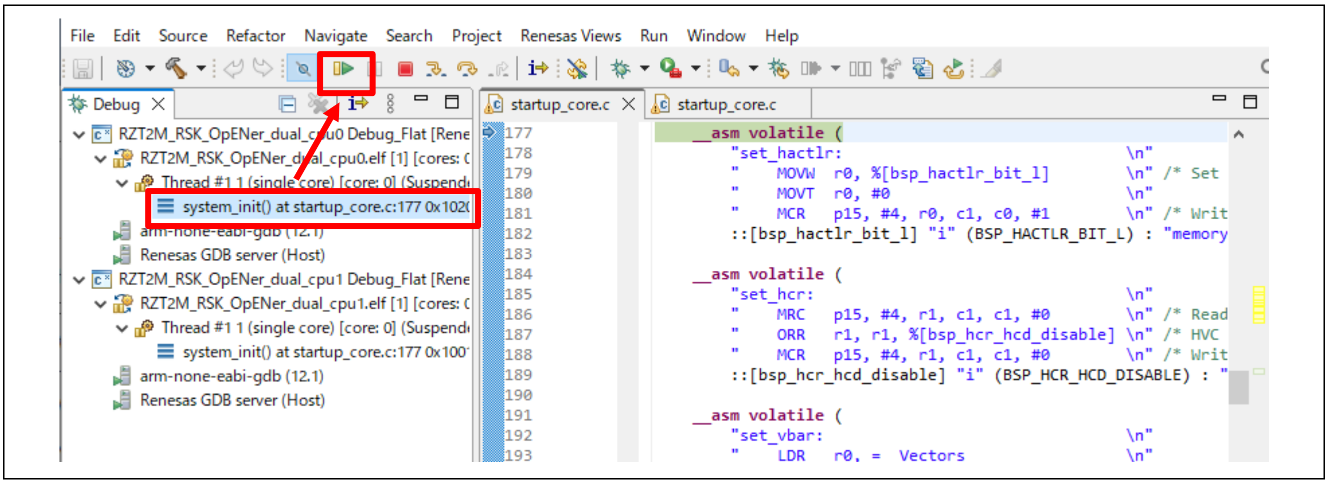


Figure 6-19 Run the CPU0 project

H) Run the CPU1 project.

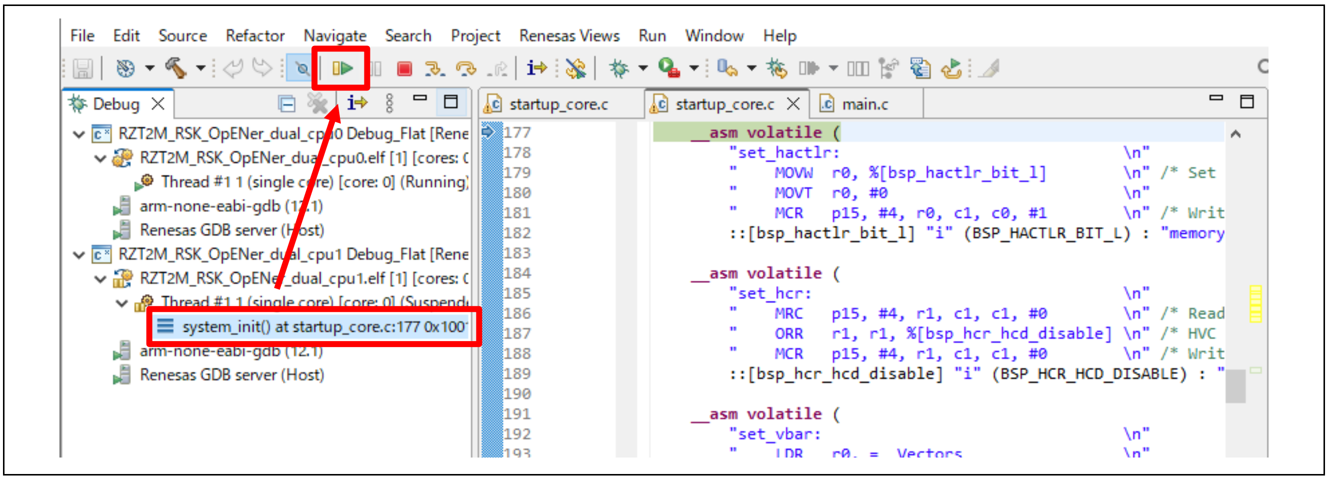


Figure 6-20 Run the CPU1 project

6.2.1.3 Flash boot mode

(1) How to generate source code

A) Click the configuration.xml in the CPU0 project.

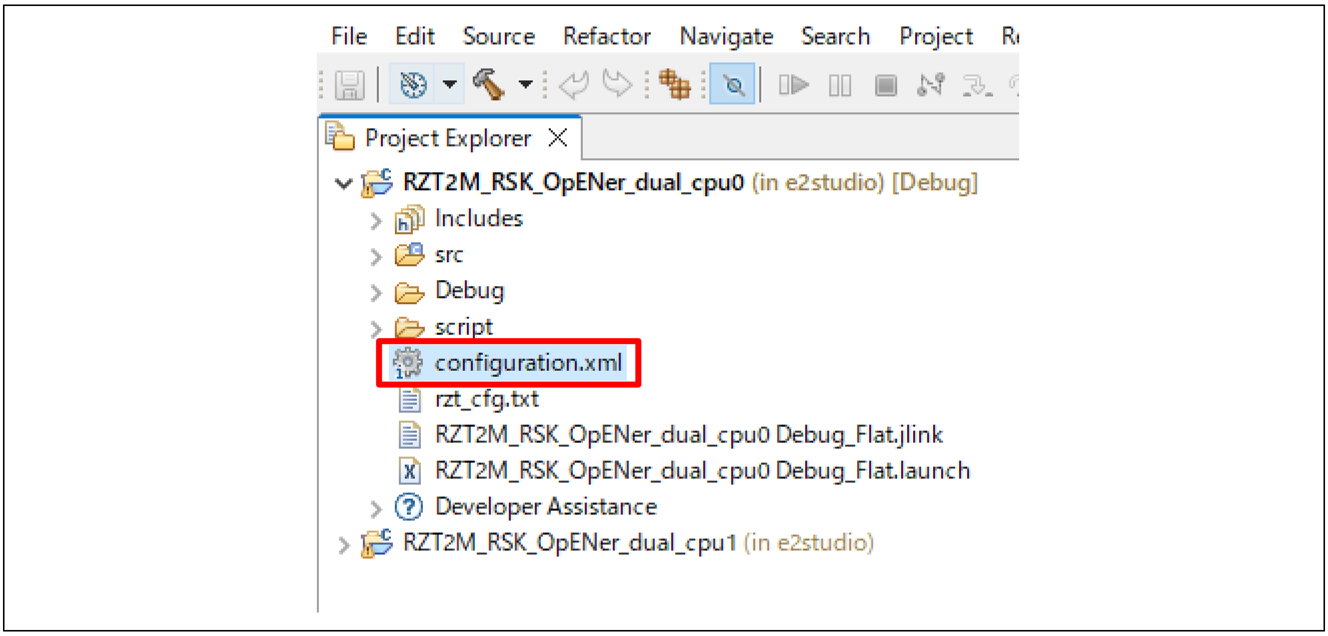


Figure 6-21 Configuration

B) Change the Boot Mode according to "Appendix. How to Change Boot Mode of FSP Project" in "RZ/T2, RZ/N2 Getting Started with Flexible Software Package (r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf)". When selecting the boot mode on the BSP tab, select the following settings.

- RZ/T2M
  - ・Board : RSK+RZT2M (xSPI0 x1 boot mode)

C) After changing the boot mode, go to the "Pins" tab and change the settings of the following pins from "Output mode (Low & Not into Input)" to "Output mode (Low & Into Input)".

Table 6-1 Port number of Pin to be changed in RZ/T2M

Port Number			
P08_2	P19_3	P19_4	P19_6
P19_7	P20_0	P20_1	P20_2
P20_3	P20_4	P20_7	P21_0
P23_4	-	-	-

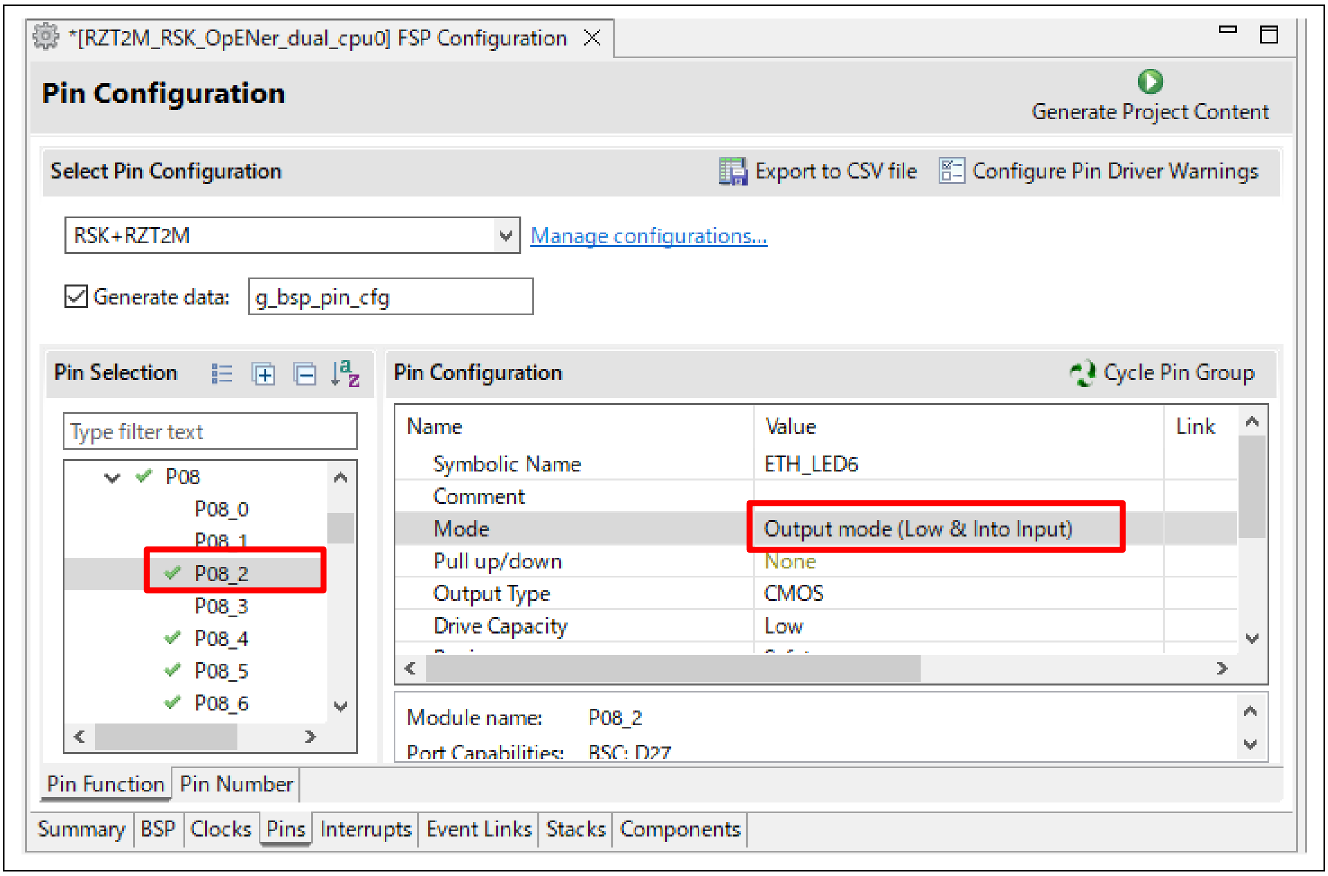


Figure 6-22 Change pin settings

D) Click “Generate Project Content” button.

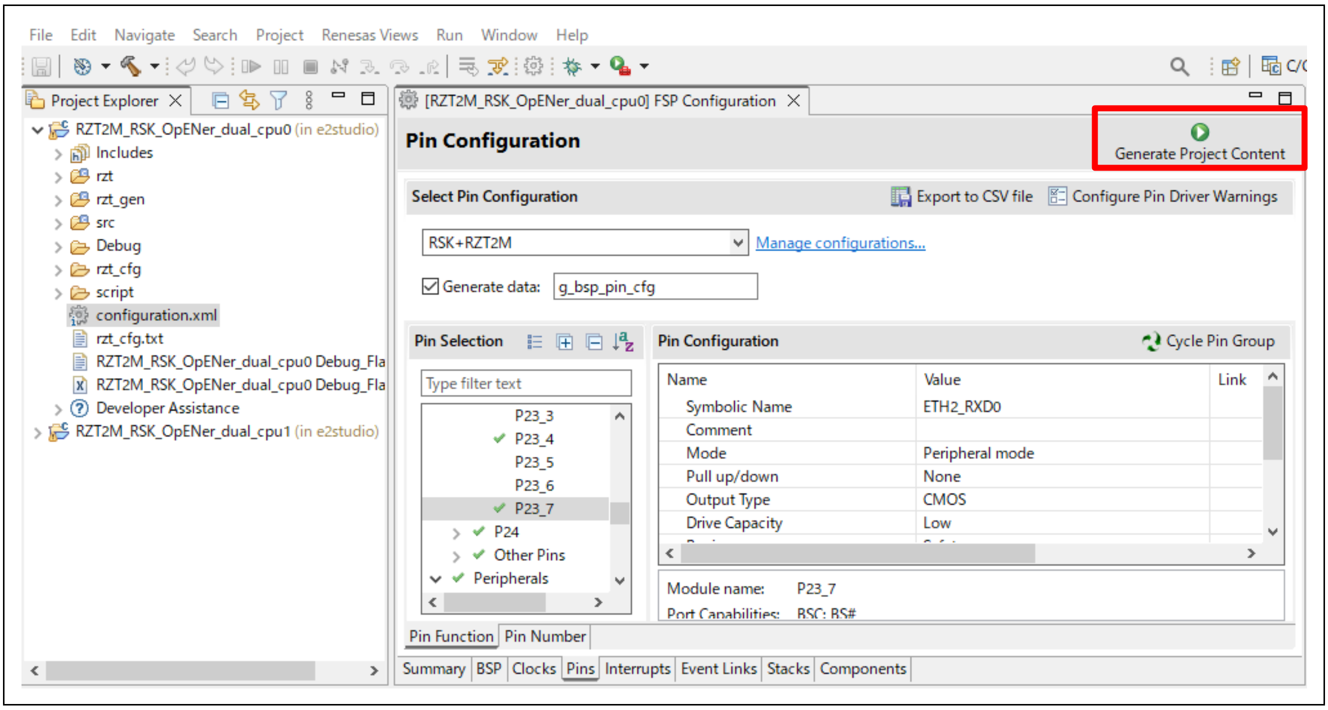


Figure 6-23 Generate Project Content, the CPU0 project

E) Click the configuration.xml in the CPU1 project and click “Generate Project Content” button.

Note: It is not necessary to change the board settings on the BSP tab and pin settings.

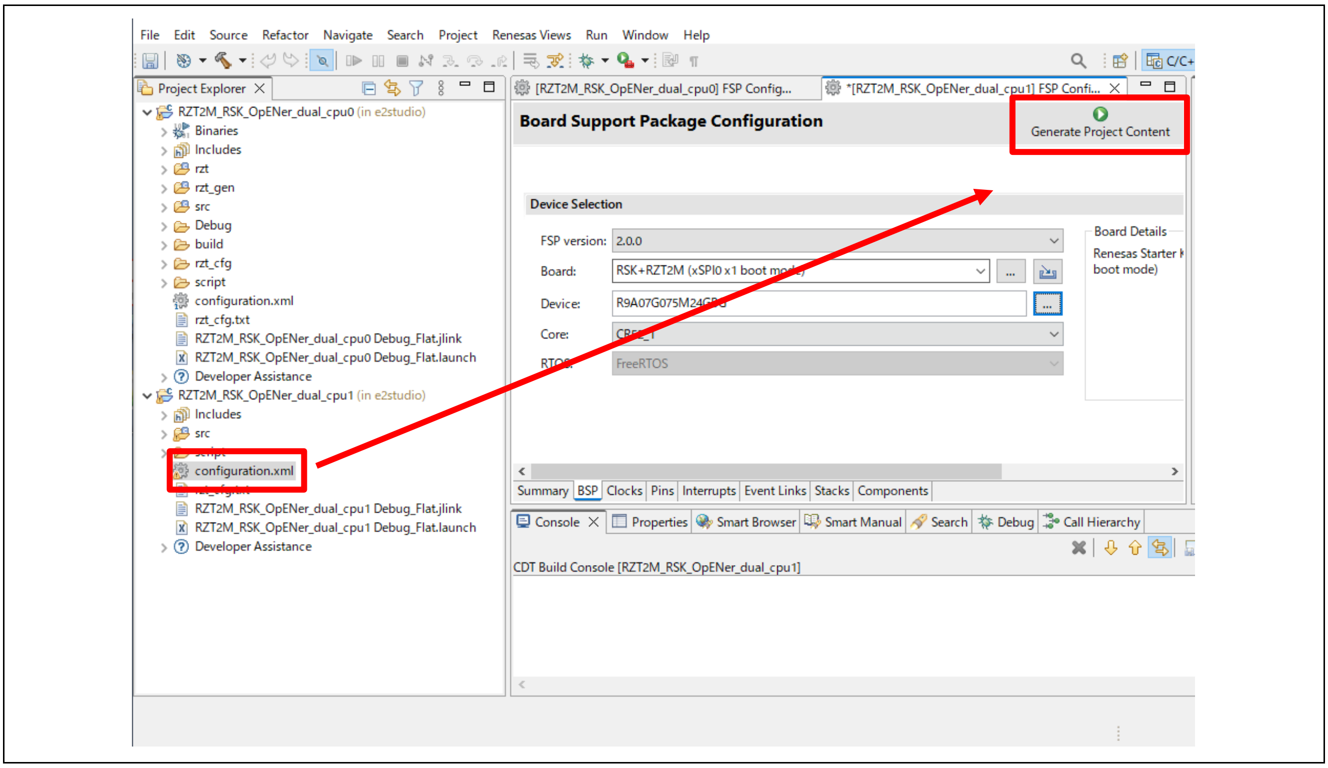


Figure 6-24 Generate Project Content, the CPU1 project

(2) Build and writing the projects to flash

A) Write the project according to “Appendix. How to Create and Debug FSP Projects for Multiprocessing in All Cases for e² studio” in “RZ/T2, RZ/N2 Getting Started with Flexible Software Package (r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf)”.

B) After writing the projects, restart the device by turning the power on/off.

Note:

- Be aware of the limitations when rewriting the program. Please refer to the section 1.3.
- If necessary to enable Type-0 Reset and Type-1 Reset, modify the macro definition “`SAMPLE_EXECUTE_SYSTEM_RESET_PROCESS`” (opener_port_main.c) to “1” enable the Reset process.

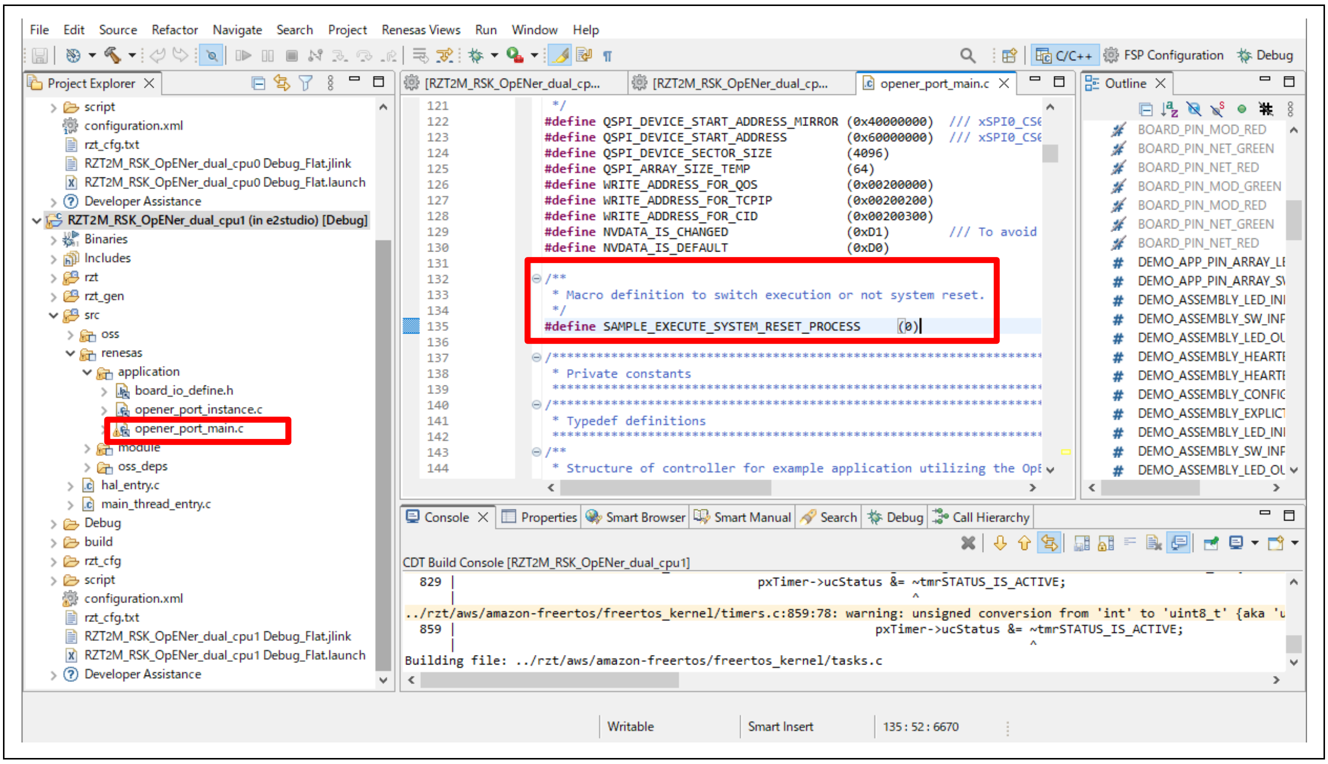


Figure 6-25 Reset process macro definition

6.2.2 Setup sample project for EWARM

The setup differs depending on the boot mode.

- When executing the sample program with RAM, refer to section 6.2.2.2.
- When executing the sample program with flash boot mode, refer to section 6.2.2.3.

6.2.2.1 Startup EWARM

1. Open the EWARM.
2. Click “Open Workspace…” in the File tab.

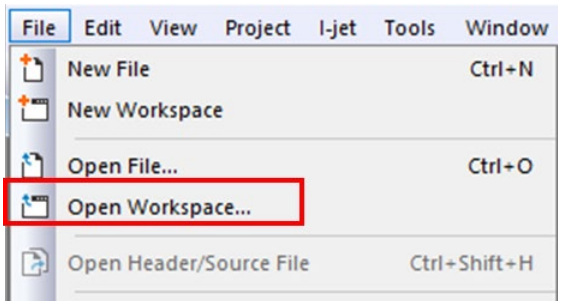


Figure 6-26 EWARM file tab

3. Select the CPU0 Workspace File(.eww) and click the Open button.
   “\project\rzt2m_rsk\opener_dual_cpu0\ewarm\RZT2M_RSK_OpENer_dual_cpu0.eww”

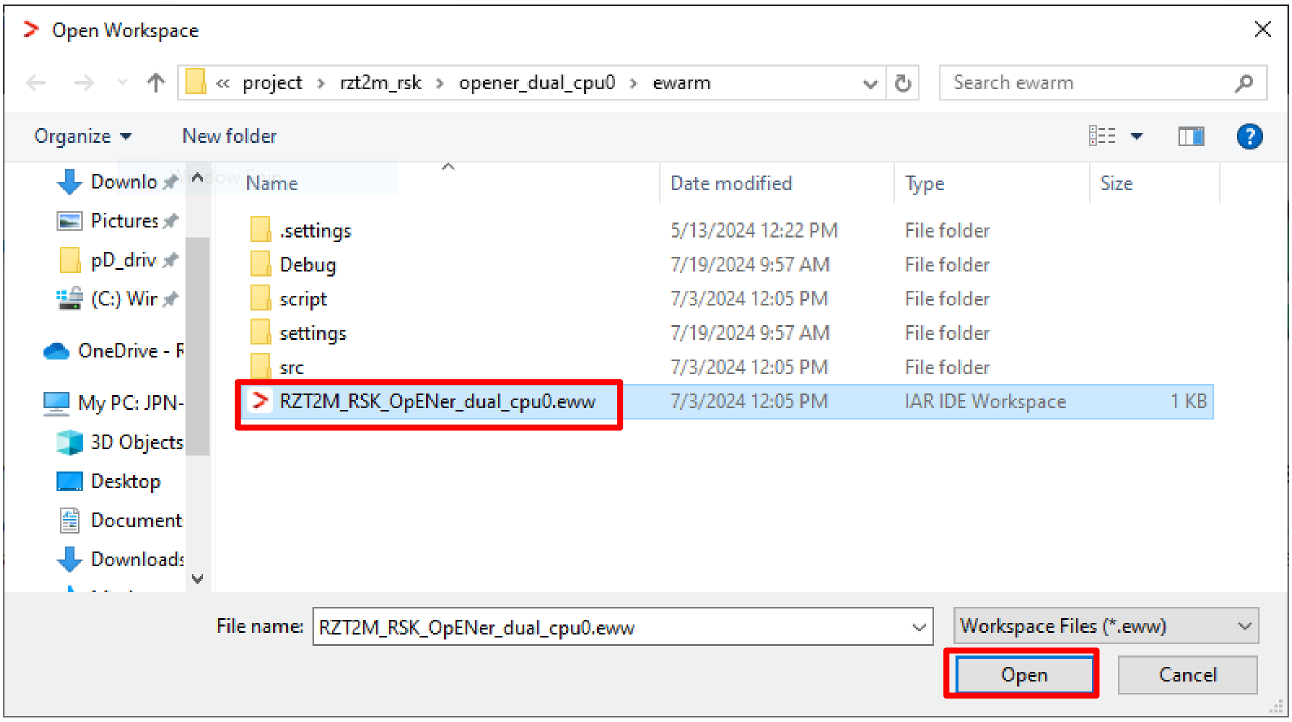


Figure 6-27 Open the CPU0 Workspace

4. Open another EWARM window.
5. Click "Open Workspace…" in the File tab, and open the CPU1 Workspace File(.eww), same as the CPU0 Workspace.

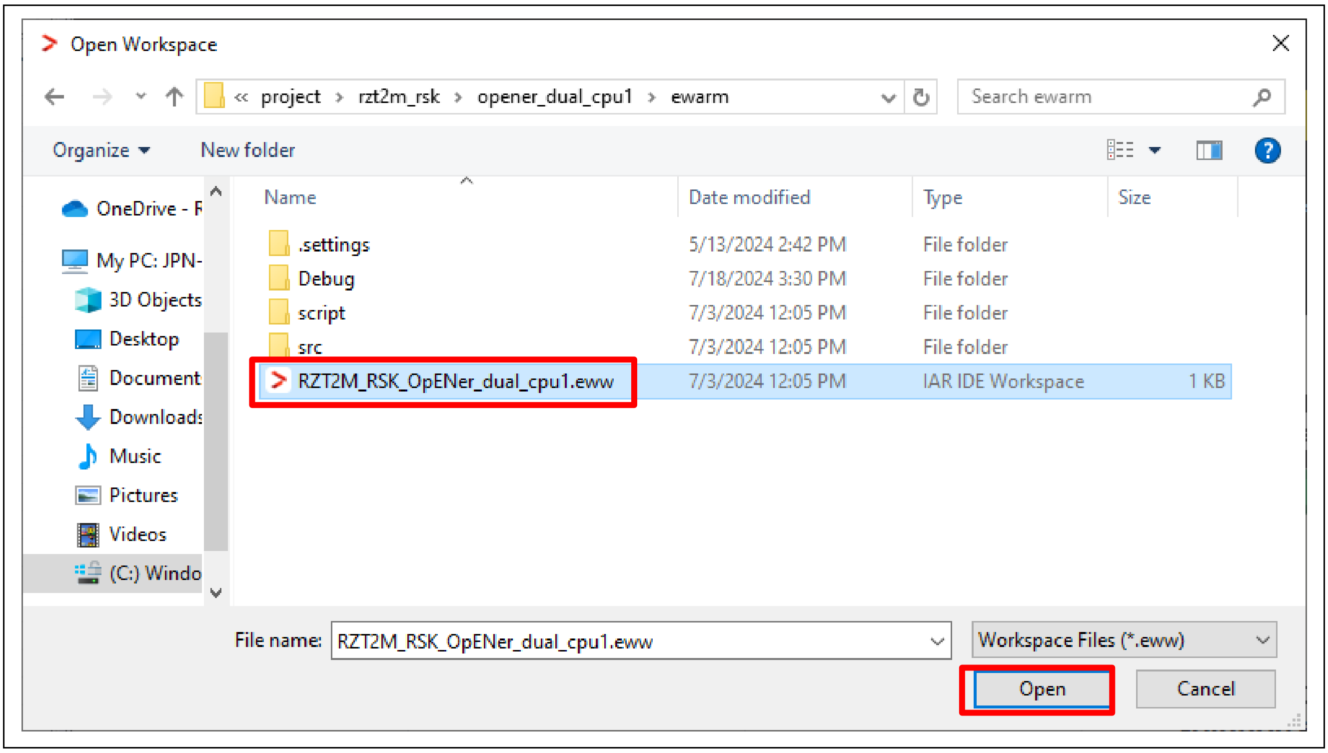


Figure 6-28 Open the CPU1 Workspace

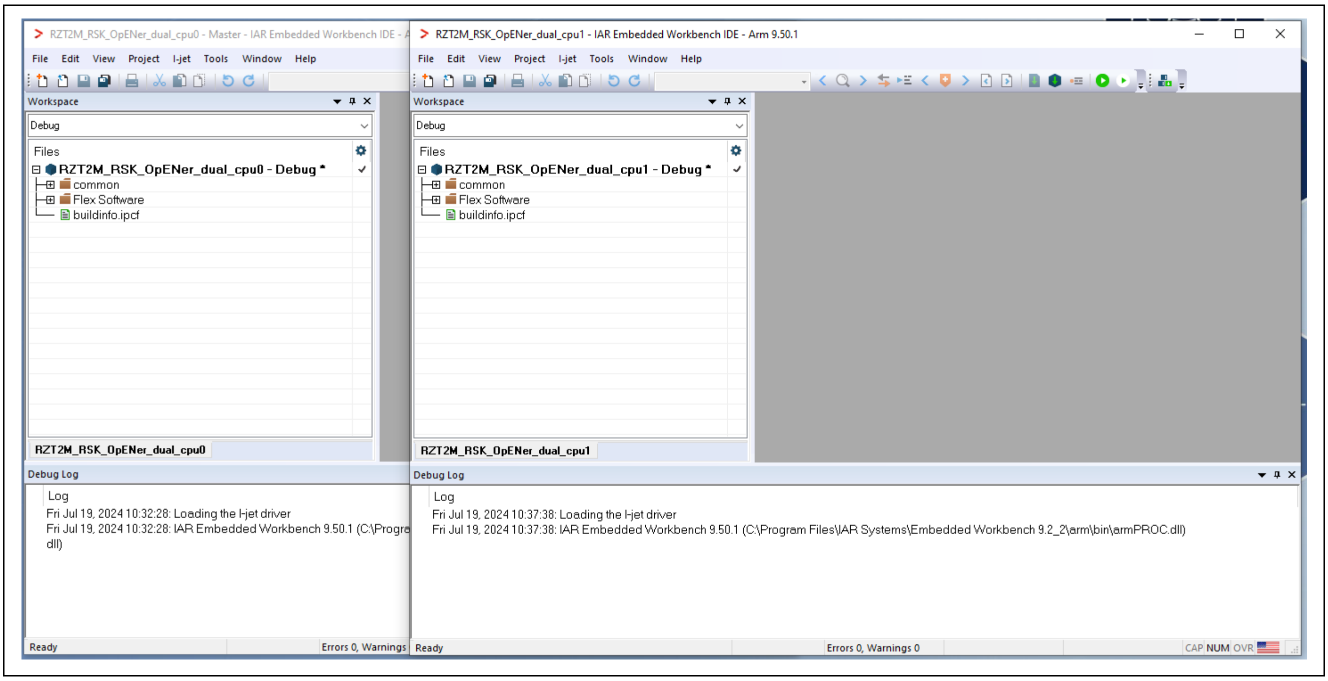


Figure 6-29 EWARM windows

6.2.2.2 RAM execution

(1) How to generate source code and how to build

A) Click the “Tools” -> “FSP Smart Configurator” on the CPU0 Workspace.

Note: If you have not set up FSP Smart Configurator yet on EWARM, refer to r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf in which section 5.4 describes how to set up it.

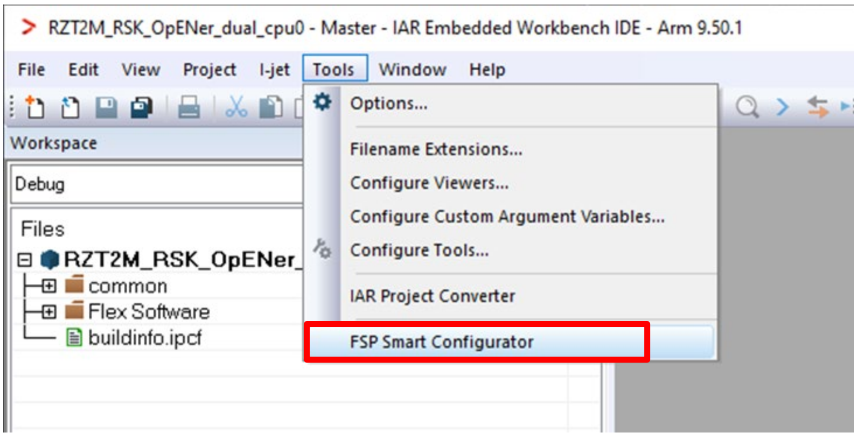


Figure 6-30 Tools tab on the CPU0 Workspace

B) Click “Generate Project Content” button then generate rzt, rzt_gen, rzt_cfg folder.

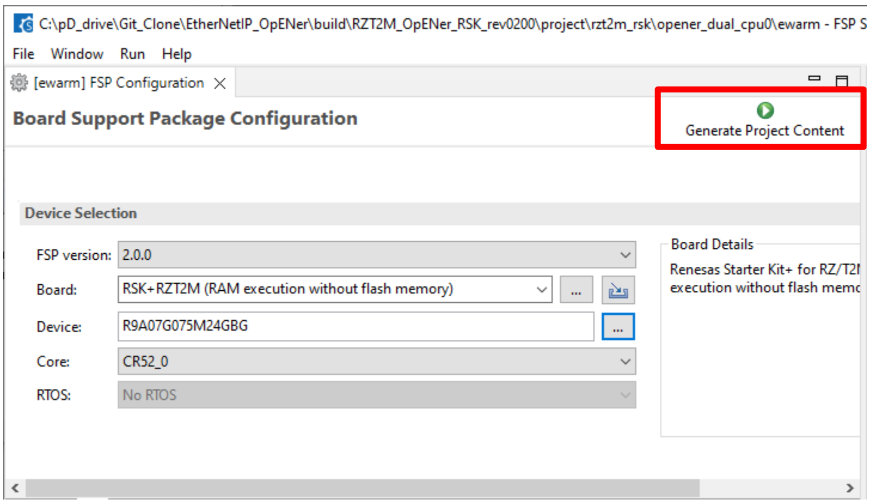


Figure 6-31 Generate Project Content

C) Click the Make button to build the CPU0 project. Once the build is completed, the build message is displayed in the Build Console window that displays compilation target files and the number of error/warnings.

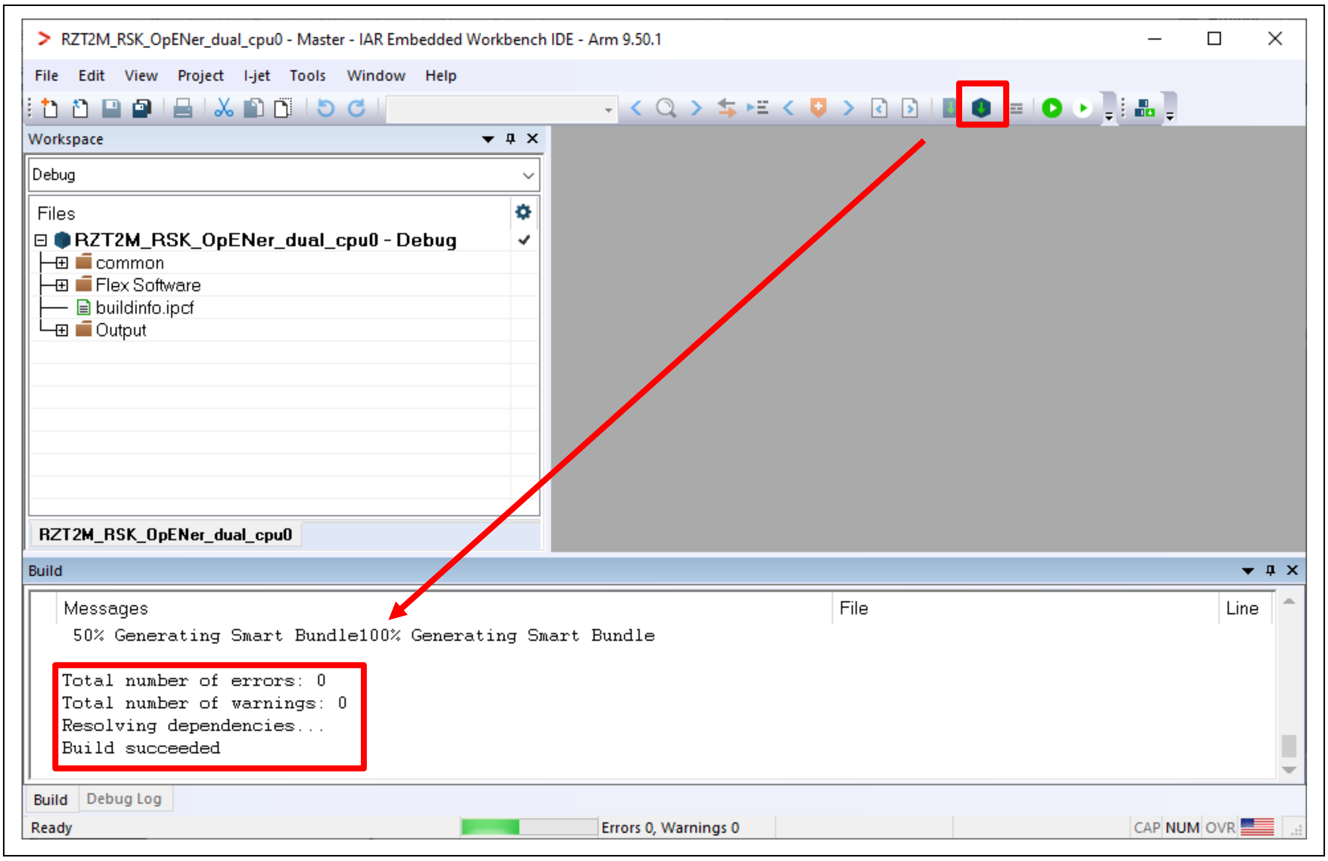


Figure 6-32 Make button on the CPU0 Workspace

D) Click the “Tools” -> “FSP Smart Configurator” on the CPU1 Workspace.

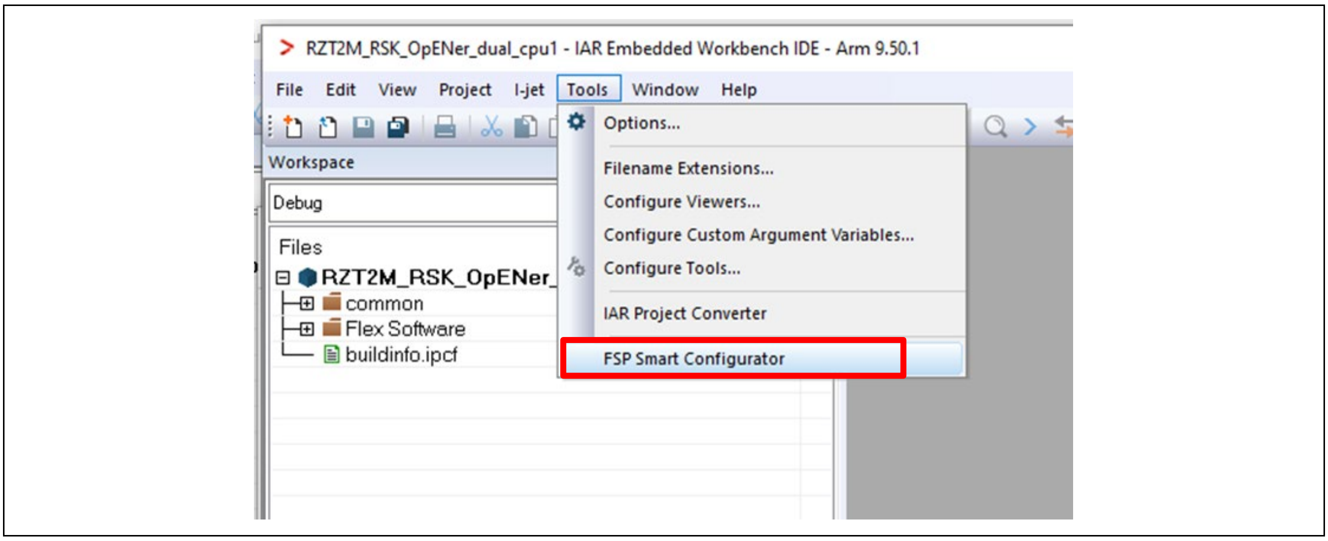


Figure 6-33 Tools tab on the CPU1 Workspace

E) Click "Generate Project Content" button then generate rzt, rzt_gen, rzt_cfg folder.

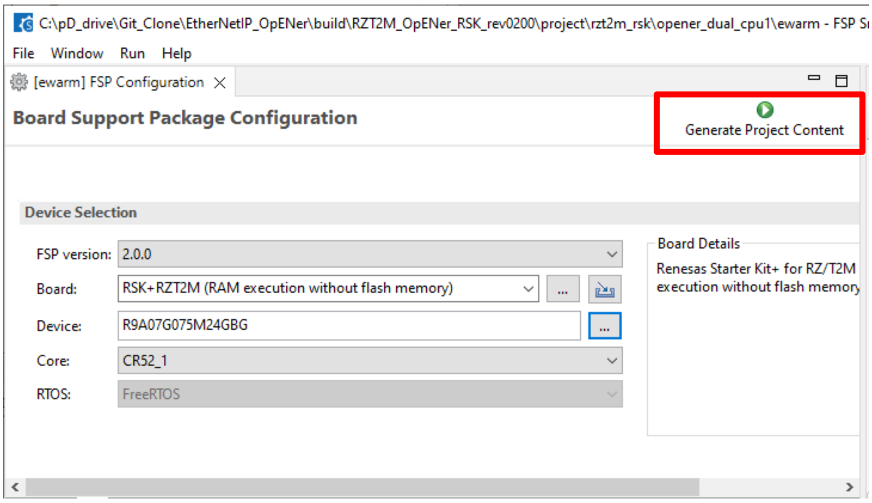


Figure 6-34 Generate Project Content

F) Click the Make button to build the CPU1 project. Once the build is completed, the build message is displayed in the Build Console window that displays compilation target files and the number of error/warnings.

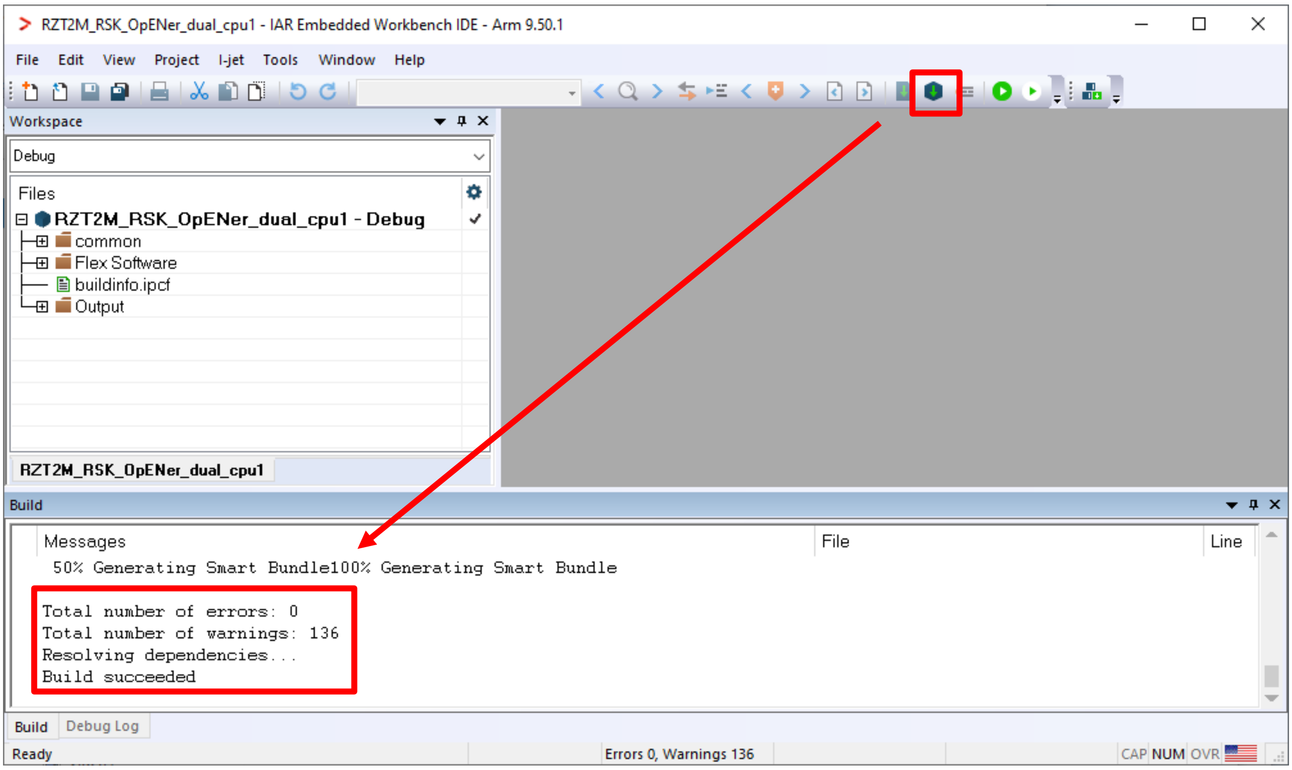


Figure 6-35 Make button on the CPU1 Workspace

G) Close the CPU1 Workspace.

(2) Download application and run debugger

A) Click the Debug button to download the CPU0 application program and launch the debugger. Then automatically the CPU1 Workspace will open.

Figure 6-36 Debug button

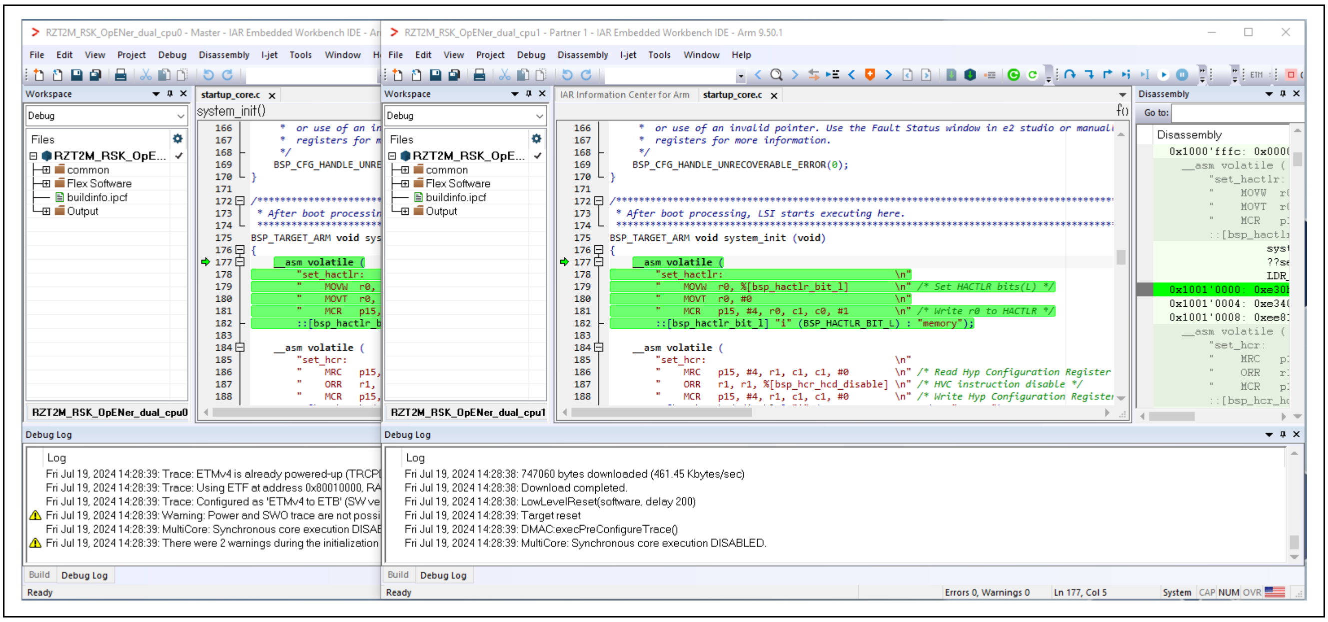


Figure 6-37 The CPU1 Workspace automatically opens

B) Click the Go button on the CPU0 Workspace.

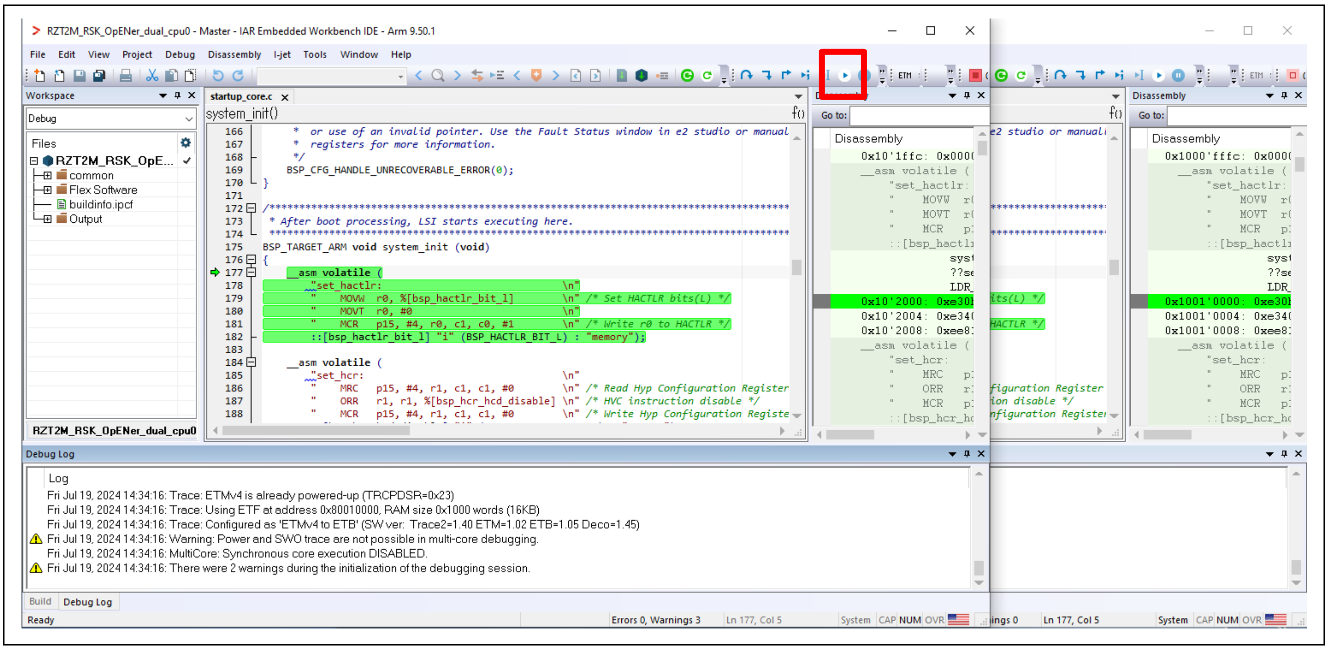


Figure 6-38 Go button on the CPU0 Workspace

C) Click the Go button on the CPU1 Workspace.

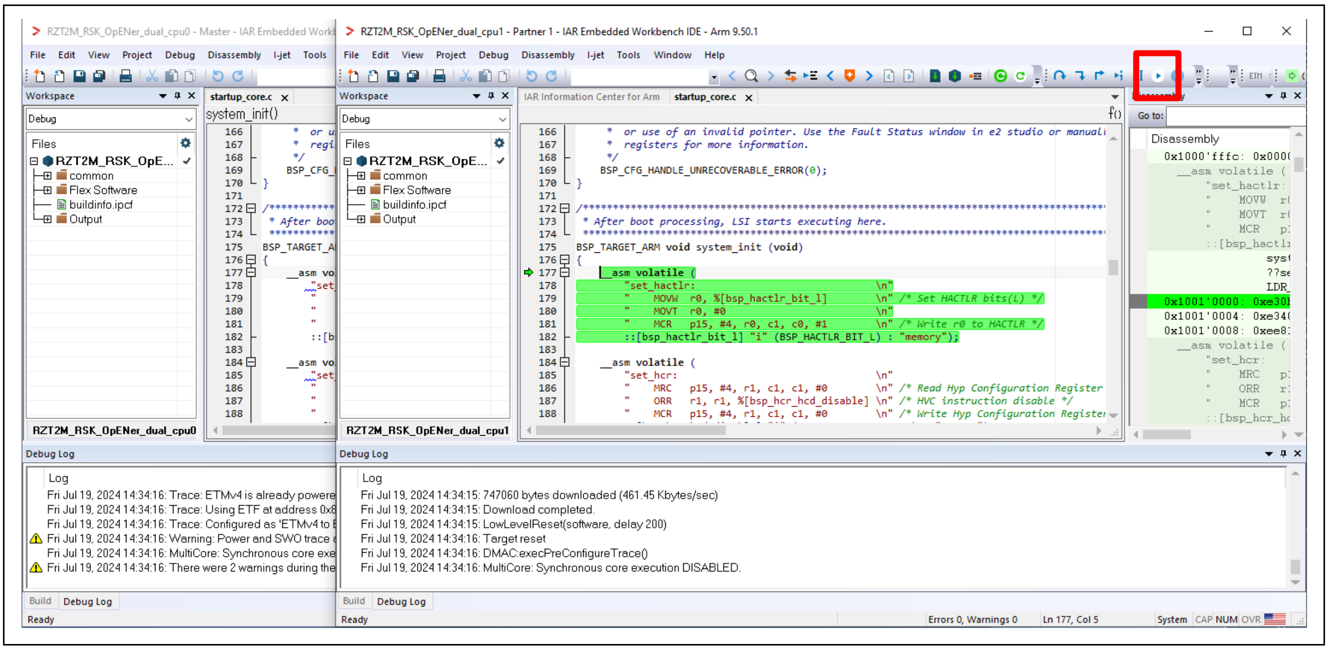


Figure 6-39 Go button on the CPU1 Workspace

6.2.2.3 Flash boot mode

(1) How to generate source code and how to build

A) Set the Asymmetric multicore setting to “Disabled” in the property of the CPU0 Workspace.

a-1. Click Project > Options….

a-2. Click Debugger > Multicore.

a-3. Select Disable in Asymmetric multicore.

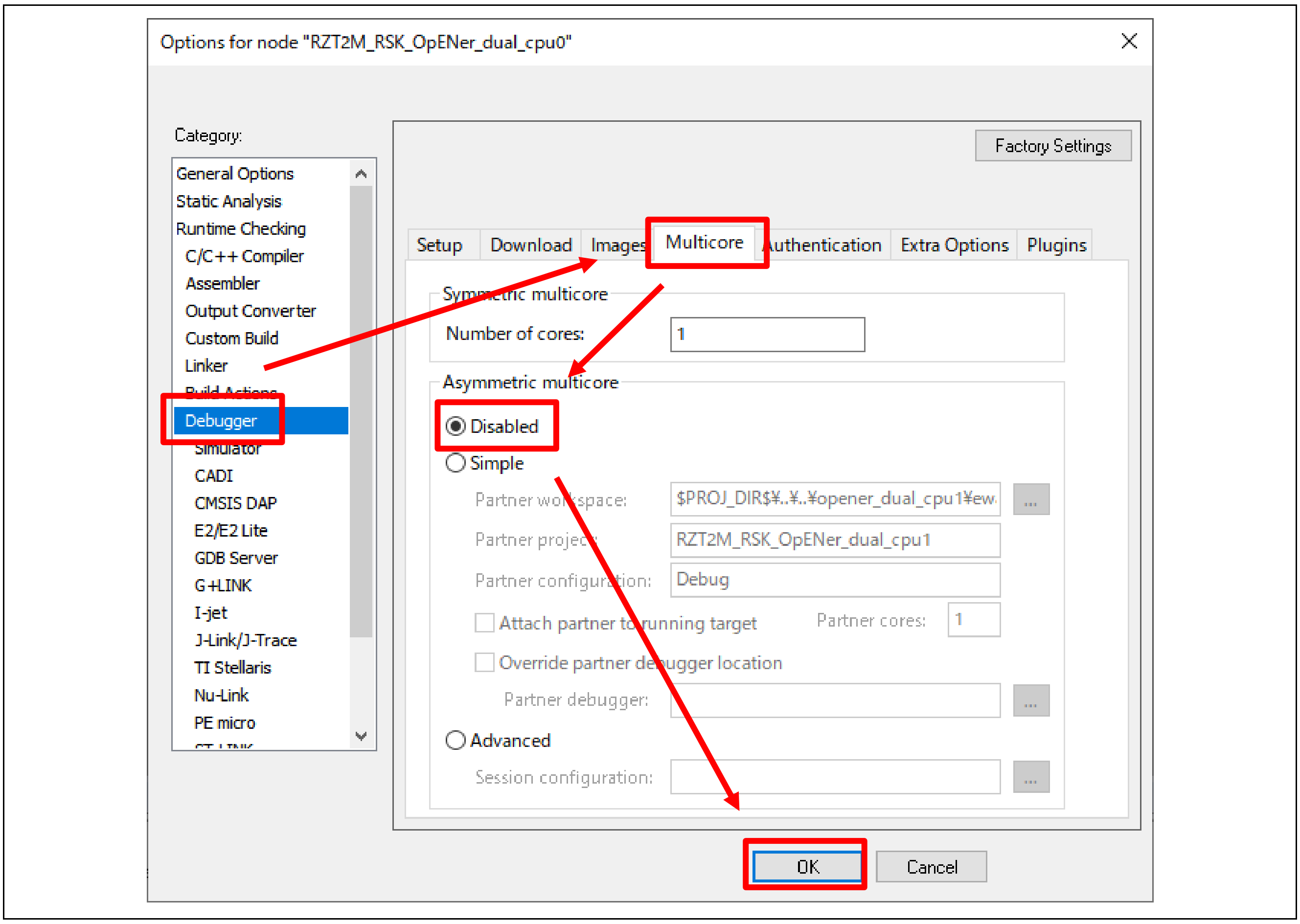


Figure 6-40 Disable multicore debug mode in CPU0 project property

B) Click the “Tools” -> “FSP Smart Configurator” on the CPU0 Workspace.

Note: If you have not set up FSP Smart Configurator yet on EWARM, refer to r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf in which section 5.4 describes how to set up it.

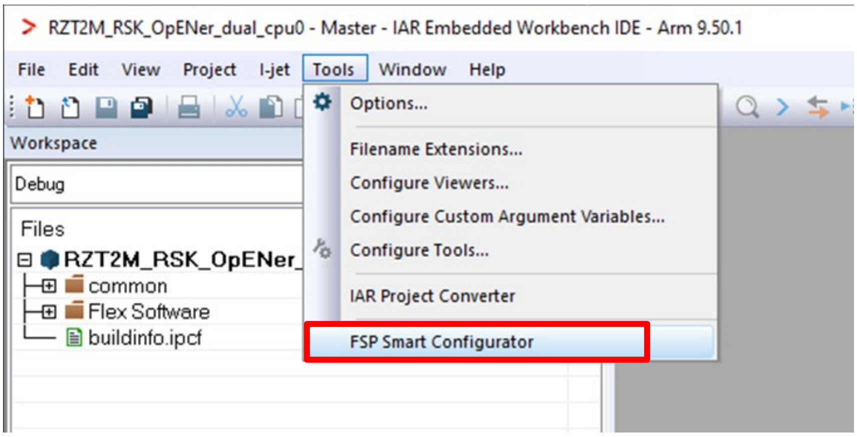


Figure 6-41 Tools tab on the CPU0 Workspace

C) Change the Boot Mode according to “Appendix. How to Change Boot Mode of FSP Project” in “RZ/T2, RZ/N2 Getting Started with Flexible Software Package (r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf)”. When selecting the boot mode on the BSP tab, select the following settings.

➢ RZ/T2M
  ・Board : RSK+RZT2M (xSPI0 x1 boot mode)

D) After changing the boot mode, go to the "Pins" tab and change the settings of the following pins from "Output mode (Low & Not into Input)" to "Output mode (Low & Into Input)".

Table 6-2 Port number of Pin to be changed in RZ/T2M

Port Number			
P08_2	P19_3	P19_4	P19_6
P19_7	P20_0	P20_1	P20_2
P20_3	P20_4	P20_7	P21_0
P23_4	-	-	-

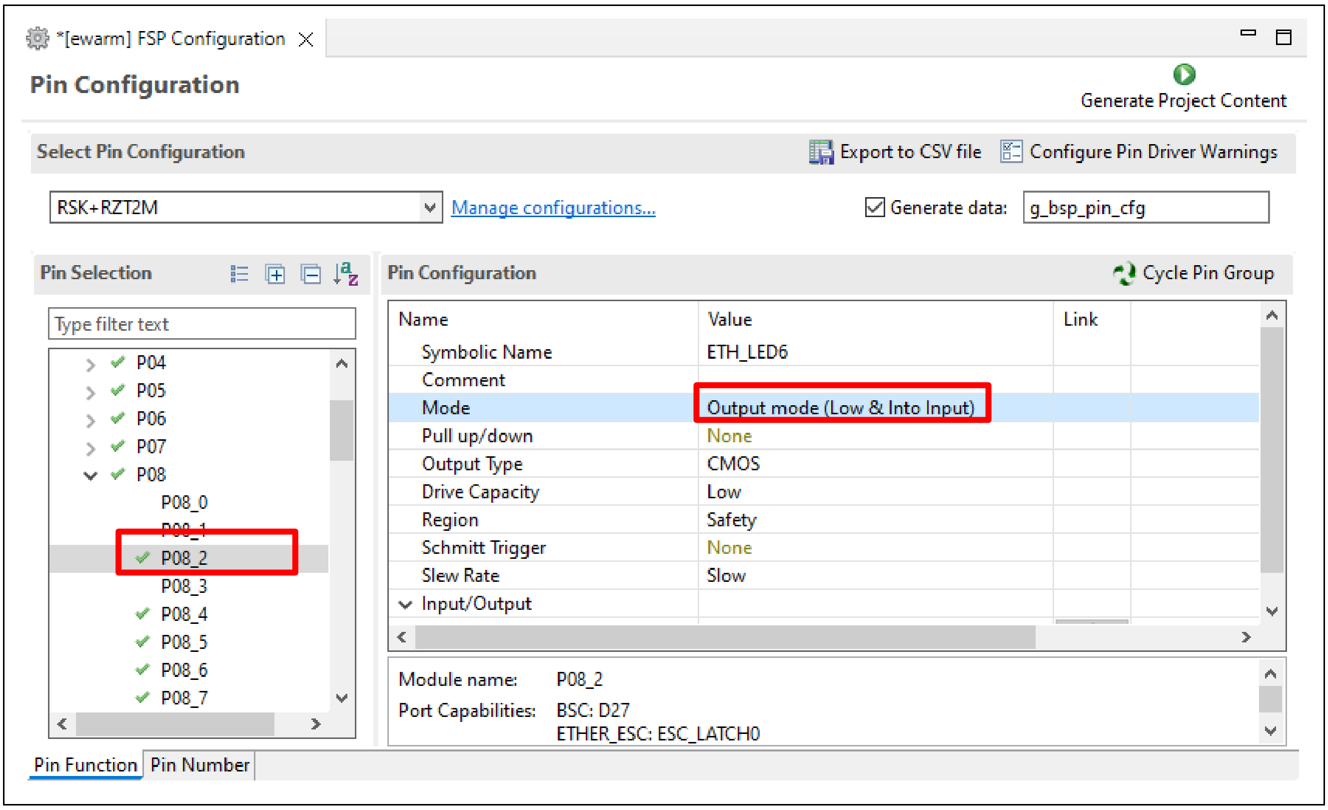


Figure 6-42 Change Pin Settings

E) Click “Generate Project Content” button then generate rzt, rzt_gen, rzt_cfg folder.

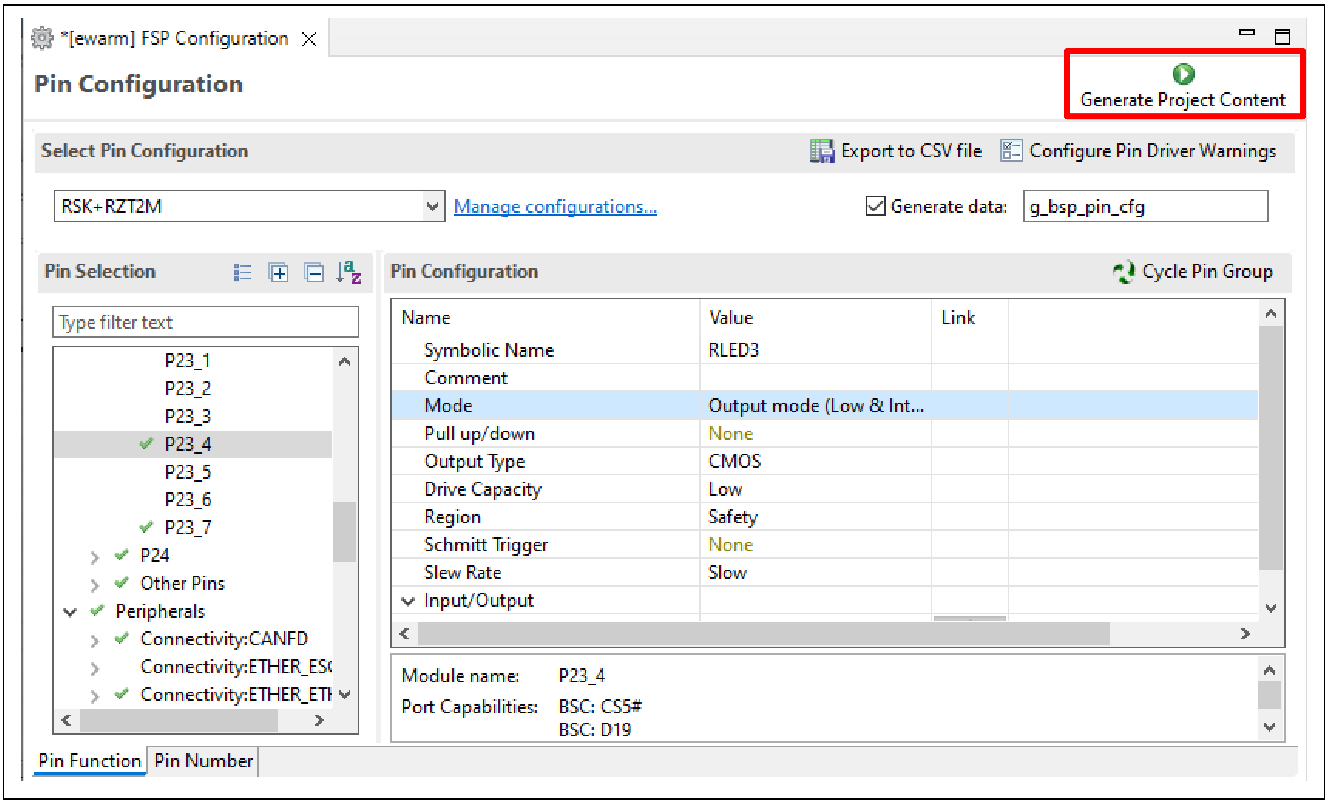


Figure 6-43 Generate Project Content

F) Build the CPU0 project.

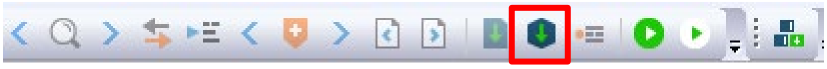


Figure 6-44 Build the CPU0 project

G) Click the “Tools” -> “FSP Smart Configurator” on the CPU1 Workspace.

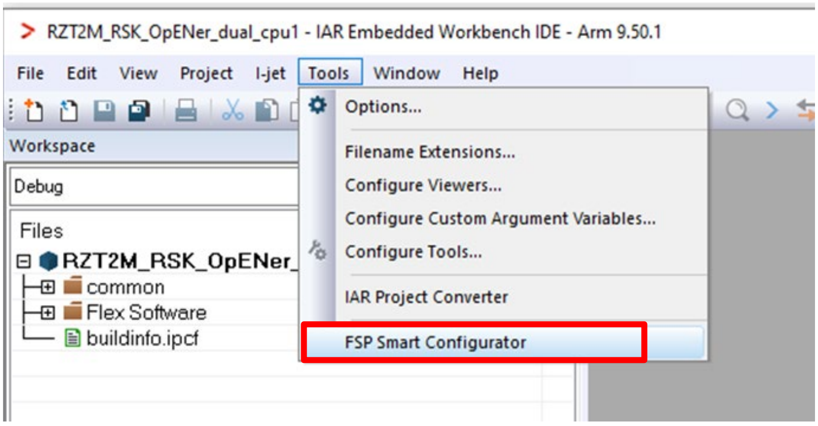


Figure 6-45 Tools tab on the CPU1 Workspace

H) Click “Generate Project Content” button then generate rzt, rzt_gen, rzt_cfg folder. It is not necessary to change pin settings.

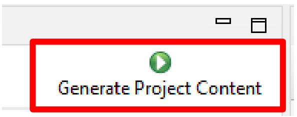


Figure 6-46 Generate Project Content

(2) Build and writing the projects to flash

A) Write the project according to “Appendix. How to Create and Debug FSP Projects for Multiprocessing in All Cases for IAR EWARM” in “RZ/T2, RZ/N2 Getting Started with Flexible Software Package (r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf)”.

B) After writing the projects, restart the device by turning the power on/off.

Note1: You may get the following error (Error [Lp011]: section placement failed) when building the CPU0 project. In this case, please try to change the icf file (Linker Script file) by the following guidance.

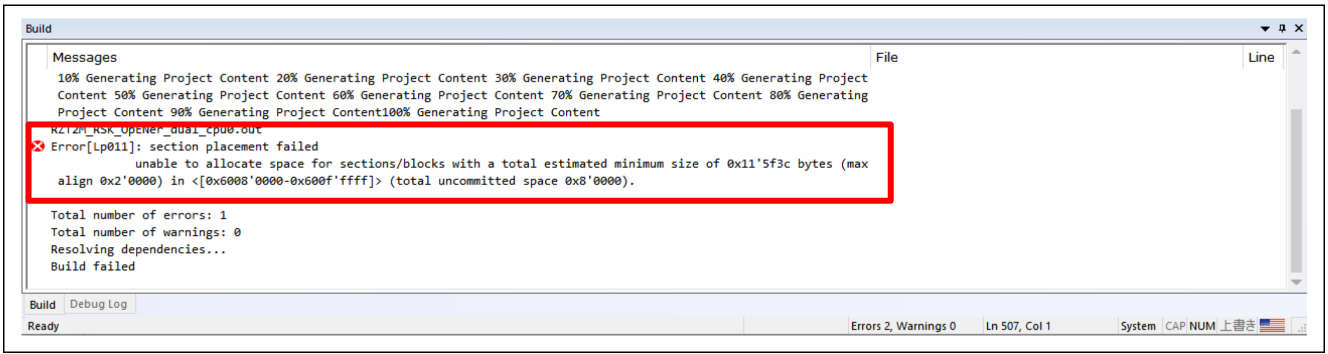


Figure 6-47 The error message, Error [Lp011]: section placement failed

1. On the CPU0 project, click “Project” -> “Options…”
2. Click “Linker” -> “Config”

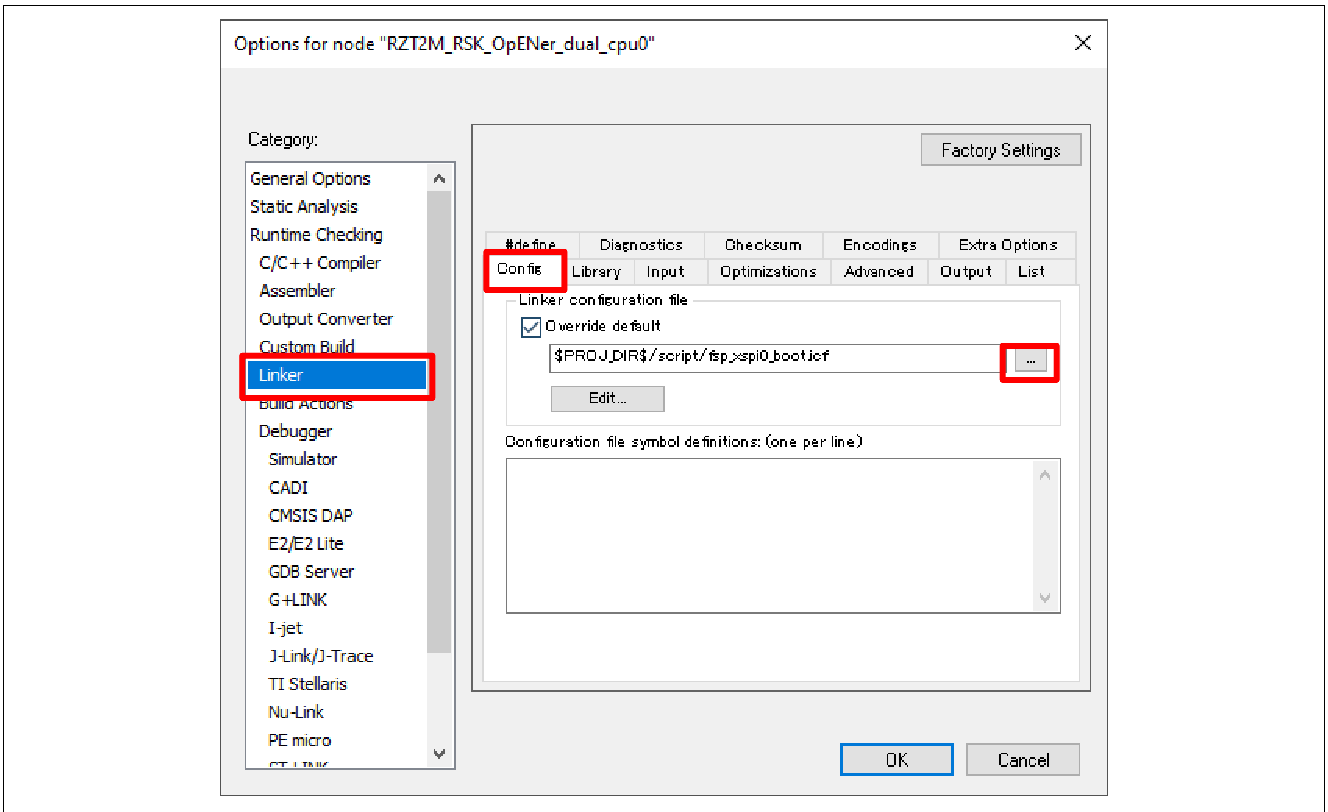


Figure 6-48 Linker -> Config in the CPU0 project Option

3. Change the icf file from default to “fsp_xspi0_boot_for_v0210.icf”.

Directory: “\project\rzt2m_rsk\opener_dual_cpu0\ewarm\script\ fsp_xspi0_boot_for_v0210.icf”

4. Click the Make button to build the CPU0 project again.

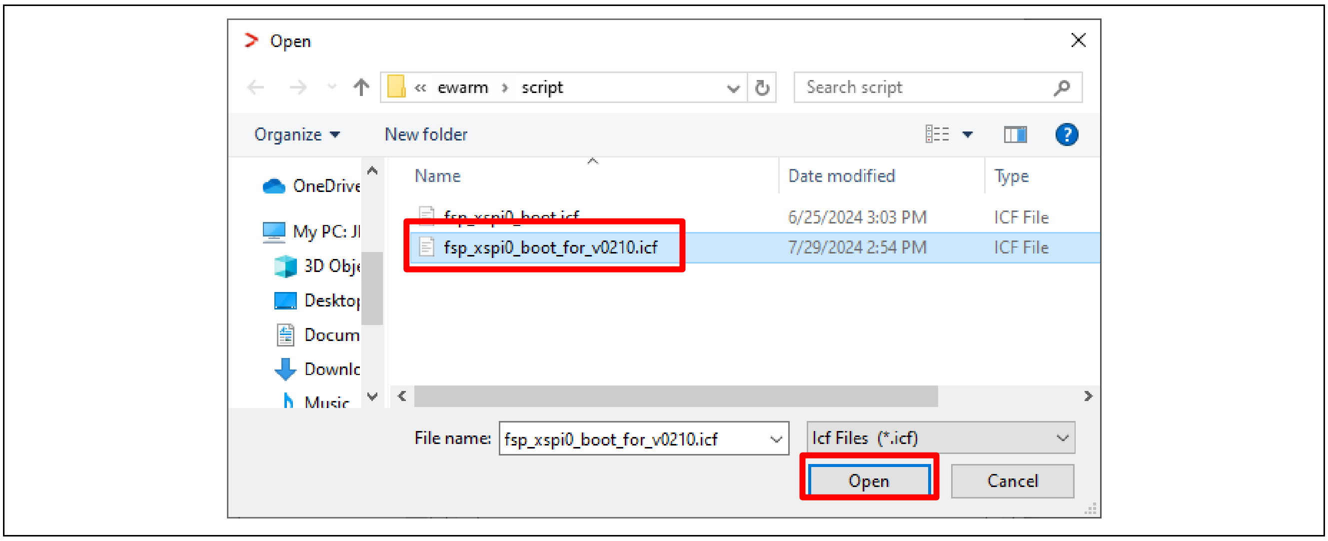


Figure 6-49 Open the icf file for FSP v2.1.0

Note 2:

- Be aware of the limitations when rewriting the program. Please refer to the section 1.3.
- If necessary to enable Type-0 Reset and Type-1 Reset, modify the macro definition “`SAMPLE_EXECUTE_SYSTEM_RESET_PROCESS`” (opener_port_main.c) to “1” enable the Reset process.

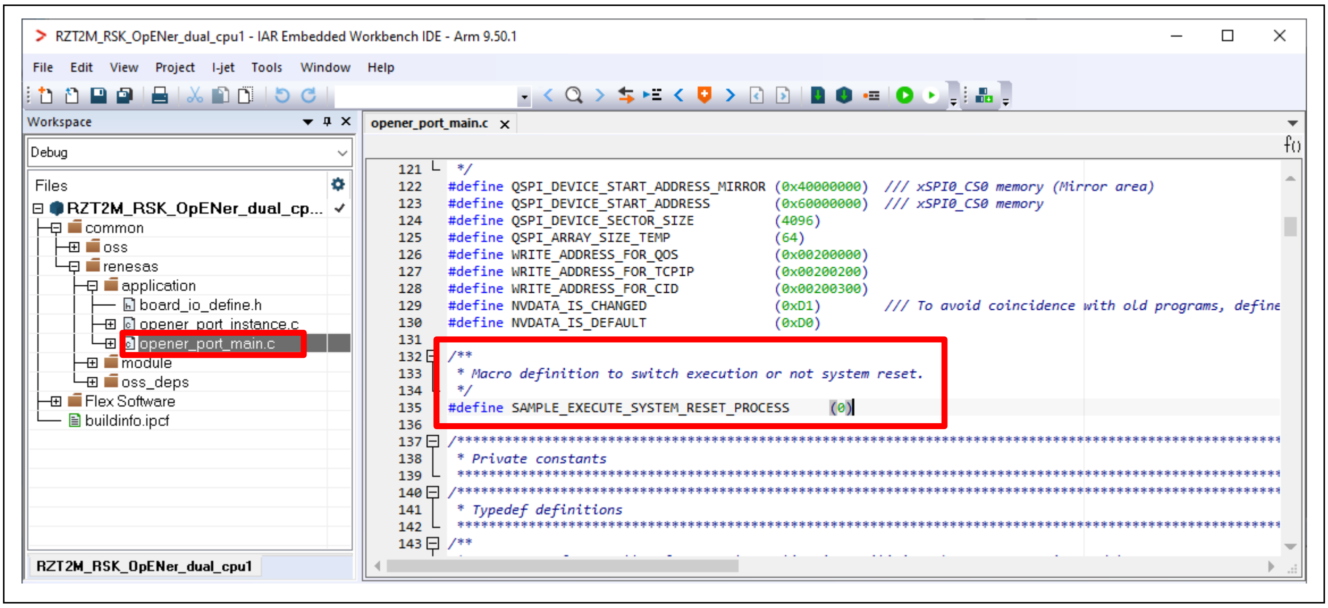


Figure 6-50 Reset process macro definition

6.3 For RZ/T2ME (Dual-core project)

Please use FSP for Renesas RZ/T series v2.1.0 or later version to setup.

The setup differs depending on the IDE.

- When using e² studio, refer to section 6.3.1.
- When using EWARM, refer to section 6.3.2.

6.3.1 Setup sample project for e² studio

The setup differs depending on the boot mode.

- When executing the sample program with RAM, refer to section 6.3.1.2.
- When executing the sample program with flash boot mode, refer to section 6.3.1.3.

6.3.1.1 Startup e² studio

1. Open the e² studio and select a directory as workspace.
2. Click "Import" in File tab.

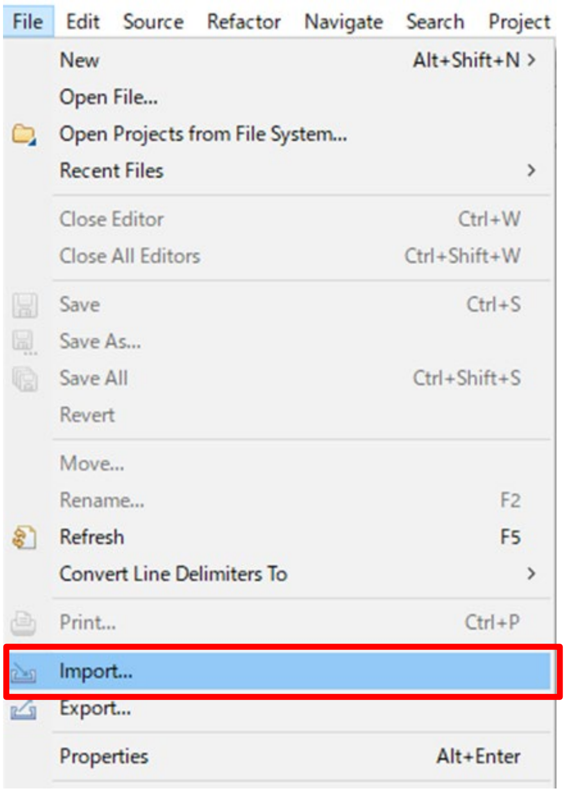


Figure 6-51 e² studio File tab

3. Click "Existing Projects into Workspace" in "General" and Click "Next".

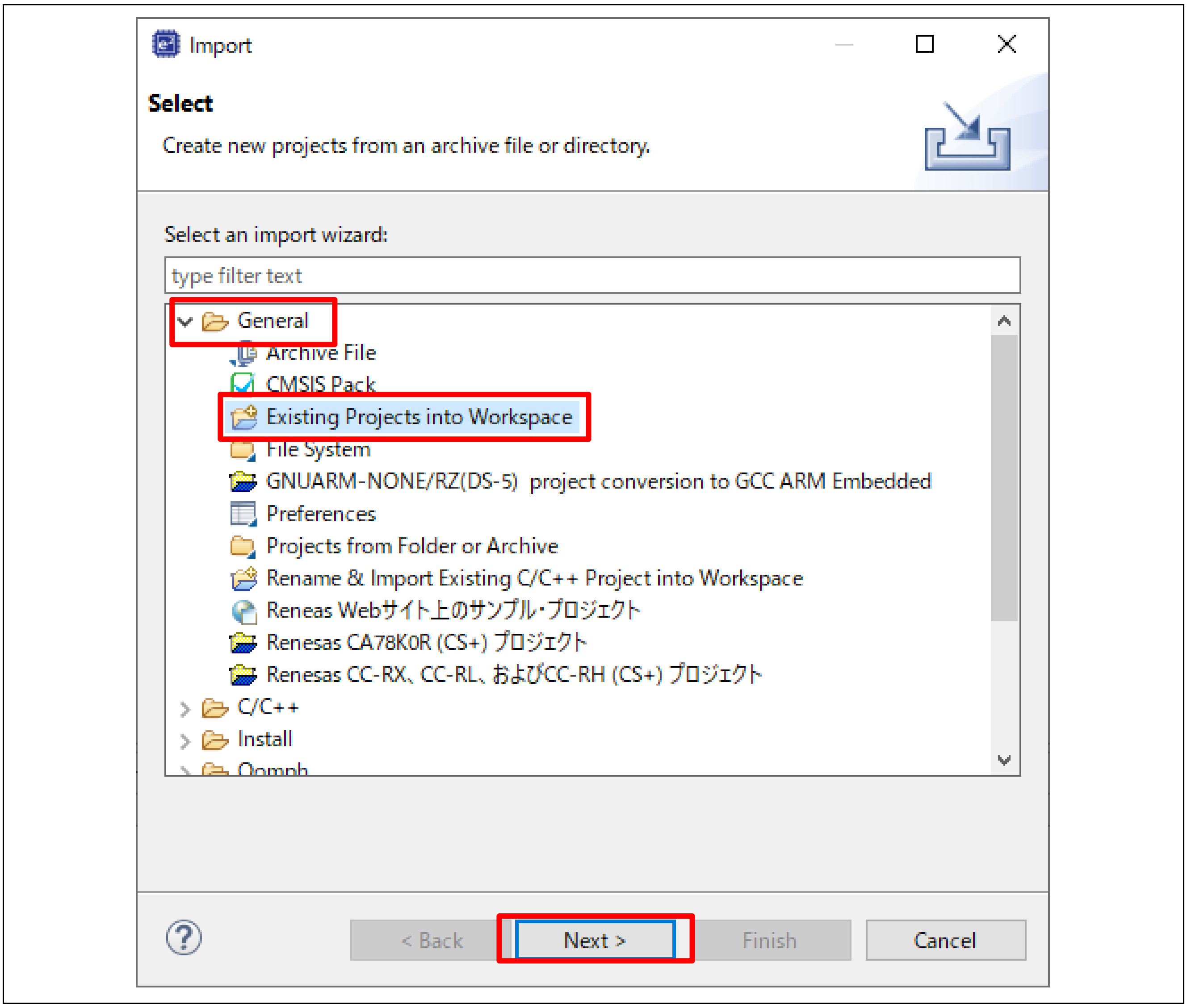


Figure 6-52 Select import type

4. Import project by the following steps.
   A) Set the root directory.
      "(Extracted zip folder path)\project\rzt2m_rsk"
   B) Check 2 projects.
   C) Click "Finish"

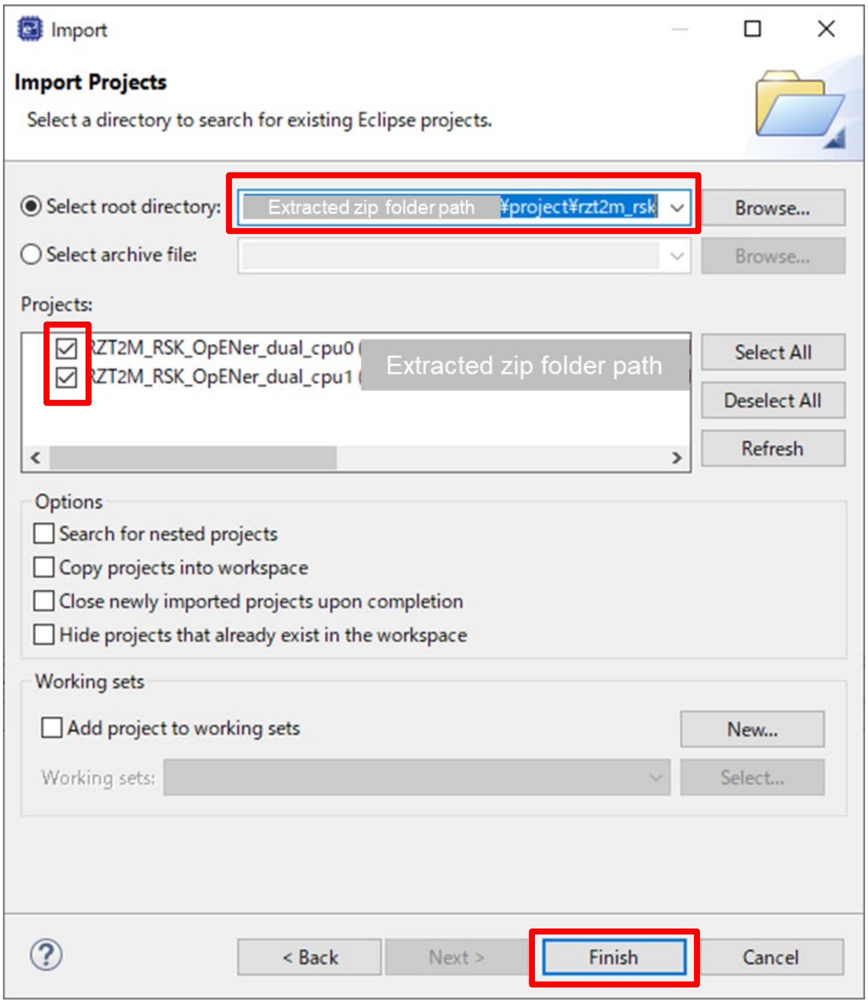


Figure 6-53 Import two projects on e² studio

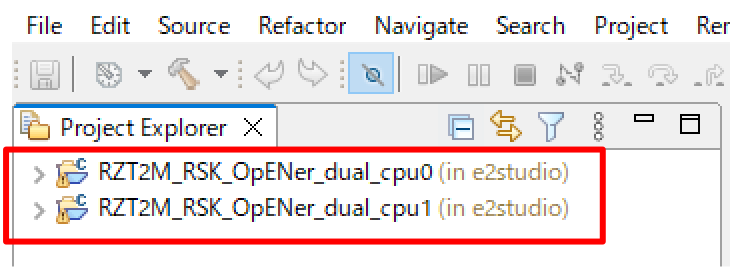


Figure 6-54 Open 2 projects

6.3.1.2 RAM execution

(1) How to generate source code and how to build

A) Click the Configuration.xml of the CPU0 project.

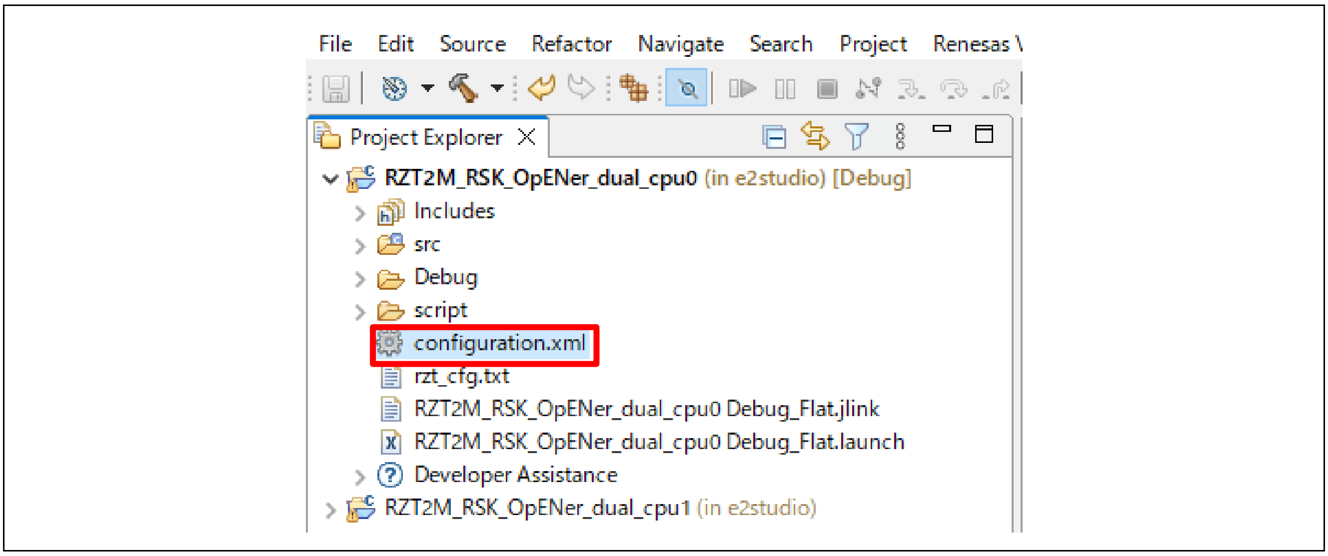


Figure 6-55 Configuration

B) Click the “BSP” tab, and change the Board setting.

➢ RZ/T2ME
  ・Board : RSK+RZT2ME (RAM execution without flash memory)

Note: You can ignore the warning beside the “Device” setting.

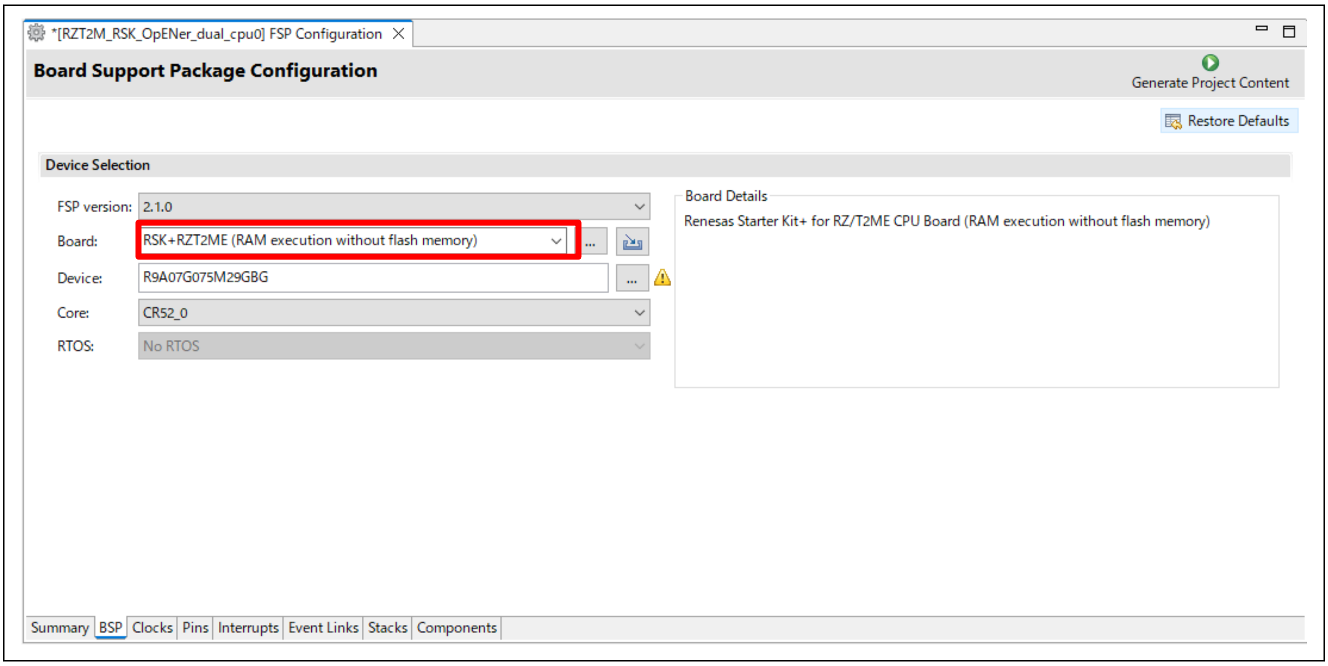


Table 6-3 Change the board in the BSP tab

C) Reselect “FSP Version” from the drop-down list.

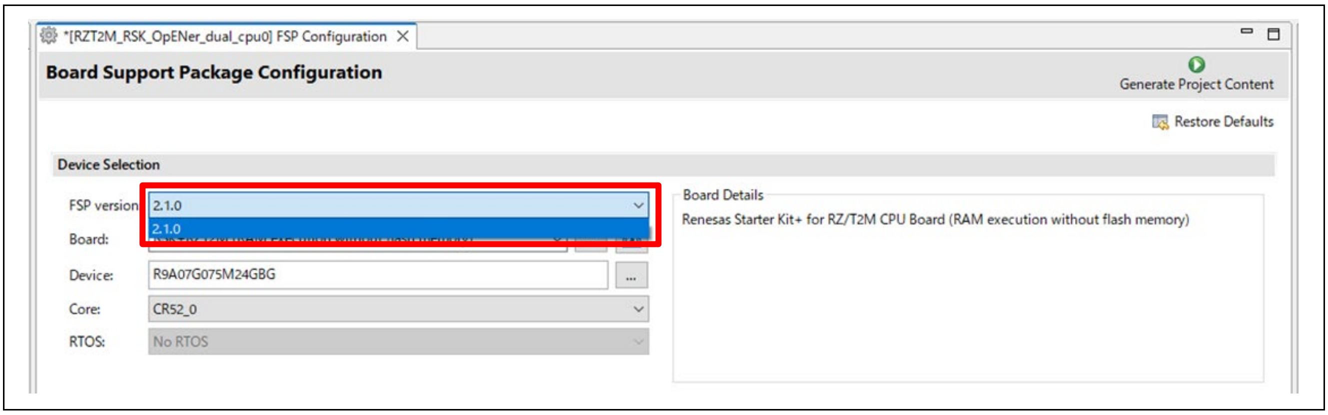


Table 6-4 Reselect “FSP Version” from the drop-down list

D) Set clock settings in the “Clock” tab as a following table.

・CPU1CLK : Mul x4
・CKIO : Div /2

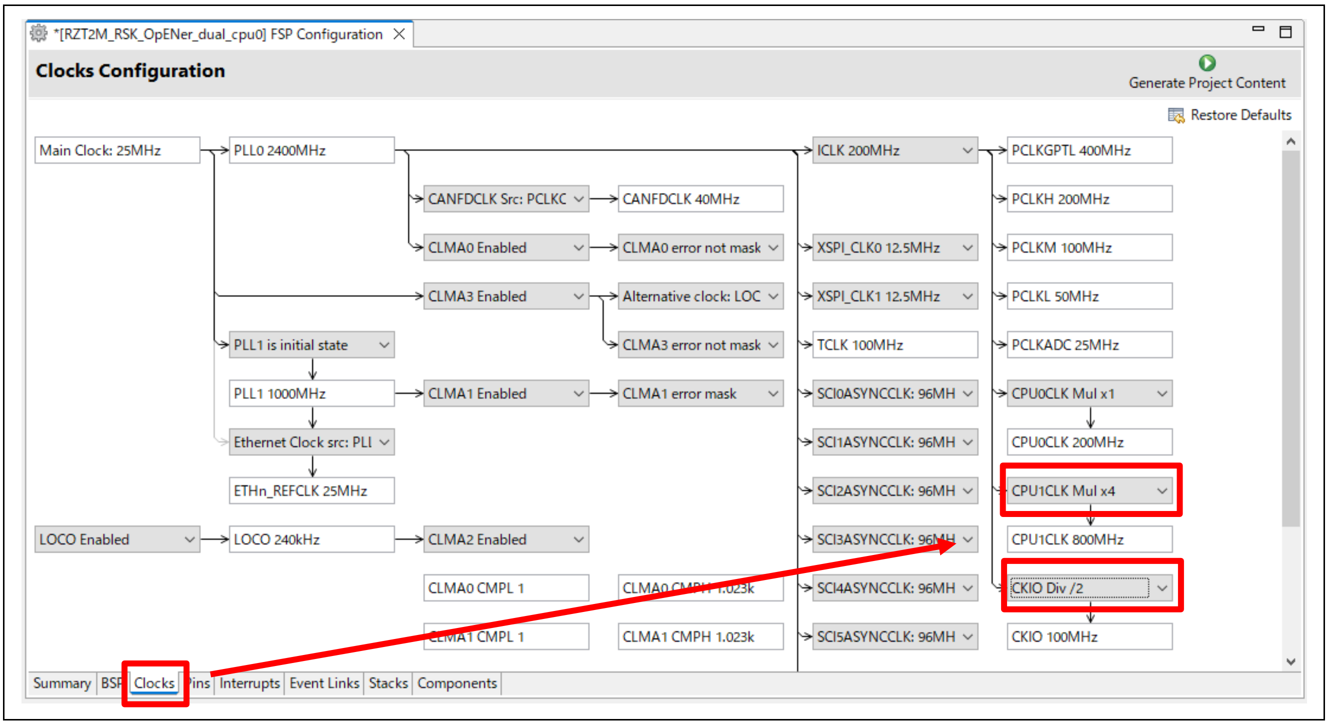


Table 6-5 Set clock settings

E) Click “Generate Project Content” button then generate rzt, rzt_gen, rzt_cfg folder.

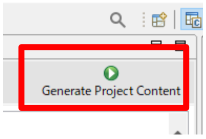


Figure 6-56 Generate Project Content

F) Click the Build button in tool bar to build the project and confirm that there is no error message in the build message log.

Figure 6-57 Build button

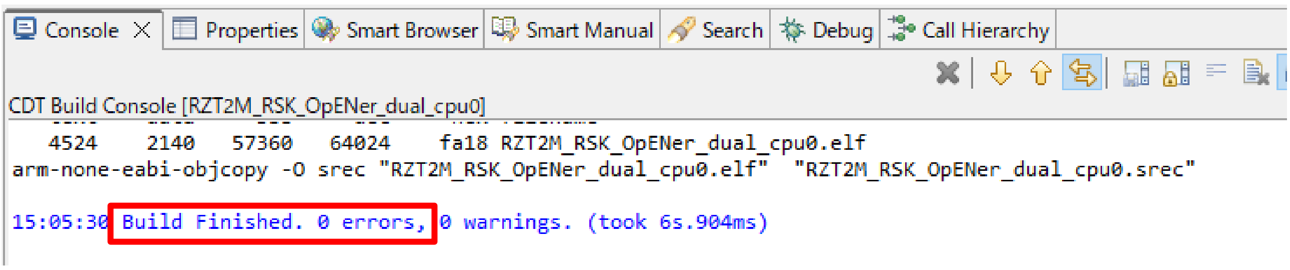


Figure 6-58 Build message

G) For the CPU1 project, generate the folders and build the project, same as the CPU0 project.

Note: You can ignore Problem Occurred dialogs with the error message “Cannot invoke "java.util.Map.entrySet()" because "securePinMap" is null”.

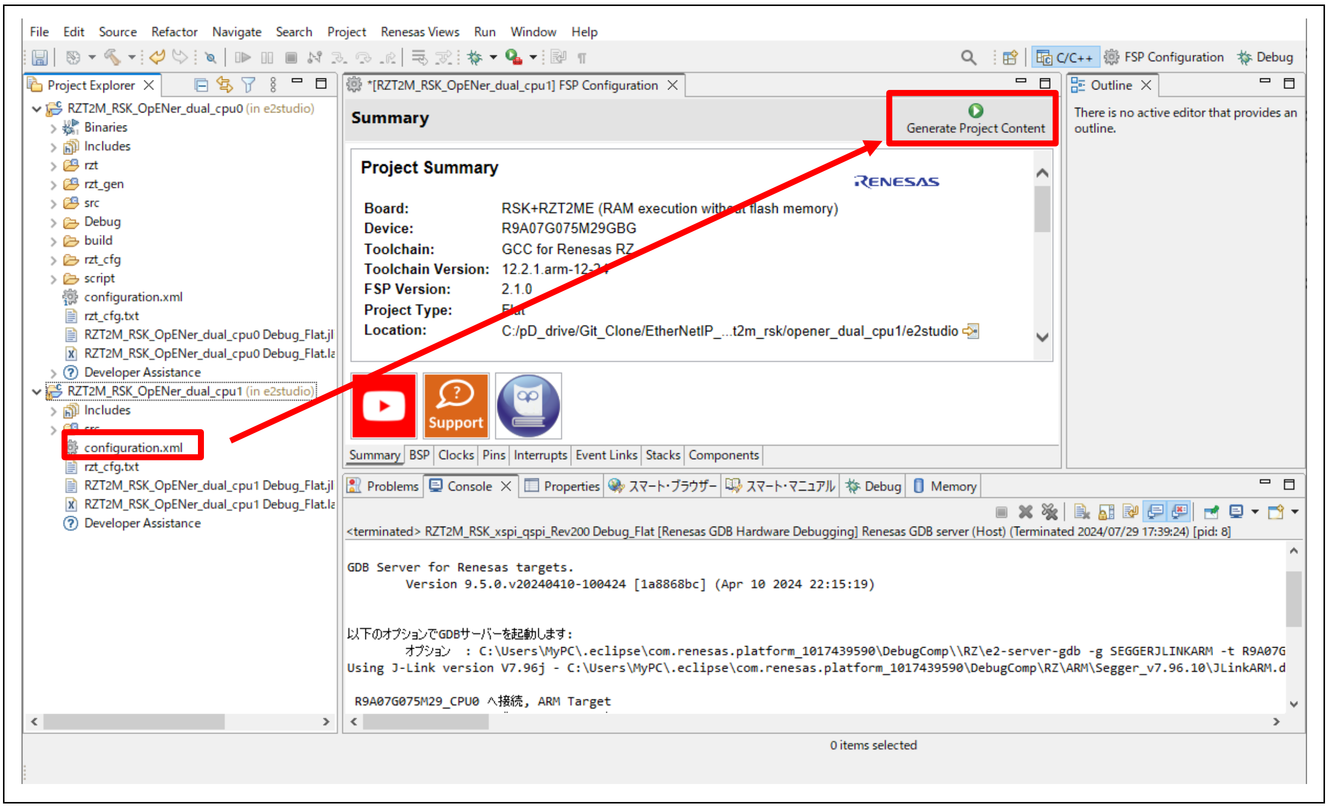


Figure 6-59 Generate folders for the CPU1 project

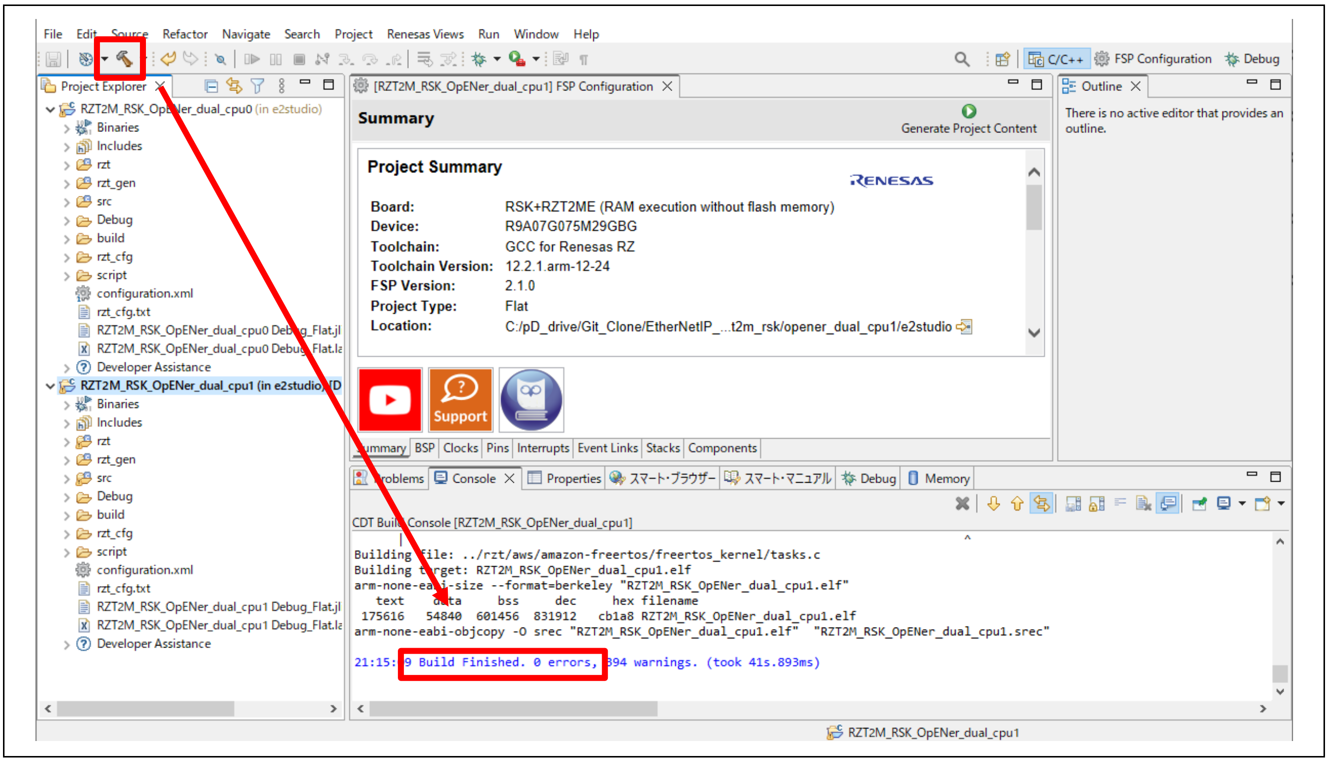


Figure 6-60 Build the CPU1 project

(2) Download application and run debugger

A) (For FSP for Renesas RZ/T series v2.1.0) Change Target Device to the CPU1 device as following steps.

a-1: Click “Debug Configurations” in the “Run” tab.

a-2: Click “RZT2M_RSK_OpENer_dual_cpu1 Debug_Flat” -> Debugger

a-3: Select the CR52_1 “R9A07G075M29_CR52_1” as Target Device.

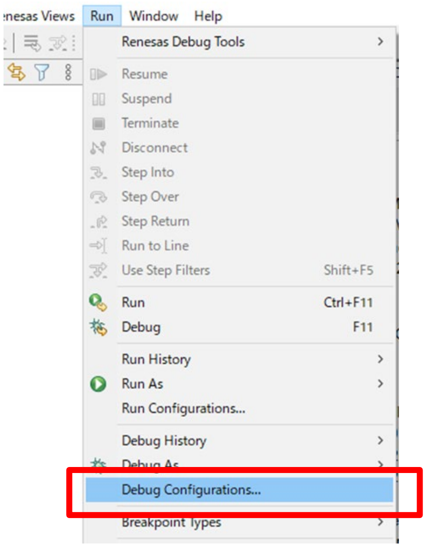


Figure 6-61 Open Debug Configuration

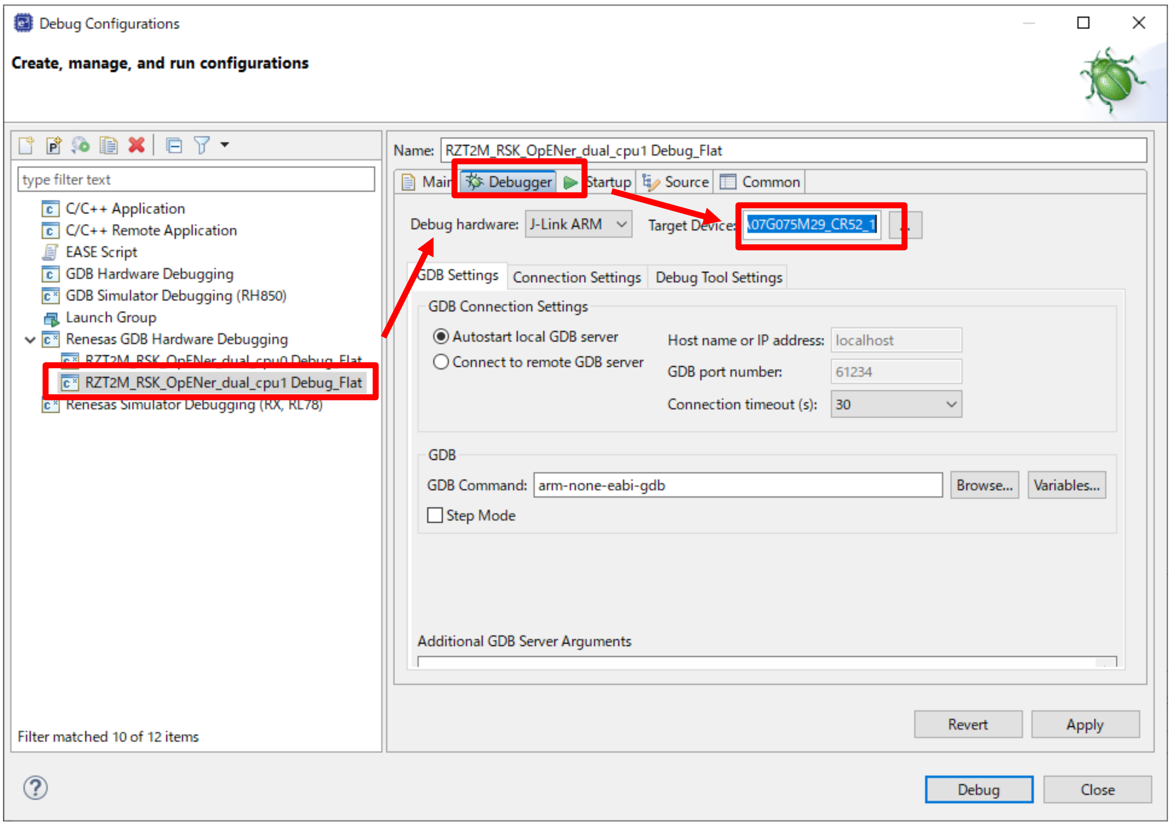


Figure 6-62 Change Target Device in Debug Configuration

B) Select the CPU0 project and click the Debug button to download the CPU0 program.

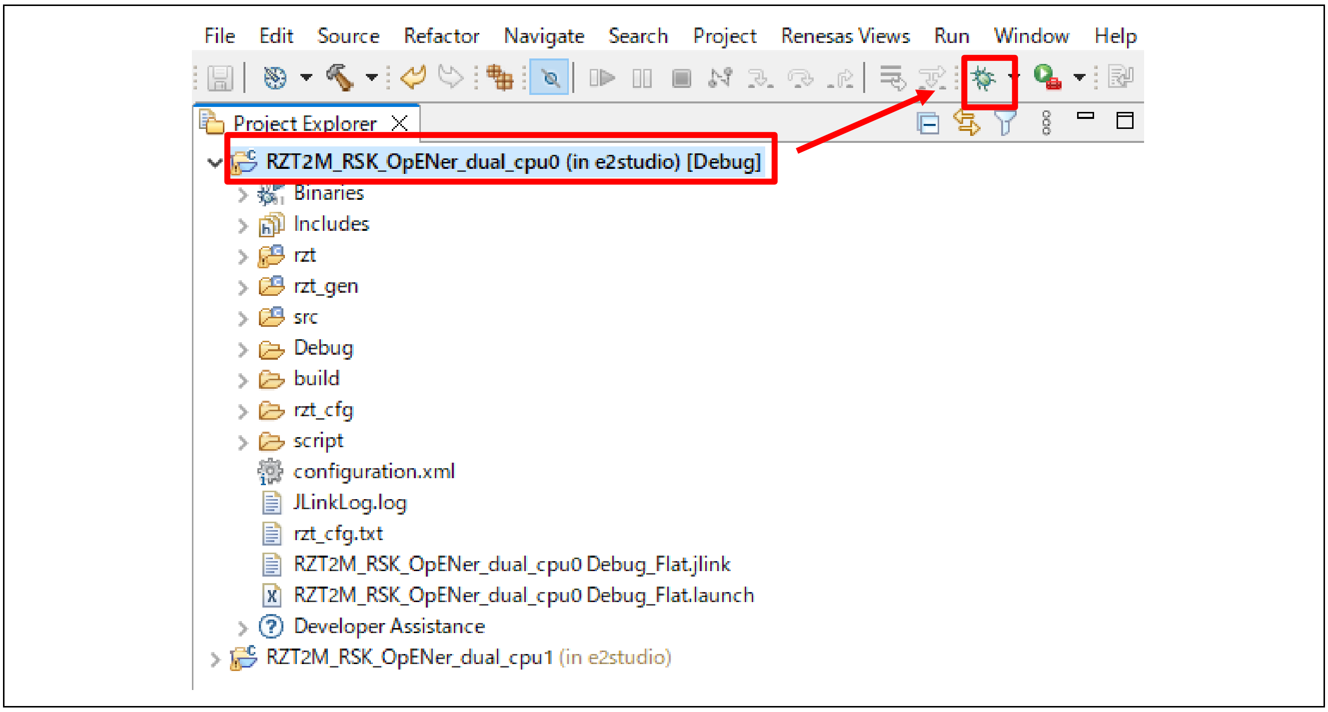


Figure 6-63 Debug button, the CPU0 project

C) Click the "No" button on the Perspective Switch dialog box.

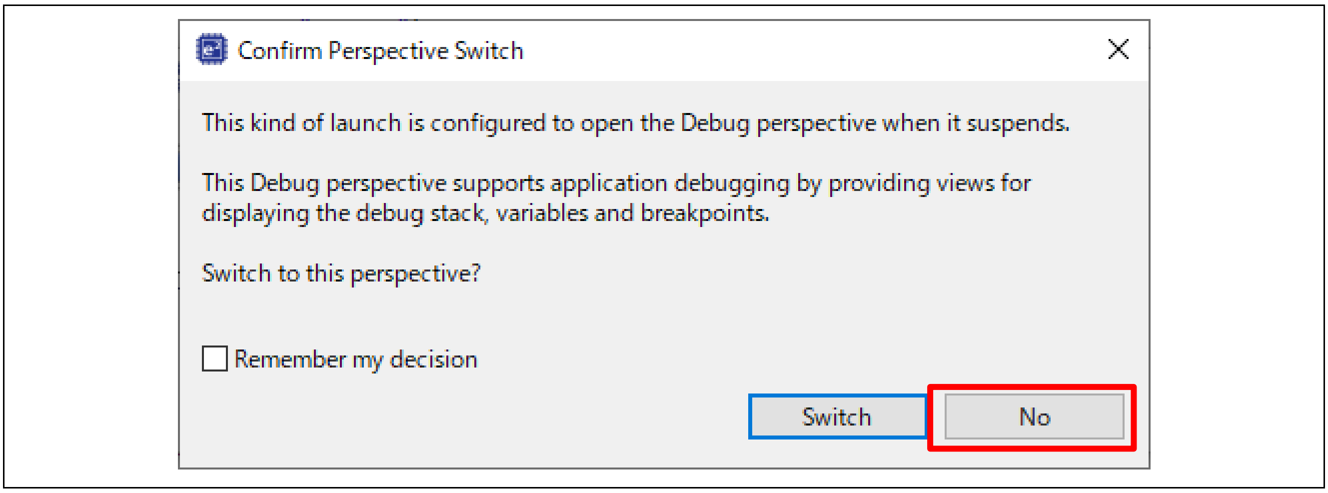


Figure 6-64 Confirm Perspective Switch

D) Select the CPU1 project and click the Debug button to download the CPU1 program.

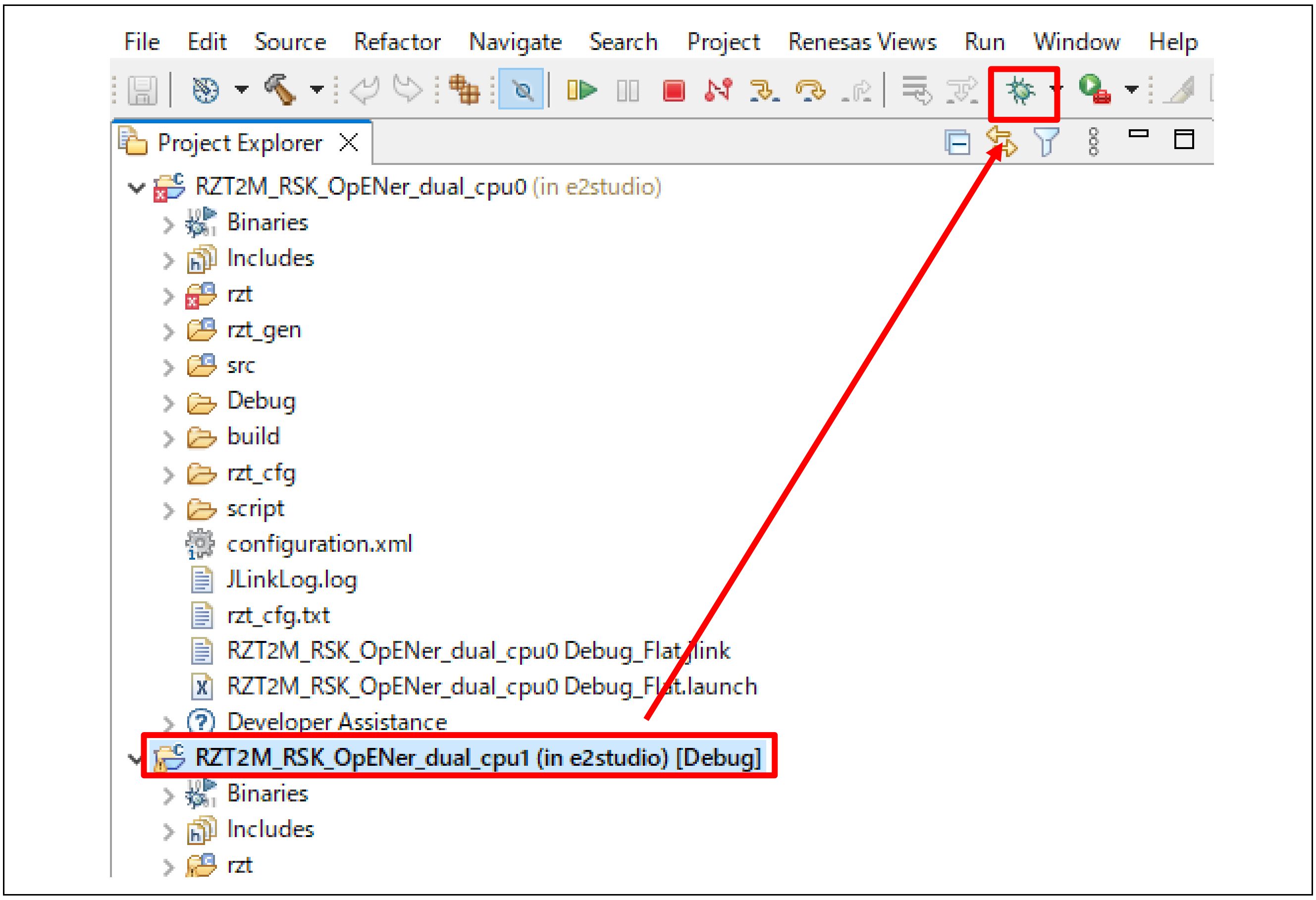


Figure 6-65 Debug button, the CPU1 project

E) Click the “No” button on the Launcher dialog box.

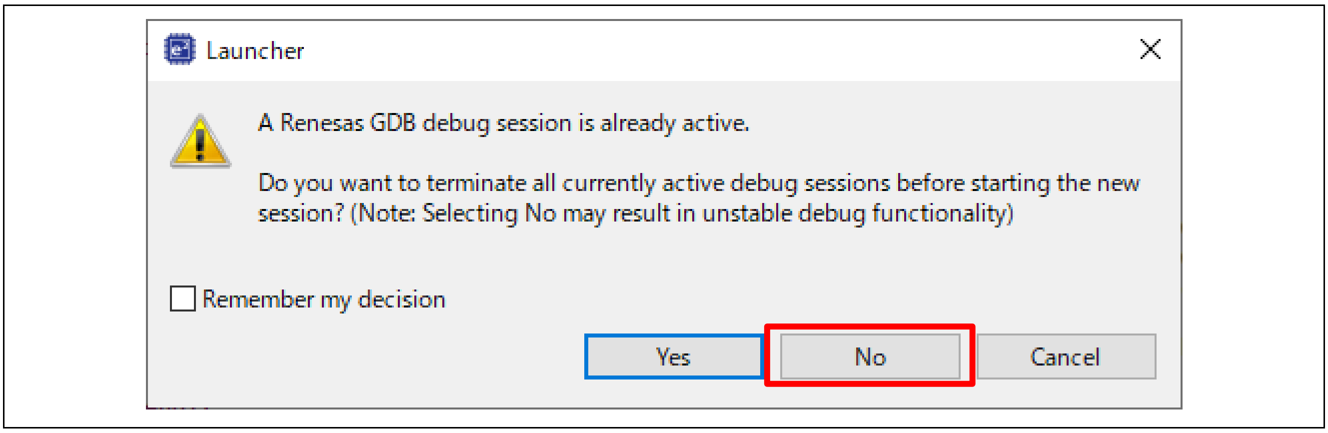


Figure 6-66 Launcher dialog box

F) Click to “Yes” button on the Proceed confirmation dialog box.

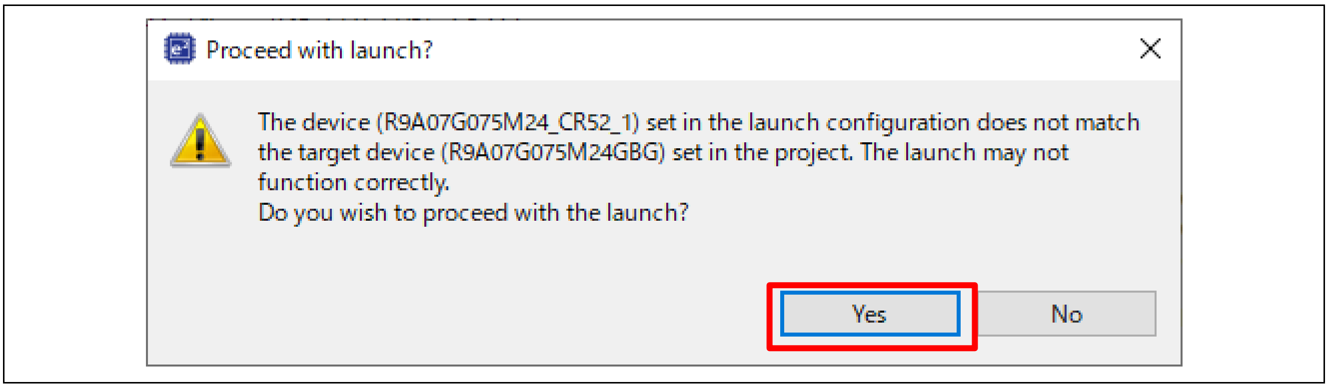


Figure 6-67 Proceed confirmation dialog box

G) Click to “Yes” button on the Perspective Switch dialog box.

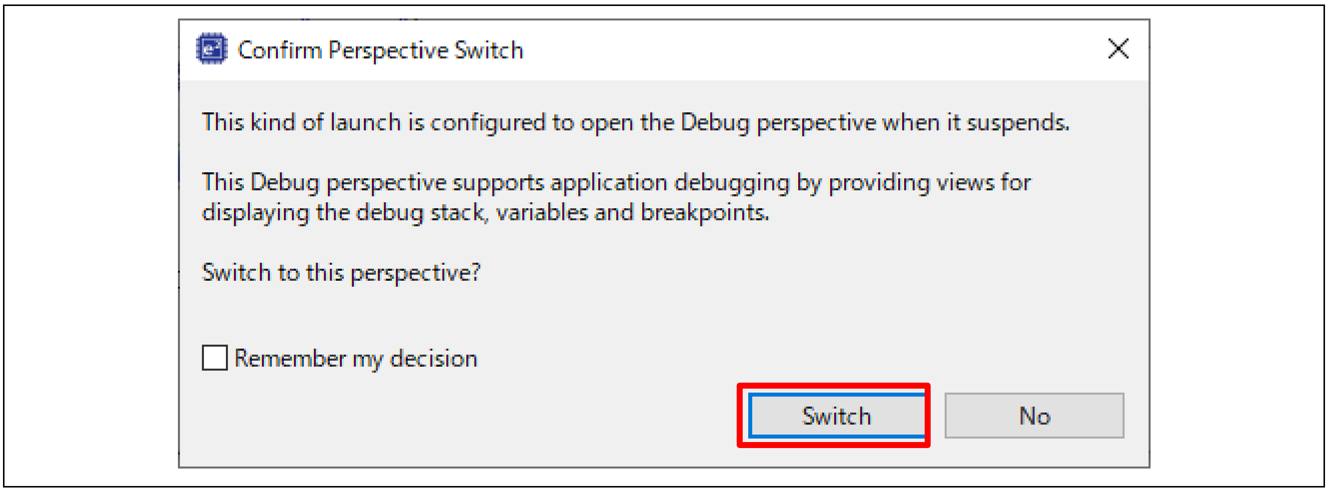


Figure 6-68 Confirm Perspective Switch

H) Run the CPU0 project.

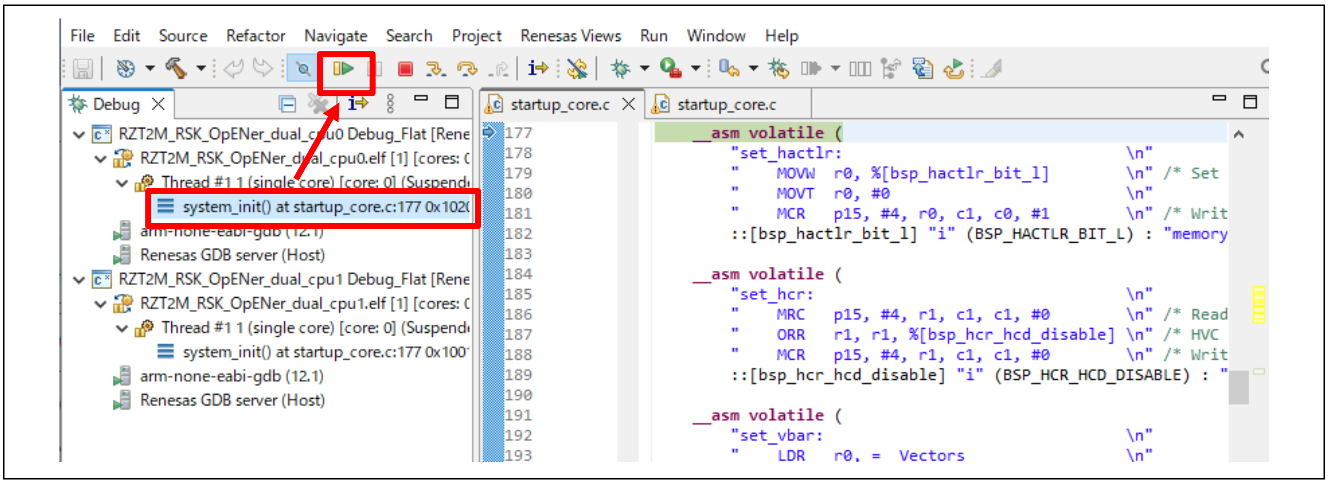


Figure 6-69 Run the CPU0 project

I) Run the CPU1 project.

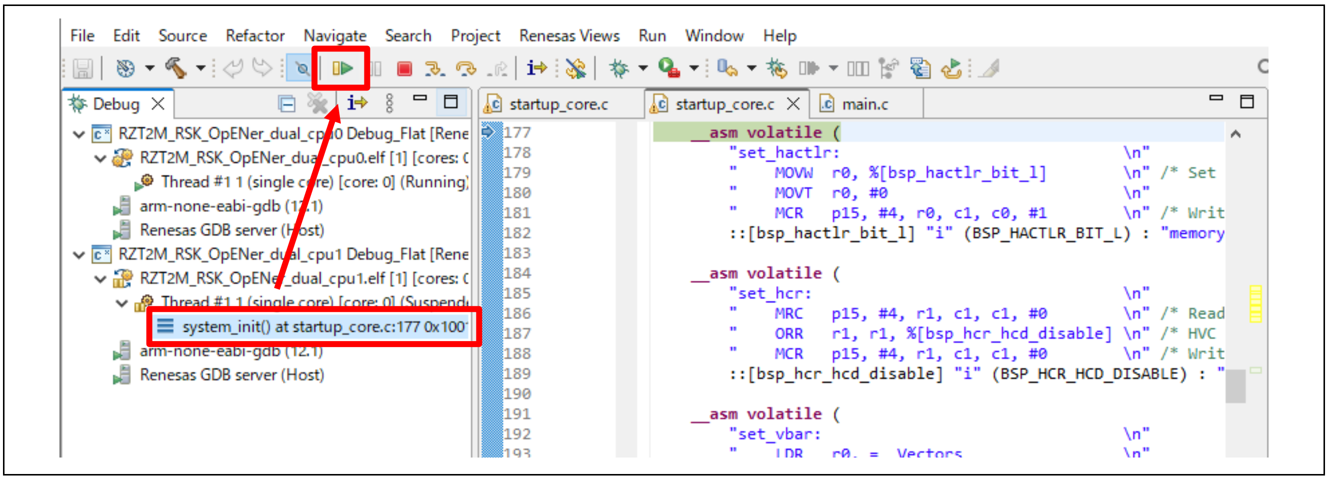


Figure 6-70 Run the CPU1 project

6.3.1.3 Flash boot mode

(1) How to generate source code

A) Click the configuration.xml in the CPU0 project.

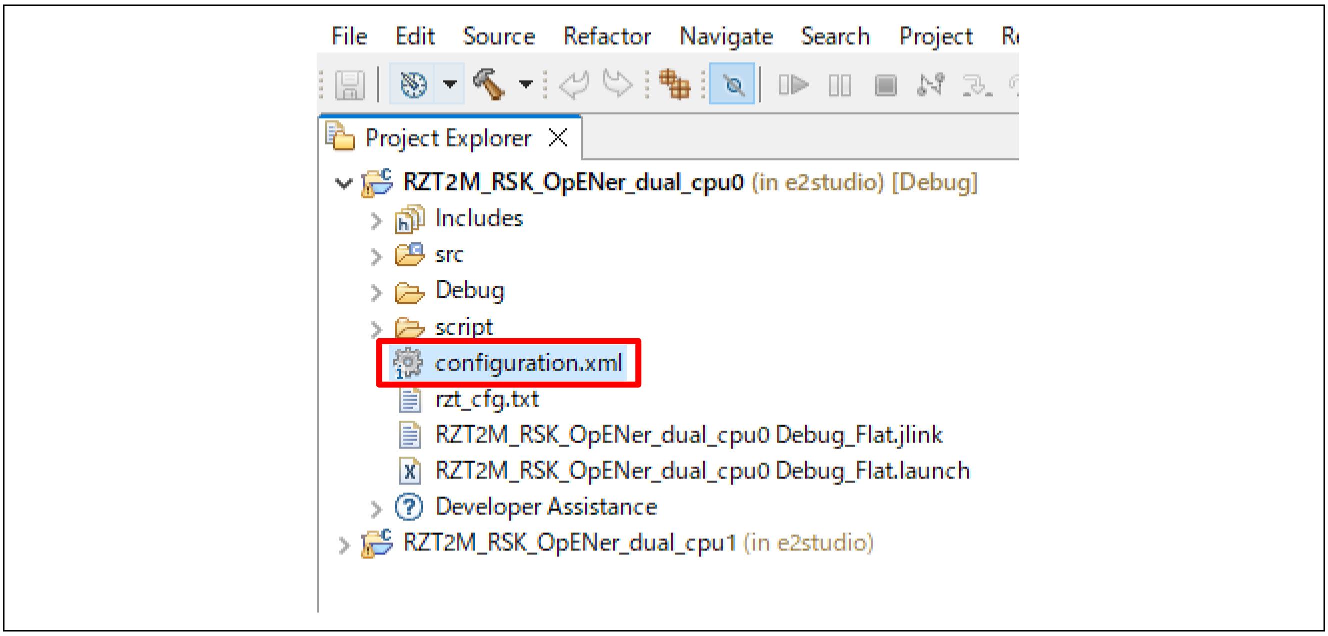


Figure 6-71 Configuration

B) Change the Boot Mode according to "Appendix. How to Change Boot Mode of FSP Project" in "RZ/T2, RZ/N2 Getting Started with Flexible Software Package (r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf)". When selecting the boot mode on the BSP tab, select the following settings.

- ➢ RZ/T2ME
  - ・Board : RSK+RZT2ME (xSPI0 x1 boot mode)

C) After changing the boot mode, go to the "Pins" tab and change the settings of the following pins from "Output mode (Low & Not into Input)" to "Output mode (Low & Into Input)".

Table 6-6 Port number of Pin to be changed in RZ/T2ME

Port Number			
P08_2	P19_3	P19_4	P19_6
P19_7	P20_0	P20_1	P20_2
P20_3	P20_4	P20_7	P21_0
P23_4	-	-	-

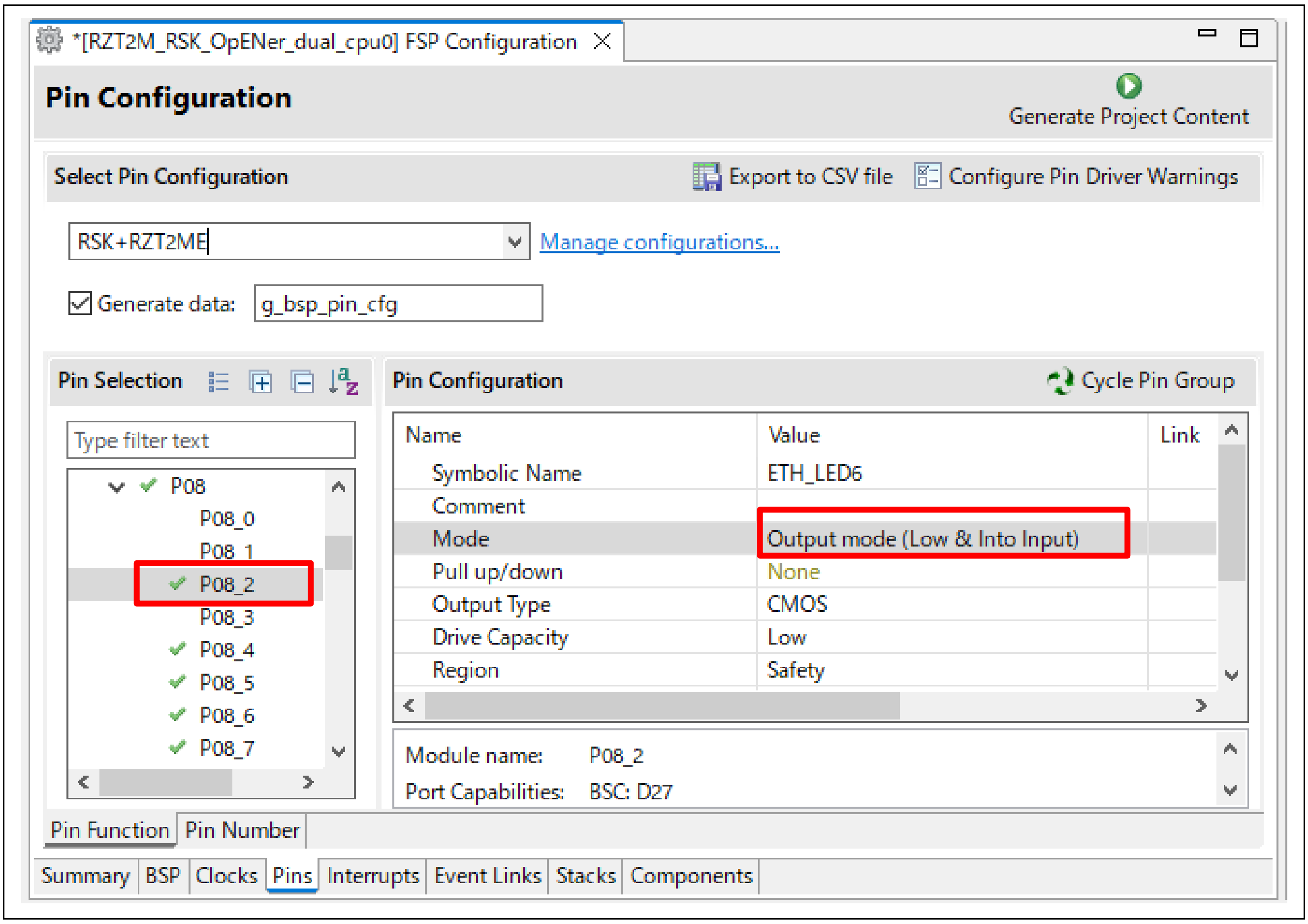


Figure 6-72 Change pin settings

D) Set clock settings in the “Clock” tab as a following table.

- CPU1CLK : Mul x4
- CKIO : Div /2

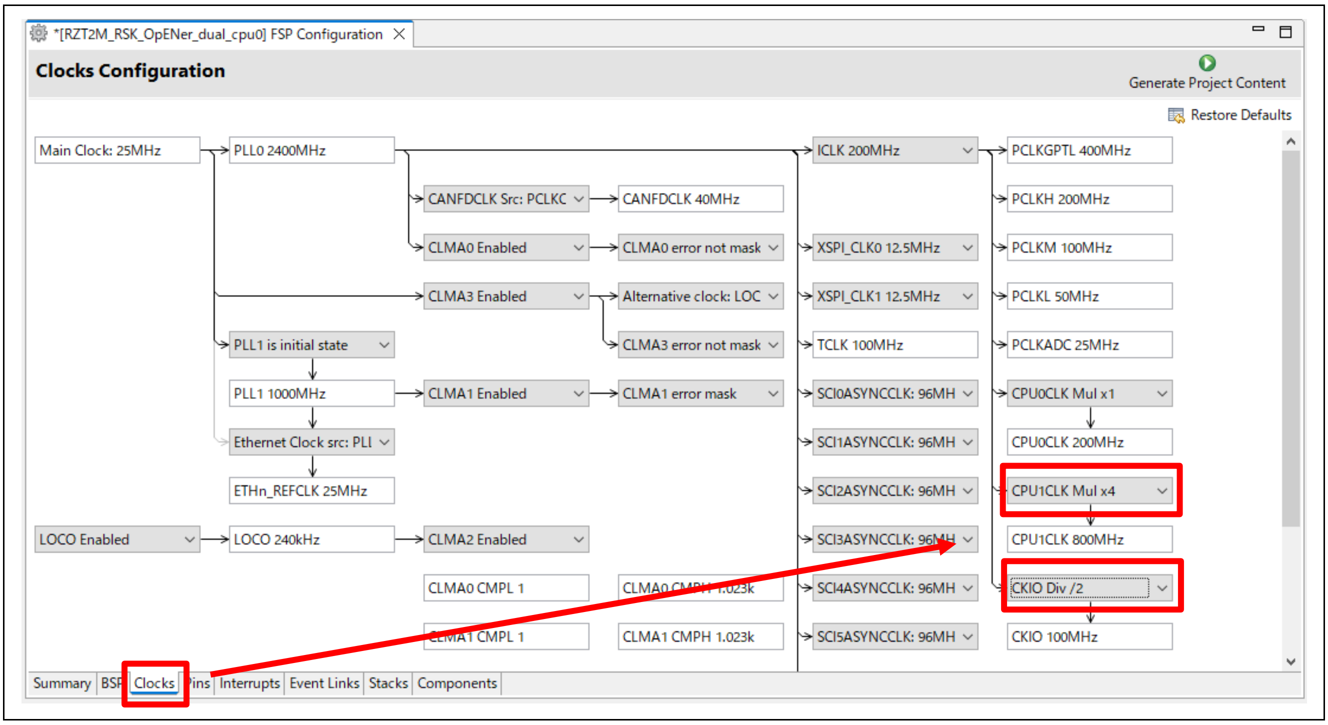


Table 6-7 Set clock settings

E) Click “Generate Project Content” button.

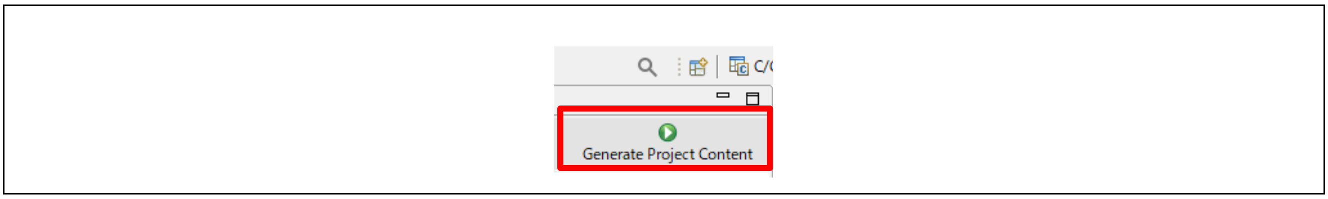


Figure 6-73 Generate Project Content, the CPU0 project

F) Build the CPU0 project.

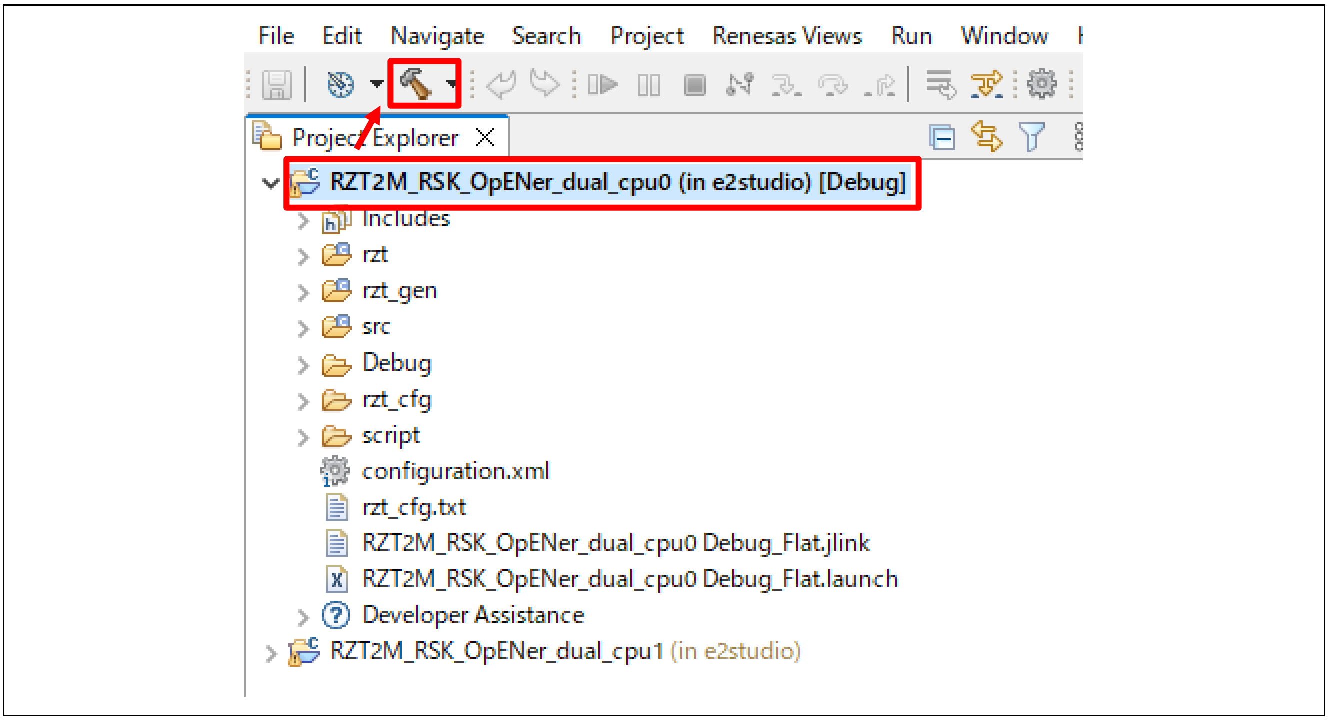


Figure 6-74 Build the CPU0 project

G) Click the configuration.xml in the CPU1 project and click “Generate Project Content” button.

Note: You can ignore Problem Occurred dialogs with the error message “Cannot invoke "java.util.Map.entrySet()" because "securePinMap" is null”.

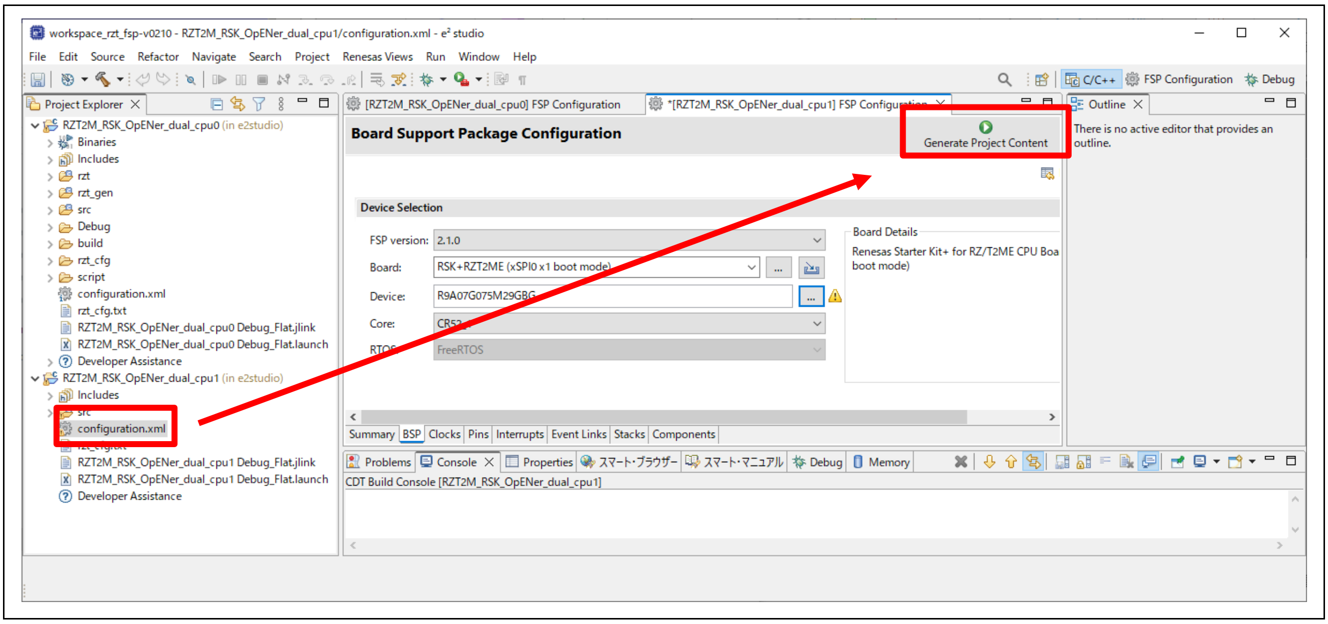


Figure 6-75 Generate Project Content, the CPU1 project

(2) Build and writing the projects to flash

A) Write the project according to “Appendix. How to Create and Debug FSP Projects for Multiprocessing in All Cases for e[2] studio” in “RZ/T2, RZ/N2 Getting Started with Flexible Software Package (r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf)”.

B) After writing the projects, restart the device by turning the power on/off.

Note:

- Be aware of the limitations when rewriting the program. Please refer to the section 1.3.
- If necessary to enable Type-0 Reset and Type-1 Reset, modify the macro definition “`SAMPLE_EXECUTE_SYSTEM_RESET_PROCESS`” (opener_port_main.c) to “1” enable the Reset process.

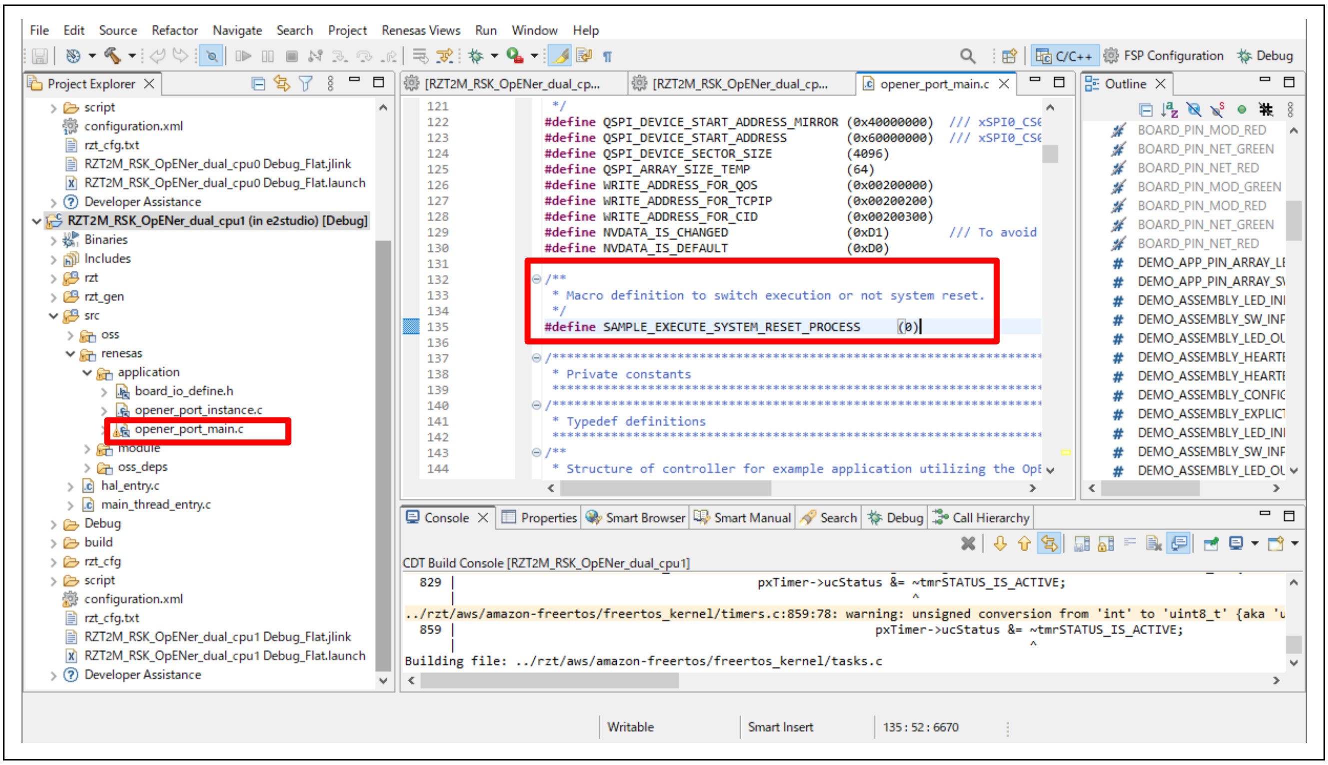


Figure 6-76 Reset process macro definition

6.3.2 Setup sample project for EWARM

The setup differs depending on the boot mode.

- When executing the sample program with RAM, refer to section 6.3.2.2.
- When executing the sample program with flash boot mode, refer to section6.3.2.3.

6.3.2.1 Startup EWARM

1. Open the EWARM.
2. Click “Open Workspace…” in the File tab.

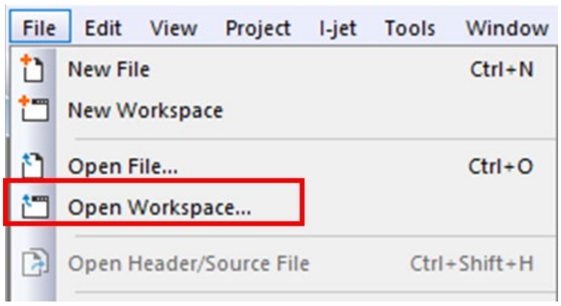


Figure 6-77 EWARM file tab

3. Select the CPU0 Workspace File(.eww) and click the Open button.
   “\project\rzt2m_rsk\opener_dual_cpu0\ewarm\RZT2M_RSK_OpENer_dual_cpu0.eww”

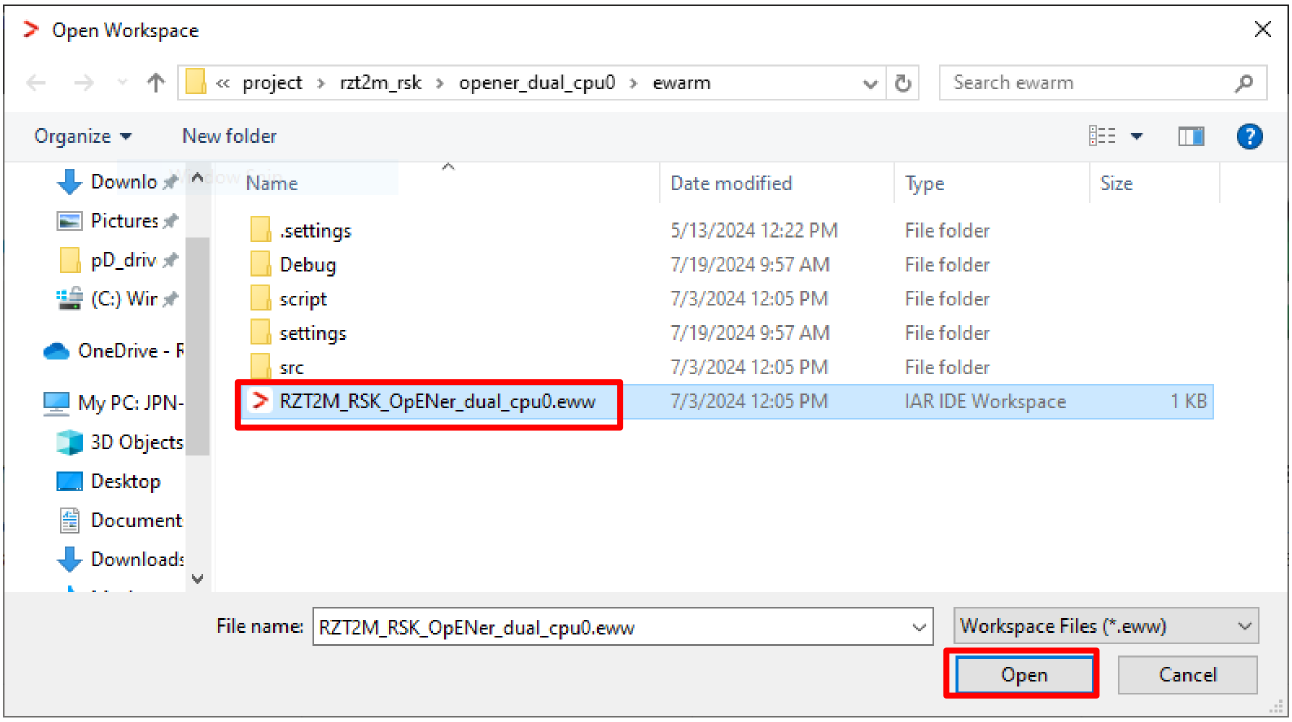


Figure 6-78 Open the CPU0 Workspace

4. Open another EWARM window.

5. Click "Open Workspace…" in the File tab, and open the CPU1 Workspace File(.eww), same as the CPU0 Workspace.

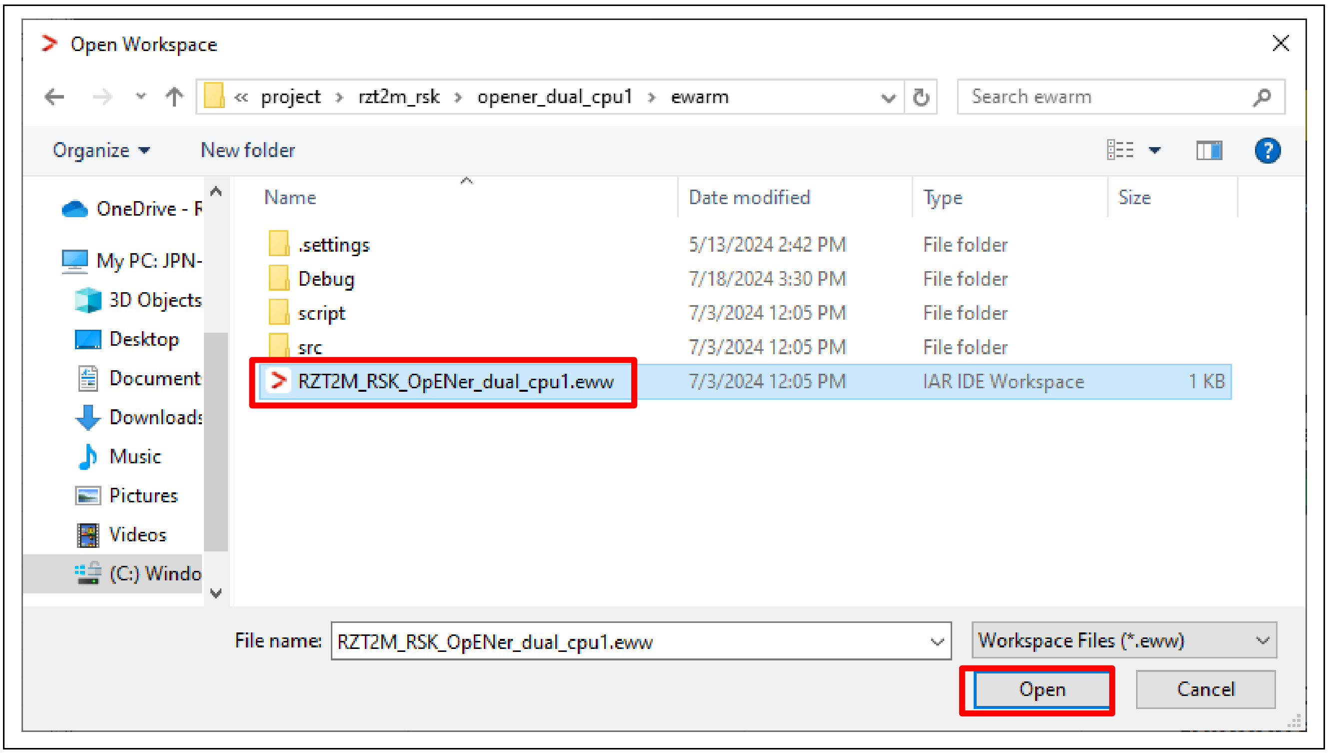


Figure 6-79 Open the CPU1 Workspace

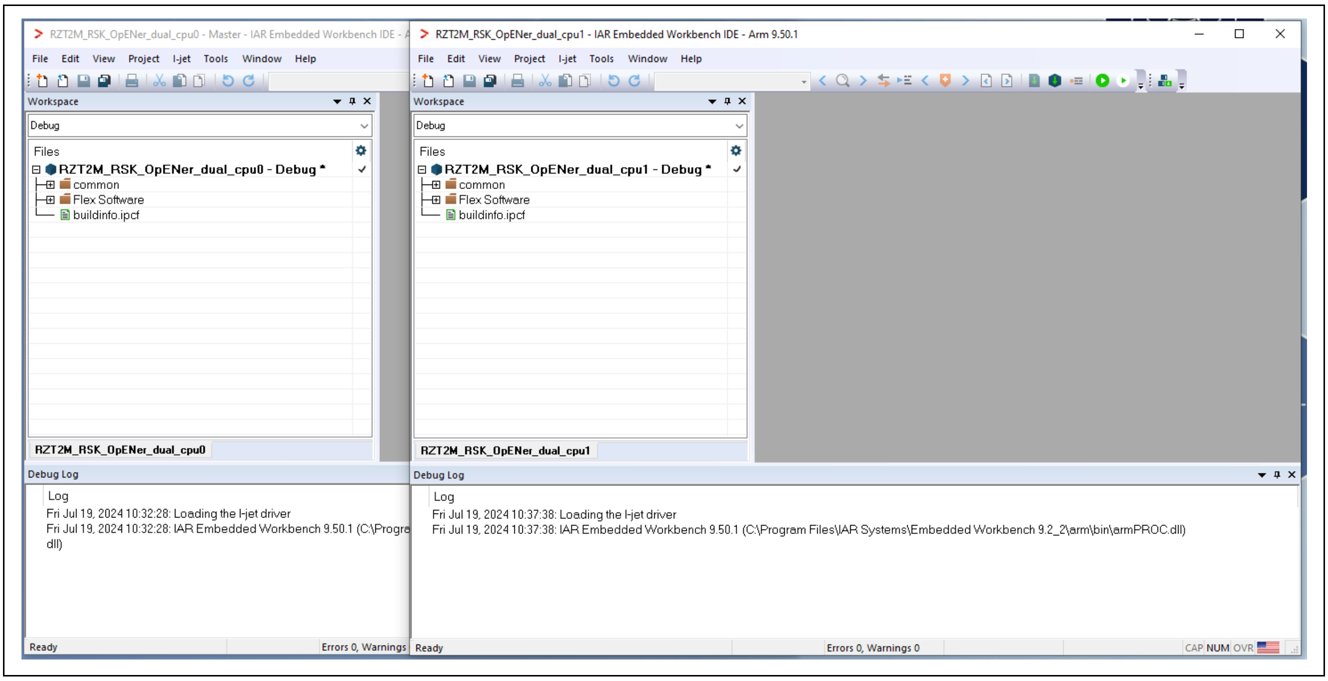


Figure 6-80 EWARM windows

6.3.2.2 RAM execution

(1) How to generate source code and how to build

A) Open FSP Smart Configurator (v.2.1.0 or later version)

B) Click “Open” in the “File” tab.

Note: For V.2.1.0 or later versions of FSP, it must be opened once from FSP Smart Configurator window.

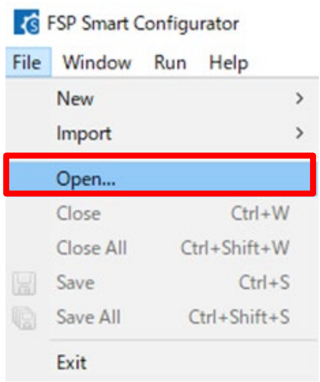


Figure 6-81 Click Open in the File tab

C) Open the configuration.xml of the CPU0 project.

“\project\rzt2m_rsk\opener_dual_cpu0\ewarm\configuration.xml”

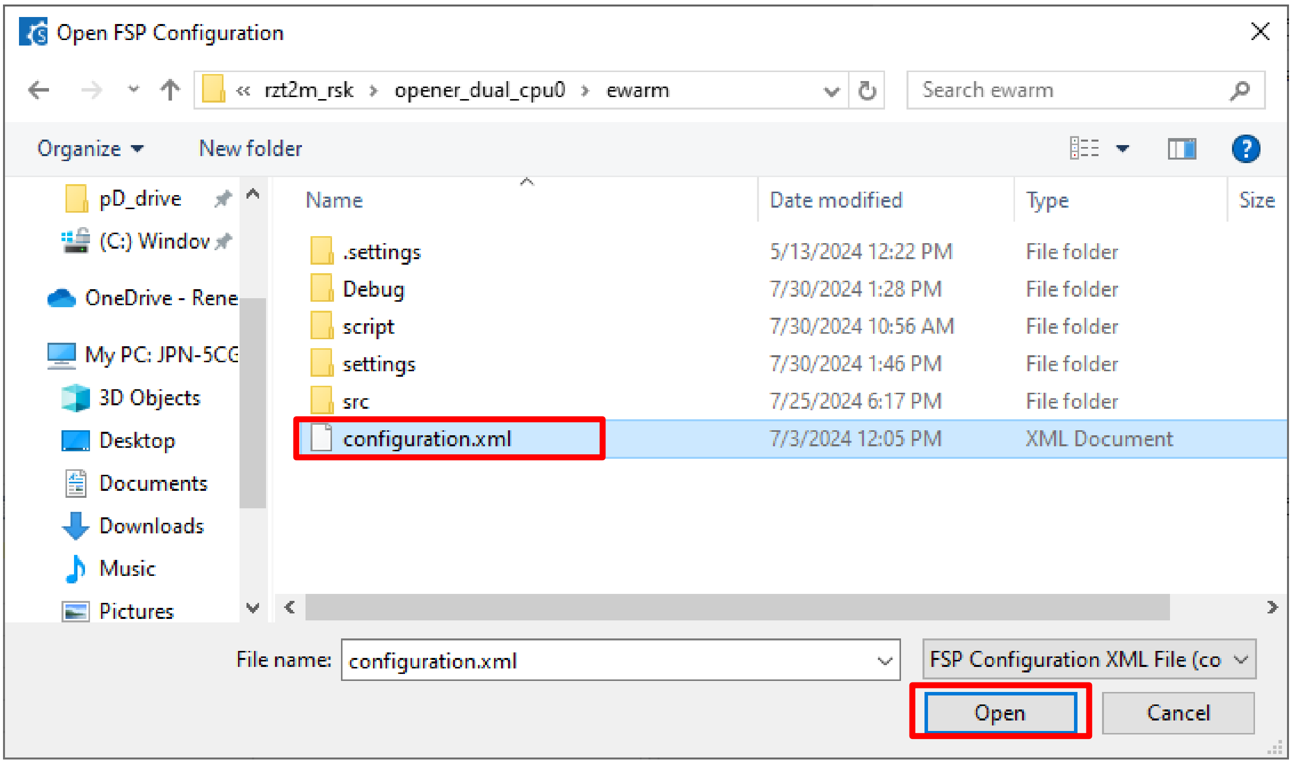


Figure 6-82 Open configuration.xml of the CPU0 project.

D) Click the “BSP” tab, and change the Board setting.

➢ RZ/T2ME

・Board : RSK+RZT2ME (RAM execution without flash memory)

Note: You can ignore the warning beside the “Device” setting.

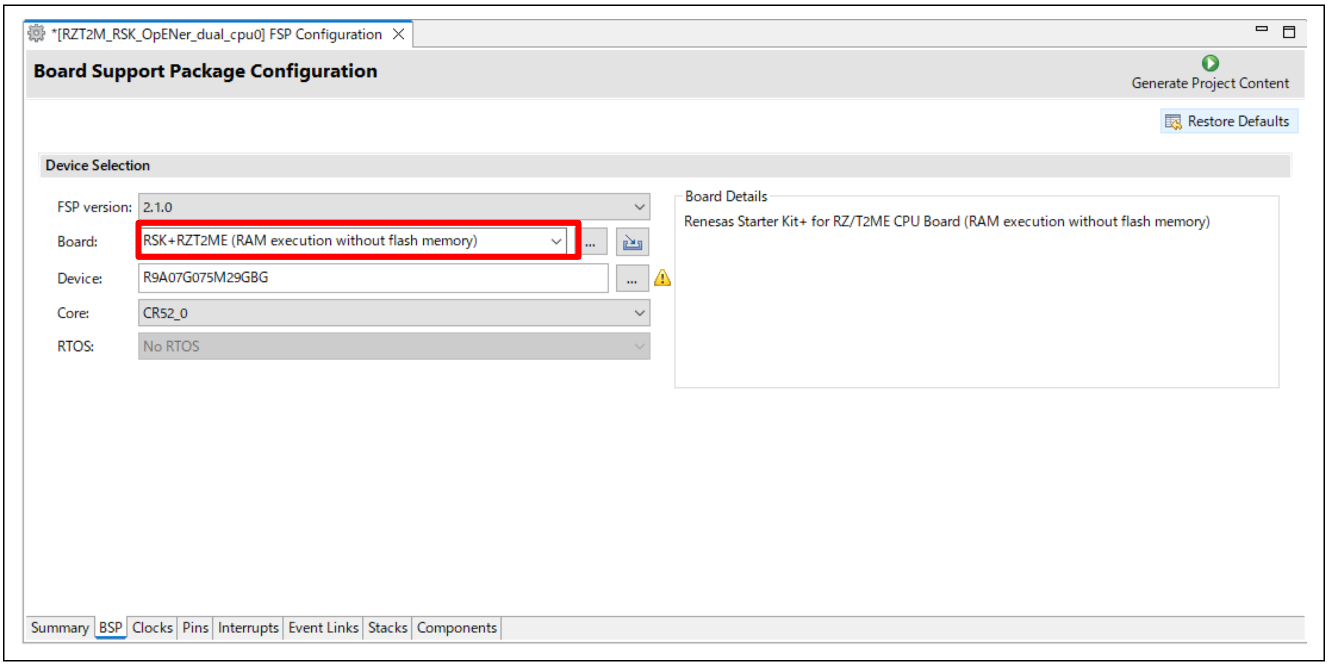


Table 6-8 Change the board in the BSP tab

E) Reselect “FSP Version” from the drop-down list.

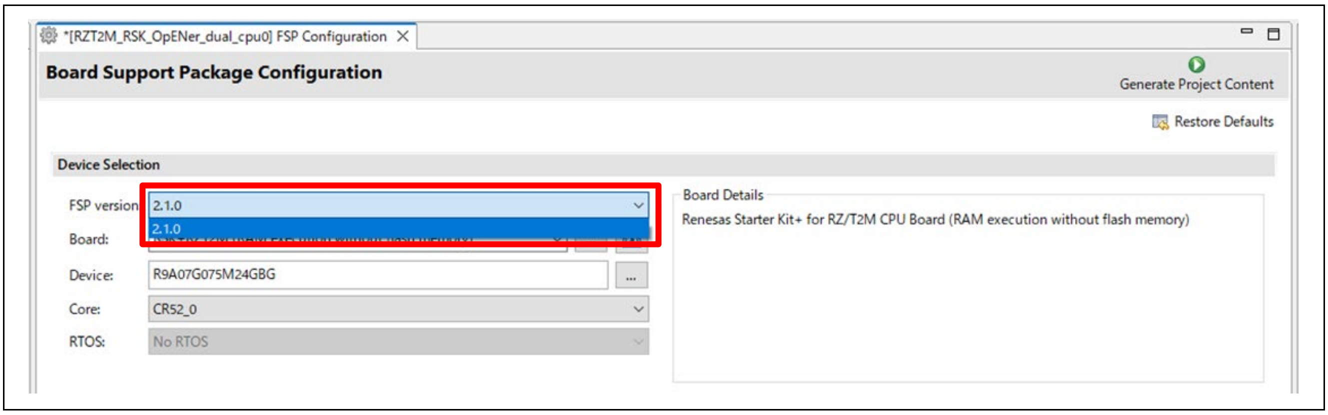


Table 6-9 Reselect “FSP Version” from the drop-down list

F) Set clock settings in the "Clock" tab as a following table.

- CPU1CLK : Mul x4
- CKIO : Div /2

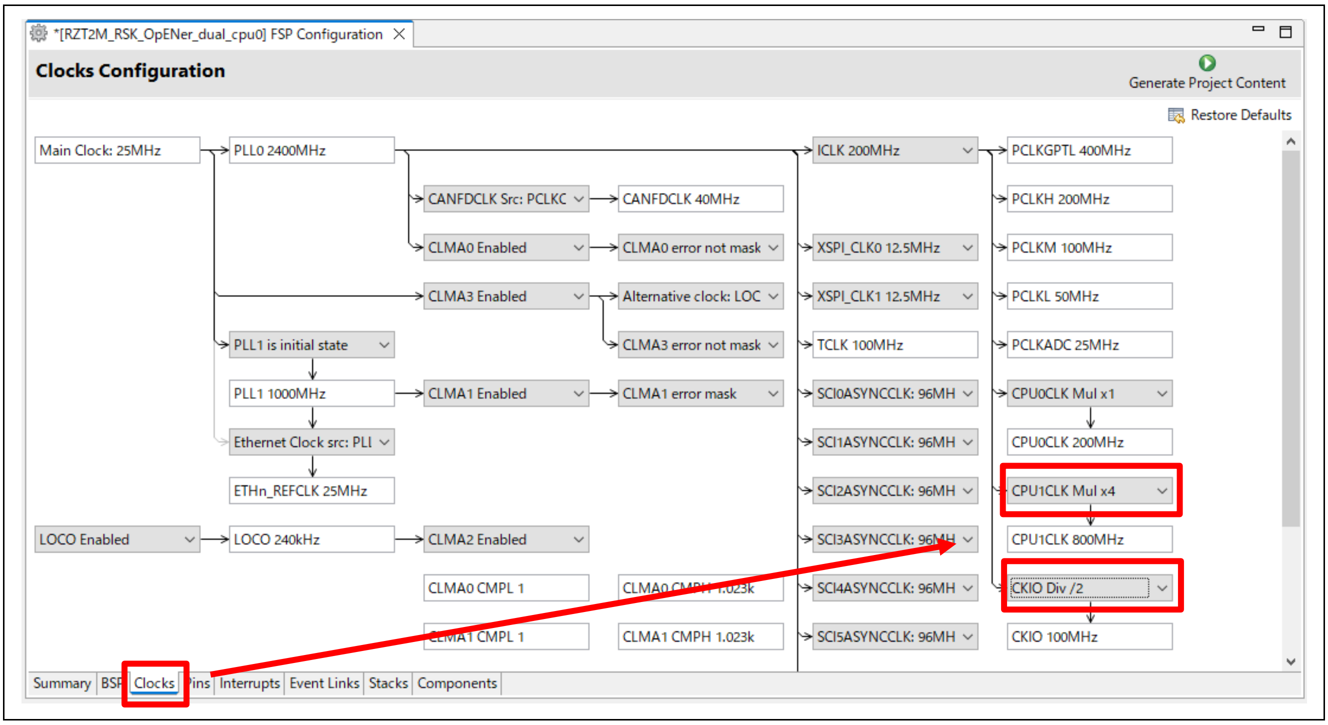


Table 6-10 Set clock settings

G) Click "Generate Project Content" button then generate rzt, rzt_gen, rzt_cfg folder.

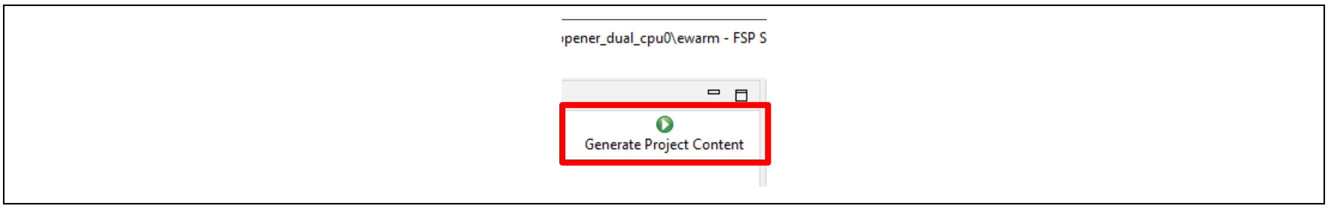


Figure 6-83 Generate Project Content

H) Click the Make button to build the CPU0 project.

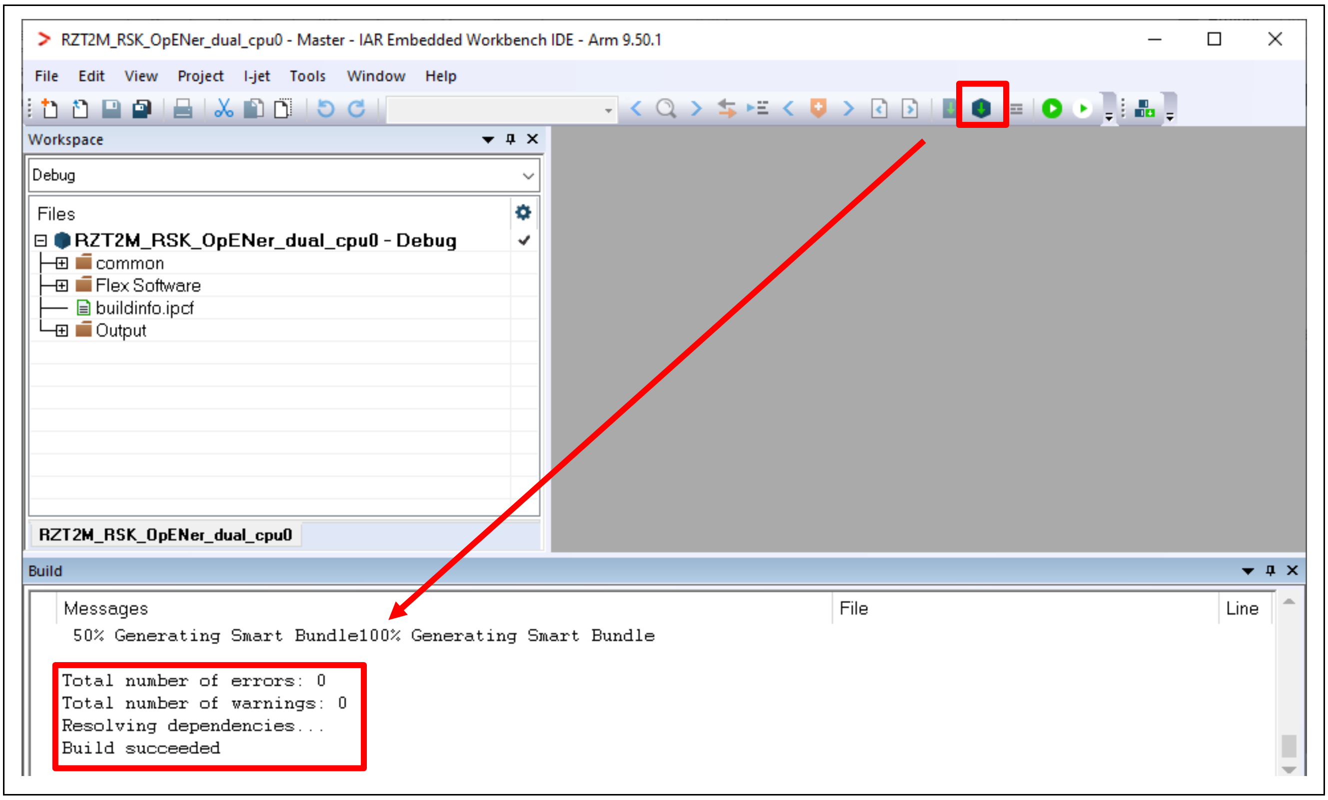


Figure 6-84 Make button on the CPU0 Workspace

I) Open another FSP Smart Configurator window

J) Click "Open" in the "File" tab.

Note: For v2.1.0 or later versions of FSP, it must be opened once from FSP Smart Configurator window.

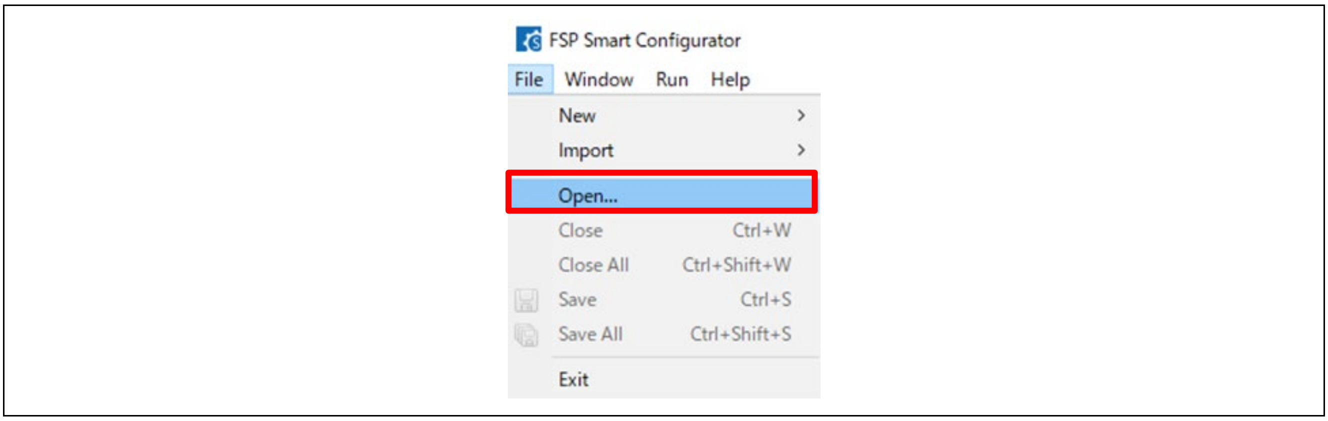


Figure 6-85 Click Open in the File tab

K) Open the configuration.xml of the CPU1 project.

“\project\rzt2m_rsk\opener_dual_cpu1\ewarm\configuration.xml”

Note: You can ignore Problem Occurred dialogs with the error message “Cannot invoke "java.util.Map.entrySet()" because "securePinMap" is null”.

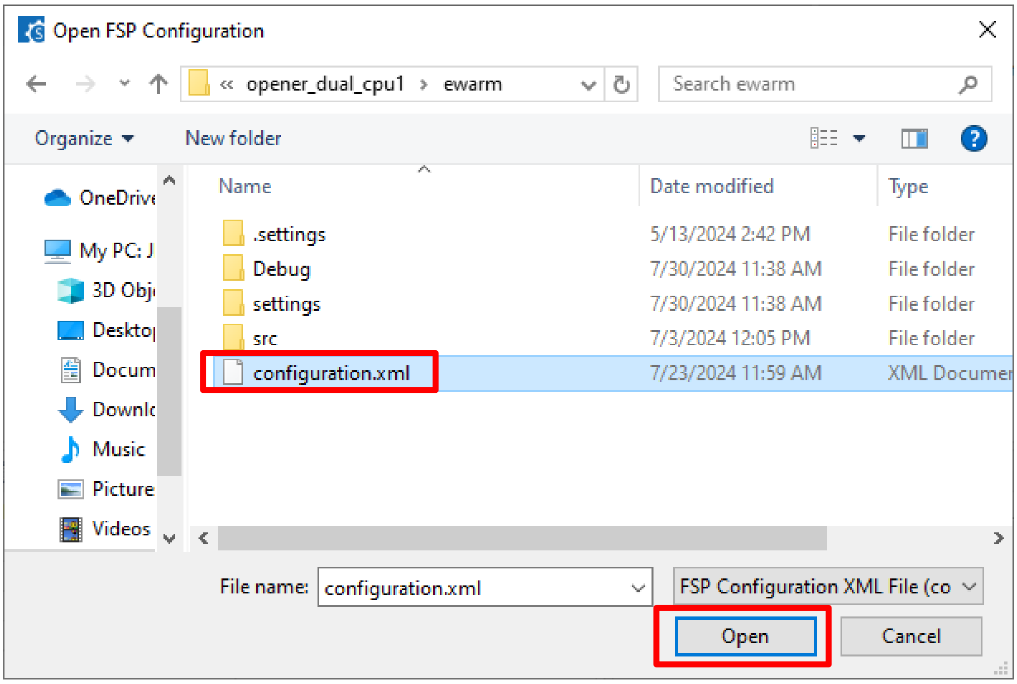


Figure 6-86 Open configuration.xml of the CPU1 project.

L) Click “Generate Project Content” button then generate rzt, rzt_gen, rzt_cfg folder.

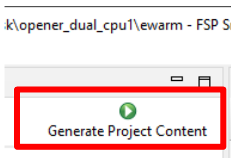


Figure 6-87 Generate Project Content

M) Add a command line option to the CPU1 project property according to No.11 EWARM 9.60.1 RAM execution limitation in the “Appendix. Tool Software Limitations” in “RZ/T2, RZ/N2 Getting Started with Flexible Software Package (r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf)”.

N) Click the Make button to build the CPU1 project. The build message is displayed that no errors and “Build succeeded”.

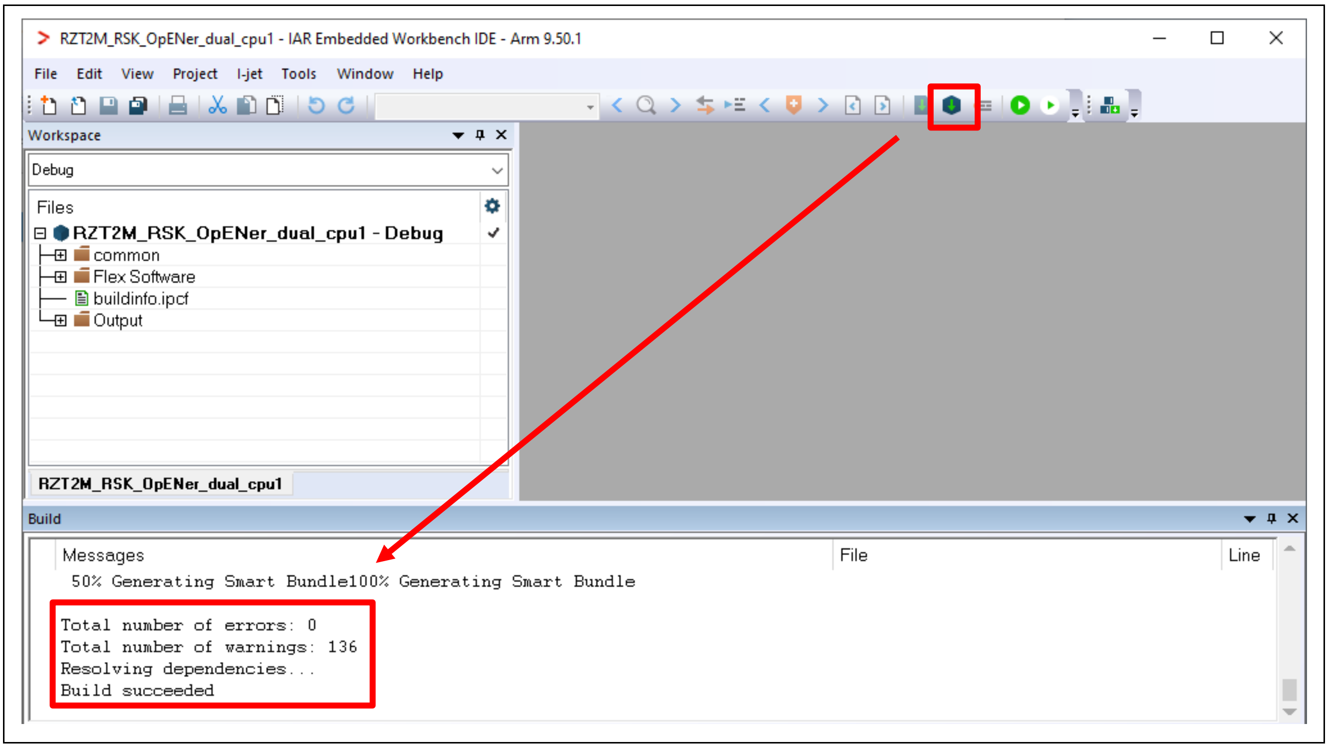


Figure 6-88 Make button on the CPU1 Workspace

O) Close the CPU1 Workspace.

(2) Download application and run debugger

A) Click the Debug button to download the CPU0 application program and launch the debugger. Then automatically the CPU1 Workspace will open.

Figure 6-89 Debug button

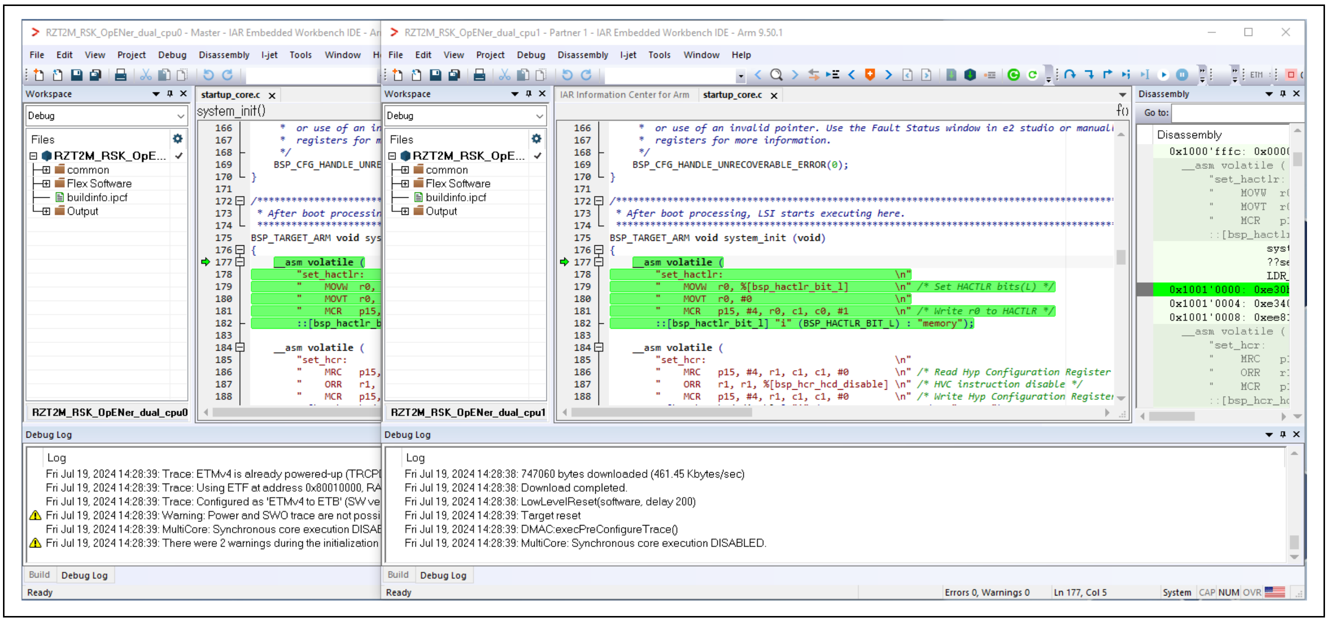


Figure 6-90 The CPU1 Workspace automatically opens

B) Click the Go button on the CPU0 Workspace.

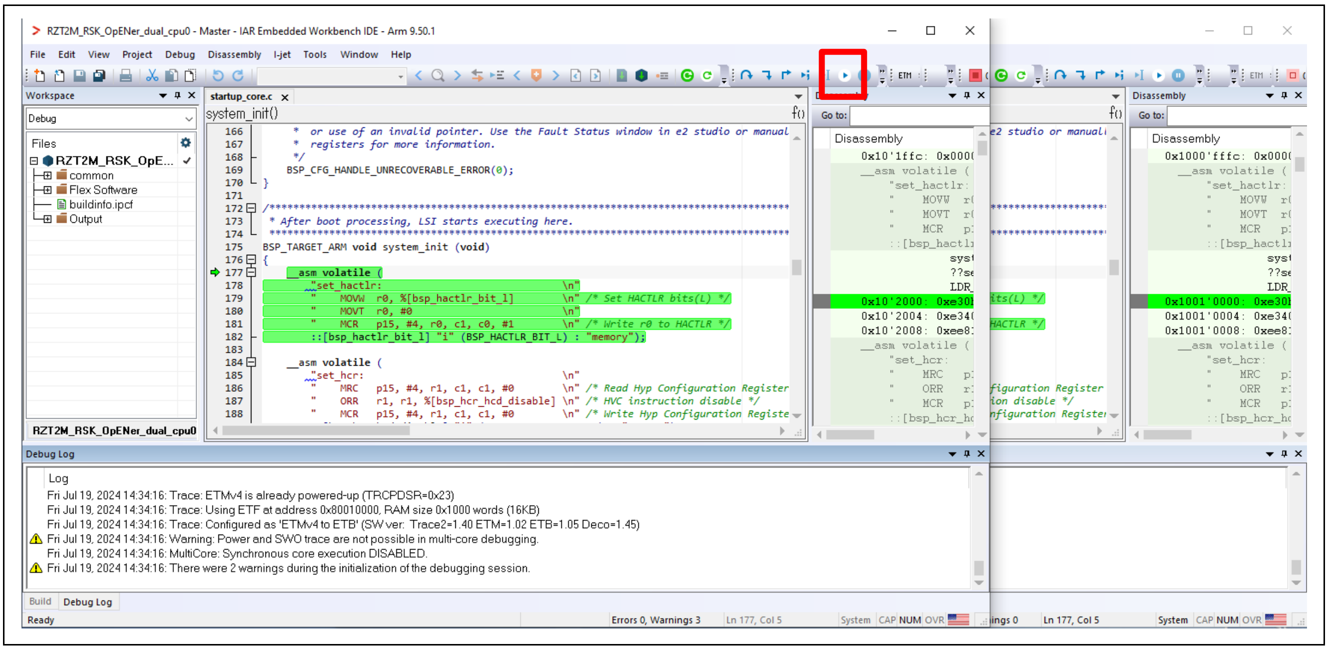


Figure 6-91 Go button on the CPU0 Workspace

C) Click the Go button on the CPU1 Workspace.

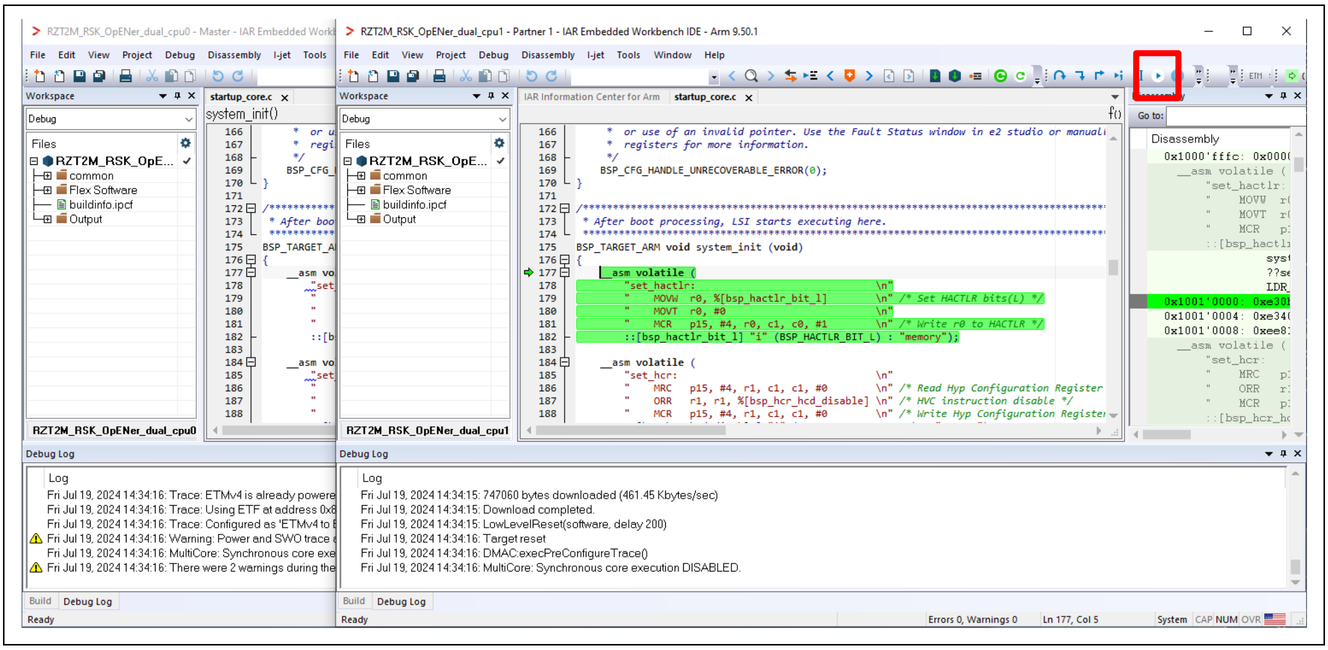


Figure 6-92 Go button on the CPU1 Workspace

6.3.2.3 Flash boot mode

(1) How to generate source code and how to build

A) Set the Asymmetric multicore setting to “Disabled” in the property of the CPU0 Workspace.

a-1. Click Project > Options….

a-2. Click Debugger > Multicore.

a-3. Select Disable in Asymmetric multicore.

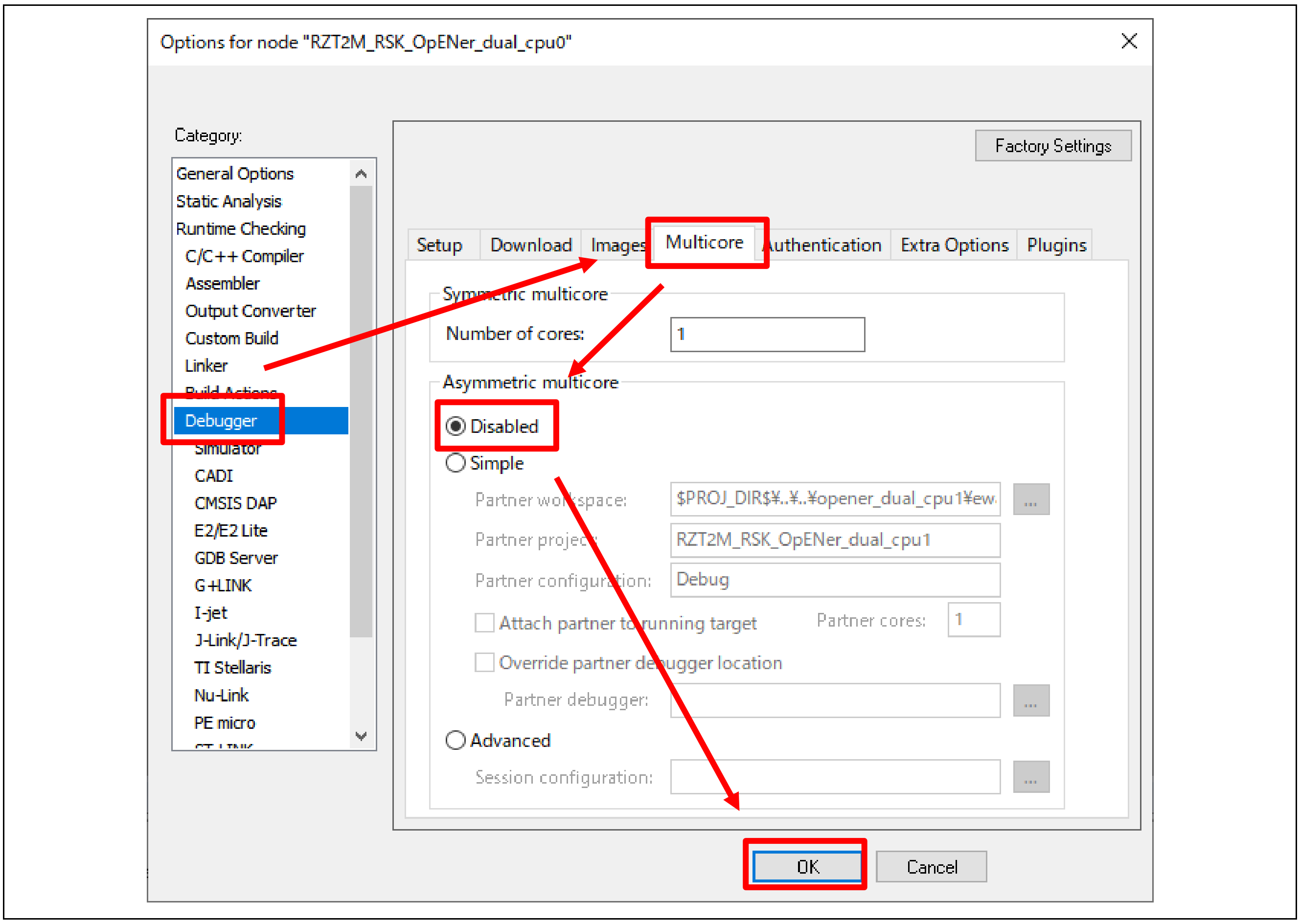


Figure 6-93 Disable multicore debug mode in CPU0 project property

B) Open FSP Smart Configurator (v.2.1.0 or later version)

C) Click “Open” in the “File” tab.

Note: For v2.1.0 or later versions of FSP, it must be opened once from FSP Smart Configurator window.

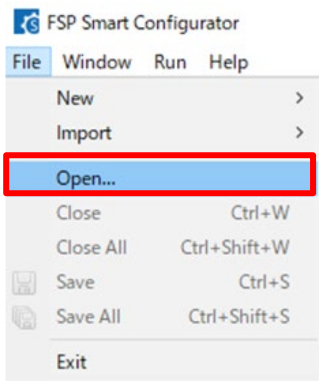


Figure 6-94 Click Open in the File tab

D) Open the configuration.xml of the CPU0 project.

“\project\rzt2m_rsk\opener_dual_cpu0\ewarm\configuration.xml”

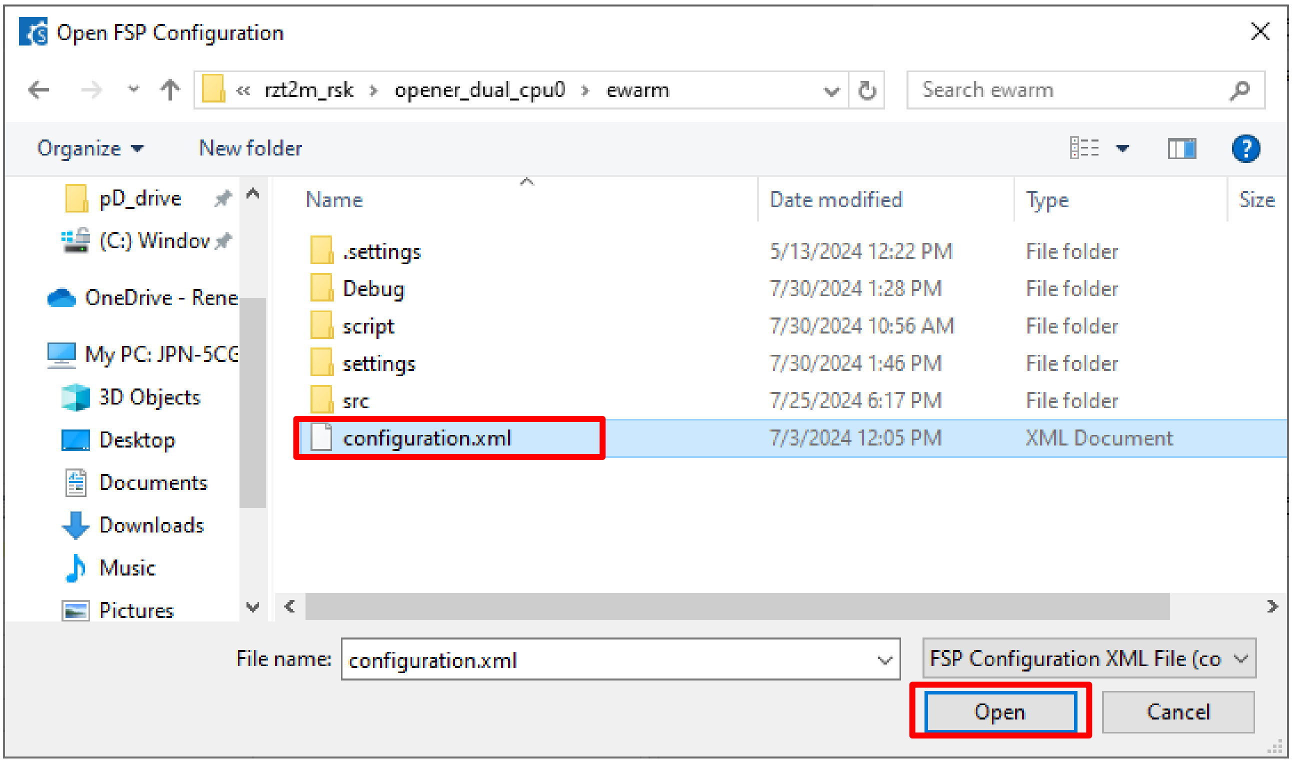


Figure 6-95 Open configuration.xml of the CPU0 project.

E) Change the Boot Mode according to “Appendix. How to Change Boot Mode of FSP Project” in “RZ/T2, RZ/N2 Getting Started with Flexible Software Package (r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf)”. When selecting the boot mode on the BSP tab, select the following settings.

➢ RZ/T2ME
・Board : RSK+RZT2ME (xSPI0 x1 boot mode)

F) After changing the boot mode, go to the "Pins" tab and change the settings of the following pins from "Output mode (Low & Not into Input)" to "Output mode (Low & Into Input)".

Table 6-11 Port number of Pin to be changed in RZ/T2ME

Port Number			
P08_2	P19_3	P19_4	P19_6
P19_7	P20_0	P20_1	P20_2
P20_3	P20_4	P20_7	P21_0
P23_4	-	-	-

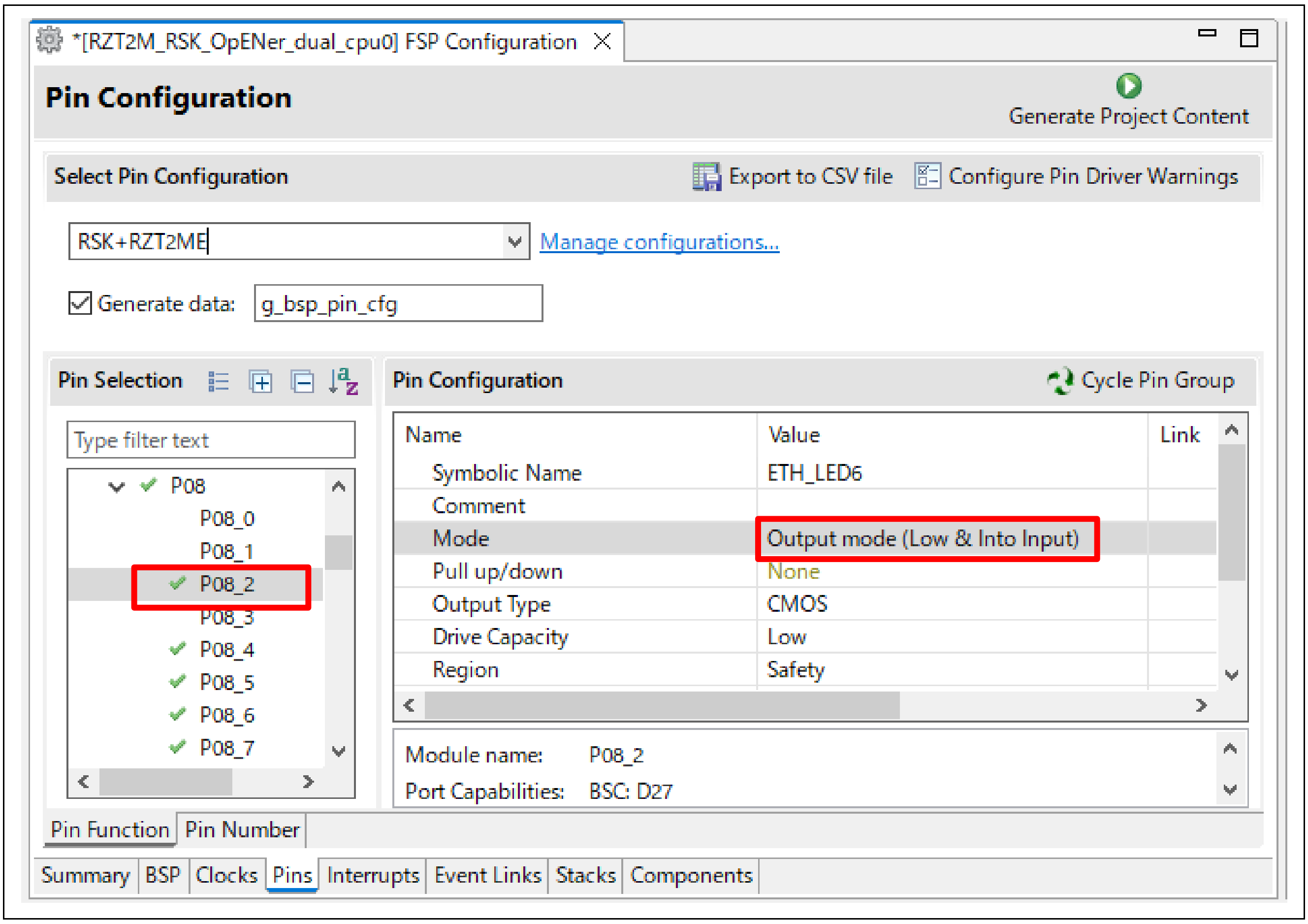


Figure 6-96 Generate Project Content

G) Set clock settings in the "Clock" tab as a following table.

- CPU1CLK : Mul x4
- CKIO : Div /2

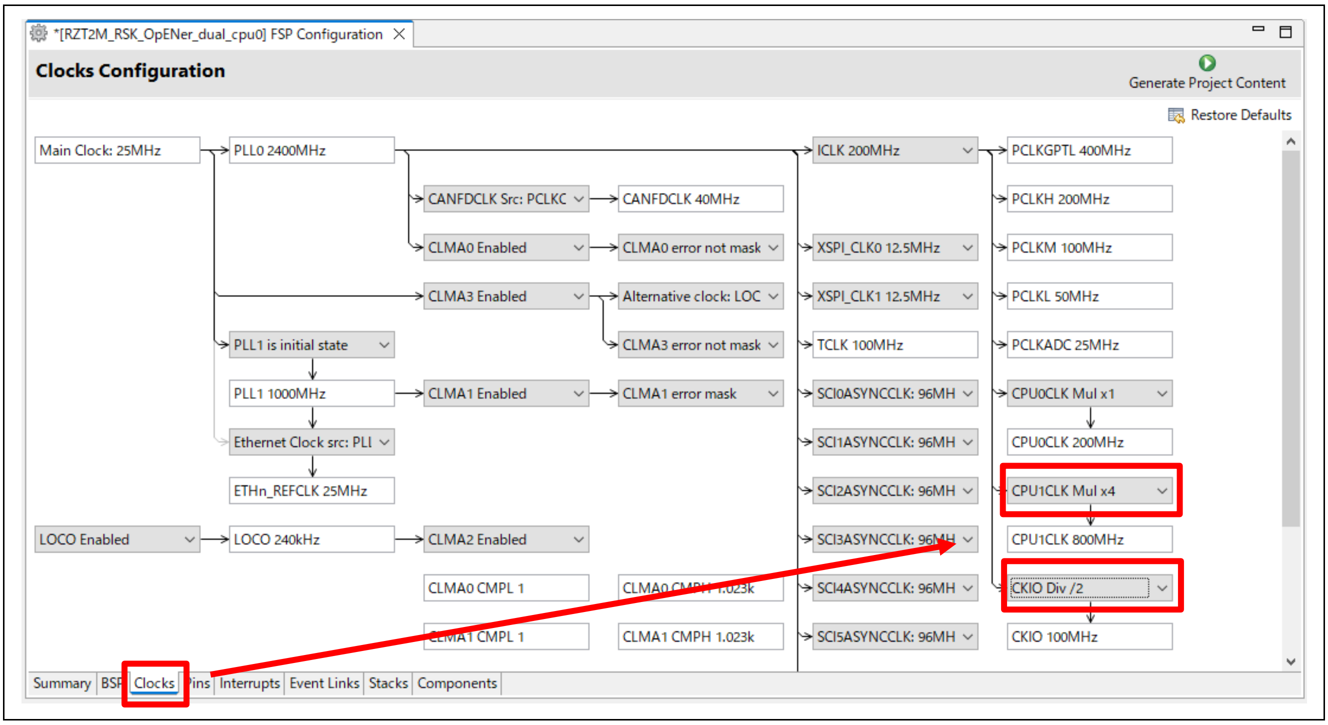


Table 6-12 Set clock settings

H) Click "Generate Project Content" button then generate rzt, rzt_gen, rzt_cfg folder.

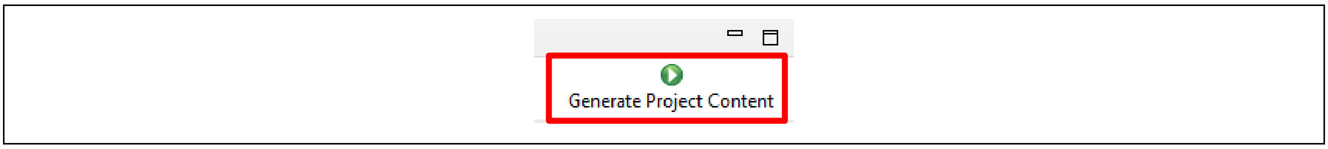


Figure 6-97 Generate Project Content

I) Click the Make button to build the CPU0 project.

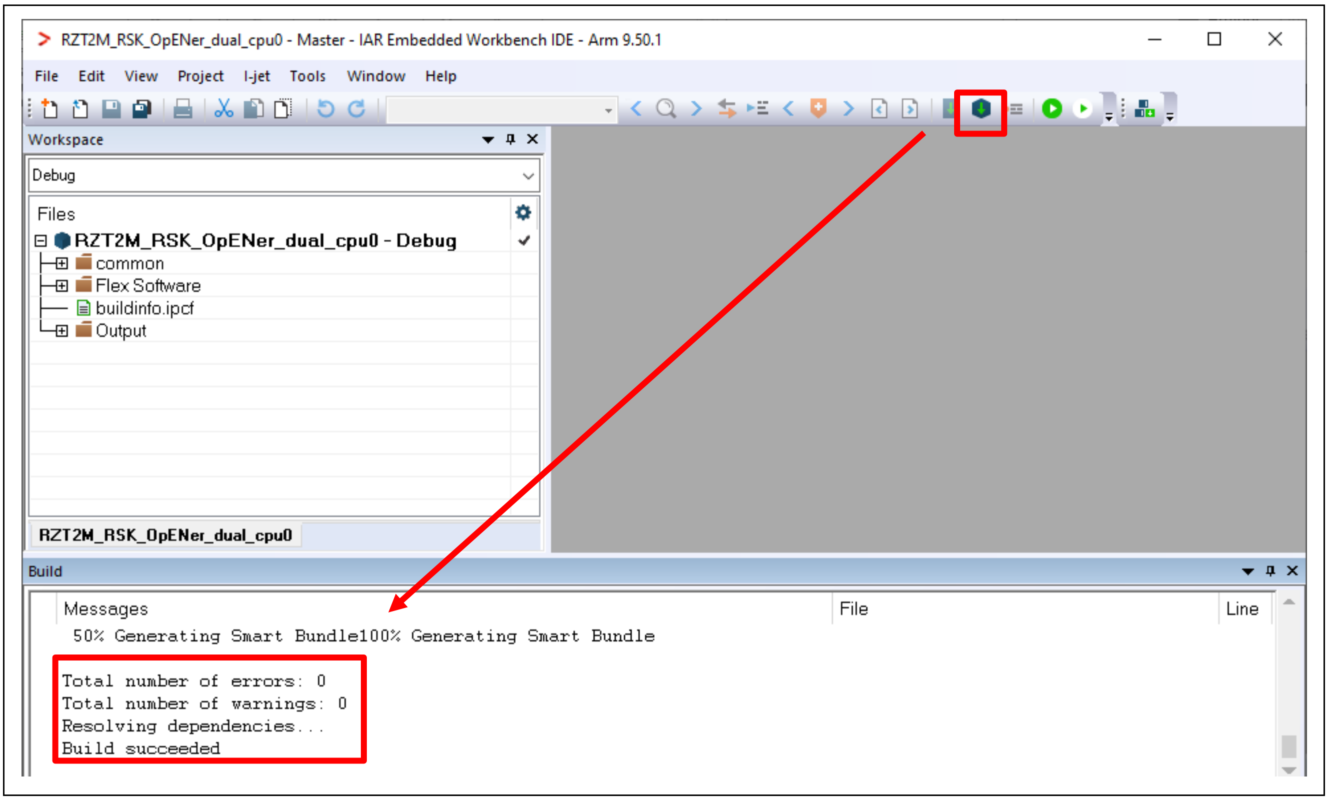


Figure 6-98 Make button on the CPU0 Workspace

J) Open FSP Smart Configurator (v.2.1.0 or later version)

K) Click “Open” in the “File” tab.
Note: For v2.1.0 or later versions of FSP, it must be opened once from FSP Smart Configurator window.

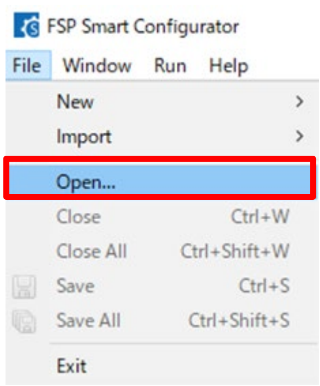


Figure 6-99 Click Open in the File tab

L) Open the configuration.xml of the CPU1 project.

"\project\rzt2m_rsk\opener_dual_cpu1\ewarm\configuration.xml"

Note: You can ignore Problem Occurred dialogs with the error message "Cannot invoke "java.util.Map.entrySet()" because "securePinMap" is null".

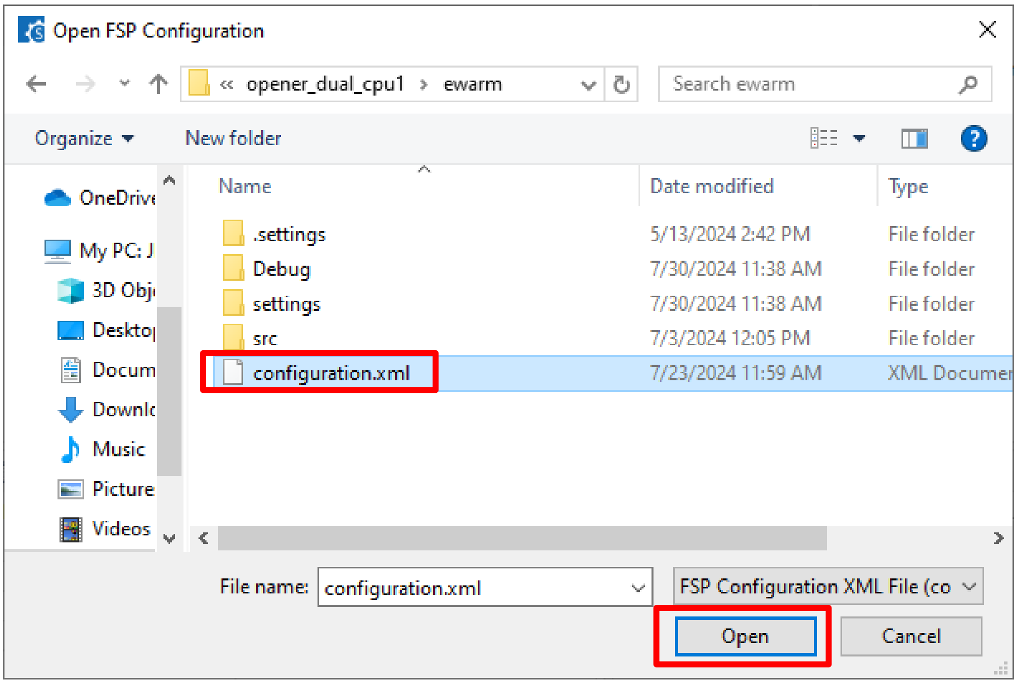


Figure 6-100 Open configuration.xml of the CPU1 project.

M) Click "Generate Project Content" button then generate rzt, rzt_gen, rzt_cfg folder. It is not necessary to change pin settings.

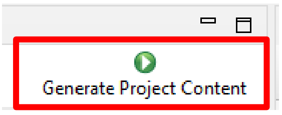


Figure 6-101 Generate Project Content

(2) Build and writing the projects to flash

A) Write the project according to “Appendix. How to Create and Debug FSP Projects for Multiprocessing in All Cases for IAR EWARM” in “RZ/T2, RZ/N2 Getting Started with Flexible Software Package (r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf)”.

B)

C) After writing the projects, restart the device by turning the power on/off.

Note1: You may get the following error (Error [Lp011]: section placement failed) when building the CPU0 project. In this case, please try to change the icf file (Linker Script file) by the following guidance.

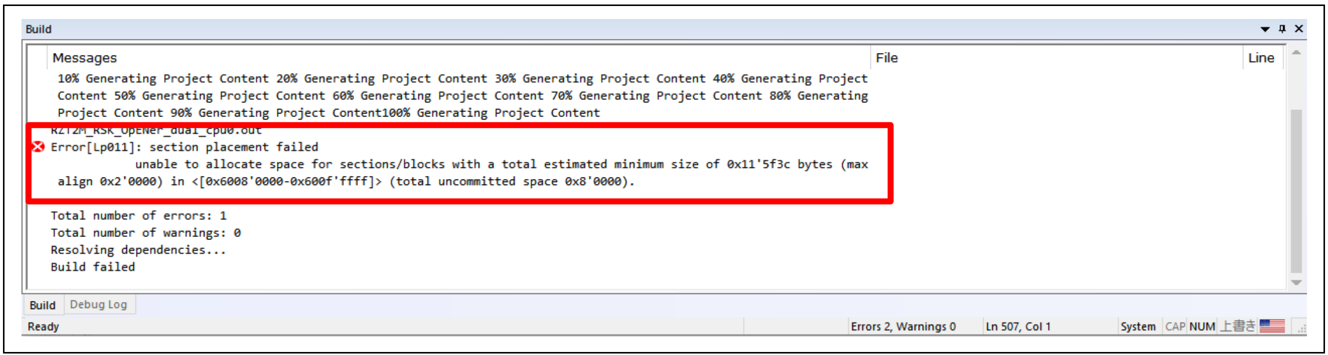


Figure 6-102 The error message, Error [Lp011]: section placement failed

5. On the CPU0 project, click “Project” -> “Options…”
6. Click “Linker” -> “Config”

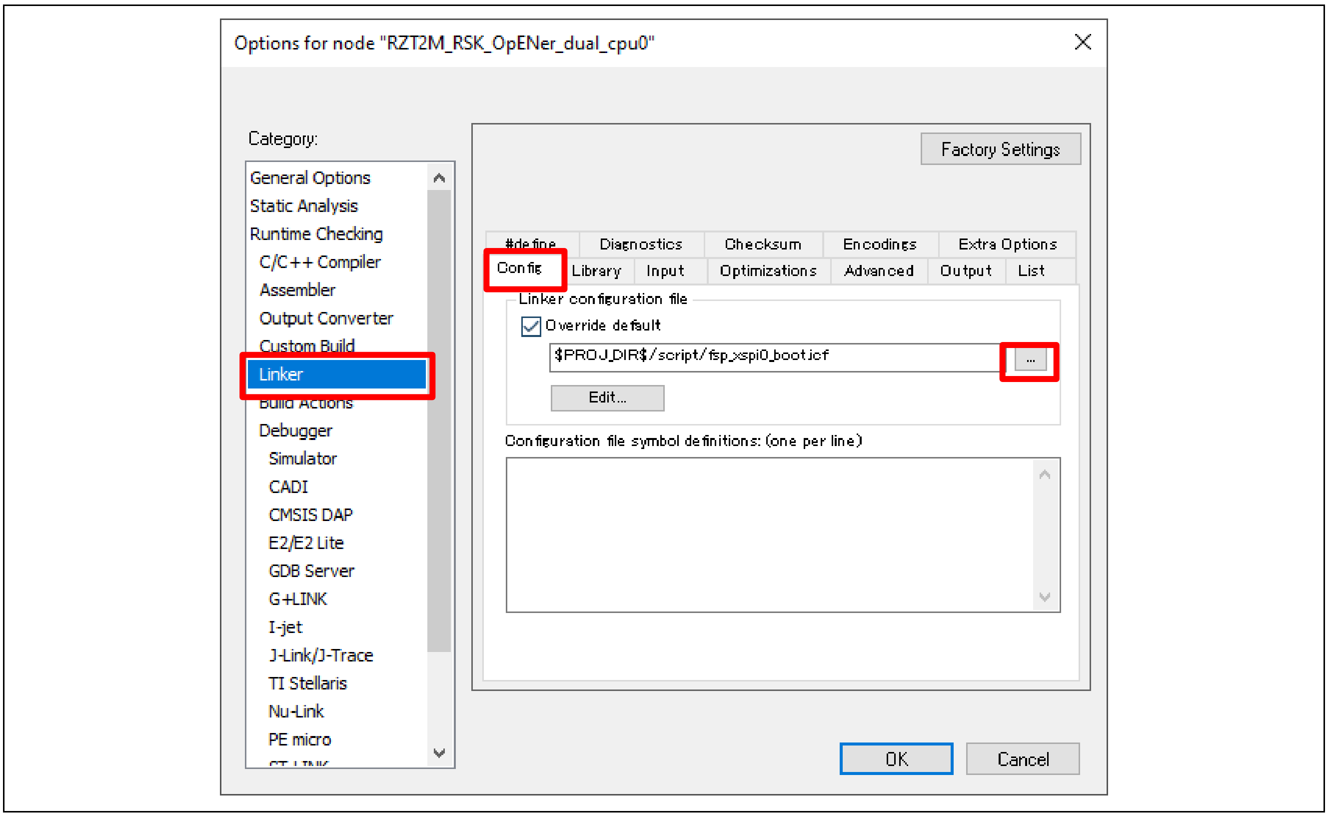


Figure 6-103 Linker -> Config in the CPU0 project Option

7. Change the icf file from default to “fsp_xspi0_boot_for_v0210.icf”.

   Directory: “\project\rzt2m_rsk\opener_dual_cpu0\ewarm\script\ fsp_xspi0_boot_for_v0210.icf”

8. Click the Make button to build the CPU0 project again.

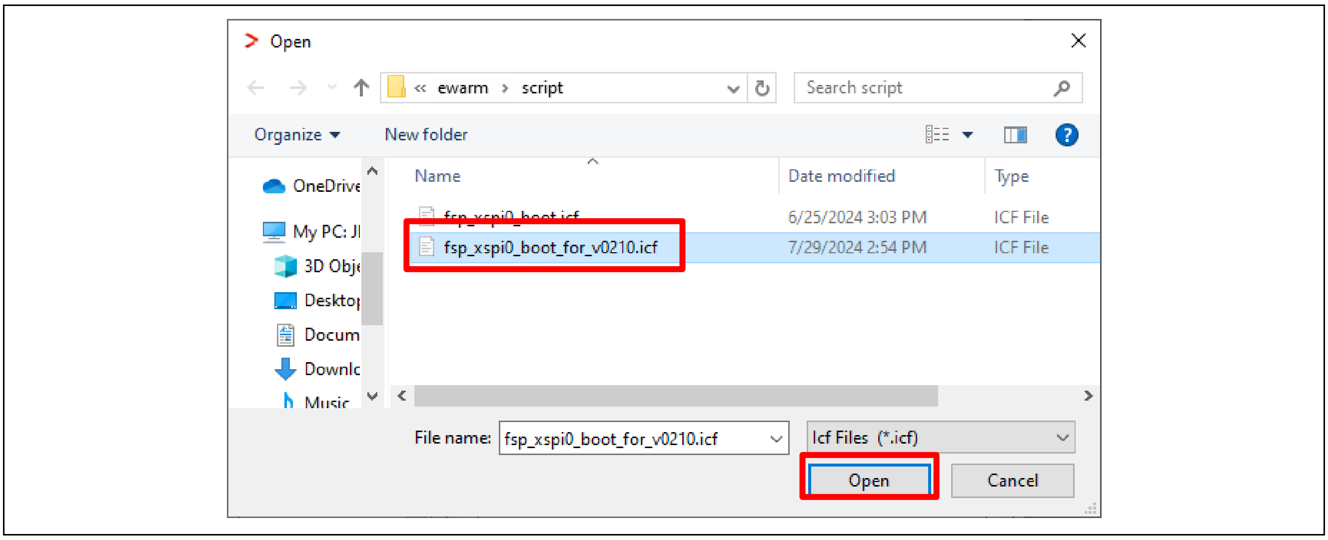


Figure 6-104 Open the icf file for FSP v2.1.0

Note 2:

- Be aware of the limitations when rewriting the program. Please refer to the section 1.3.

- If necessary to enable Type-0 Reset and Type-1 Reset, modify the macro definition "`SAMPLE_EXECUTE_SYSTEM_RESET_PROCESS`" (opener_port_main.c) to "1" enable the Reset process.

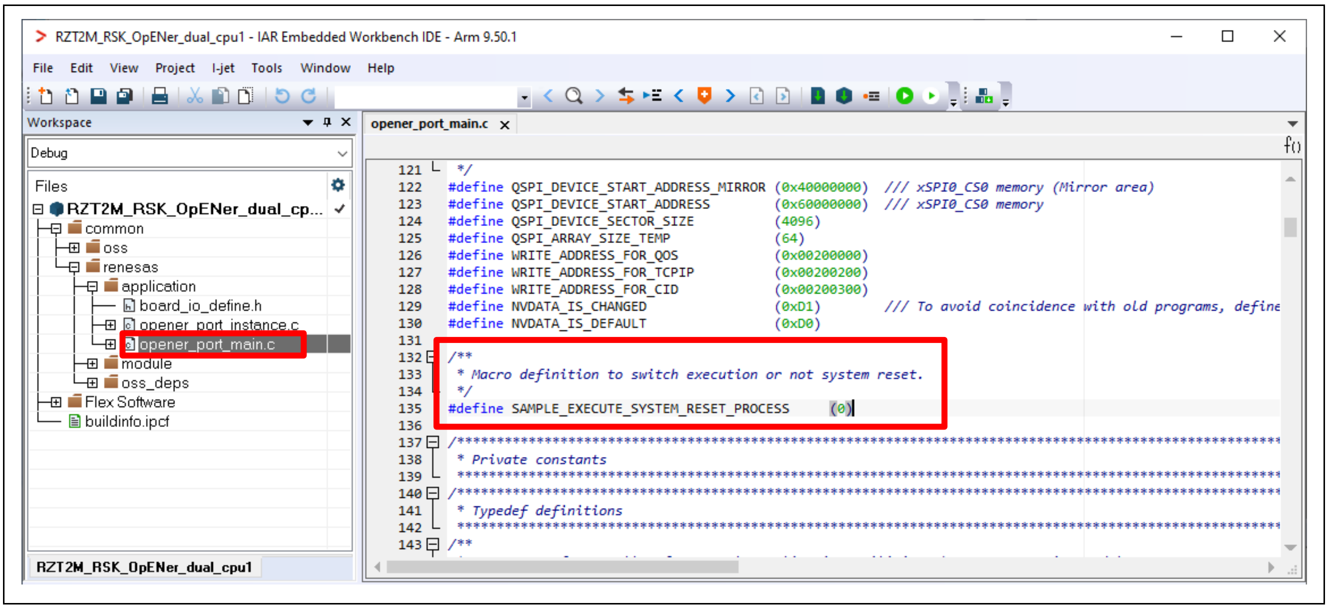


Figure 6-105 Reset process macro definition

6.4 For RZ/T2H CA55 Core (Single-core project)

The setup differs depending on the IDE.

- When using e[2] studio, refer to section 6.4.1
- When using EWARM, refer to section 6.4.2

6.4.1 Setup sample project for e[2] studio

This section shows a procedure using e[2] studio.

6.4.1.1 Startup e[2] studio

1. Open the e[2] studio and select a directory as workspace.
2. Click “Import” in File tab.

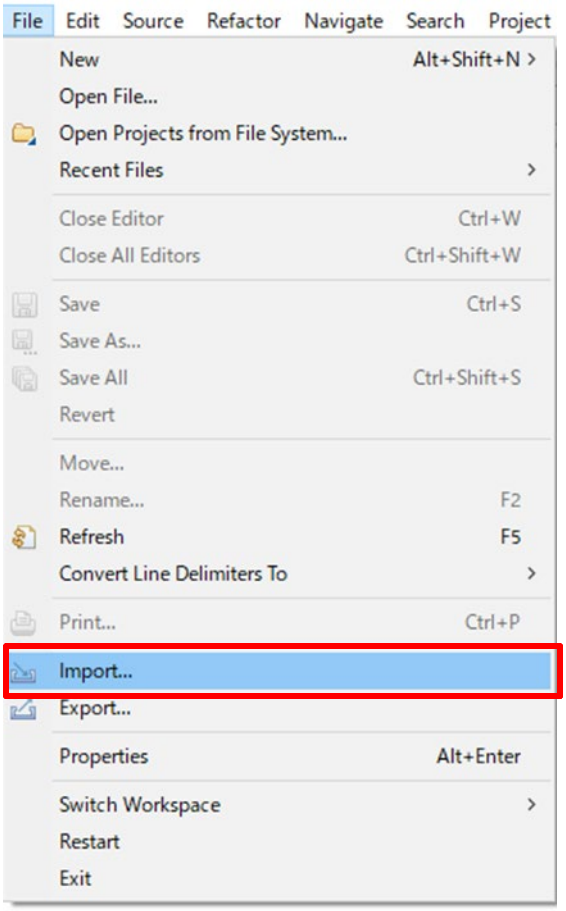


Figure 6-106 e[2] studio File tab

3. Click “Existing Projects into Workspace” in “General” and Click “Next”.

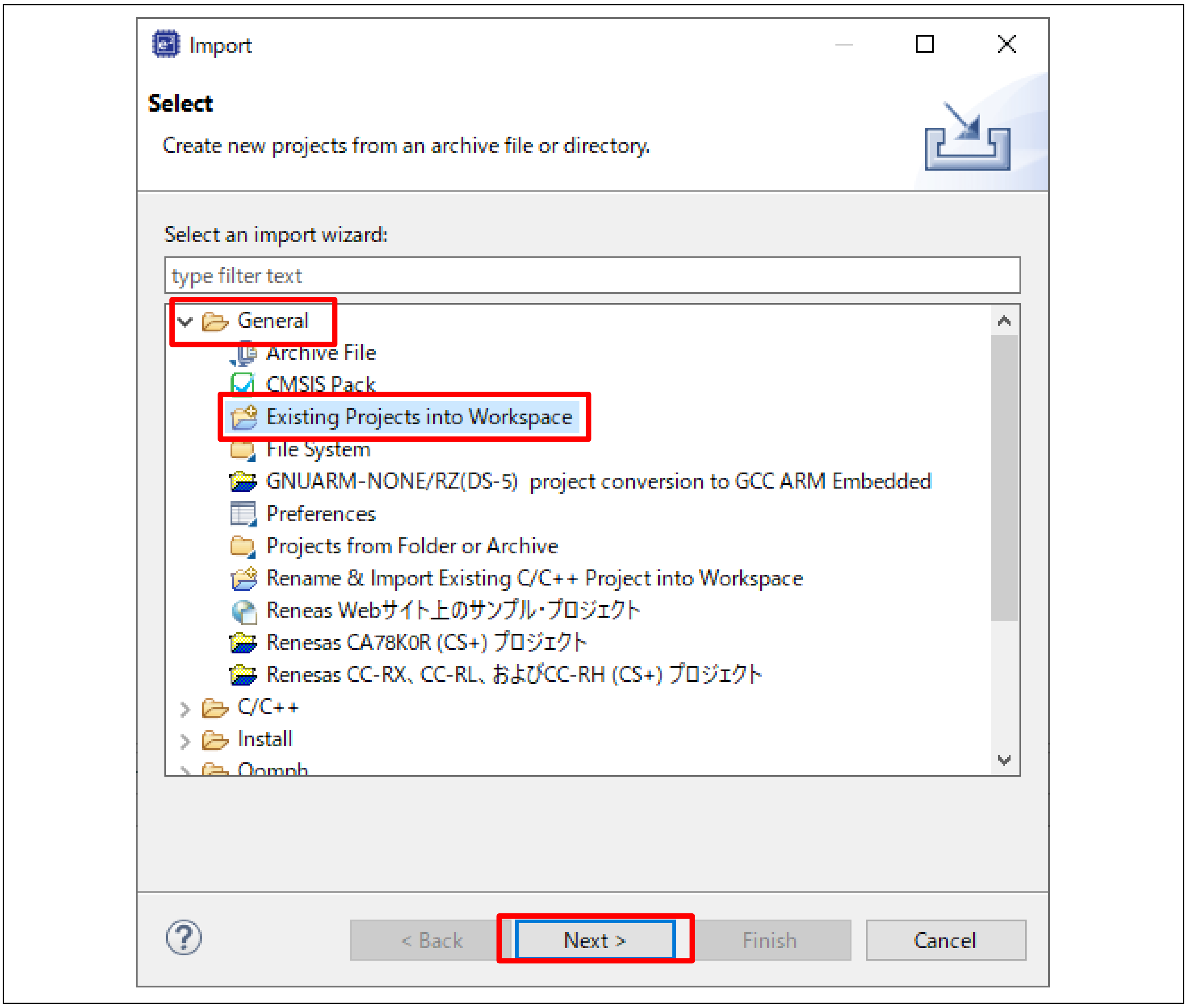


Figure 6-107 Select import type

4. Import project by the following steps.
   A) Set the root directory.
      When you use the RAM-execution projects, please set the root directory.
      ➔ “(Extracted zip folder path)\project\rzt2h_evb\CA55\e2studio\RAM”
      When you use the flash boot mode projects, please set the root directory.
      ➔ “(Extracted zip folder path)\project\rzt2h_evb\CA55\e2studio\xSPI1”
   B) Select the project.
   C) Click “Finish”

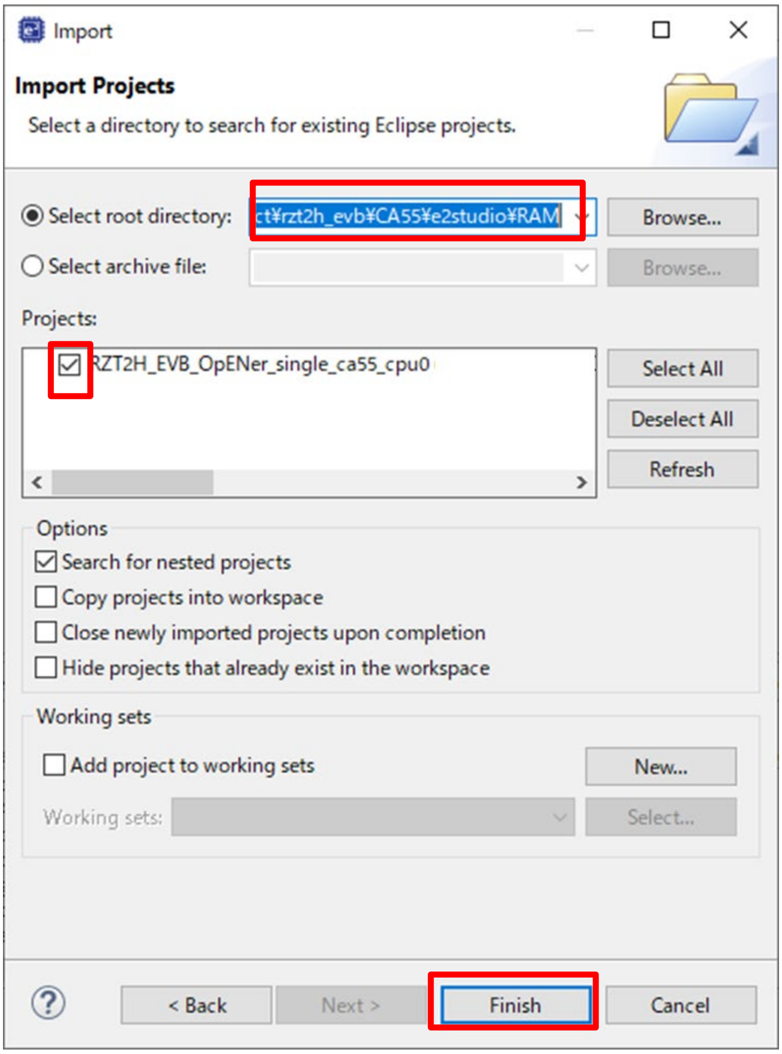


Figure 6-108 Import the project on e² studio

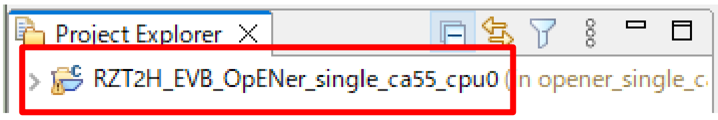


Figure 6-109 Open the project

6.4.1.2 RAM execution / Flash boot mode

(1) How to generate source code and how to build

A) Click the configuration.xml.

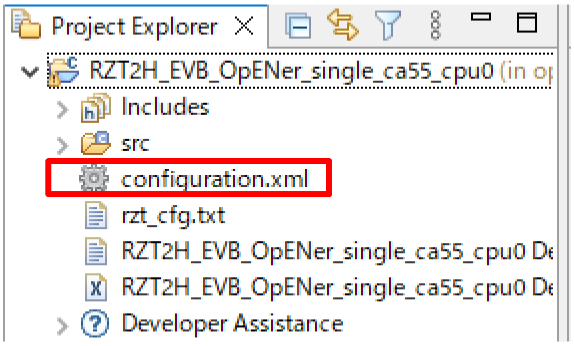


Figure 6-110 Configuration

B) Click “Generate Project Content” button then generate rzt, rzt_gen, rzt_cfg folder.

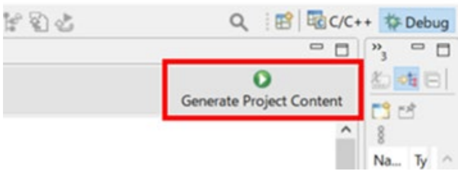


Figure 6-111 Generate Project Content

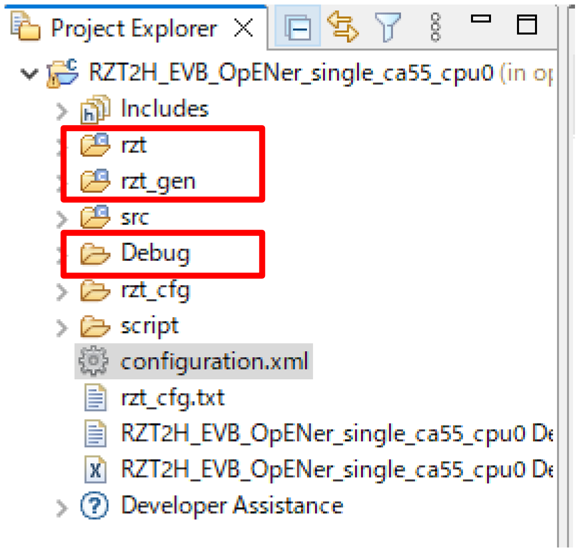


Figure 6-112 Generate project folder

C) (**Flash boot mode only**): - If necessary to enable Type-0 Reset and Type-1 Reset, modify the macro definition "`SAMPLE_EXECUTE_SYSTEM_RESET_PROCESS`" (opener_port_main.c) to "1" enable the Reset process.

D) Click the Build button in tool bar to build the project and confirm that there is no error message in the build message log.

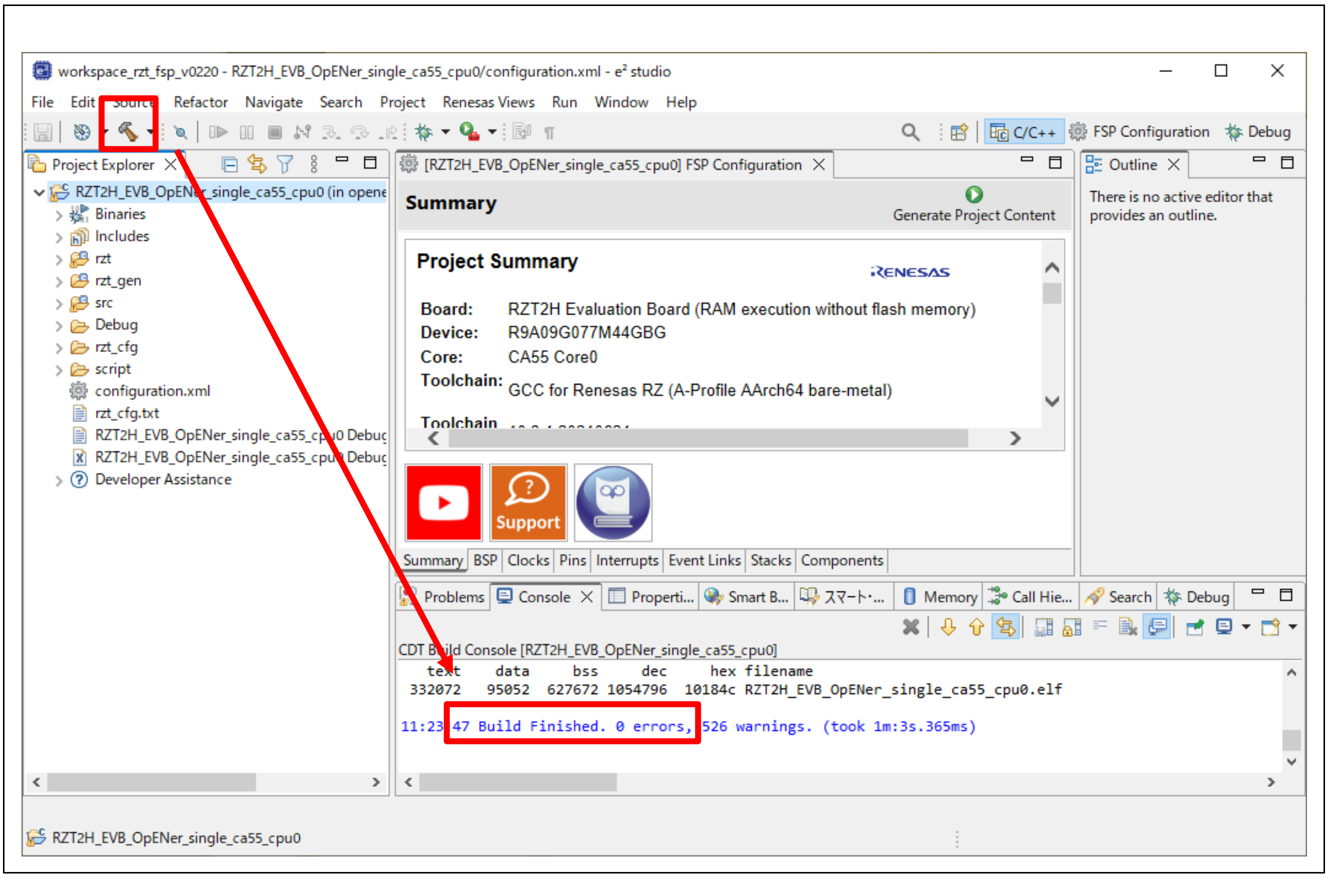


Figure 6-113 Build button and Build message

(2) Download application and run debugger

A) Click the Debug button in tool bar to download the built application program and launch the debugger.

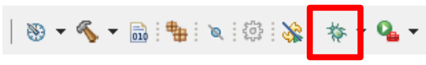


Figure 6-114 Debug button

B) Click to "Yes" button on the Proceed confirmation dialog box.

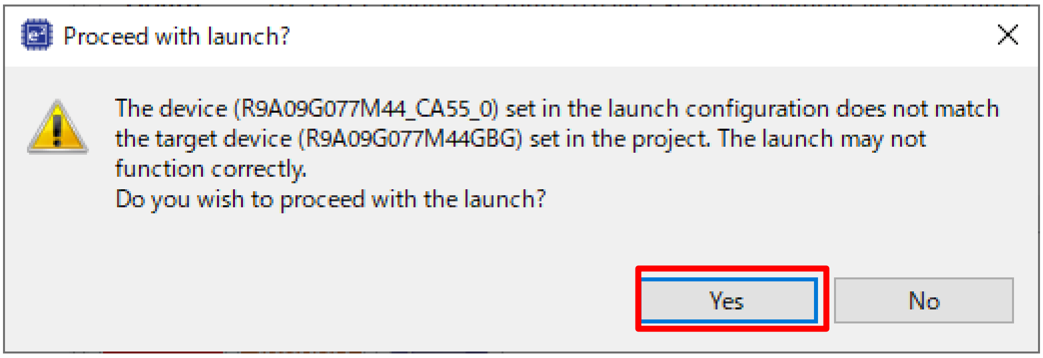


Figure 6-115 Proceed confirmation dialog box

C) Click the "Switch" button.

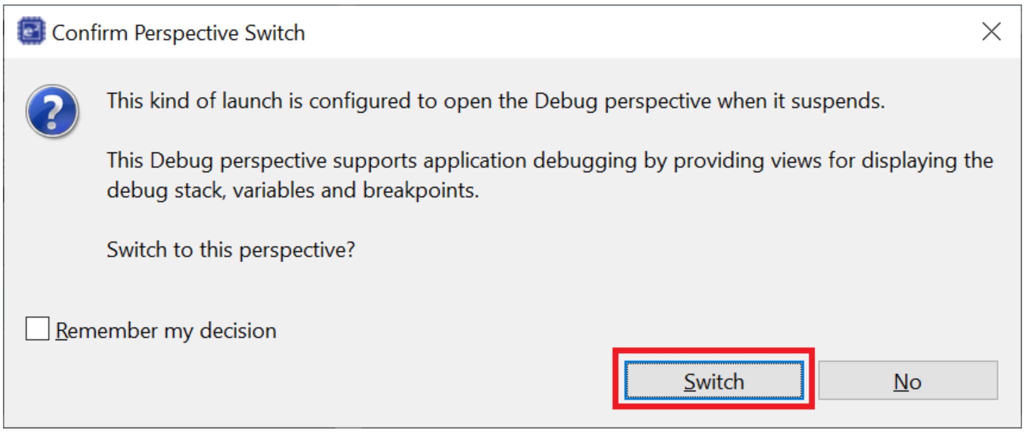


Figure 6-116 Confirm Perspective Switch

D) (**Flash boot mode only**): Disconnect the debugger, and re-power up the board.

Note: For the RZT FSP v2.2.0, there is the limitation "Unable to debug CA55 flash boot project with e^2 studio." (r01an6434ej0112-rzt2-rzn2-fsp-getting-started.pdf, Tool Software Limitations No.13)

E) (**RAM-execution only**): Click the Resume button to execute the program.

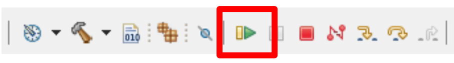


Figure 6-117 Resume button

6.4.2 Setup sample project for EWARM

This section shows a procedure using EWARM.

Some files in the CA55 project with flash boot mode were modified to address the issue of the RZT FSP v2.2.0. (Known Issues No.39 in the FSP quick start guide r01an6434ej0113-rzt2-rzn2-fsp-getting-started.pdf) Therefore, please note the following.

- Modified files:
  - ➢ The system.c (\opener_single_ca55_cpu0\rzt\fsp\src\bsp\cmsis\Device\RENESAS\Source\)
  - ➢ The fsp_xspi1_boot.icf file (\opener_single_ca55_cpu0\script)
- Note:
  - ➢ If you change the boot mode, these files will be overwritten at the time of code generation, so please save it in advance and restore it before building.

6.4.2.1 Startup EWARM

1. Open the EWARM.
2. Click “Open Workspace…” in the File tab.

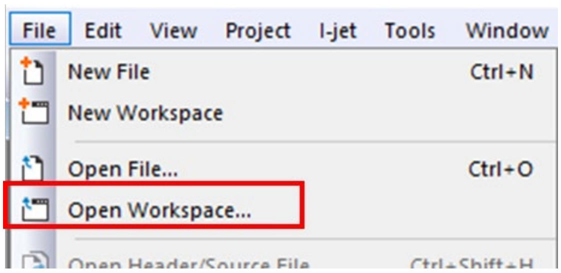


Figure 6-118 EWARM file tab

3. Select the Workspace File(.eww) and click the Open button.

   When you use the RAM-execution projects, please select the following folder.

   ➔ “(Extracted folder path)\project\rzt2h_evb\CA55\ewarm**RAM**\”

   When you use the flash boot mode projects, please select the following folder.

   ➔ “(Extracted folder path) \project\rzt2h_evb\CA55\ewarm**xSPI1**\”

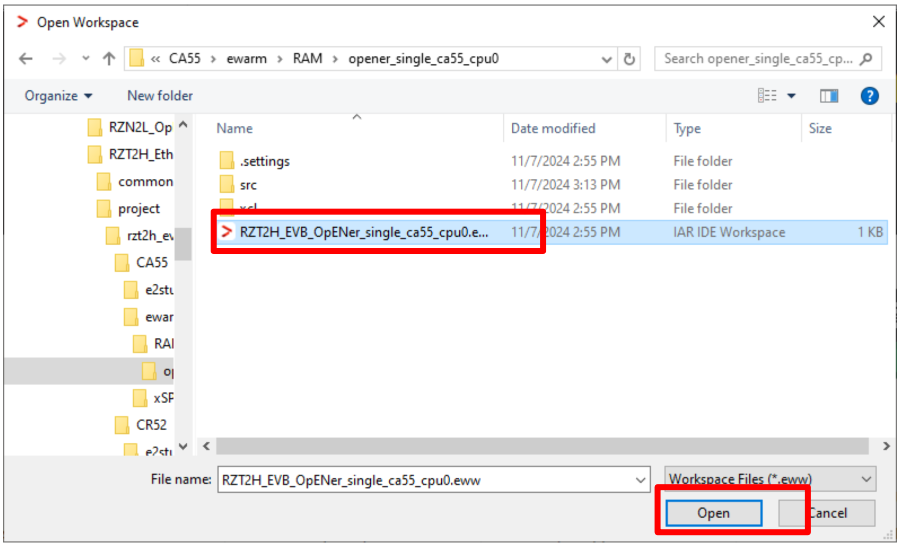


Figure 6-119 Open project file

6.4.2.2 RAM execution / Flash boot mode

(1) How to generate source code and how to build

A) Click the “Tools” -> “FSP Smart Configurator” on tool bar. If you have not set up FSP Smart Configurator yet on EWARM, refer to r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf in which section 5.4 describes how to set up it.

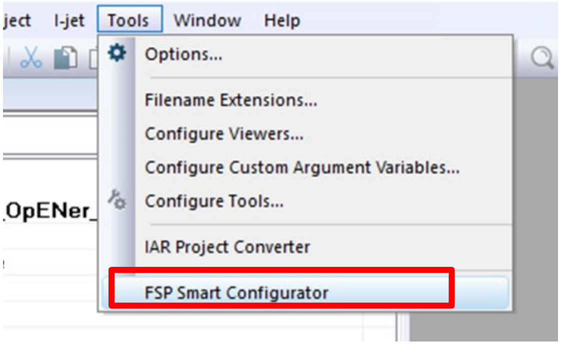


Figure 6-120 Tools tab

B) Click “Generate Project Content” button then generate rzt, rzt_gen, rzt_cfg folder.

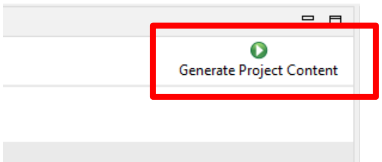


Figure 6-121 Generate Project Content

C) Change the device name in the buildinfo.ipcf file.

R9A09G077M44_CA55_0 -> R9A09G077M44_A55

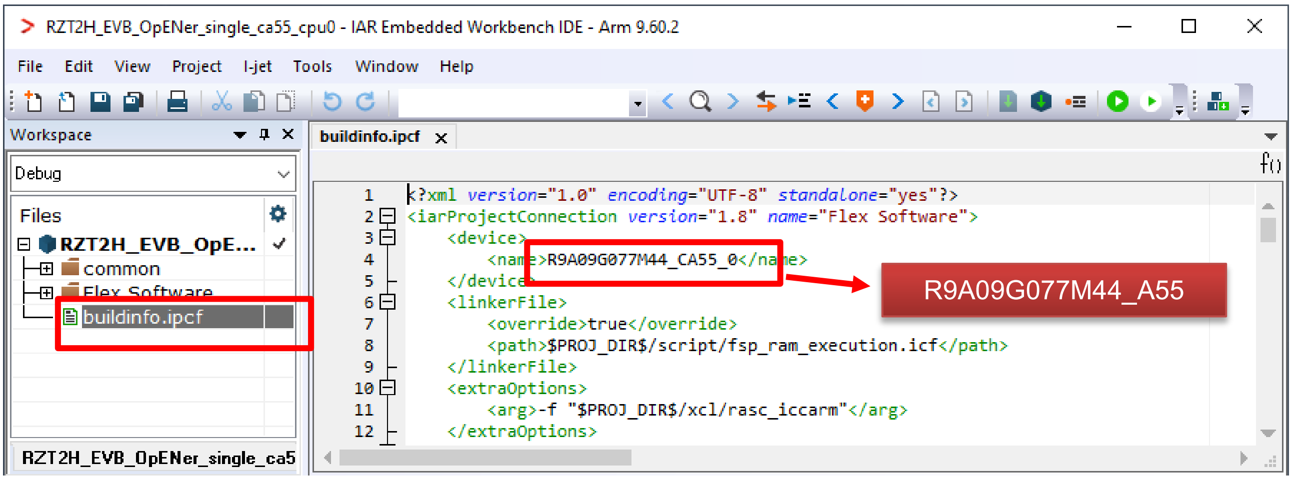


Figure 6-122 Change the device name in the ipcf file

D) Click the Make button on the tool bar to build. Once the build is completed, the build message is displayed in the Build Console window.

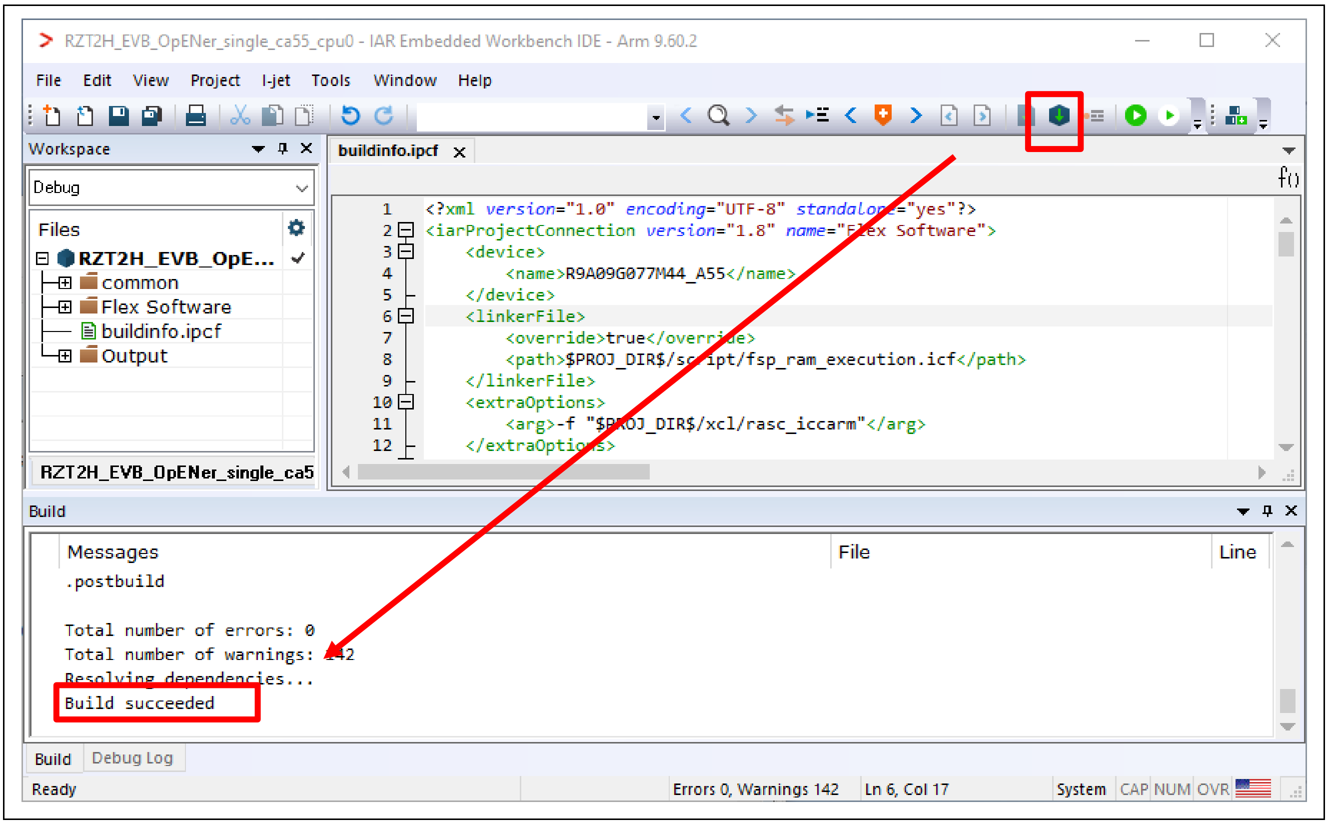


Figure 6-123 Make button and Build console

(2) Download application and run debugger

A) Click the Debug button in tool bar to download the built application program and launch the debugger.

Figure 6-124 Debug button

B) Click the Go button.

Figure 6-125 Go button

6.5 For RZ/T2H CR52 Core (Dual-core project)

The setup differs depending on the IDE.

- When using e^2 studio, refer to section 6.5.1.
- When using EWARM, refer to section 6.5.2.

6.5.1 Setup sample project for e^2 studio

This section shows a procedure using e^2 studio.

6.5.1.1 Startup e^2 studio

1. Open the e^2 studio and select a directory as workspace.
2. Click "Import" in File tab.

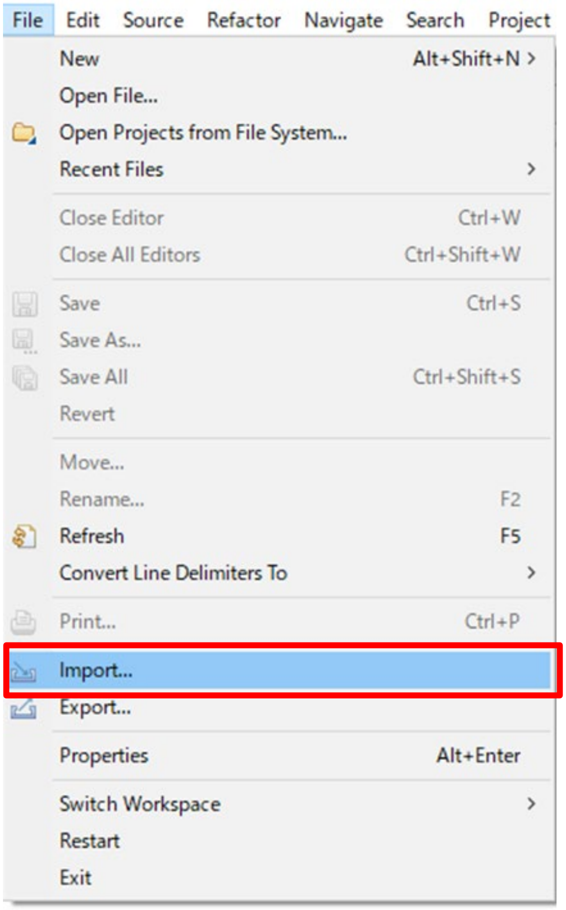


Figure 6-126 e^2 studio File tab

3. Click “Existing Projects into Workspace” in “General” and Click “Next”.

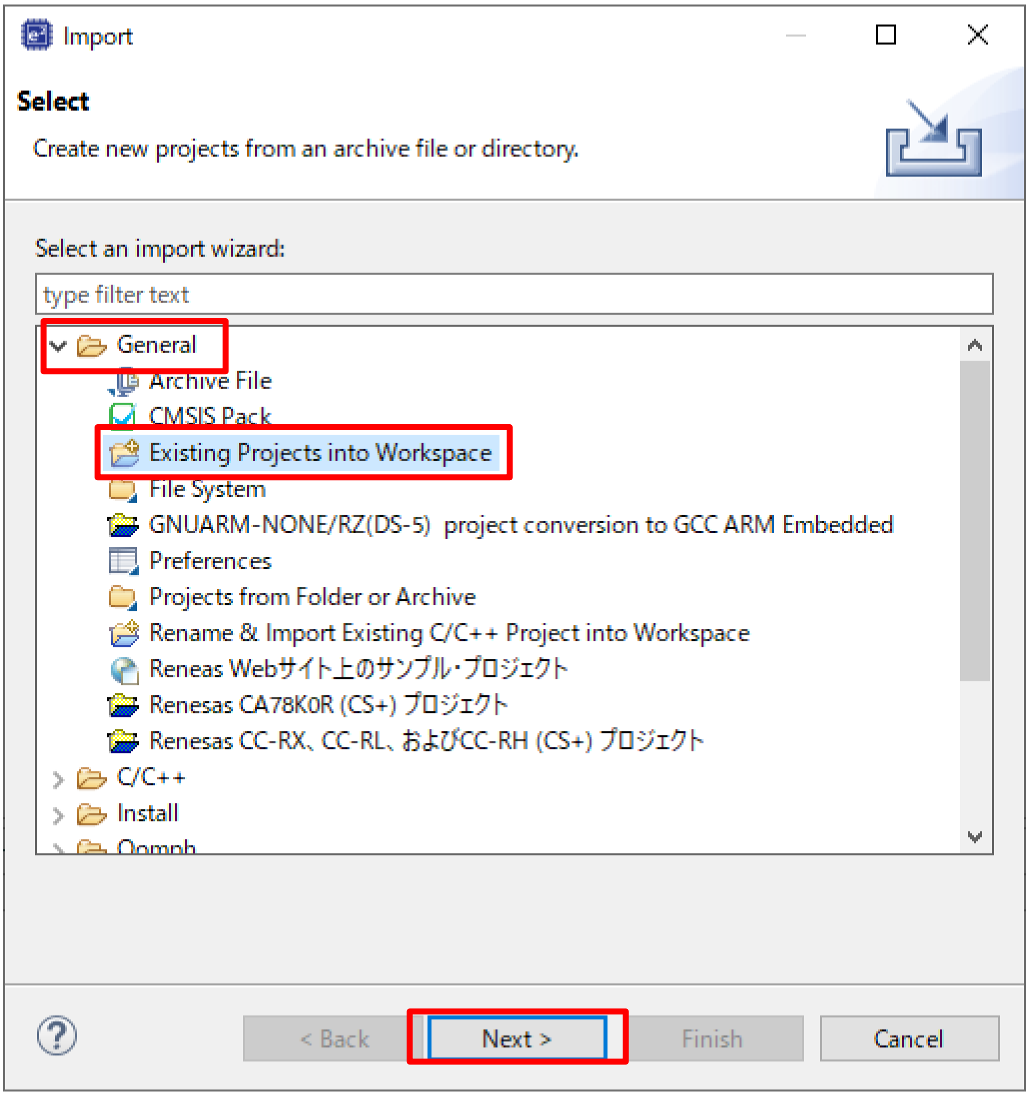


Figure 6-127 Select import type

4. Import project by the following steps.
   A) Set the root directory.
      When you use the RAM-execution projects, please set the root directory.
      ➔ "(Extracted zip folder path)\project\rzt2h_evb\CR52\e2studio**RAM**"
      When you use the flash boot mode projects, please set the root directory.
      ➔ "(Extracted zip folder path)\project\rzt2h_evb\CR52\e2studio**xSPI1**"
   B) Select 2 projects.
   C) Click "Finish"

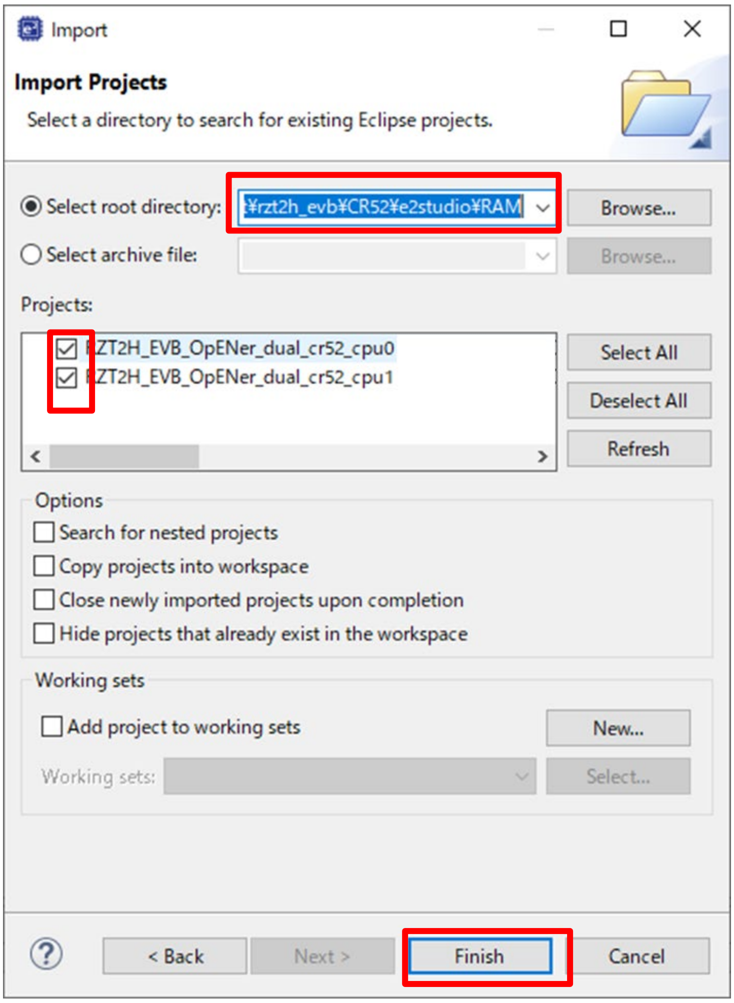


Figure 6-128 Import two projects on e² studio

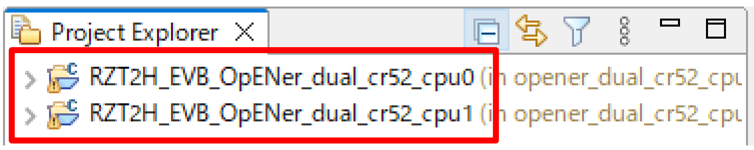


Figure 6-129 Open 2 projects

6.5.1.2 RAM execution / Flash boot mode

(1) How to generate source code and how to build

A) Click the configuration.xml of the CPU0 project.

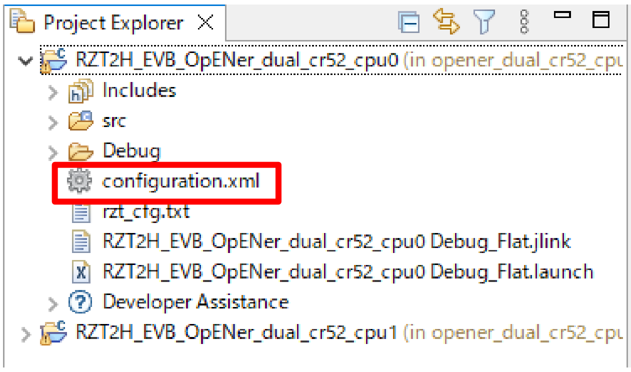


Figure 6-130 Configuration

B) Click “Generate Project Content” button then generate rzt, rzt_gen, rzt_cfg folder.

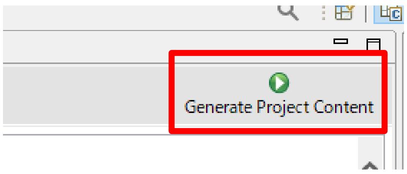


Figure 6-131 Generate Project Content

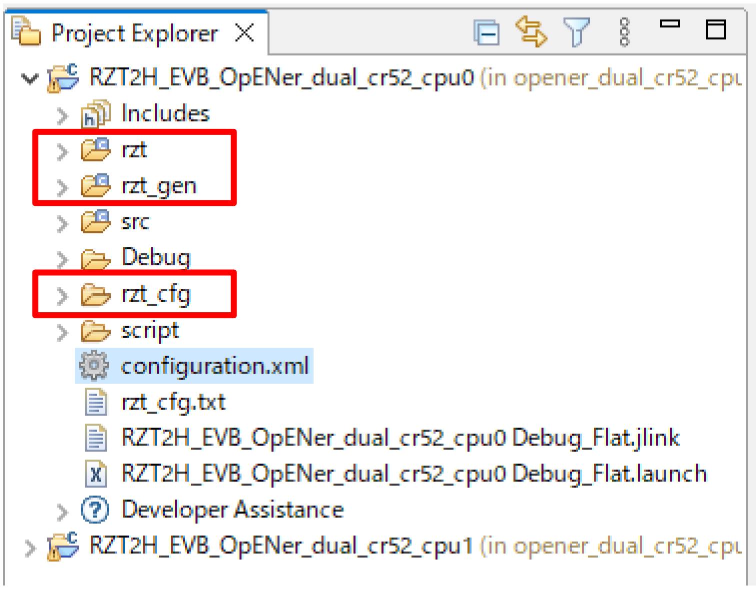


Figure 6-132 Generate project folder

C) Click the Build button in tool bar to build the project and confirm that there is no error message in the build message log.

Figure 6-133 Build button

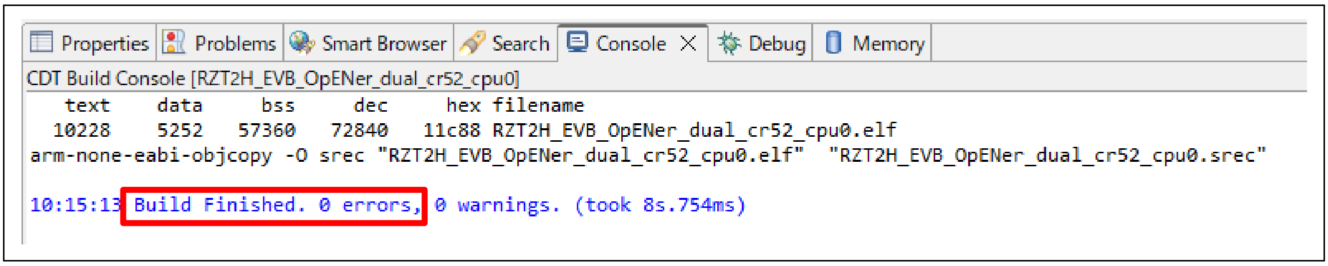


Figure 6-134 Build message

D) For the CPU1 project, generate the folders.

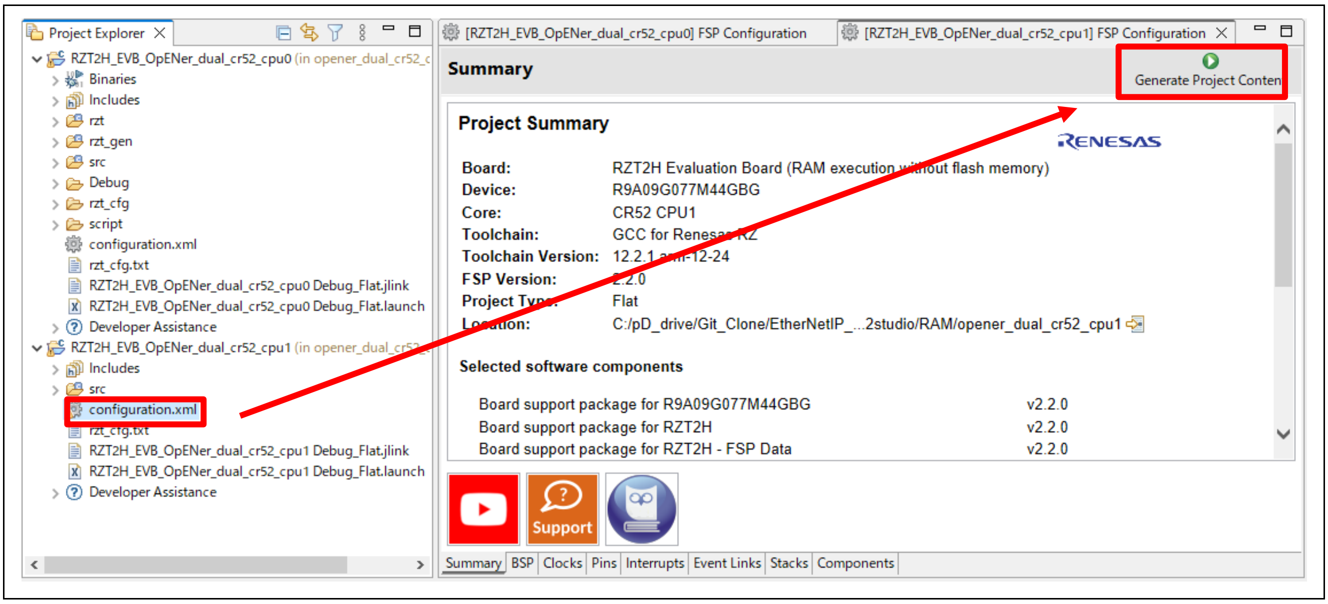


Figure 6-135 Generate folders for the CPU1 project

E) (**Flash boot mode only**): - If necessary to enable Type-0 Reset and Type-1 Reset, modify the macro definition "`SAMPLE_EXECUTE_SYSTEM_RESET_PROCESS`" (opener_port_main.c) to "1" enable the Reset process.

F) For the CPU1 project, build the project.

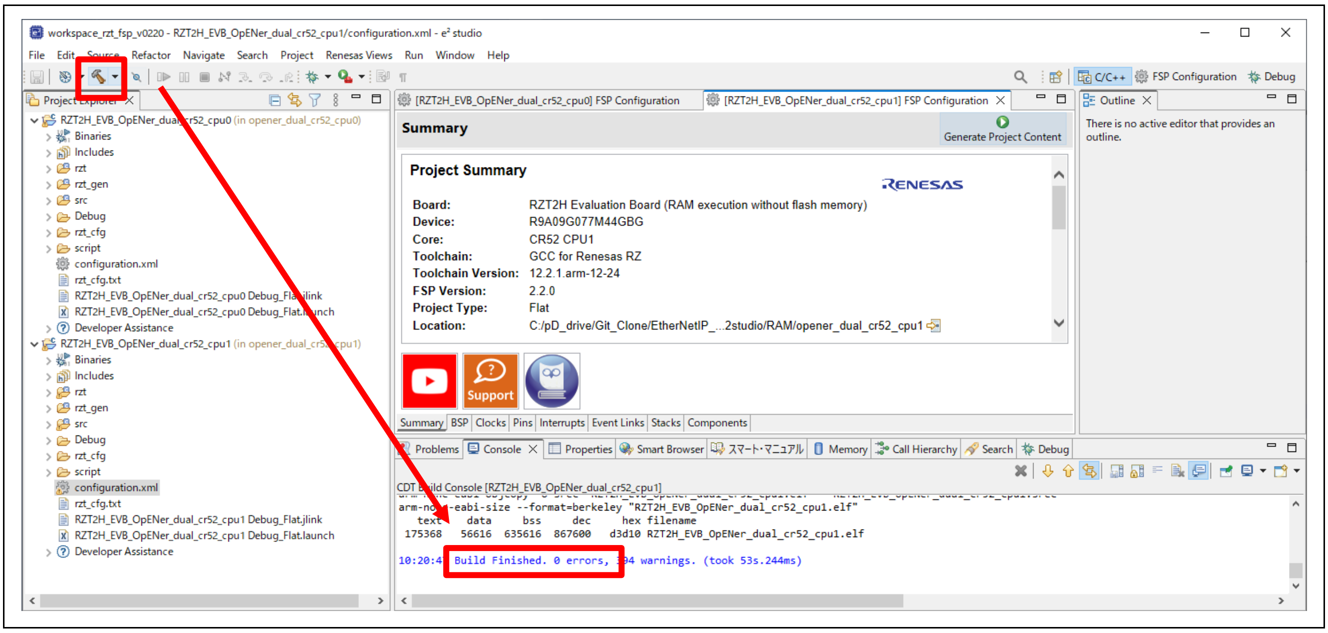
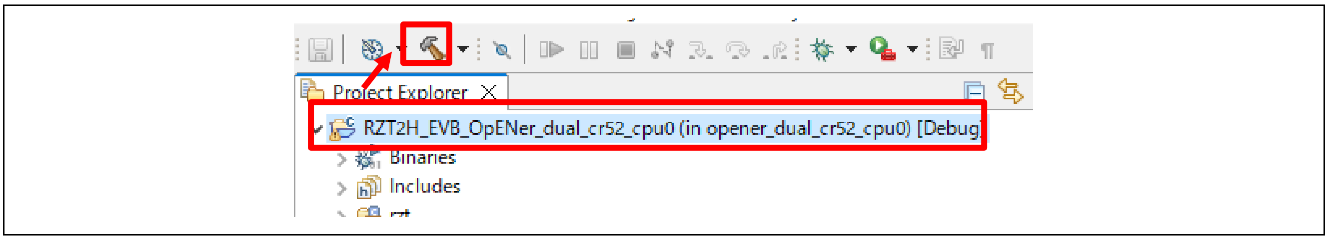


Figure 6-136 Build the CPU1 project

G) (**Flash boot mode only**): For the CPU0 project, build the project again.



(2) Download application and run debugger

A) Select the CPU0 project and click the Debug button to download the CPU0 program.

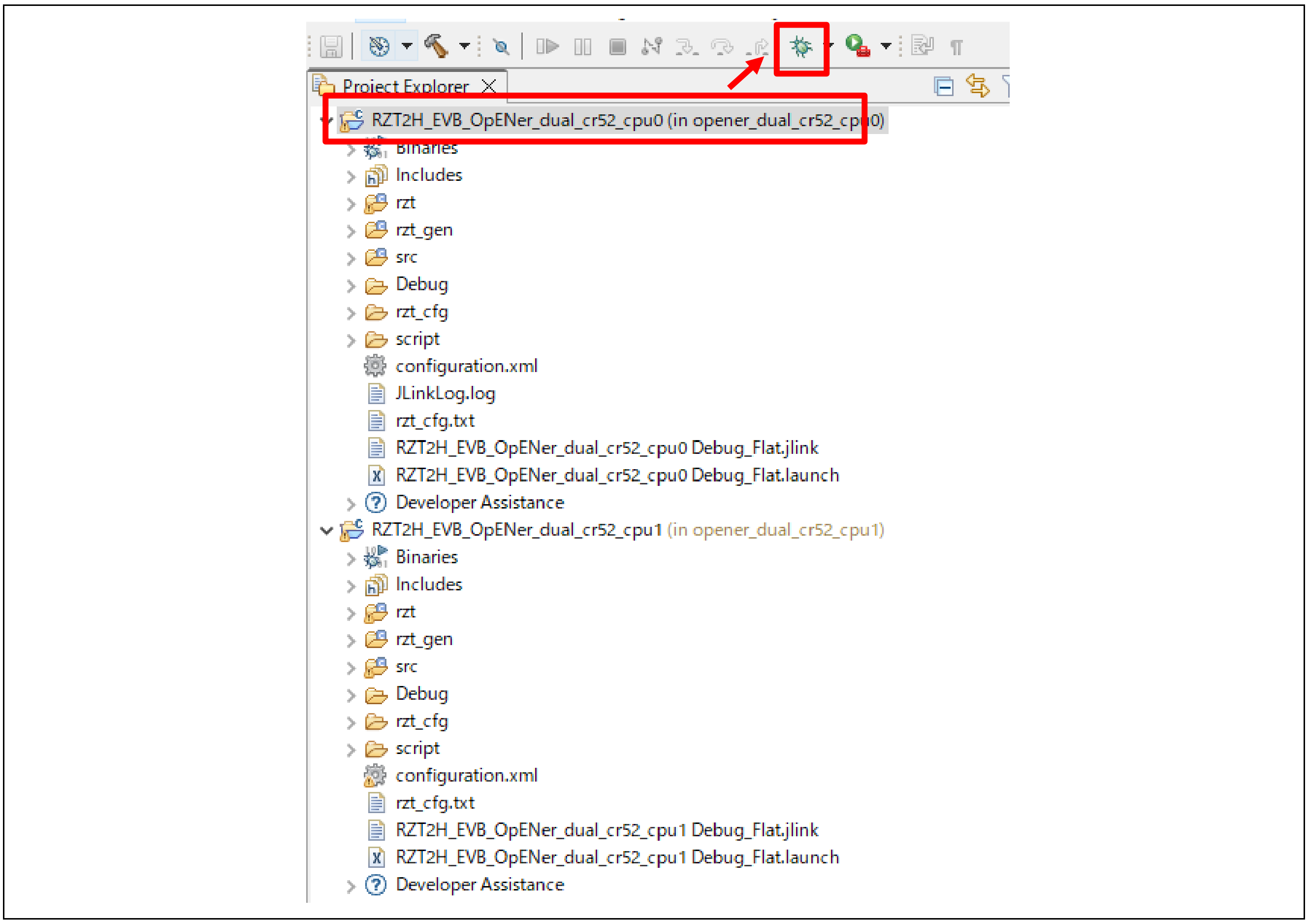


Figure 6-137 Debug button, the CPU0 project

B) Click the “No” button on the Perspective Switch dialog box.

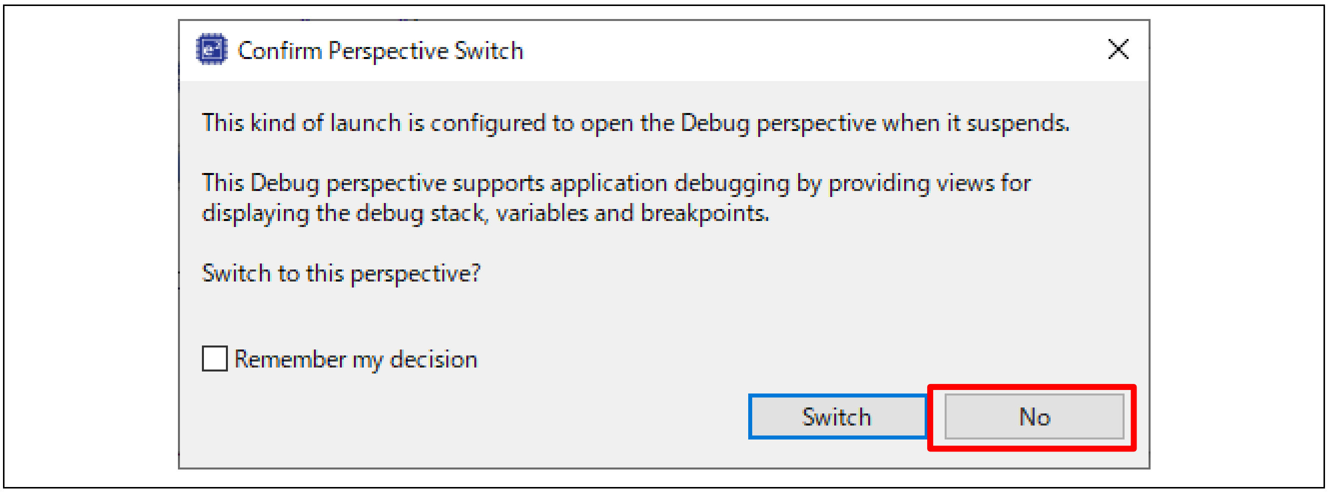


Figure 6-138 Confirm Perspective Switch

C) (**Flash boot mode only**): Disconnect the debugger, and re-power up the board.

D) (**RAM-execution only**): Select the CPU1 project and click the Debug button to download the CPU1 program.

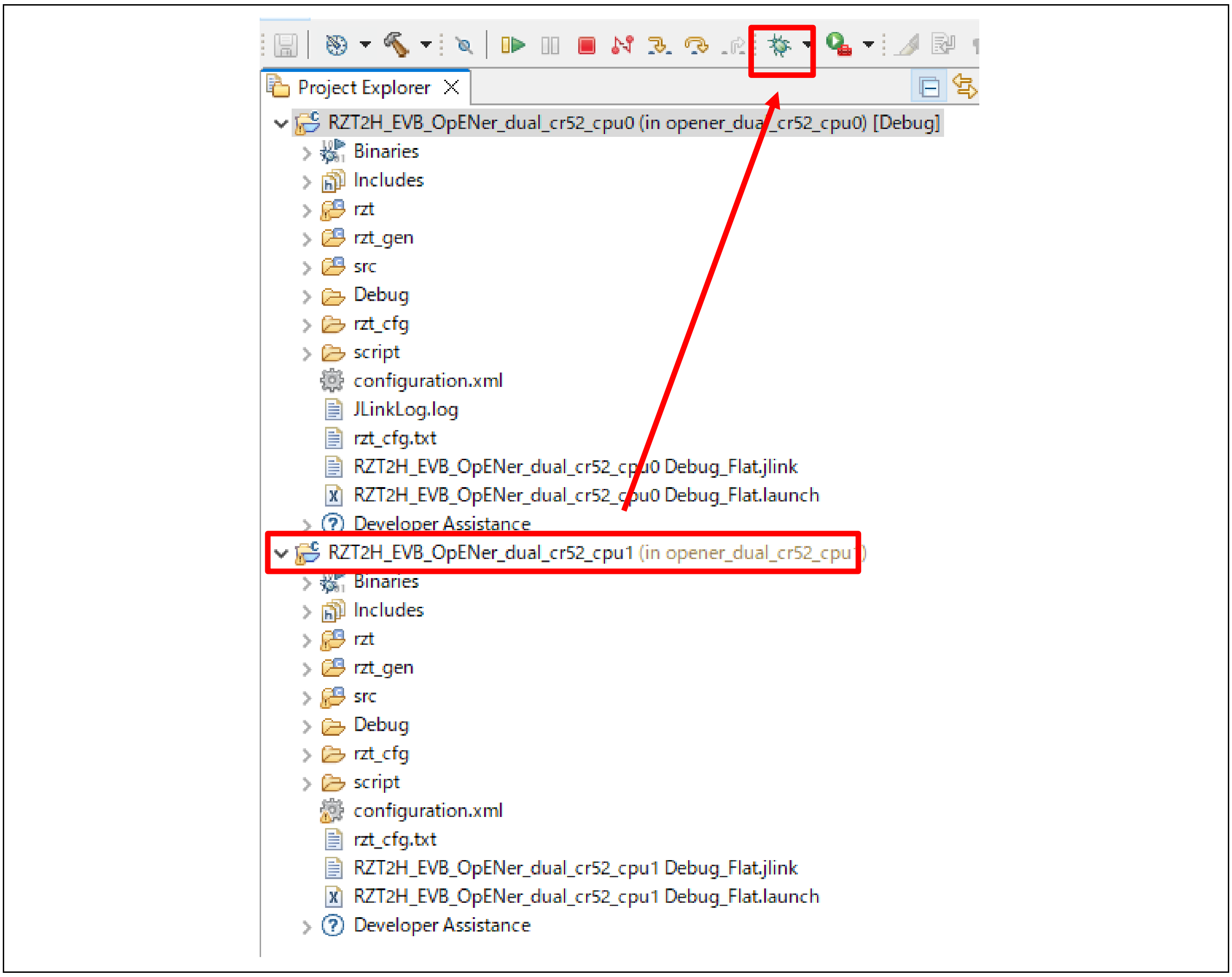


Figure 6-139 Debug button, the CPU1 project

E) (**RAM-execution only**): Click the “No” button on the Launcher dialog box.

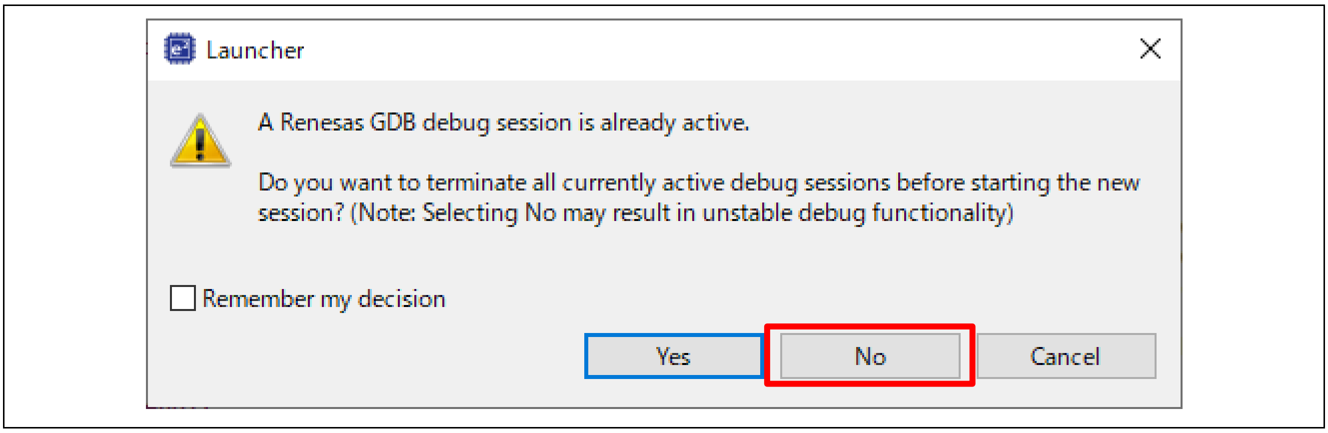


Figure 6-140 Launcher dialog box

F) (**RAM-execution only**): Click to “Yes” button on the Proceed confirmation dialog box.

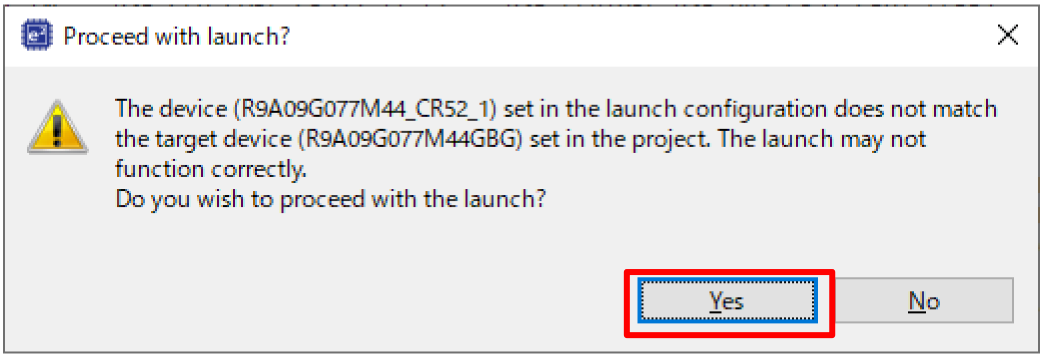


Figure 6-141 Proceed confirmation dialog box

G) (**RAM-execution only**): Click “Yes” button on the Perspective Switch dialog box.

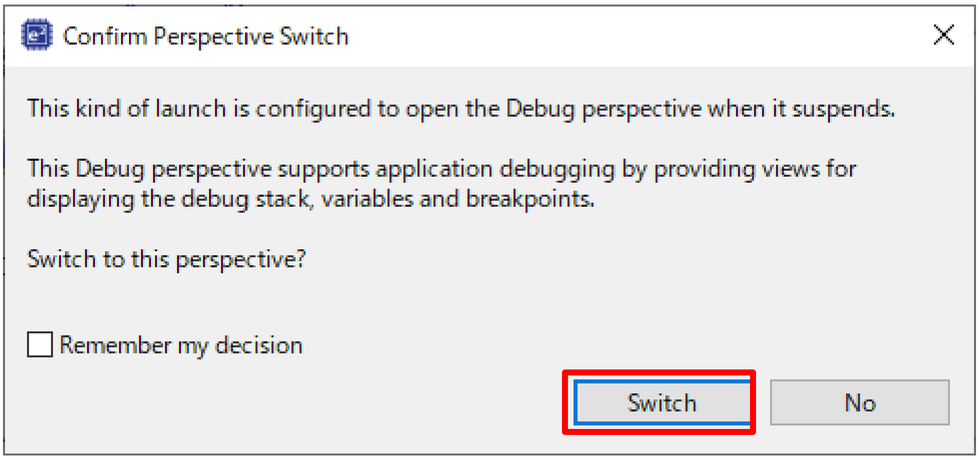


Figure 6-142 Confirm Perspective Switch

H) (**RAM-execution only**): Run the CPU0 project.

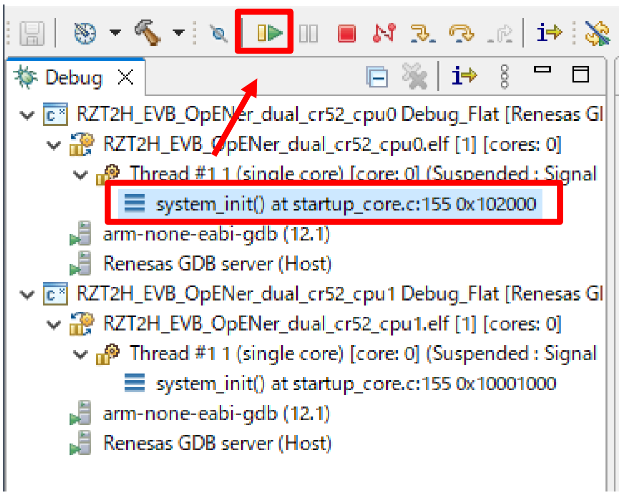


Figure 6-143 Run the CPU0 project

I) (**RAM-execution only**): Run the CPU1 project.

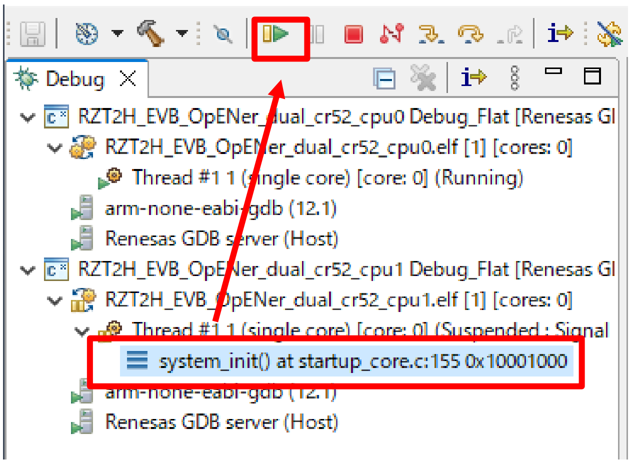


Figure 6-144 Run the CPU1 project

6.5.2 Setup sample project for EWARM

This section shows a procedure using EWARM.

6.5.2.1 Startup EWARM

1. Open the EWARM.
2. Click "Open Workspace…" in the File tab.

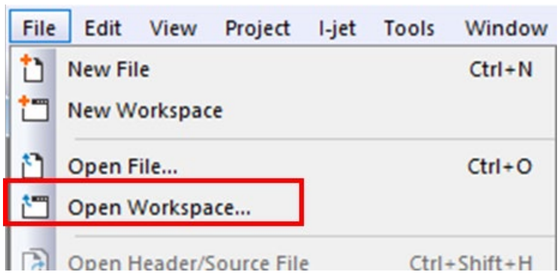


Figure 6-145 EWARM file tab

3. Select the CPU0 Workspace File(.eww) and click the Open button.

   When you use the RAM-execution projects, please select the following fodler.

   ➔ "(Extracted folder path)\project\rzt2h_evb\CR52\ewarm**RAM**\opener_dual_cr52_cpu0"

   When you use the flash boot mode projects, please select the following fodler.

   ➔ "(Extracted folder path) \project\rzt2h_evb\CR52\ewarm**xSPI1**\opener_dual_cr52_cpu0"

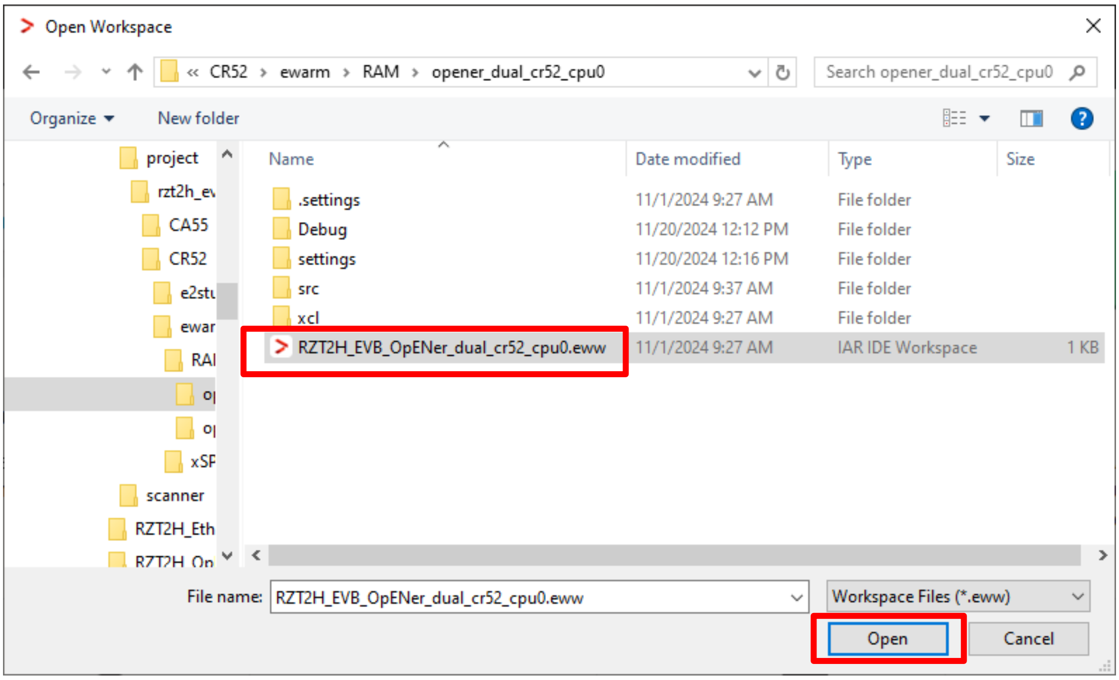


Figure 6-146 Open the CPU0 Workspace

4. Open another EWARM window.
5. Click "Open Workspace…" in the File tab, and open the CPU1 Workspace File(.eww).

When you use the RAM-execution projects, please select the following folder.

➔ "(Extracted folder path)\project\rzt2h_evb\CR52\ewarm**RAM**\opener_dual_cr52_cpu1"

When you use the flash boot mode projects, please select the following folder.

➔ "(Extracted folder path)\project\rzt2h_evb\CR52\ewarm**xSPI1**\opener_dual_cr52_cpu1"

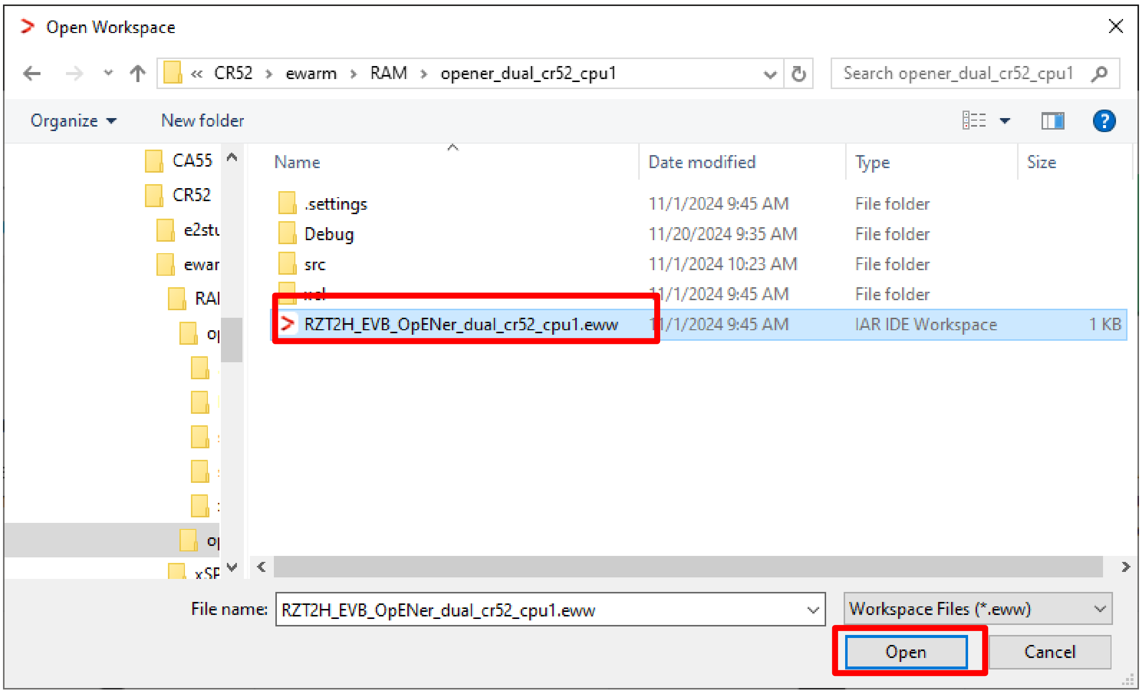


Figure 6-147 Open the CPU1 Workspace

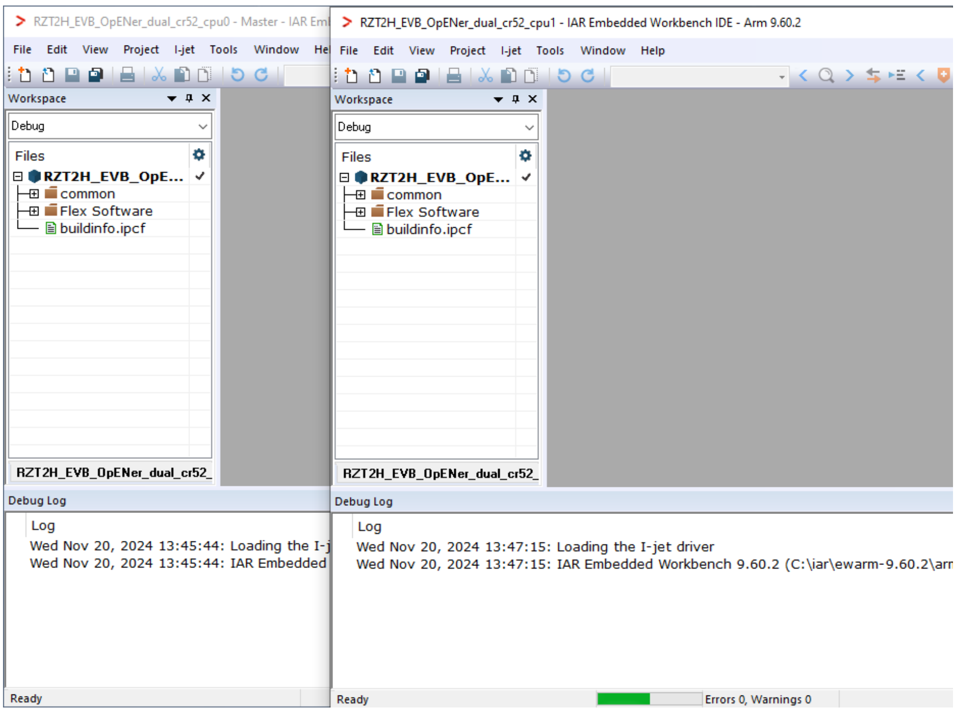


Figure 6-148 EWARM windows

6.5.2.2 RAM execution / Flash boot mode

(1) How to generate source code and how to build

A) Click the “Tools” -> “FSP Smart Configurator” on the CPU1 Workspace.

Note: If you have not set up FSP Smart Configurator yet on EWARM, refer to r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf in which section 5.4 describes how to set up it.

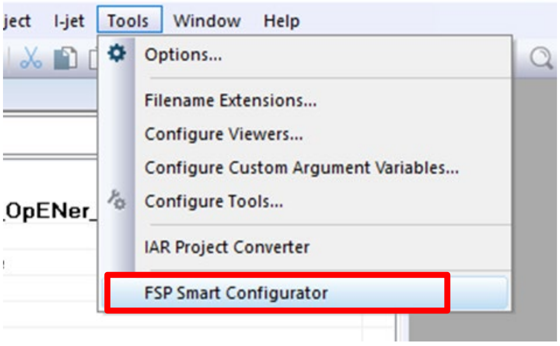


Figure 6-149 Tools tab on the CPU1 Workspace

B) Click “Generate Project Content” button then generate rzt, rzt_gen, rzt_cfg folder.

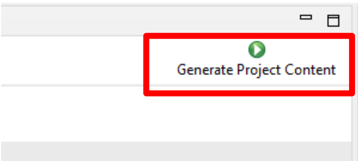


Figure 6-150 Generate Project Content

C) Change the device name in the buildinfo.ipcf file.

R9A09G077M44_CR52_1 -> R9A09G077M44_R52_1

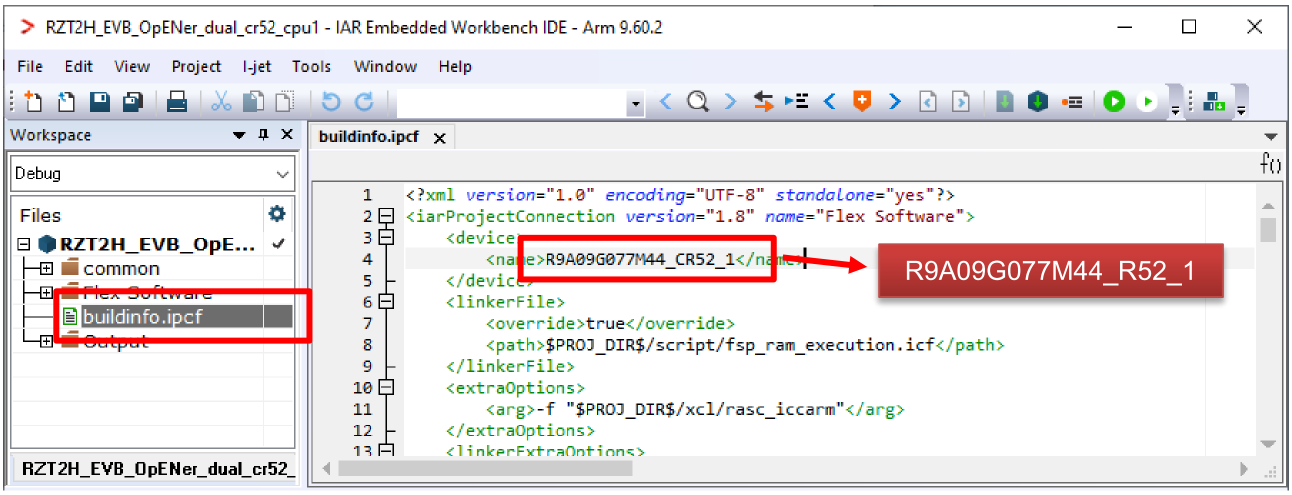


Figure 6-151 Change the device name in the ipcf file

D) (**Flash boot mode only**): - If necessary to enable Type-0 Reset and Type-1 Reset, modify the macro definition "`SAMPLE_EXECUTE_SYSTEM_RESET_PROCESS`" (opener_port_main.c) to "1" enable the Reset process.

E) Click the Make button to build the CPU1 project. Once the build is completed, the build message is displayed in the Build Console window.

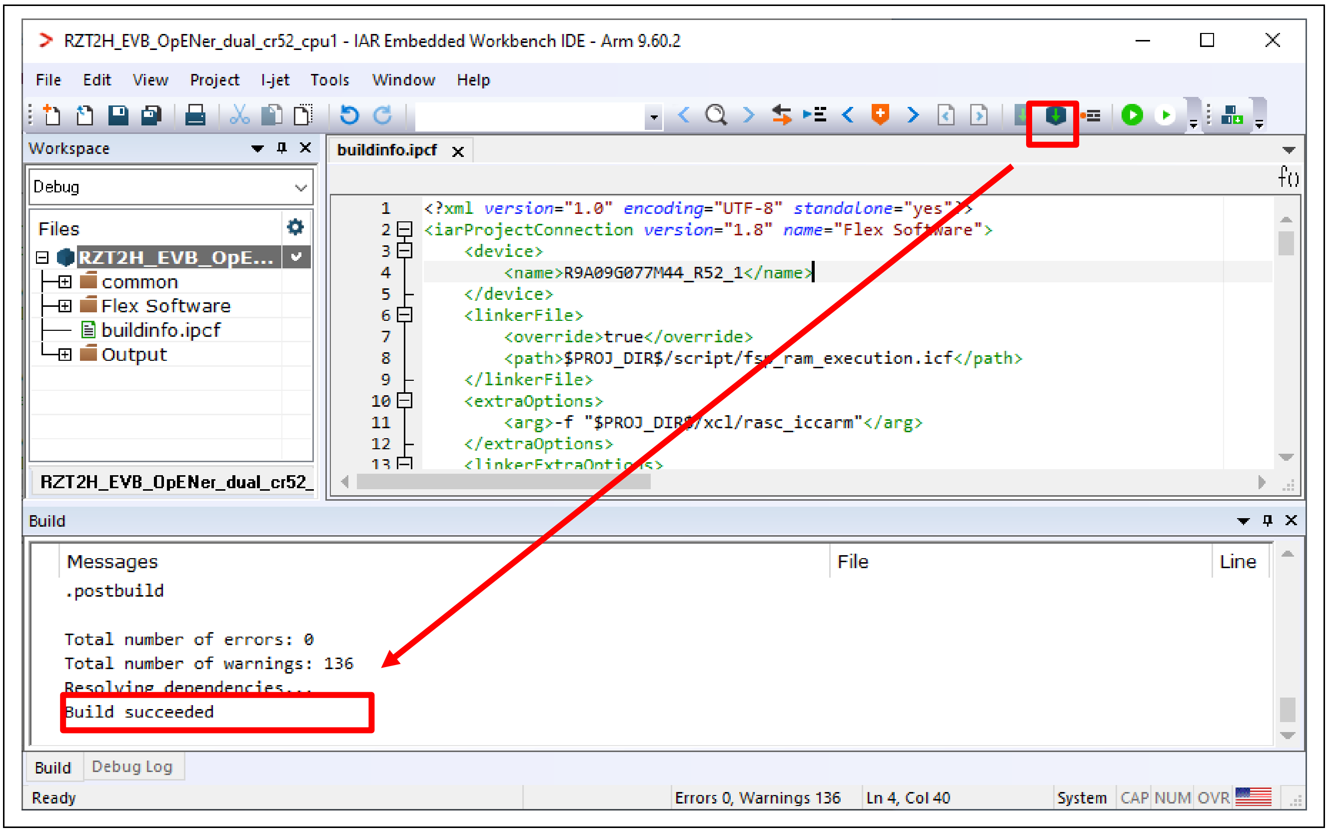


Figure 6-152 Make button on the CPU1 Workspace

F) Click the "Tools" -> "FSP Smart Configurator" on the CPU0 Workspace.

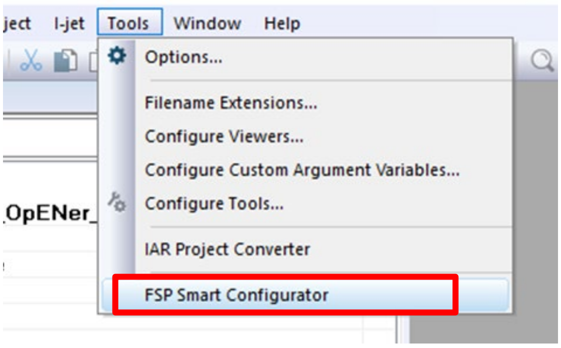


Figure 6-153 Tools tab on the CPU0 Workspace

G) Click “Generate Project Content” button then generate rzt, rzt_gen, rzt_cfg folder.

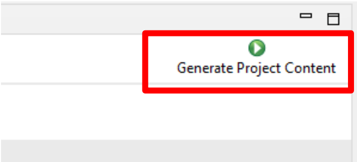


Figure 6-154 Generate Project Content

H) Change the device name in the buildinfo.ipcf file.

R9A09G077M44_CR52_0 -> R9A09G077M44_R52_0

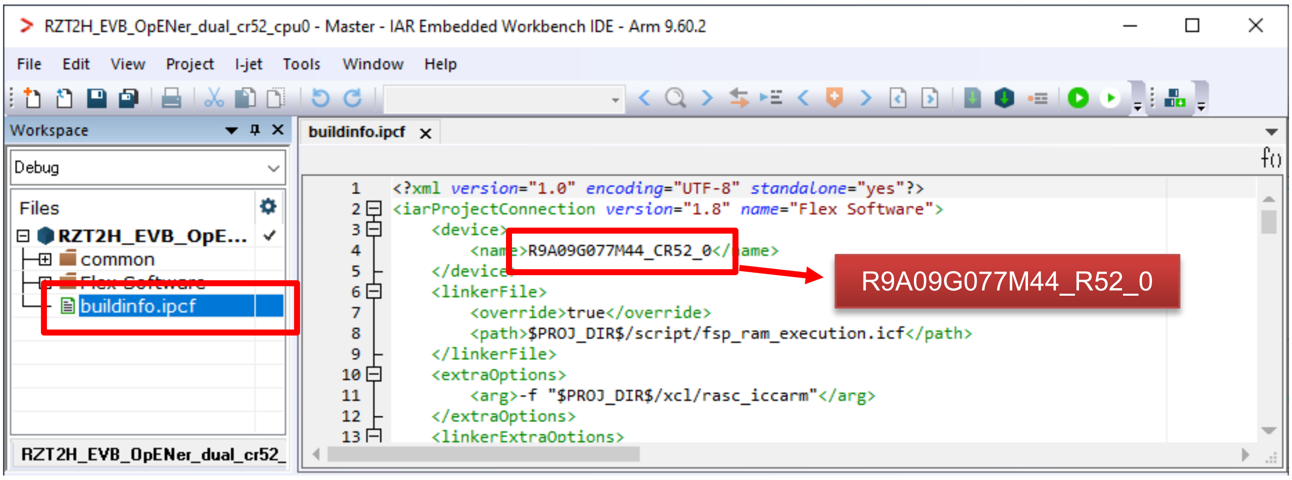


Figure 6-155 Change the device name in the ipcf file

I) Click the Make button to build the CPU0 project. Once the build is completed, the build message is displayed in the Build Console window that displays compilation target files and the number of error / warnings.

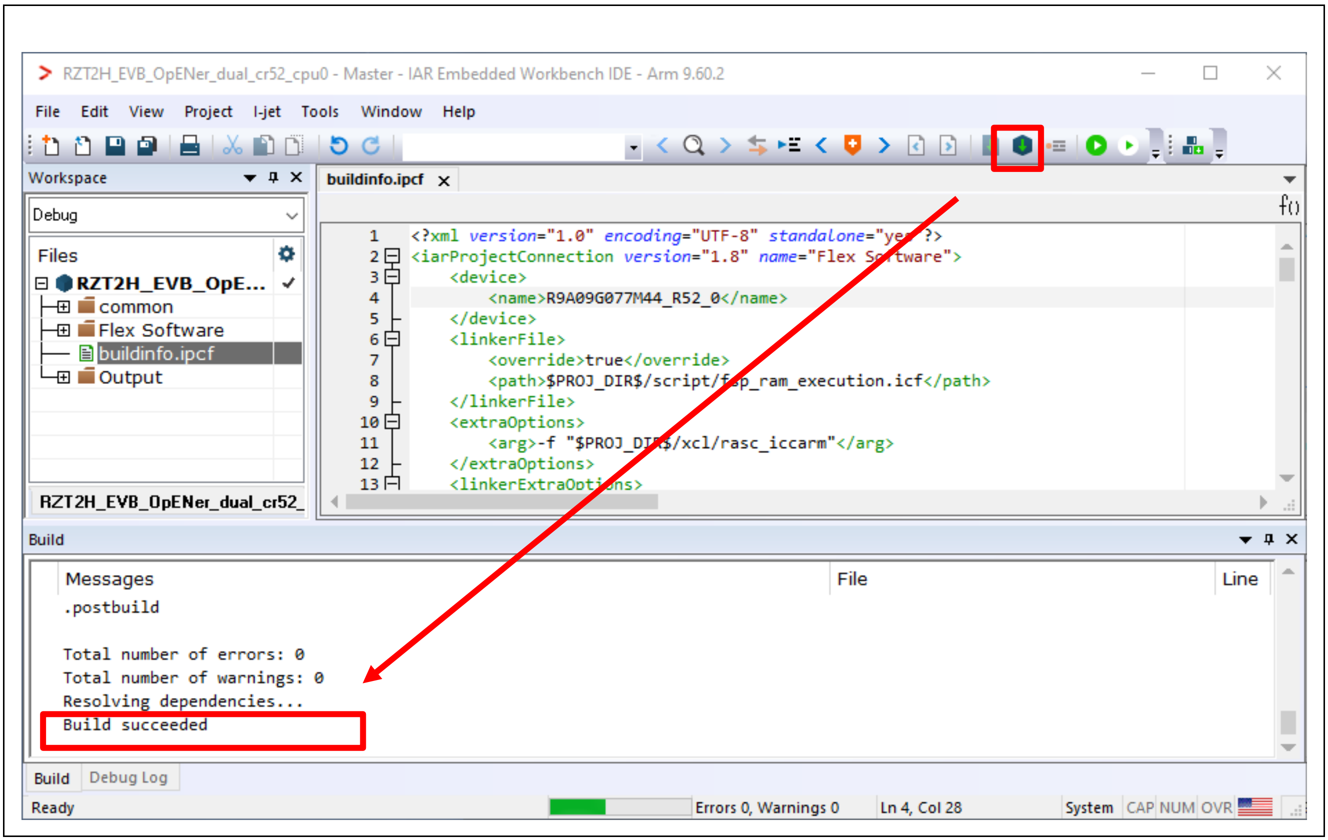


Figure 6-156 Make button on the CPU1 Workspace

J) Close the CPU1 Workspace.

(2) (For RAM-execution): Download application and run debugger

A) Click the Debug button to download the CPU0 application program and launch the debugger. Then automatically the CPU1 Workspace will open.

Figure 6-157 Debug button

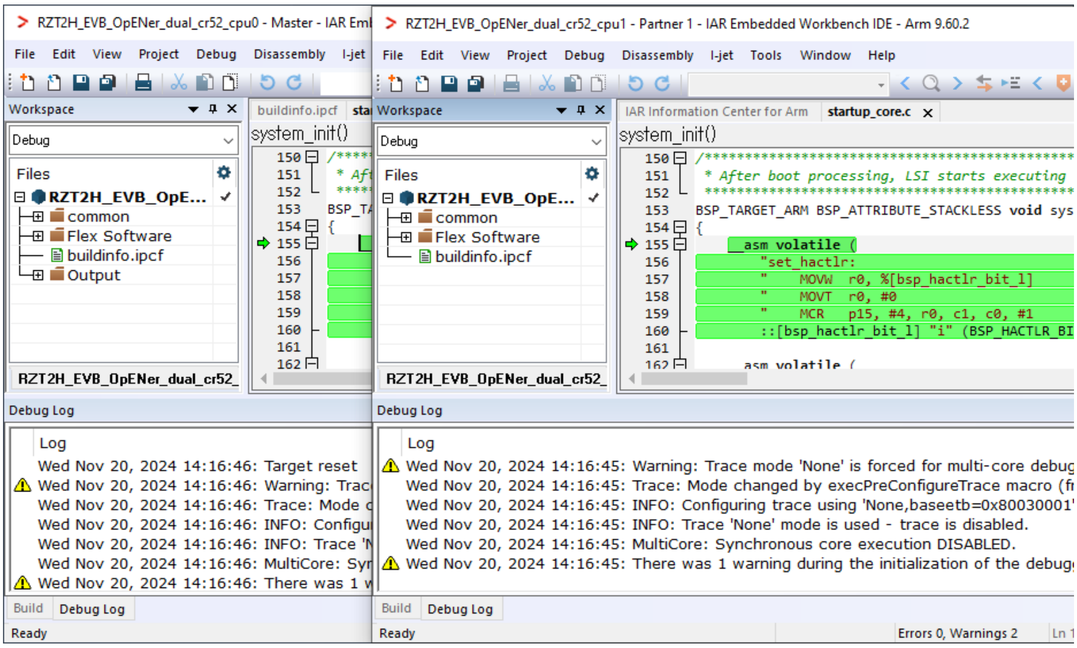


Figure 6-158 The CPU1 Workspace automatically opens

B) Click the Go button on the CPU0 Workspace.

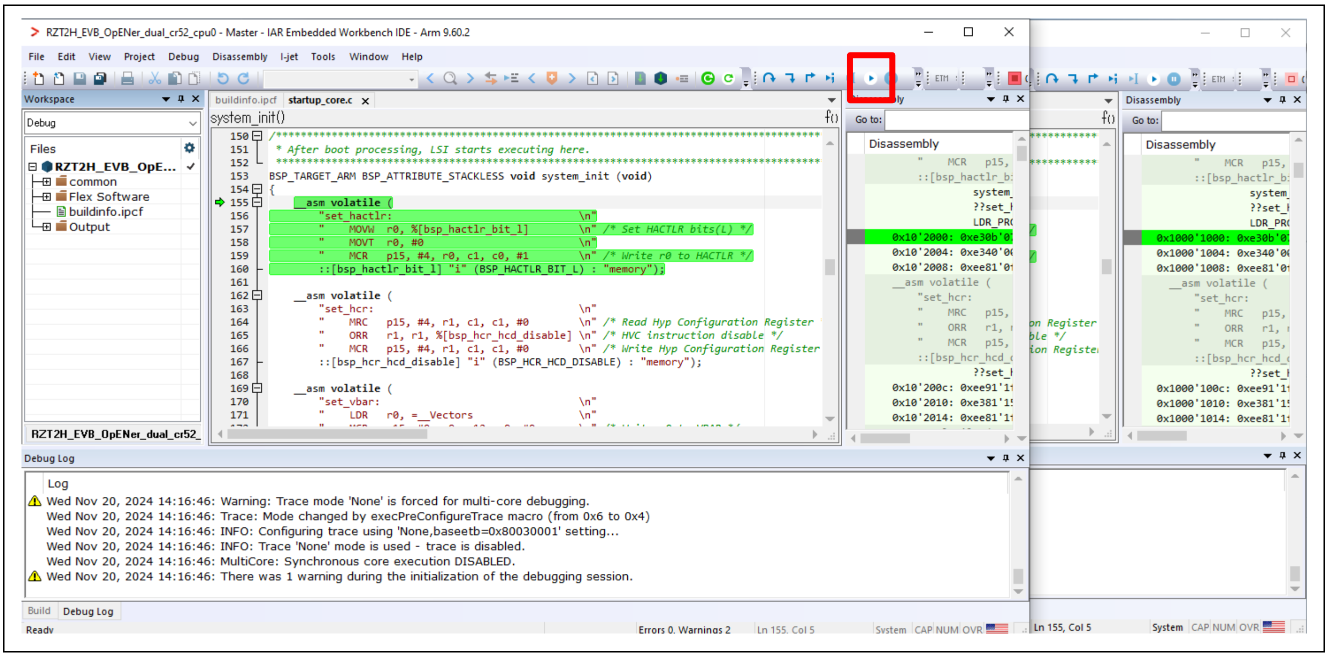


Figure 6-159 Go button on the CPU0 Workspace

C) Click the Go button on the CPU1 Workspace.

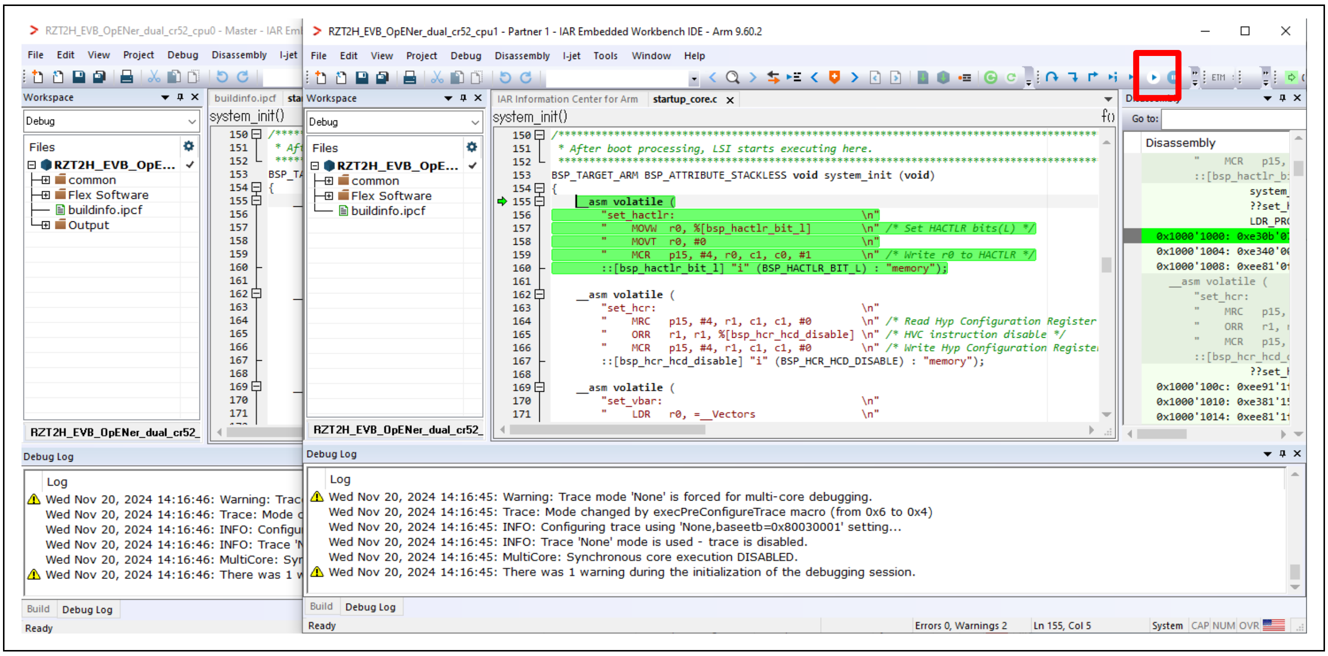


Figure 6-160 Go button on the CPU1 Workspace

Note : Please try to execute "Rebuild All" command on the CPU1 and CPU0 project before downloading if the project doesn't operate.

(3) (For Flash boot mode): Download application

A) Set the Asymmetric multicore setting to "Disabled" in the property of the CPU0 Workspace.

a-1. Click Project > Options….

a-2. Click Debugger > Multicore.

a-3. Select Disable in Asymmetric multicore.

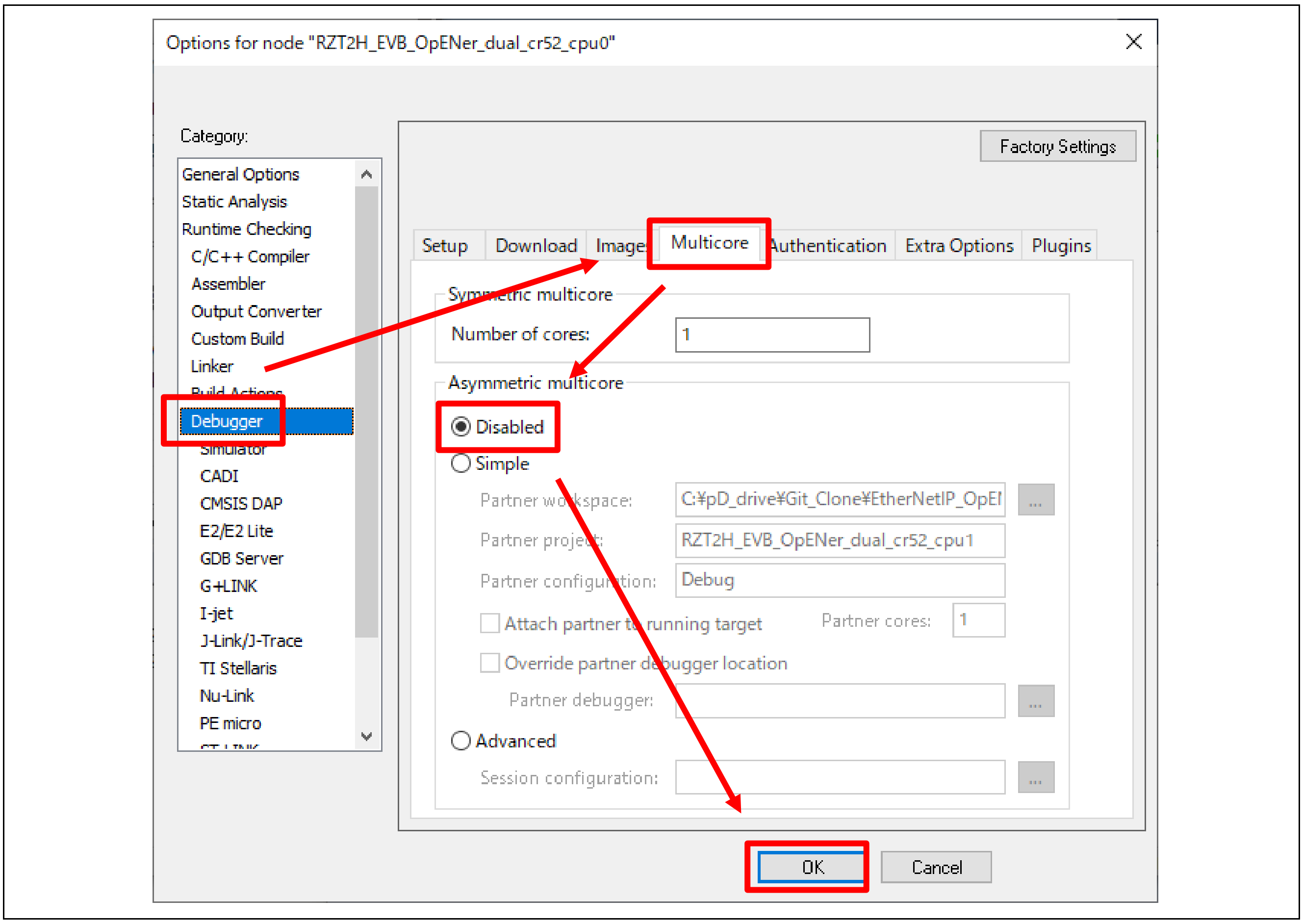


Figure 6-161 Disable multicore debug mode in CPU0 project property

B) In the CPU0 project, click Project > Download > Download file… and select out file of the primary project.

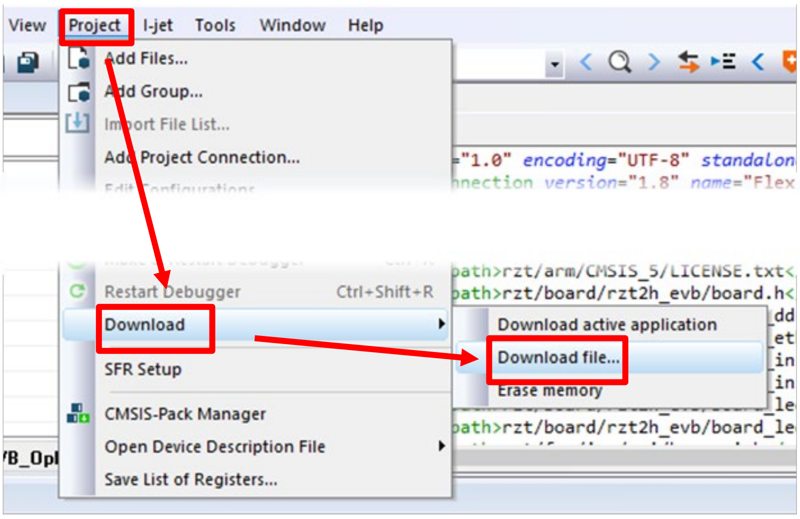
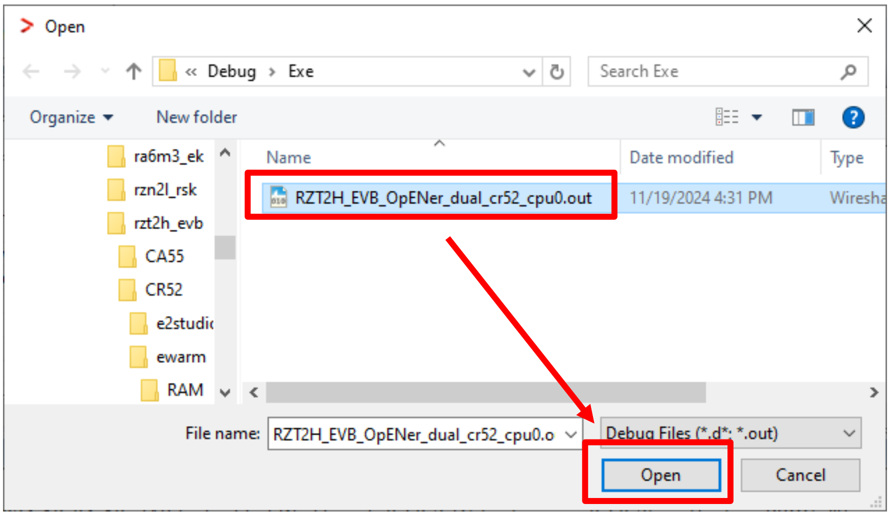


Figure 6-162 IAR EWARM Download File



C) Re-power up the board.

Note : Please try to execute "Rebuild All" command on the CPU1 and CPU0 project before downloading if the project doesn't operate.

6.6 For RZ/N2L (Single-core project)

Before following this chapter, please look at Section 4 for board setup.

The setup differs depending on the IDE.

- When using e² studio, refer to section 6.6.1.
- When using EWARM, refer to section 6.6.2.

6.6.1 Setup sample project for e² studio

This sample program changes the "ld" file in the “script” folder from the default. Therefore, note the following.

・If you use anything other than the “ld” file in the folder, unexpected results may occur.

・If you change the boot mode or FSP version, the “ld” file will be rewritten at the time of code generation, so please save it in advance and restore it to the script before building.

6.6.1.1 Startup e² studio

1. Open the e² studio and select a directory as workspace.
2. Click “Import” in File tab.

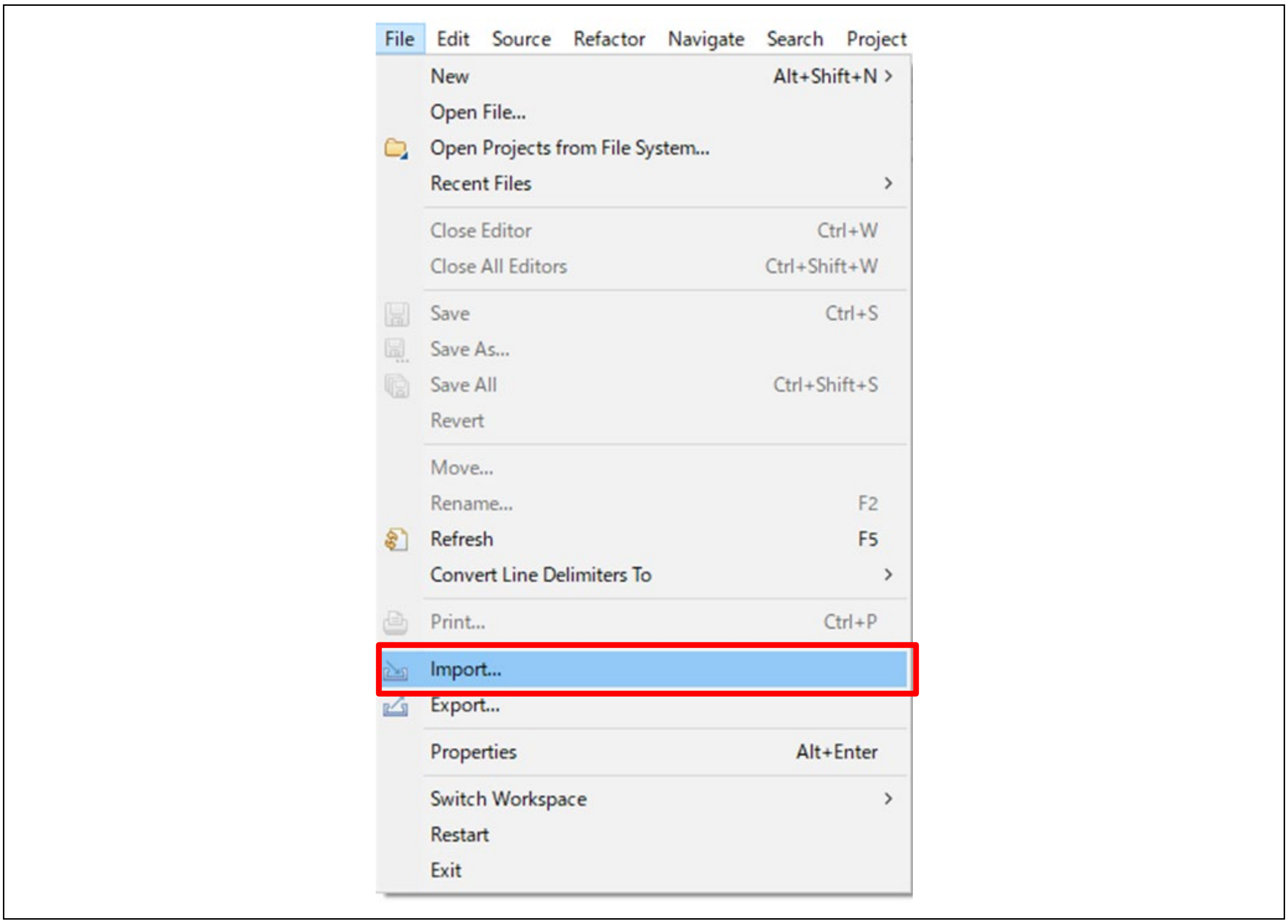


Figure 6-163 e² studio File tab

3. Click “Existing Projects into Workspace” in “General” and Click “Next”.

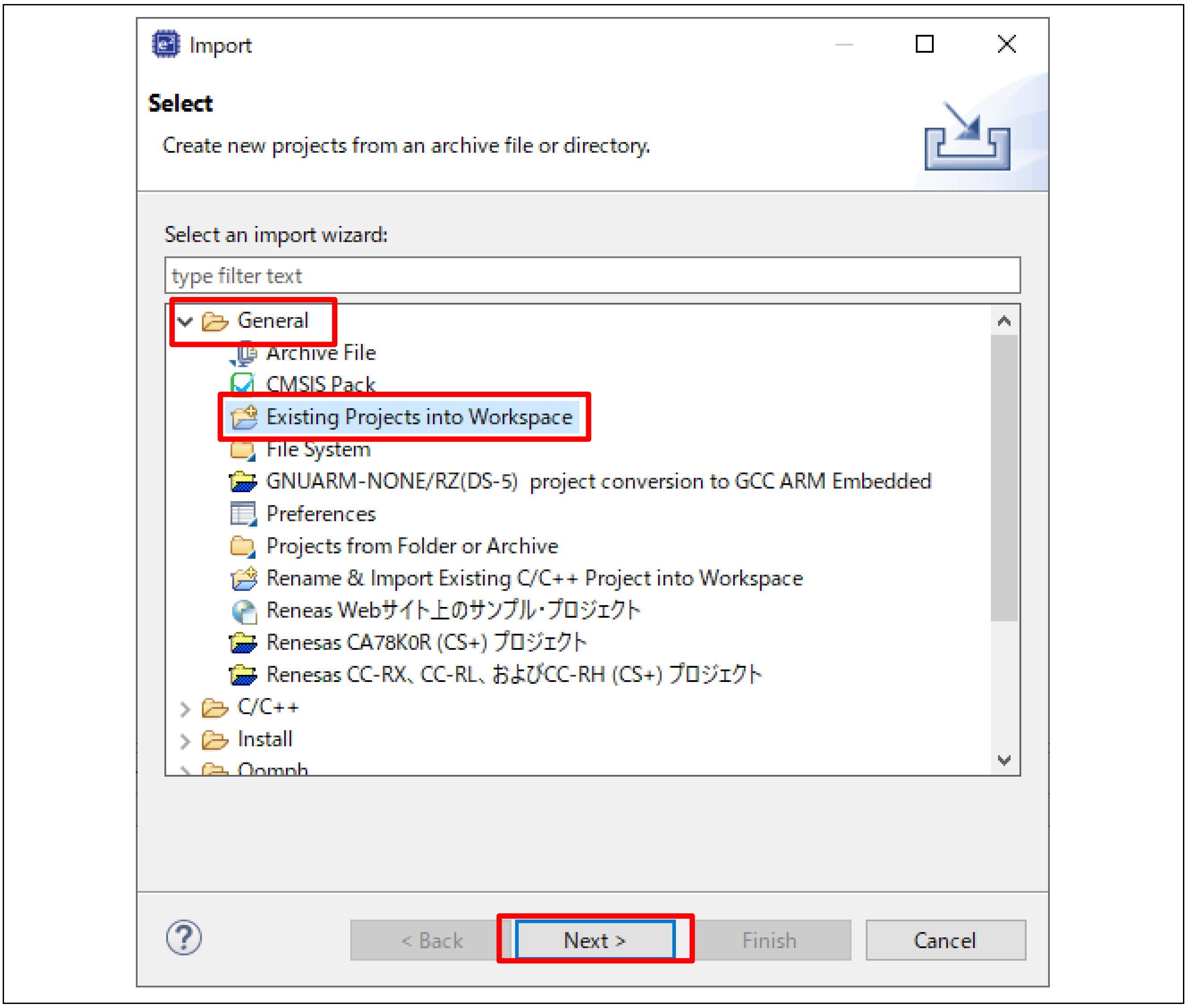


Figure 6-164 Select import type

4. Import project by the following steps.

   A) Set the root directory.

      When you use the RAM-execution projects, please set the root directory.

      ➔ “(Extracted zip folder path)\project\rzn2l_rsk\CR52\e2studio**RAM**”

      When you use the flash boot mode projects, please set the root directory.

      ➔ “(Extracted zip folder path)\project\rzn2l_rsk\CR52\e2studio**xSPI0**”

   B) Select the project.

   C) Click “Finish”

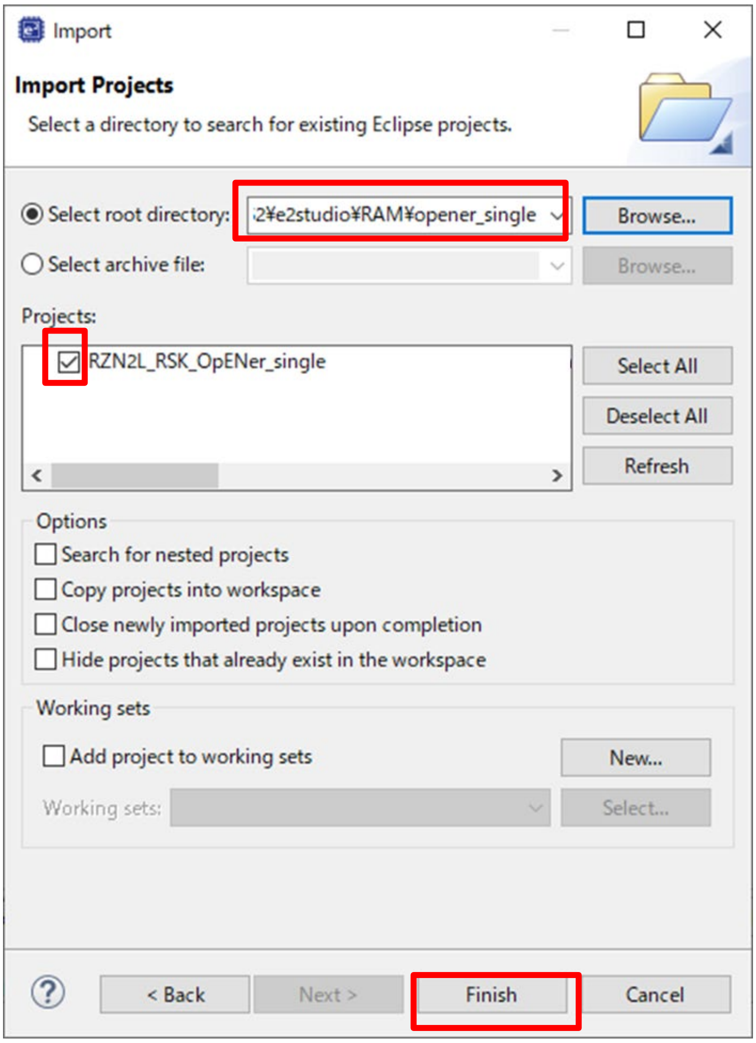


Figure 6-165 Import the project on e^2 studio

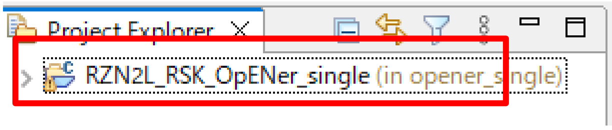


Figure 6-166 Open the project

6.6.1.2 RAM execution / Flash boot mode

(1) How to generate source code and how to build

A) Click the configuration.xml.

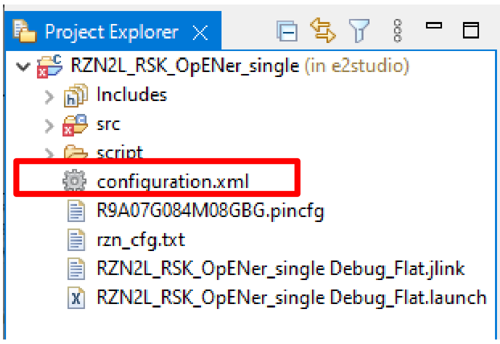


Figure 6-167 Configuration

B) Click “Generate Project Content” button then generate rzn, rzn_gen, rzn_cfg folder.

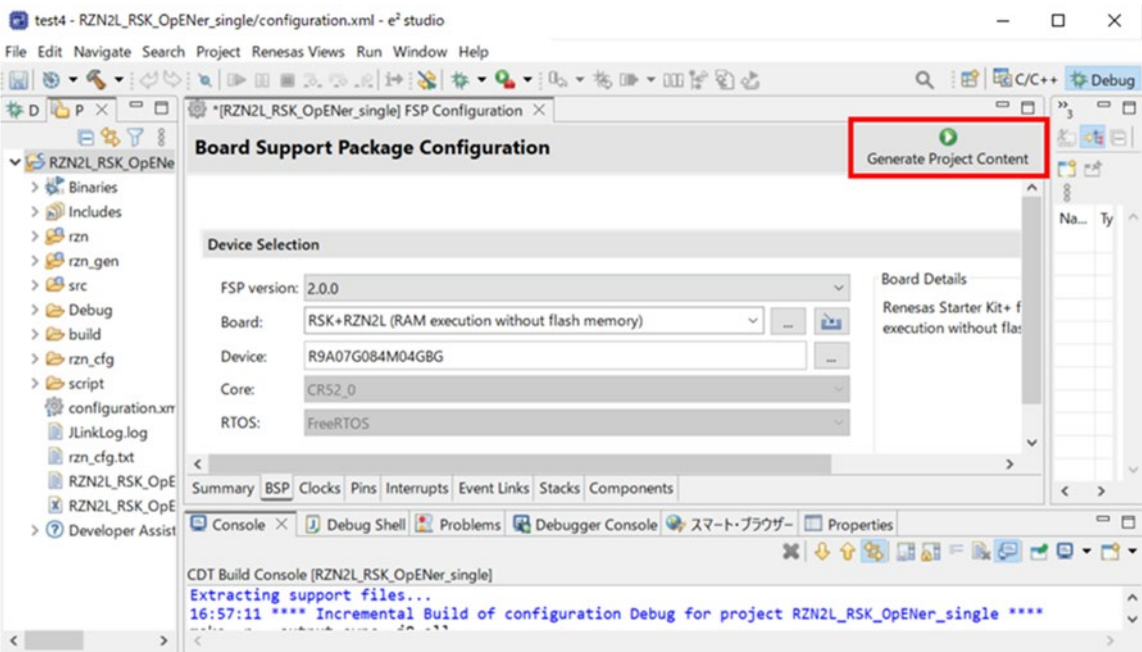


Figure 6-168 Generate Project Content

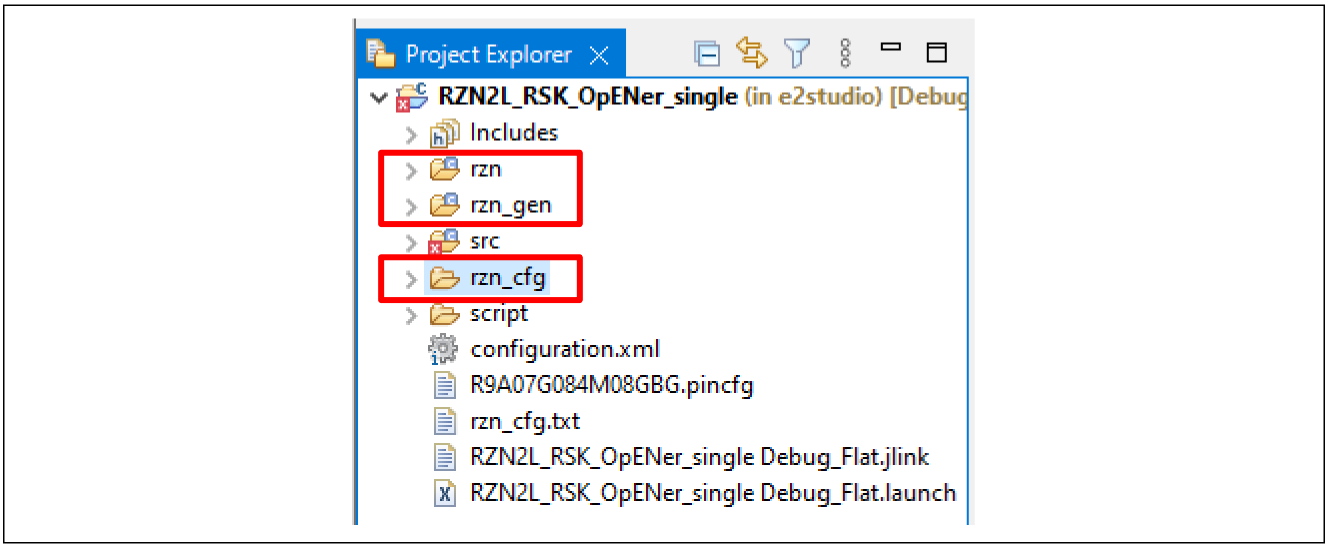


Figure 6-169 Generate project folder

C) Click the Build button in tool bar to build the project and confirm that there is no error message in the build message log.

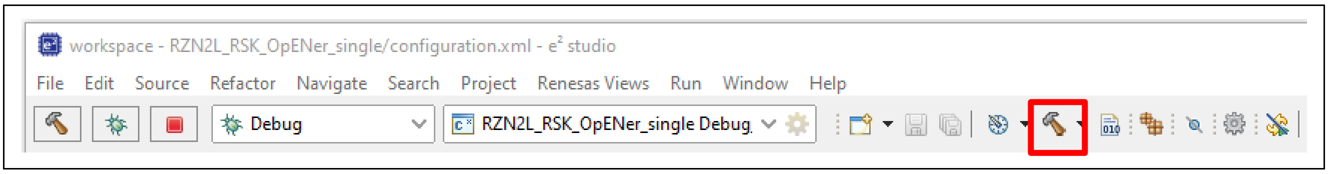


Figure 6-170 Build button

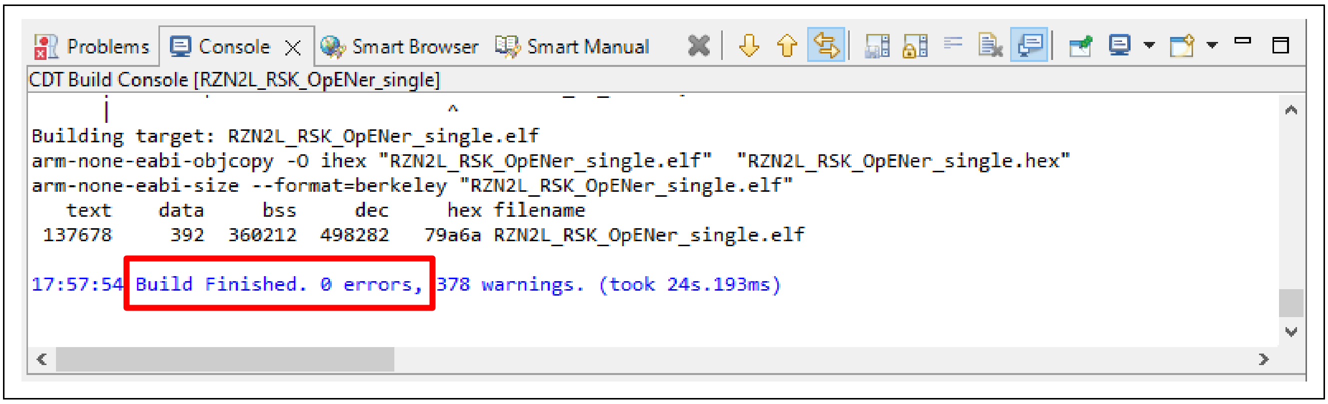


Figure 6-171 Build message

(2) Download application and run debugger

A) Click the Debug button in tool bar to download the built application program and launch the debugger.

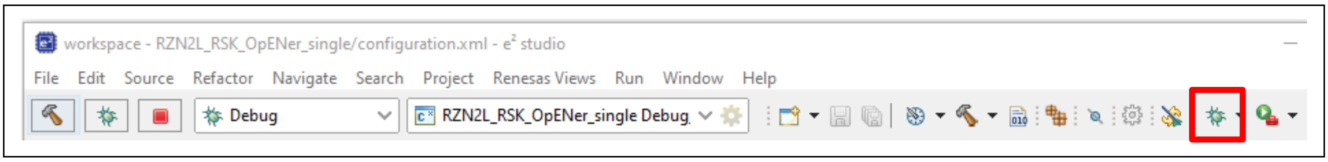


Figure 6-172 Debug button

B) Click the “Switch” button.

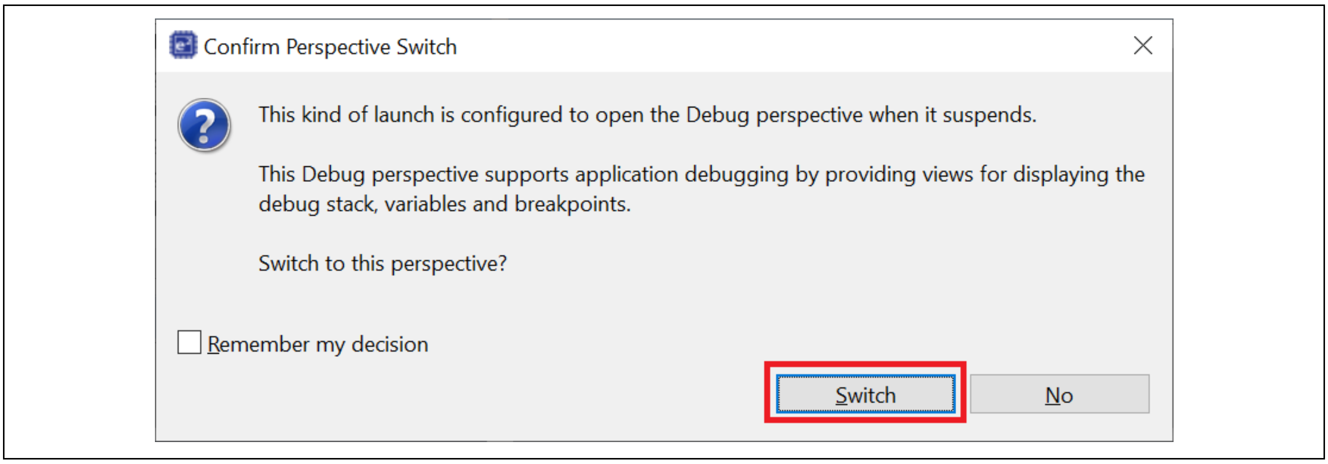


Figure 6-173 Confirm Perspective Switch

C) (**Flash boot mode only**): Disconnect the debugger, and re-power up the board.

D) (**RAM execution only**): The program will break at “system_init” for startup.

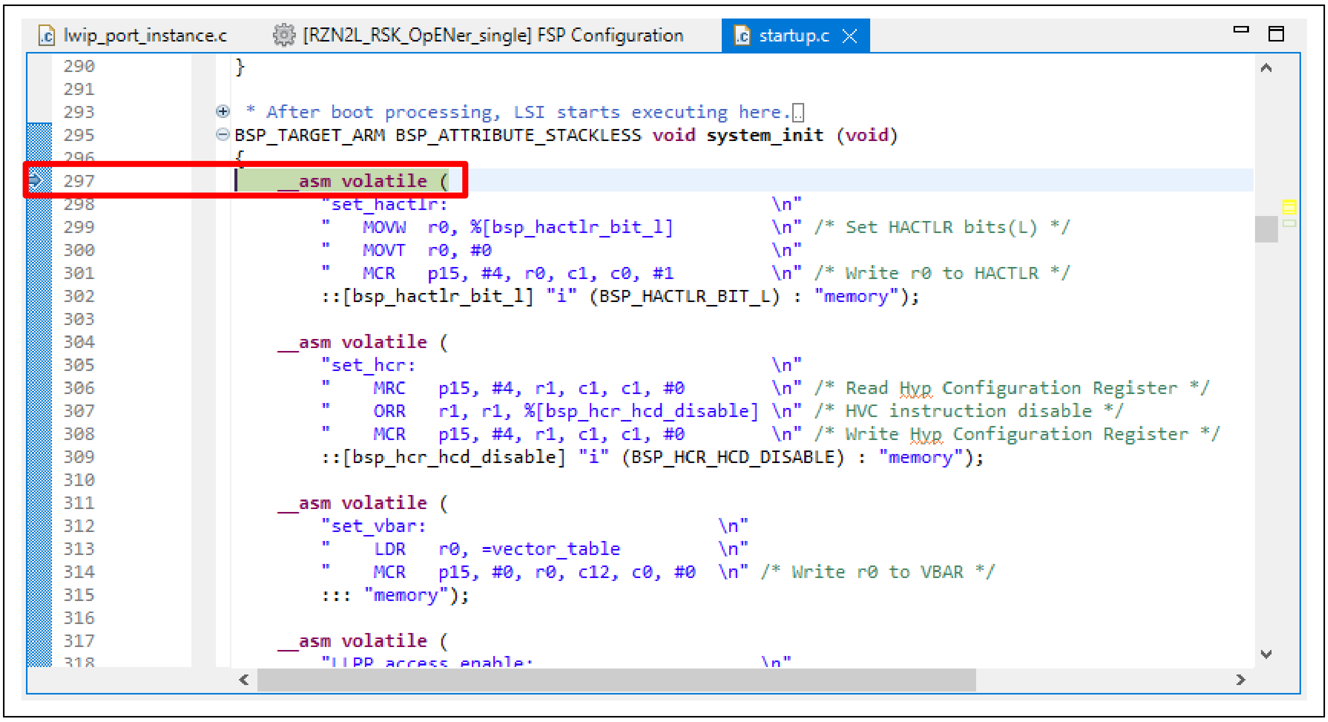


Figure 6-174 Break point 1

E) (**RAM execution only**): Click the Resume button. The program will break at the first of the main function.

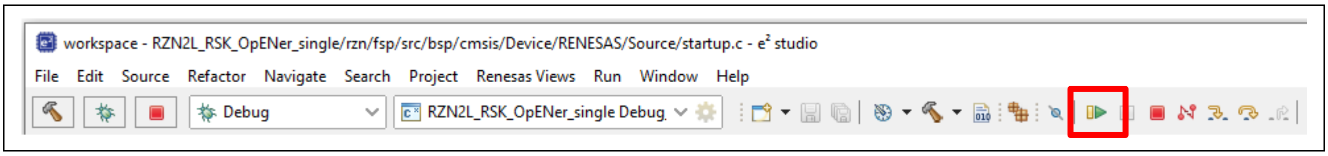


Figure 6-175 Resume button

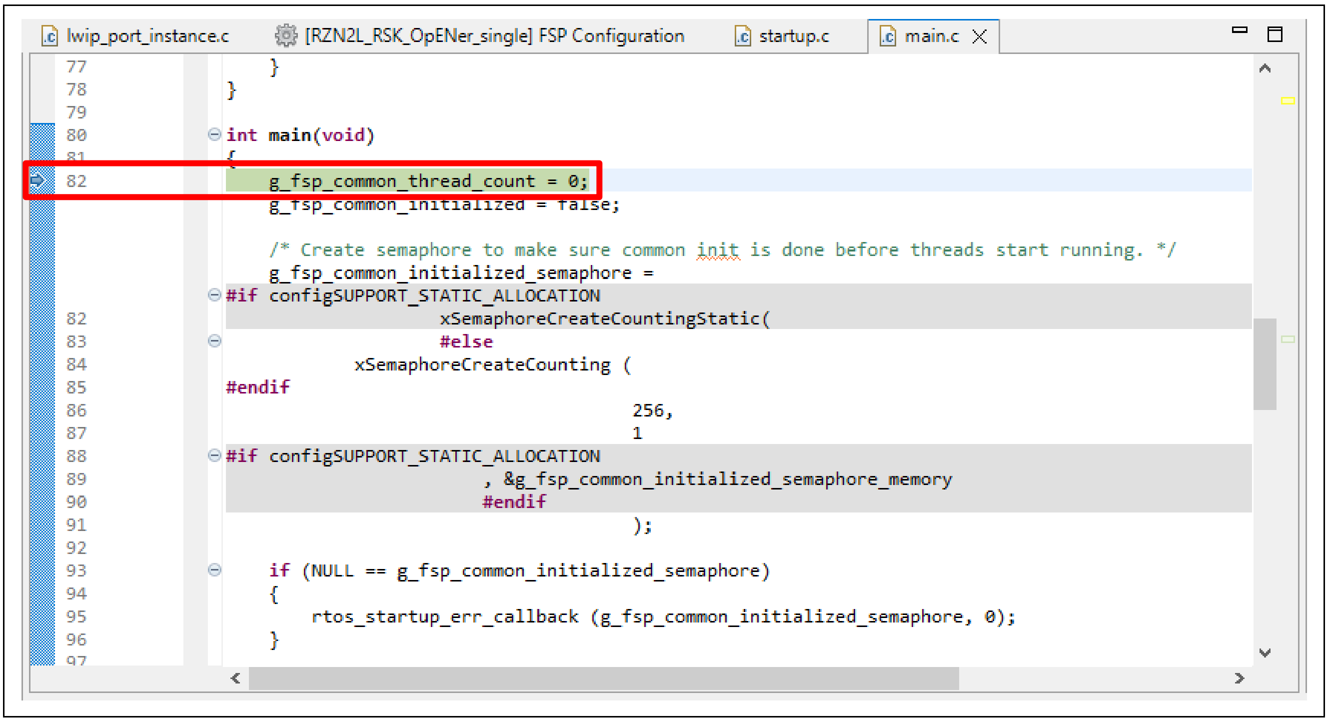


Figure 6-176 Break point 2

F) (**RAM execution only**): Click the Resume button again to execute the program.

6.6.2 Setup sample project for EWARM

This sample program changes the "icf" file in the "script" folder from the default.

・If you use anything other than the “icf” file in the folder, unexpected results may occur.

・If you change the boot mode or FSP version, the “icf” file will be rewritten at the time of code generation, so please save it in advance and restore it to the script before building.

6.6.2.1 Startup EWARM

1. Open the EWARM.
2. Click “Open Workspace…” in the File tab.

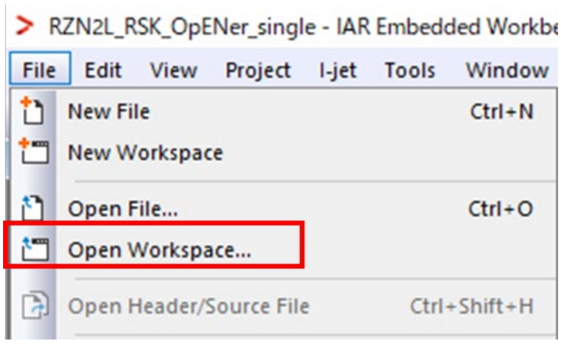


Figure 6-177 EWARM file tab

3. Select the Workspace File(.eww) and click the Open button.

   When you use the RAM-execution projects, please select the following folder.

   ➔ “(Extracted folder path)\project\rzn2l_rsk\CR52\ewarm**RAM**\”

   When you use the flash boot mode projects, please select the following folder.

   ➔ “(Extracted folder path) \project\rzn2l_rsk\CR52\ewarm**xSPI0**\”

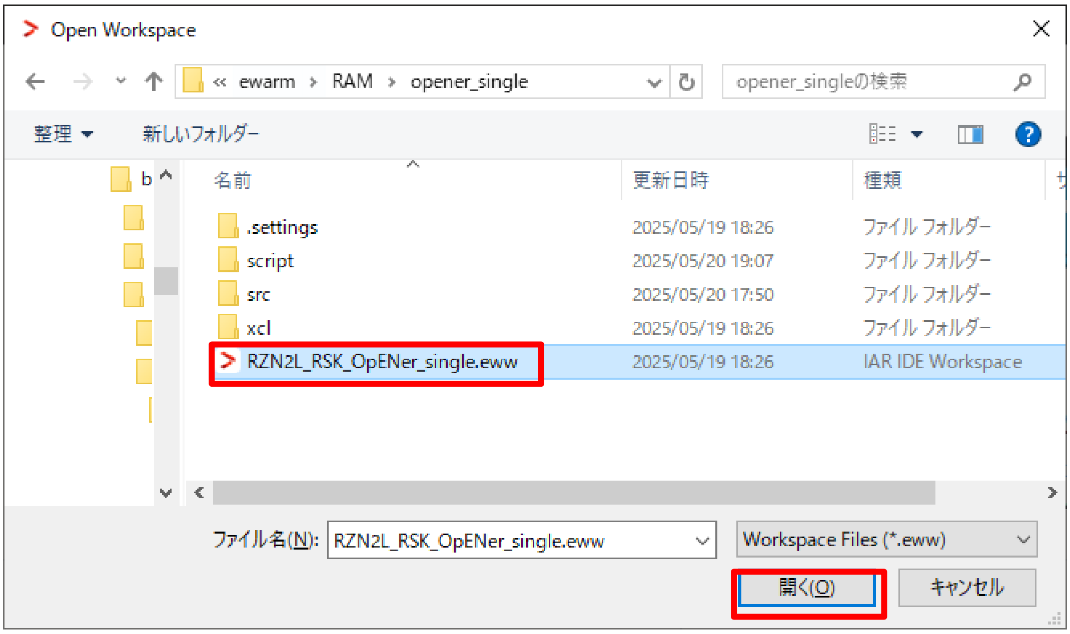


Figure 6-178 Open project file

6.6.2.2 RAM execution / Flash boot mode

(1) How to generate source code and how to build

A) Click the “Tools” -> “FSP Smart Configurator” on tool bar. If you have not set up FSP Smart Configurator yet on EWARM, refer to r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf in which section 5.4 describes how to set it up.

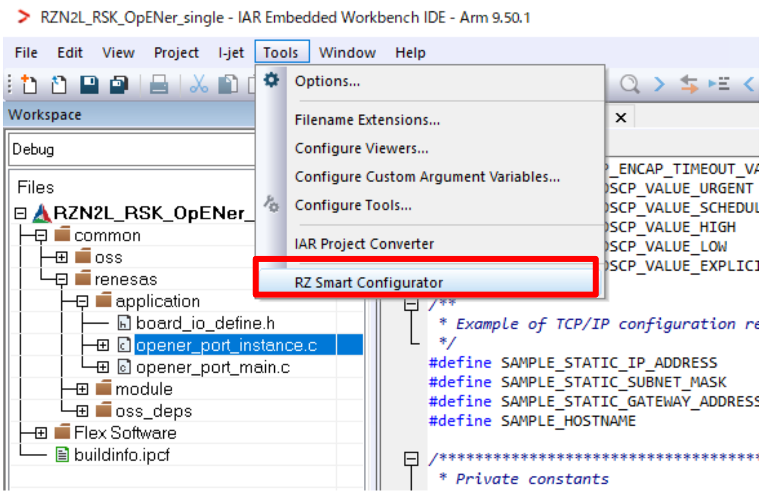


Figure 6-179 Tools tab

B) Click “Generate Project Content” button then generate rzn, rzn_gen, rzn_cfg folder.

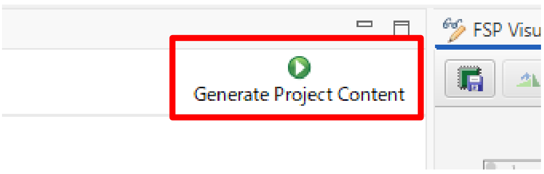


Figure 6-180 Generate Project Content

C) Click the Make button on the tool bar to build. Once the build is completed, the build message is displayed in the Build Console window that displays compilation target files and the number of error/warnings.

Figure 6-181 Make button 1

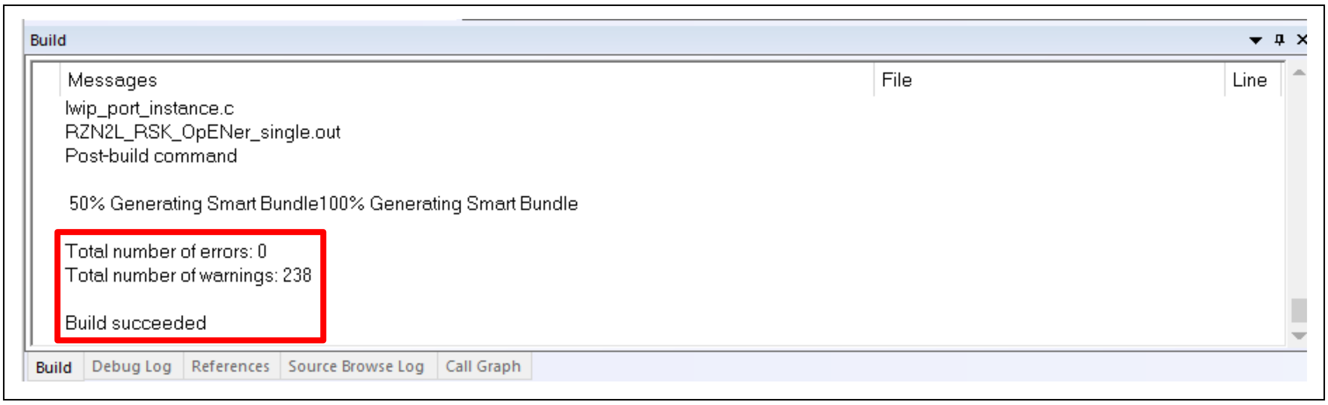


Figure 6-182 Make button 2 and Build console

(2) Download application and run debugger

A) Click the Debug button in tool bar to download the built application program and launch the debugger. The program will break at the first code in main function.

Figure 6-183 Debug button

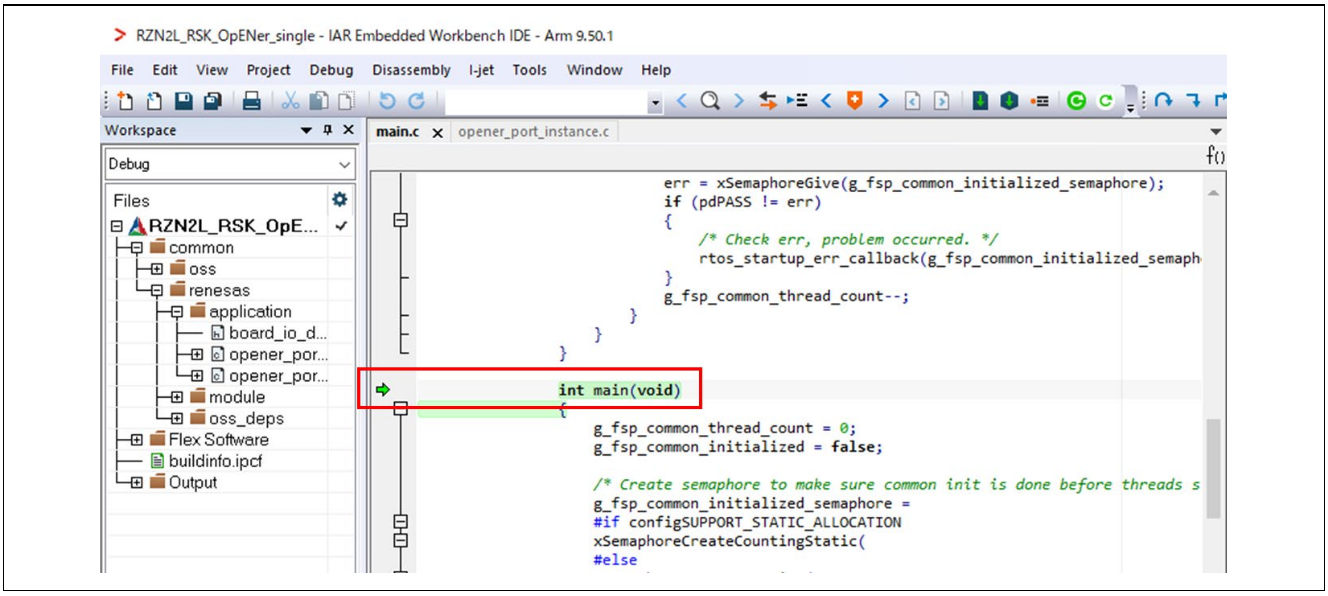


Figure 6-184 Break point

B) Click the Go button.

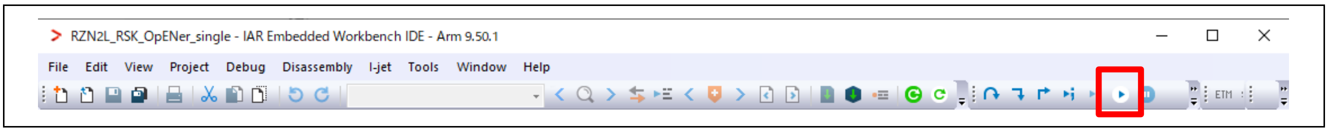


Figure 6-185 Go button

6.7 For RZ/N2H CA55 Core (Single-core project)

The setup differs depending on the IDE.

- When using e² studio, refer to section 6.7.1.
- When using EWARM, refer to section 6.7.2.

6.7.1 Setup sample project for e² studio

This section shows a procedure using e² studio.

6.7.1.1 Startup e² studio

5. Open the e² studio and select a directory as workspace.
6. Click “Import” in File tab.

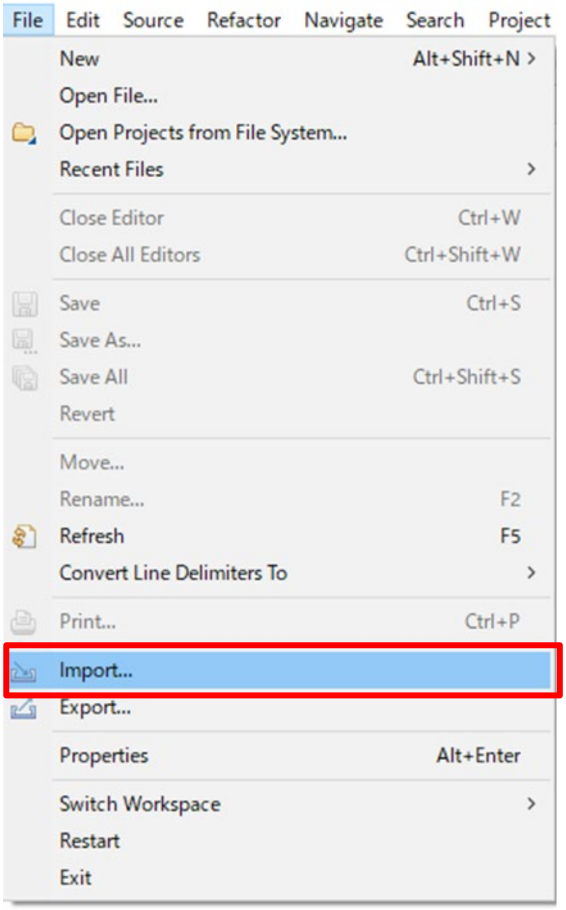


Figure 6-186 e² studio File tab

7. Click “Existing Projects into Workspace” in “General” and Click “Next”.

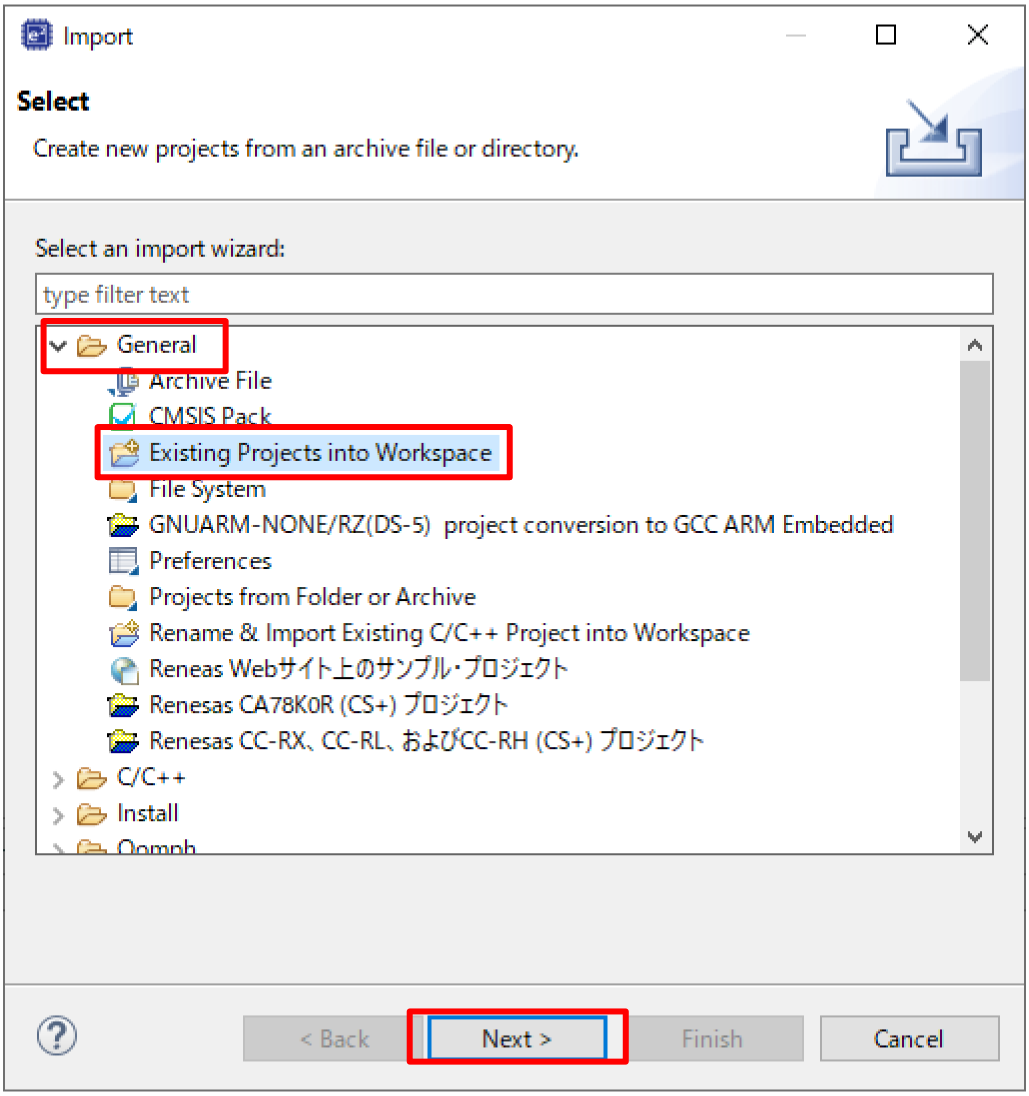


Figure 6-187 Select import type

8. Import project by the following steps.
   A) Set the root directory.
      When you use the RAM-execution projects, please set the root directory.
         ➔ “(Extracted zip folder path)\project\rzn2h_evb\CA55\e2studio\RAM”
      When you use the flash boot mode projects, please set the root directory.
         ➔ “(Extracted zip folder path)\project\rzn2h_evb\CA55\e2studio\xSPI1”
   B) Select the project.
   C) Click “Finish”

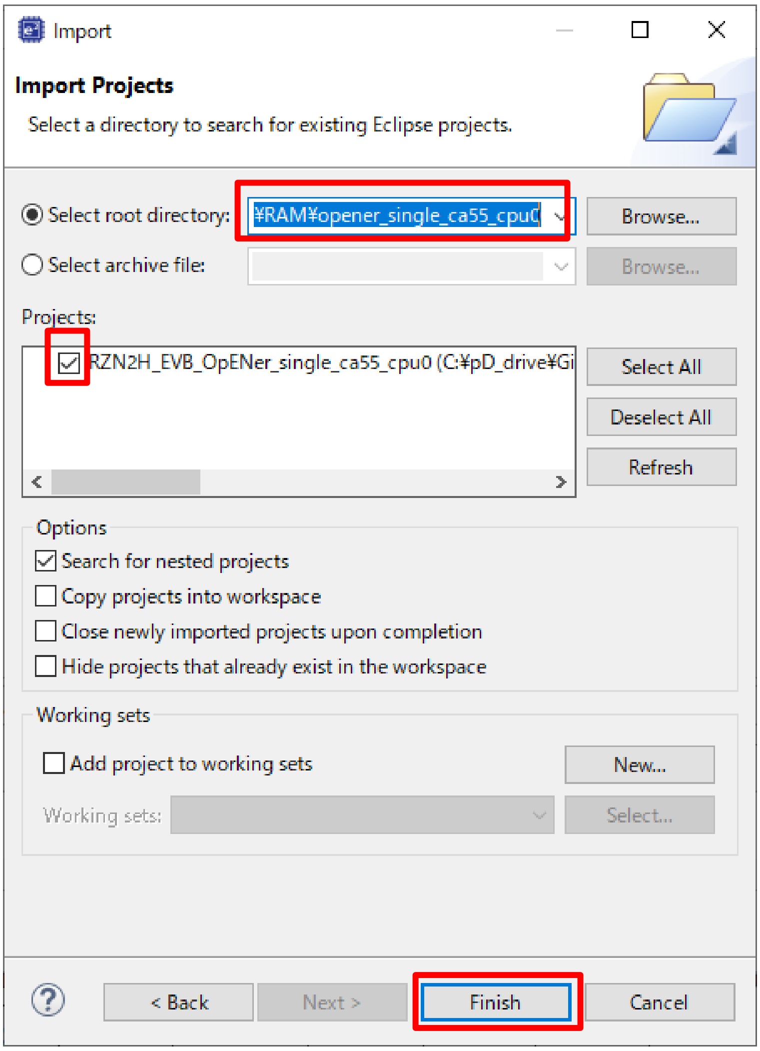


Figure 6-188 Import the project on e² studio

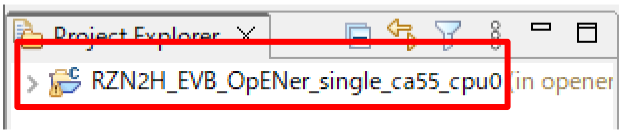


Figure 6-189 Open the project

6.7.1.2 RAM execution / Flash boot mode

(1) How to generate source code and how to build

A) Click the configuration.xml.

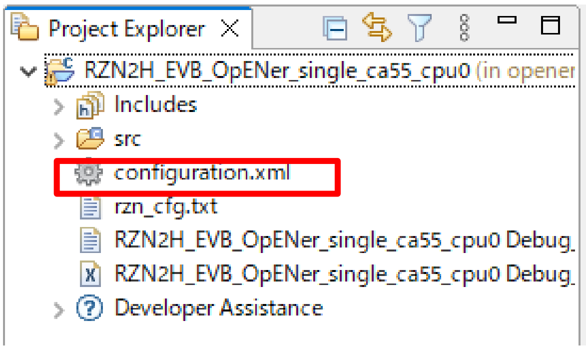


Figure 6-190 Configuration

B) Click “Generate Project Content” button then generate rzn, rzn_gen, rzn_cfg folder.

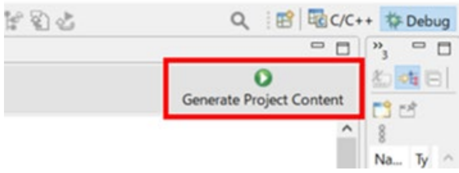


Figure 6-191 Generate Project Content

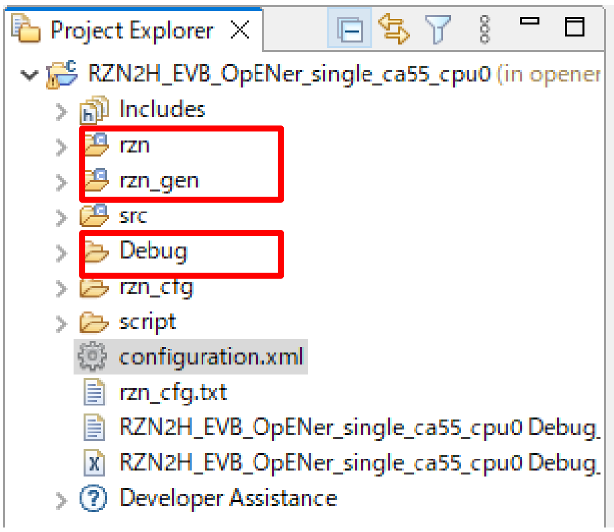


Figure 6-192 Generate project folder

C) (**Flash boot mode only**): - If necessary to enable Type-0 Reset and Type-1 Reset, modify the macro definition "`SAMPLE_EXECUTE_SYSTEM_RESET_PROCESS`" (opener_port_main.c) to "1" enable the Reset process.

D) Click the Build button in tool bar to build the project and confirm that there is no error message in the build message log.

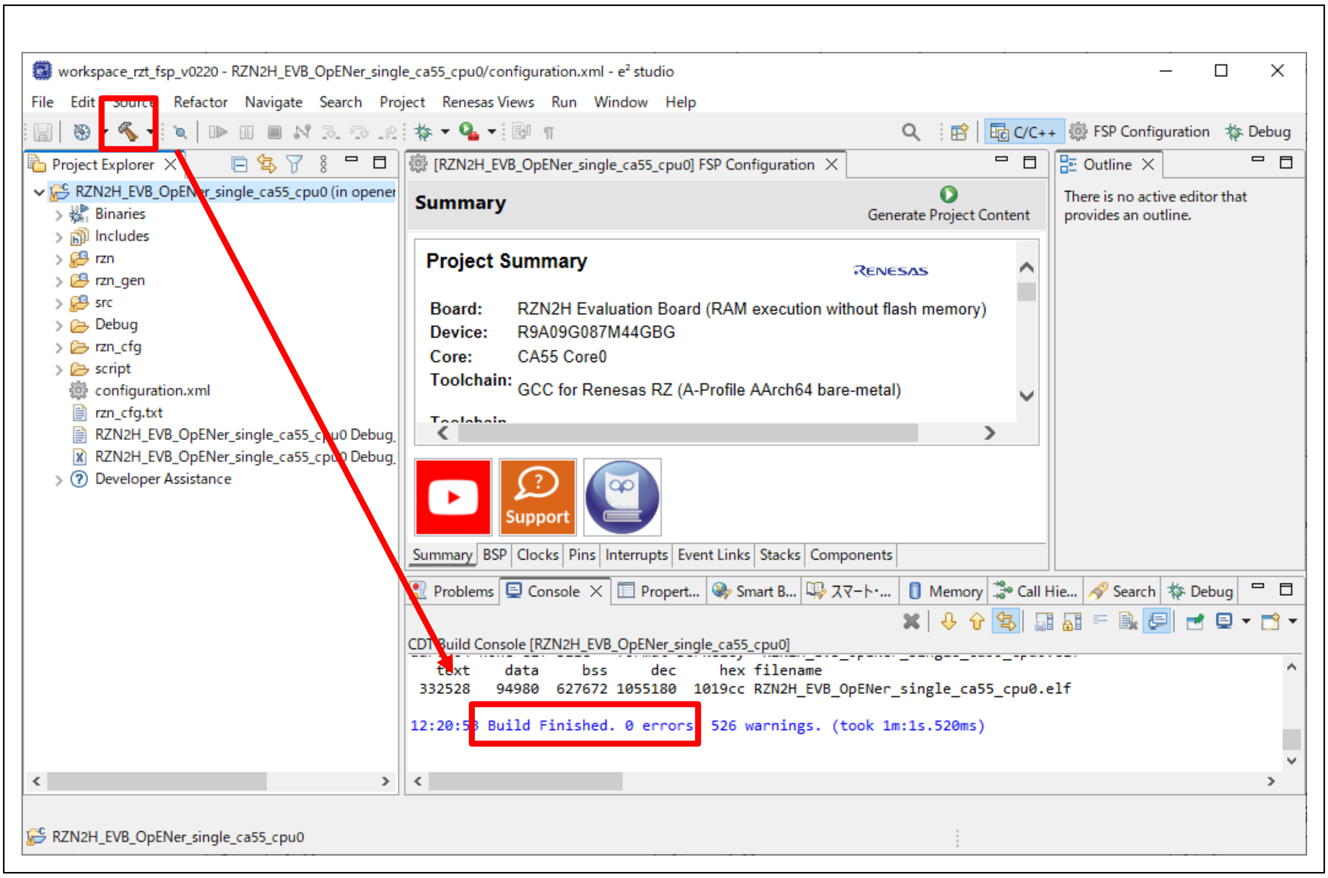


Figure 6-193 Build button and Build message

(2) Download application and run debugger

A) Click the Debug button in tool bar to download the built application program and launch the debugger.

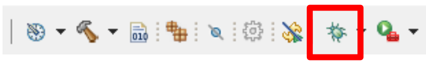


Figure 6-194 Debug button

B) Click to "Yes" button on the Proceed confirmation dialog box.

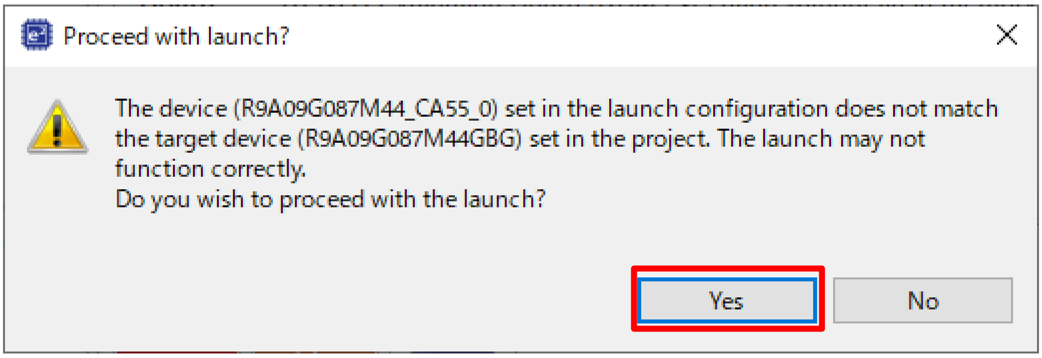


Figure 6-195 Proceed confirmation dialog box

C) Click the "Switch" button.

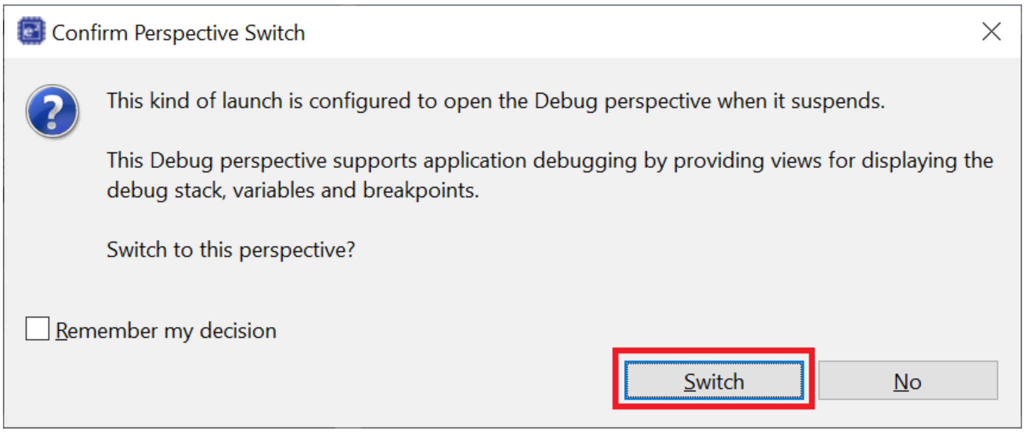


Figure 6-196 Confirm Perspective Switch

D) (**Flash boot mode only**): Disconnect the debugger, and re-power up the board.

Note: For the RZN FSP v2.1.0, there is the limitation "Unable to debug CA55 flash boot project with e^2 studio." (r01an6434ej0112-rzt2-rzn2-fsp-getting-started.pdf, Tool Software Limitations No.13)

E) (**RAM-execution only**): Click the Resume button to execute the program.

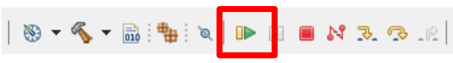


Figure 6-197 Resume button

6.7.2 Setup sample project for EWARM

This section shows a procedure using EWARM.

Some files in the CA55 project with flash boot mode were modified to address the issue of the RZN FSP v2.1.0. (Known Issues No.39 in the FSP quick start guide r01an6434ej0113-rzt2-rzn2-fsp-getting-started.pdf) Therefore, please note the following.

- Modified files:
  - ➢ The system_core.c (\opener_single_ca55_cpu0\rzn\fsp\src\bsp\cmsis\Device\RENESAS\Source\ca\)
  - ➢ The bsp_cache.c (\opener_single_ca55_cpu0\rzn\fsp\src\bsp\mcu\all\)
  - ➢ The fsp_xspi1_boot.icf file (\opener_single_ca55_cpu0\script)
- Note:
  - ➢ If you change the boot mode, these files will be overwritten at the time of code generation, so please save it in advance and restore it before building.

6.7.2.1 Startup EWARM

1. Open the EWARM.
2. Click “Open Workspace…” in the File tab.

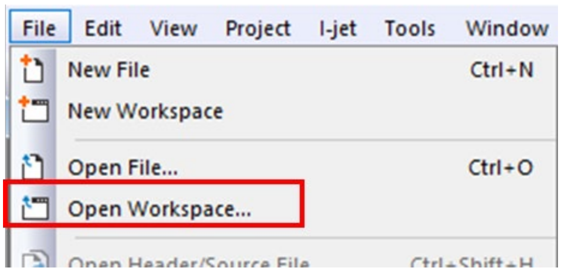


Figure 6-198 EWARM file tab

3. Select the Workspace File(.eww) and click the Open button.

   When you use the RAM-execution projects, please select the following folder.

   ➔ “(Extracted folder path)\project\rzn2h_evb\CA55\ewarm**RAM**\”

   When you use the flash boot mode projects, please select the following folder.

   ➔ “(Extracted folder path) \project\rzn2h_evb\CA55\ewarm**xSPI1**\”

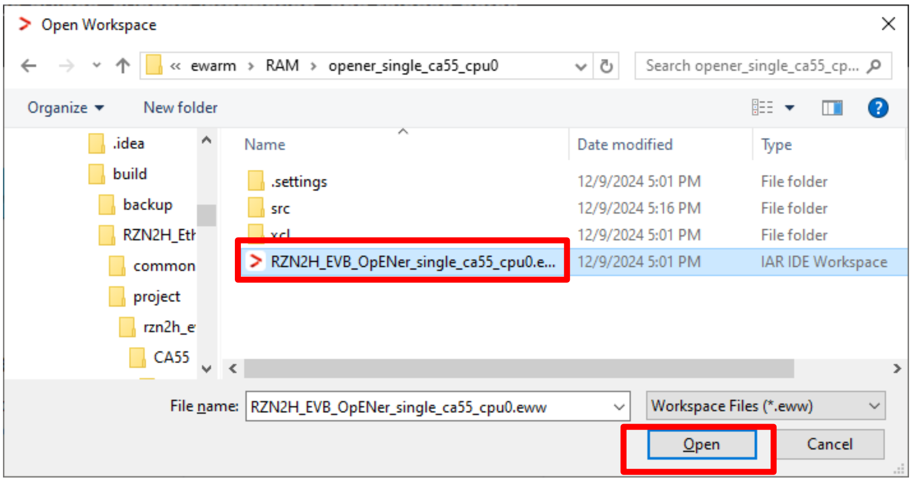


Figure 6-199 Open project file

6.7.2.2 RAM execution / Flash boot mode

(1) How to generate source code and how to build

A) Click the “Tools” -> “FSP Smart Configurator” on tool bar. If you have not set up FSP Smart Configurator yet on EWARM, refer to r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf in which section 5.4 describes how to set it up.

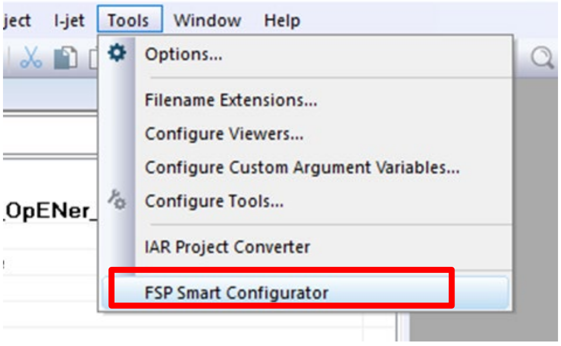


Figure 6-200 Tools tab

B) Click “Generate Project Content” button then generate rzn, rzn_gen, rzn_cfg folder.

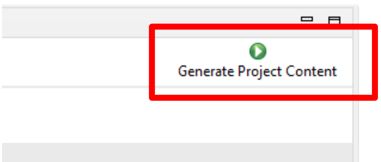


Figure 6-201 Generate Project Content

C) Change the device name in the buildinfo.ipcf file.

R9A09G087M44_CR52_0 -> R9A09G087M44_A55

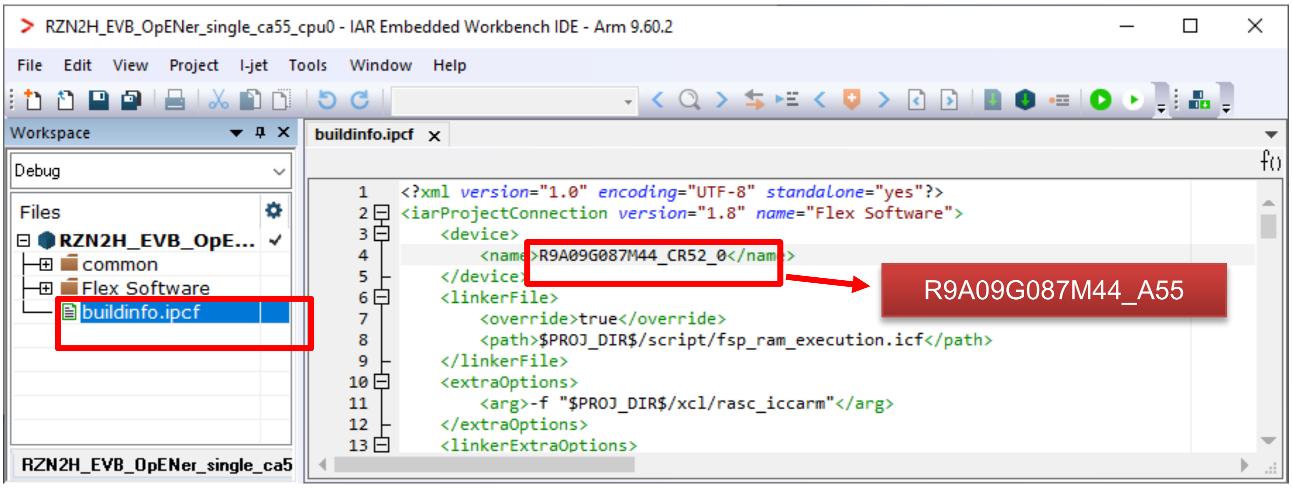


Figure 6-202 Change the device name in the ipcf file

D) Click the Make button on the tool bar to build. Once the build is completed, the build message is displayed in the Build Console window.

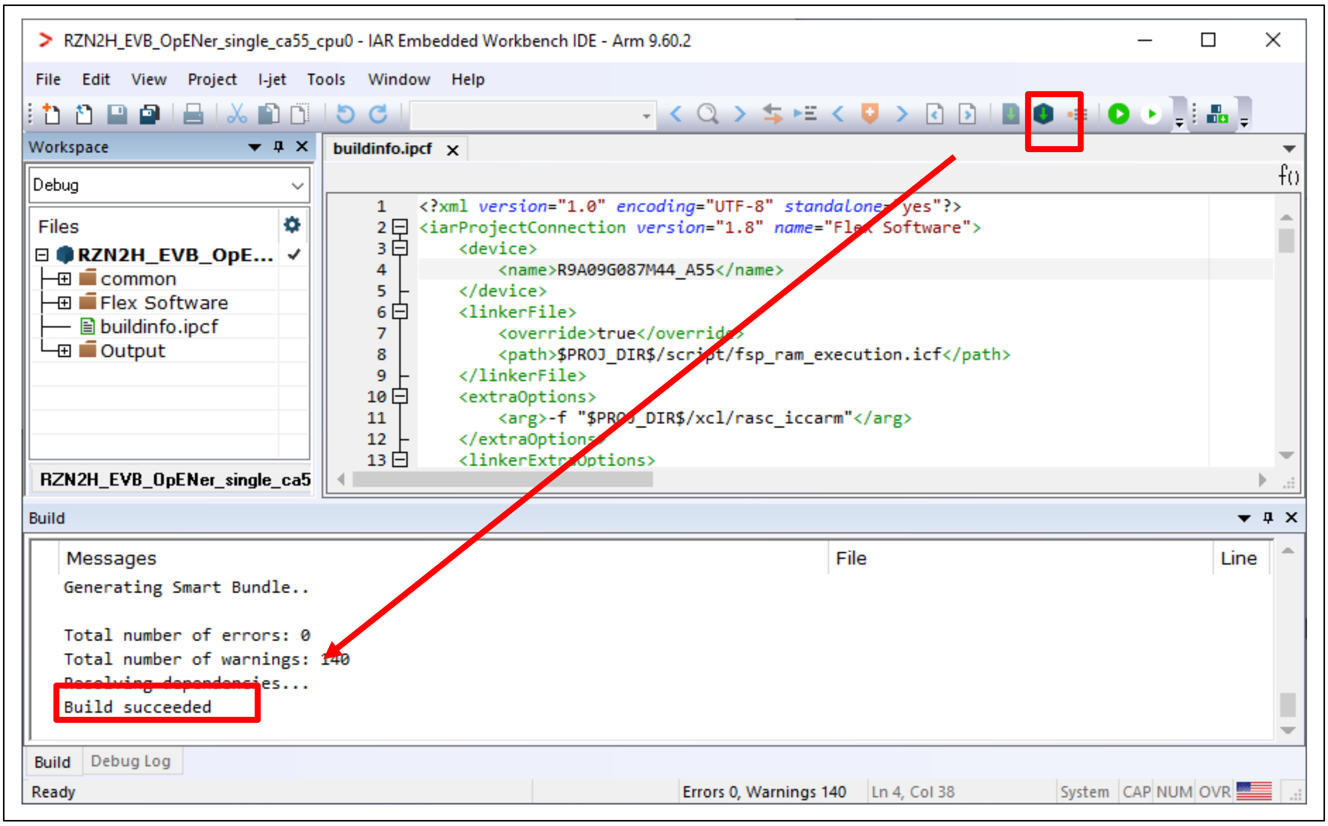


Figure 6-203 Make button and Build console

(2) Download application and run debugger

A) Click the Debug button in tool bar to download the built application program and launch the debugger.

Figure 6-204 Debug button

B) Click the Go button.

Figure 6-205 Go button

6.8 For RZ/N2H CR52 Core (Dual-core project)

The setup differs depending on the IDE.

- When using e^2 studio, refer to section 6.8.1.
- When using EWARM, refer to section 6.8.2.

6.8.1 Setup sample project for e^2 studio

This section shows a procedure using e^2 studio.

6.8.1.1 Startup e^2 studio

1. Open the e^2 studio and select a directory as workspace.
2. Click “Import” in File tab.

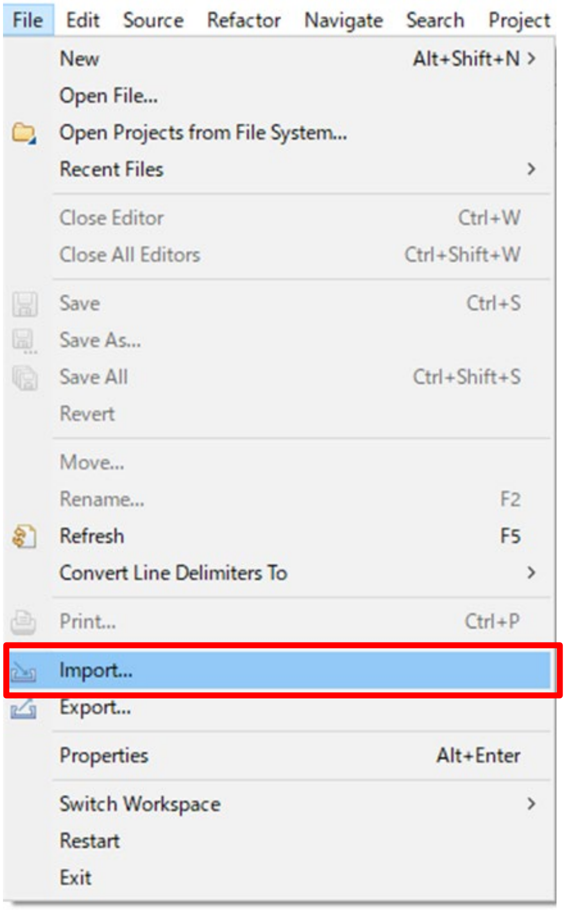


Figure 6-206 e^2 studio File tab

3. Click “Existing Projects into Workspace” in “General” and Click “Next”.

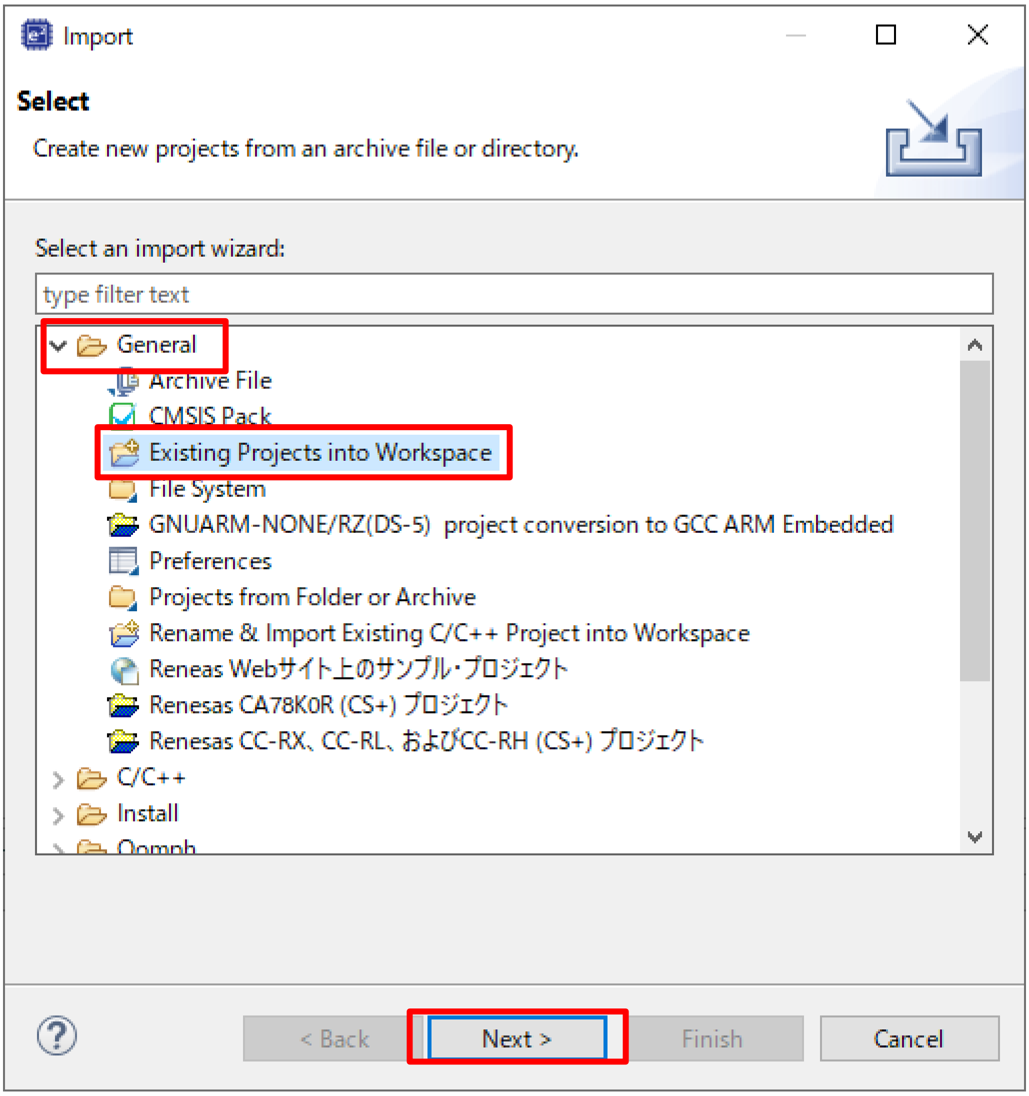


Figure 6-207 Select import type

4. Import project by the following steps.
   A) Set the root directory.
      When you use the RAM-execution projects, please set the root directory.
      ➔ "(Extracted zip folder path)\project\rzn2h_evb\CR52\e2studio**RAM**"
      When you use the flash boot mode projects, please set the root directory.
      ➔ "(Extracted zip folder path)\project\rzn2h_evb\CR52\e2studio**xSPI1**"
   B) Select 2 projects.
   C) Click "Finish"

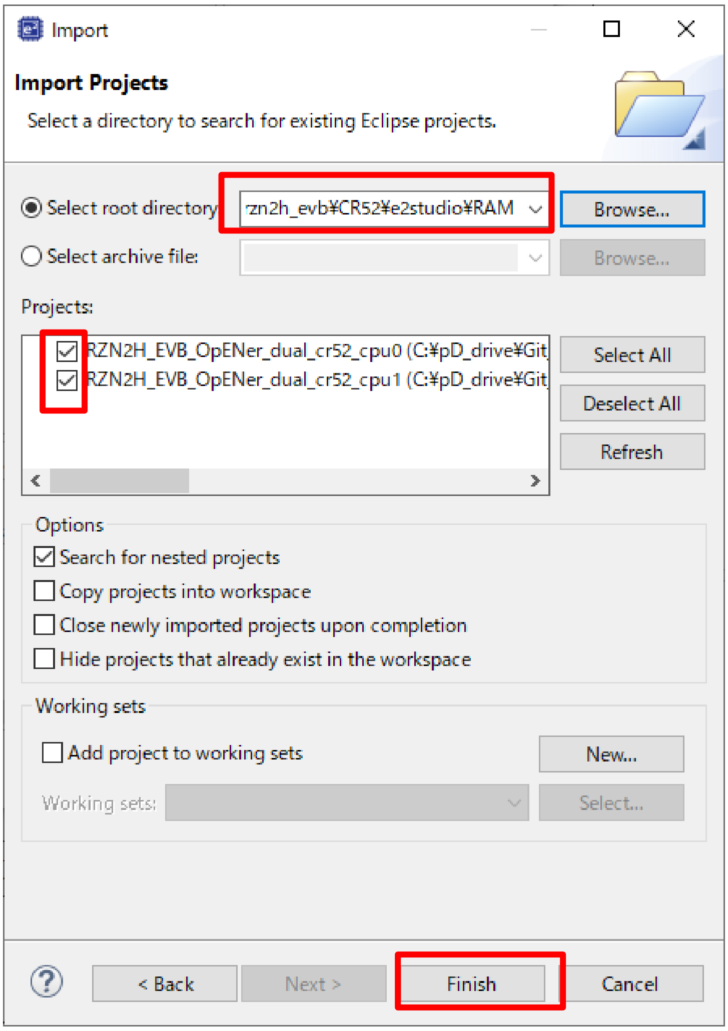


Figure 6-208 Import two projects on e² studio

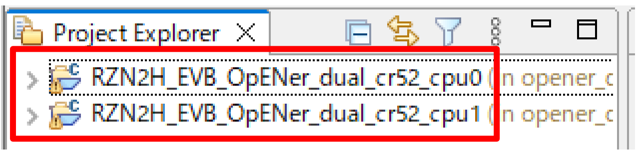


Figure 6-209 Open 2 projects

6.8.1.2 RAM execution / Flash boot mode

(1) How to generate source code and how to build

A) Click the configuration.xml of the CPU0 project.

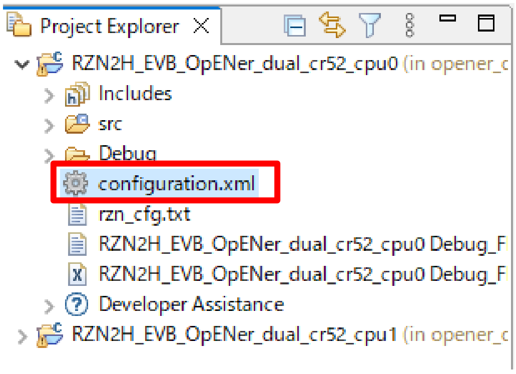


Figure 6-210 Configuration

B) Click “Generate Project Content” button then generate rzn, rzn_gen, rzn_cfg folder.

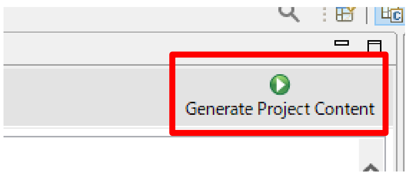


Figure 6-211 Generate Project Content

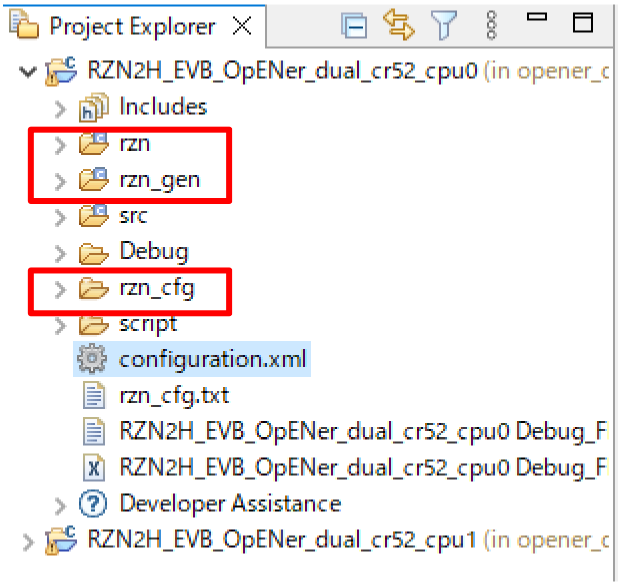


Figure 6-212 Generate project folder

C) Click the Build button in tool bar to build the project and confirm that there is no error message in the build message log.

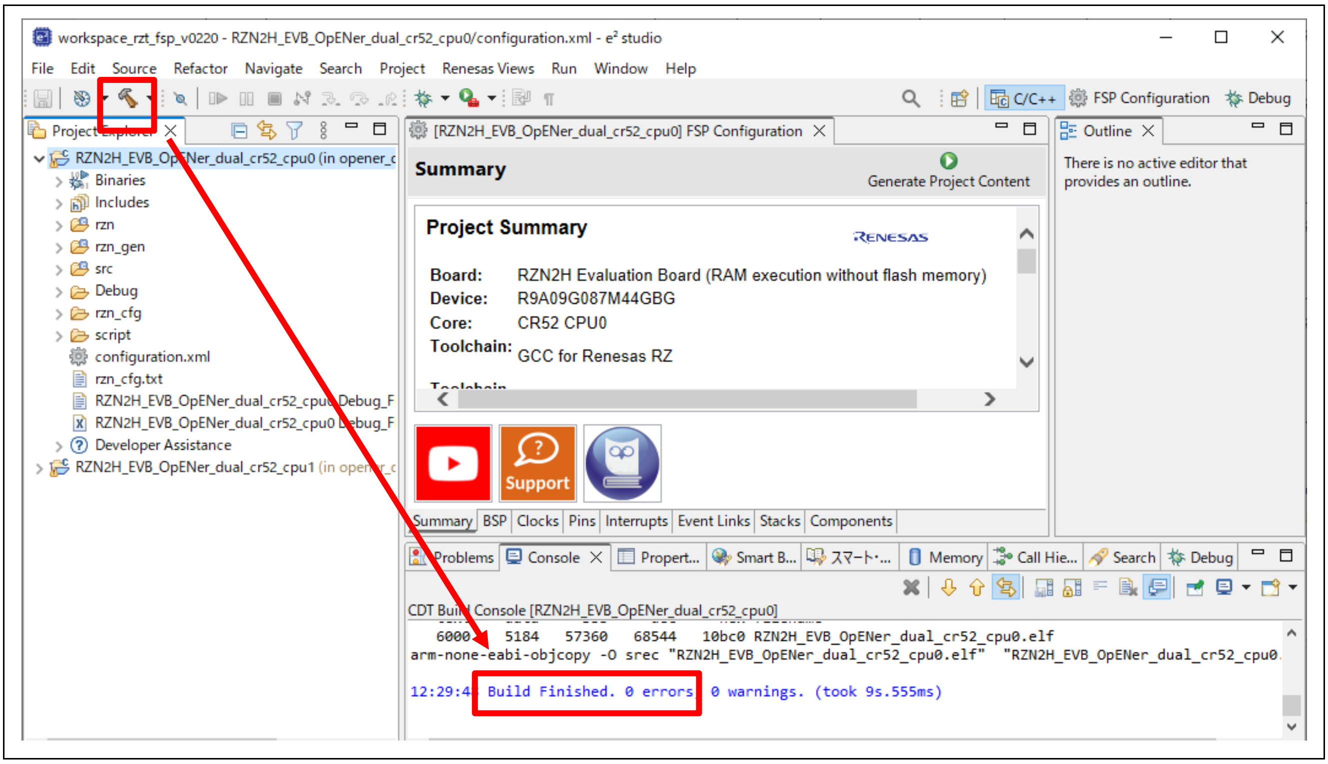


Figure 6-213 Build button and Build message

D) For the CPU1 project, generate the folders.

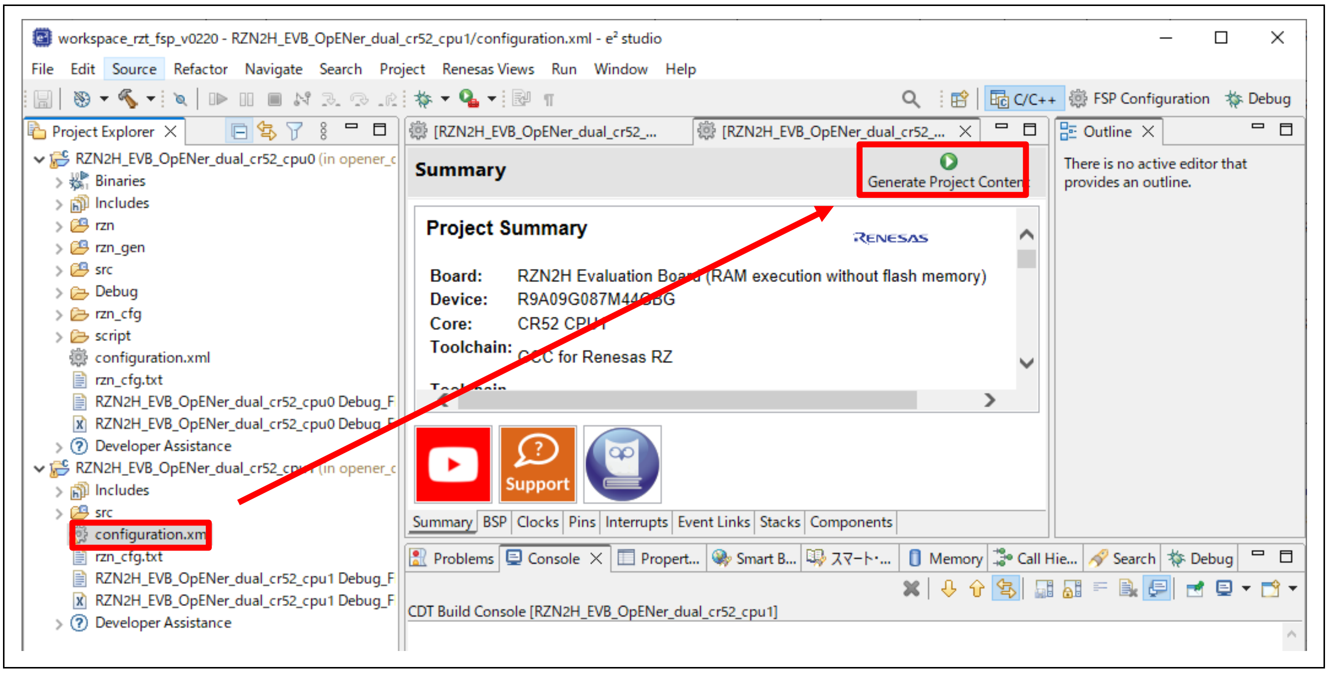


Figure 6-214 Generate folders for the CPU1 project

E) (**Flash boot mode only**): - If necessary to enable Type-0 Reset and Type-1 Reset, modify the macro definition "`SAMPLE_EXECUTE_SYSTEM_RESET_PROCESS`" (opener_port_main.c) to "1" enable the Reset process.

F) For the CPU1 project, build the project.

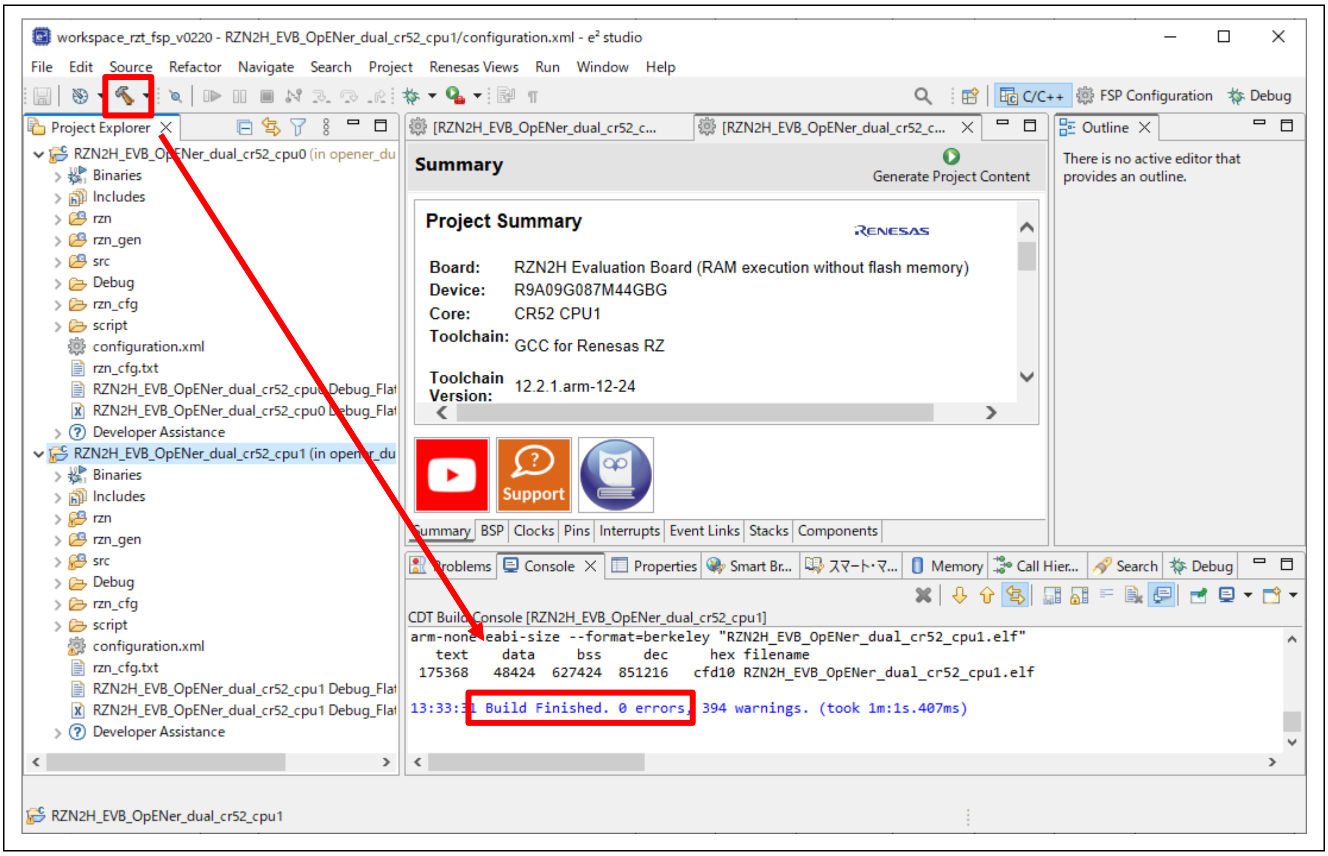
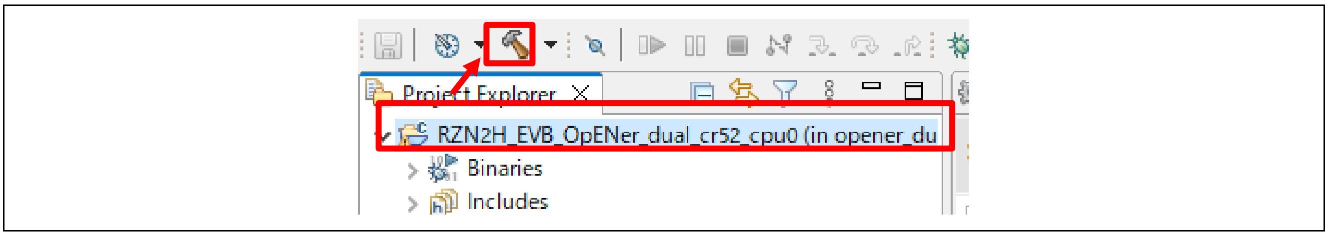


Figure 6-215 Build the CPU1 project

G) (**Flash boot mode only**): For the CPU0 project, build the project again.



(2) Download application and run debugger

A) Select the CPU0 project and click the Debug button to download the CPU0 program.

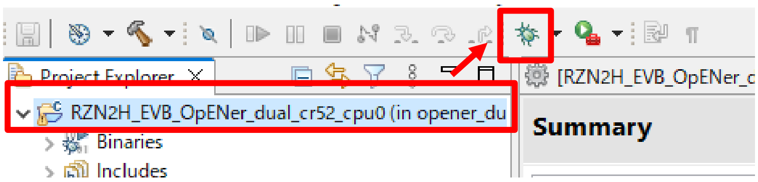


Figure 6-216 Debug button, the CPU0 project

B) Click the “No” button on the Perspective Switch dialog box.

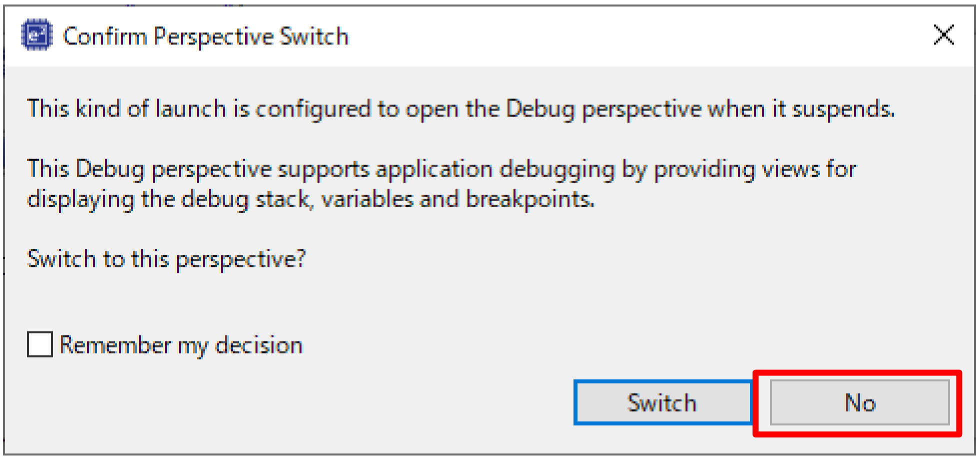


Figure 6-217 Confirm Perspective Switch

C) (**Flash boot mode only**): Disconnect the debugger, and re-power up the board.

D) (**RAM-execution only**): Select the CPU1 project and click the Debug button to download the CPU1 program.

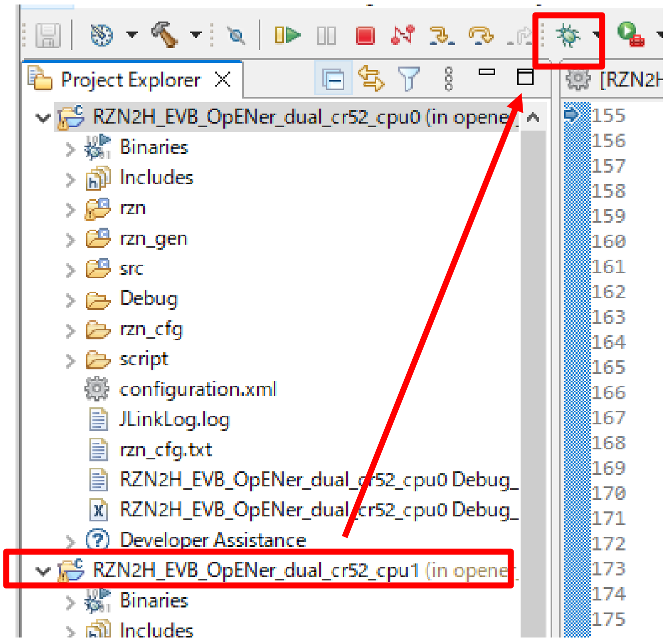


Figure 6-218 Debug button, the CPU1 project

E) (**RAM-execution only**): Click the “No” button on the Launcher dialog box.

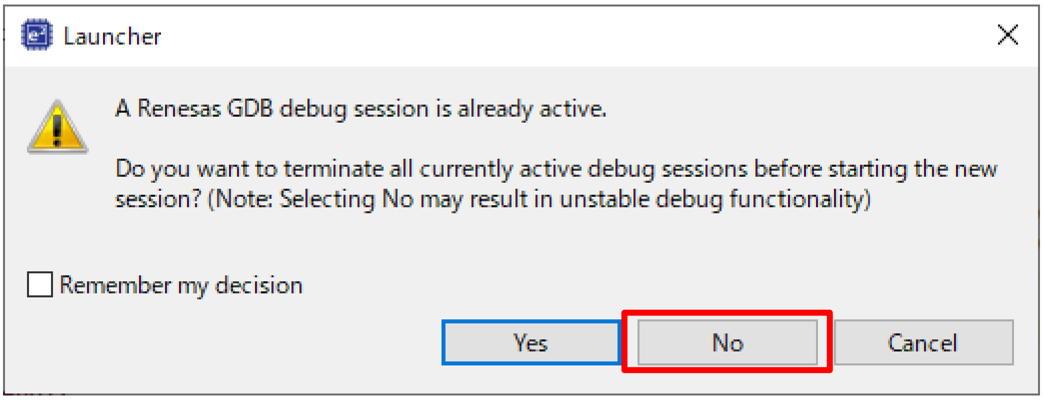


Figure 6-219 Launcher dialog box

F) (**RAM-execution only**): Click to “Yes” button on the Proceed confirmation dialog box.

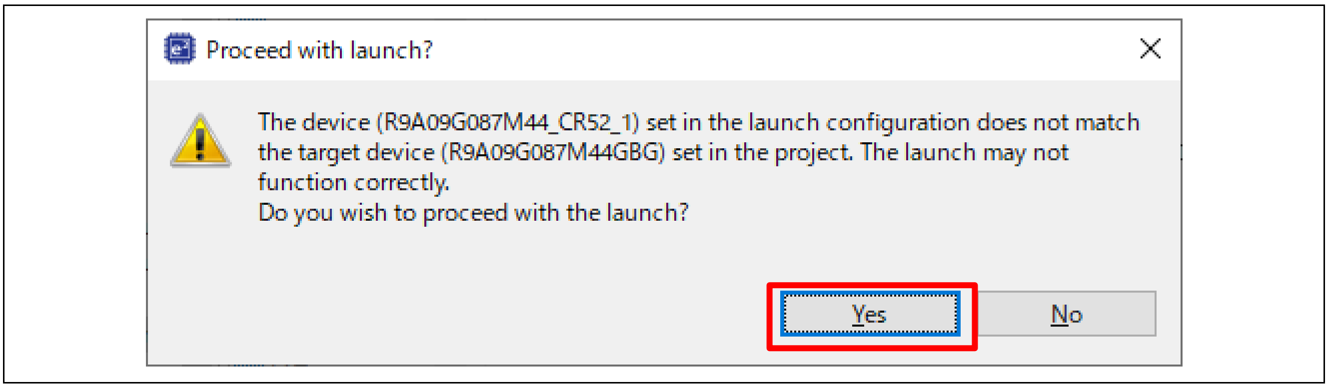


Figure 6-220 Proceed confirmation dialog box

G) (**RAM-execution only**): Click “Yes” button on the Perspective Switch dialog box.

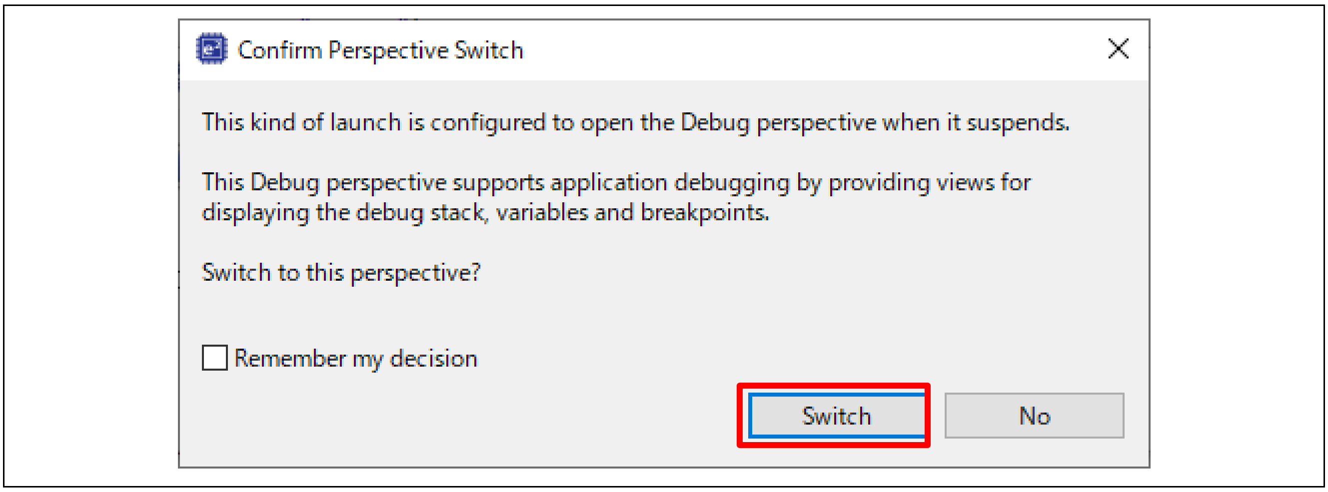


Figure 6-221 Confirm Perspective Switch

H) (**RAM-execution only**): Run the CPU0 project.

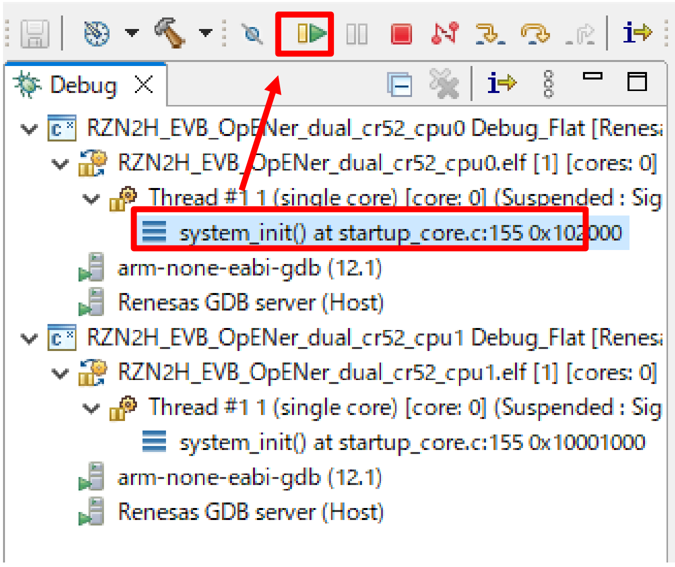


Figure 6-222 Run the CPU0 project

I) (**RAM-execution only**): Run the CPU1 project.

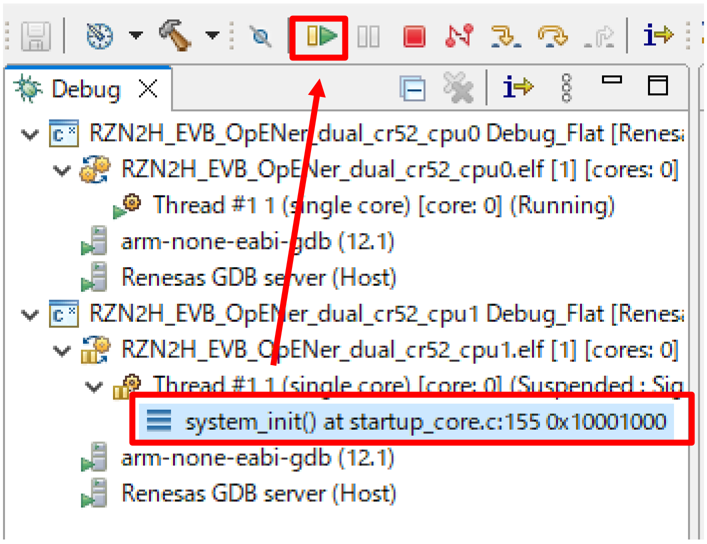


Figure 6-223 Run the CPU1 project

6.8.2 Setup sample project for EWARM

This section shows a procedure using EWARM.

6.8.2.1 Startup EWARM

1. Open the EWARM.
2. Click “Open Workspace…” in the File tab.

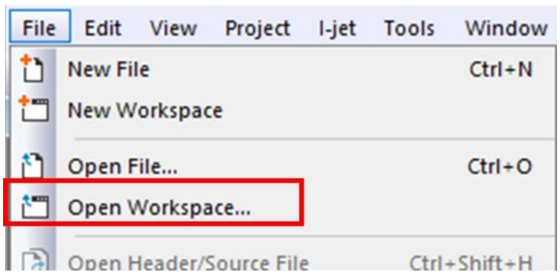


Figure 6-224 EWARM file tab

3. Select the CPU0 Workspace File(.eww) and click the Open button.

When you use the RAM-execution projects, please select the following folder.

➔ “(Extracted folder path)\project\rzn2h_evb\CR52\ewarm**RAM**\opener_dual_cr52_cpu0”

When you use the flash boot mode projects, please select the following folder.

➔ “(Extracted folder path) \project\rzn2h_evb\CR52\ewarm**xSPI1**\opener_dual_cr52_cpu0”

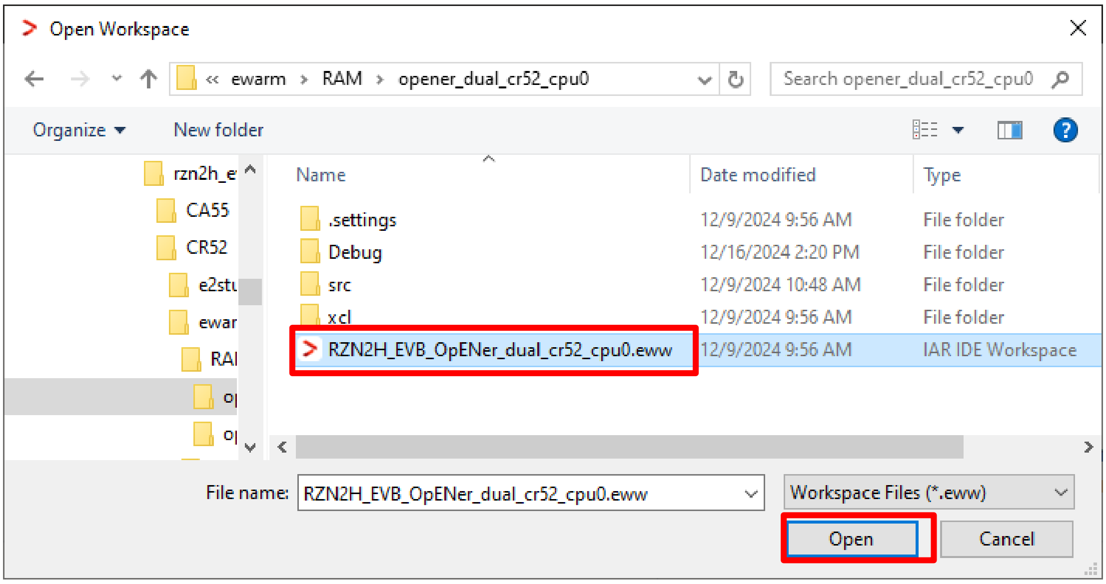


Figure 6-225 Open the CPU0 Workspace

4. Open another EWARM window.
5. Click "Open Workspace…" in the File tab, and open the CPU1 Workspace File(.eww).

   When you use the RAM-execution projects, please select the following folder.

   ➔ "(Extracted folder path)\project\rzn2h_evb\CR52\ewarm**RAM**\opener_dual_cr52_cpu1"

   When you use the flash boot mode projects, please select the following folder.

   ➔ "(Extracted folder path)\project\rzn2h_evb\CR52\ewarm**xSPI1**\opener_dual_cr52_cpu1"

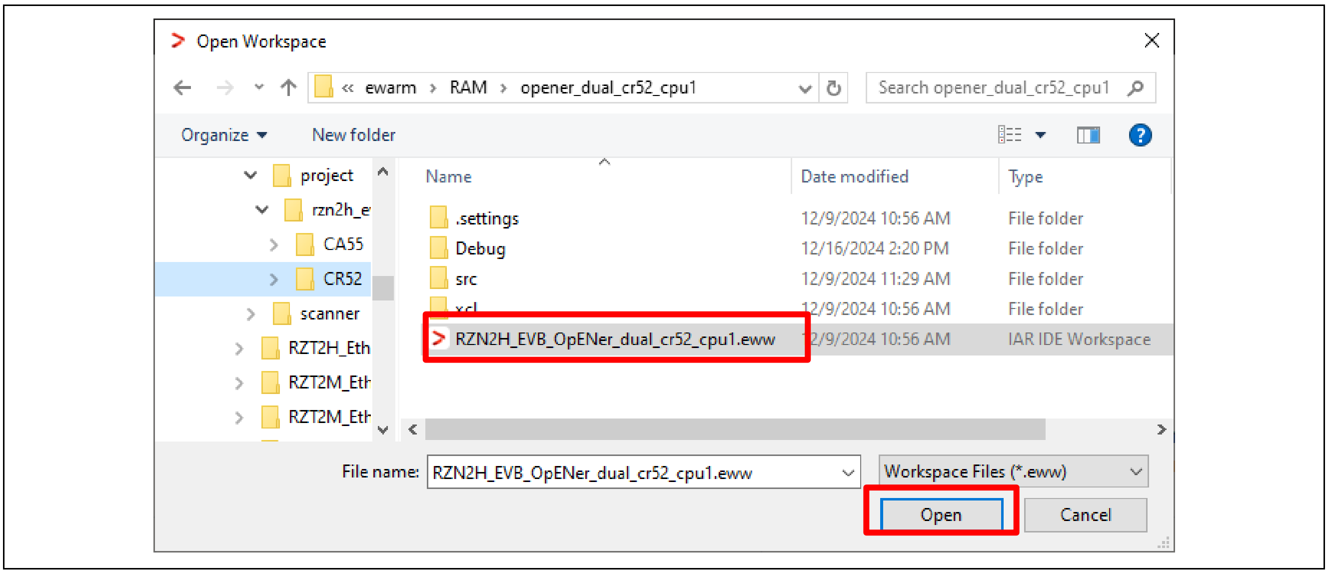


Figure 6-226 Open the CPU1 Workspace

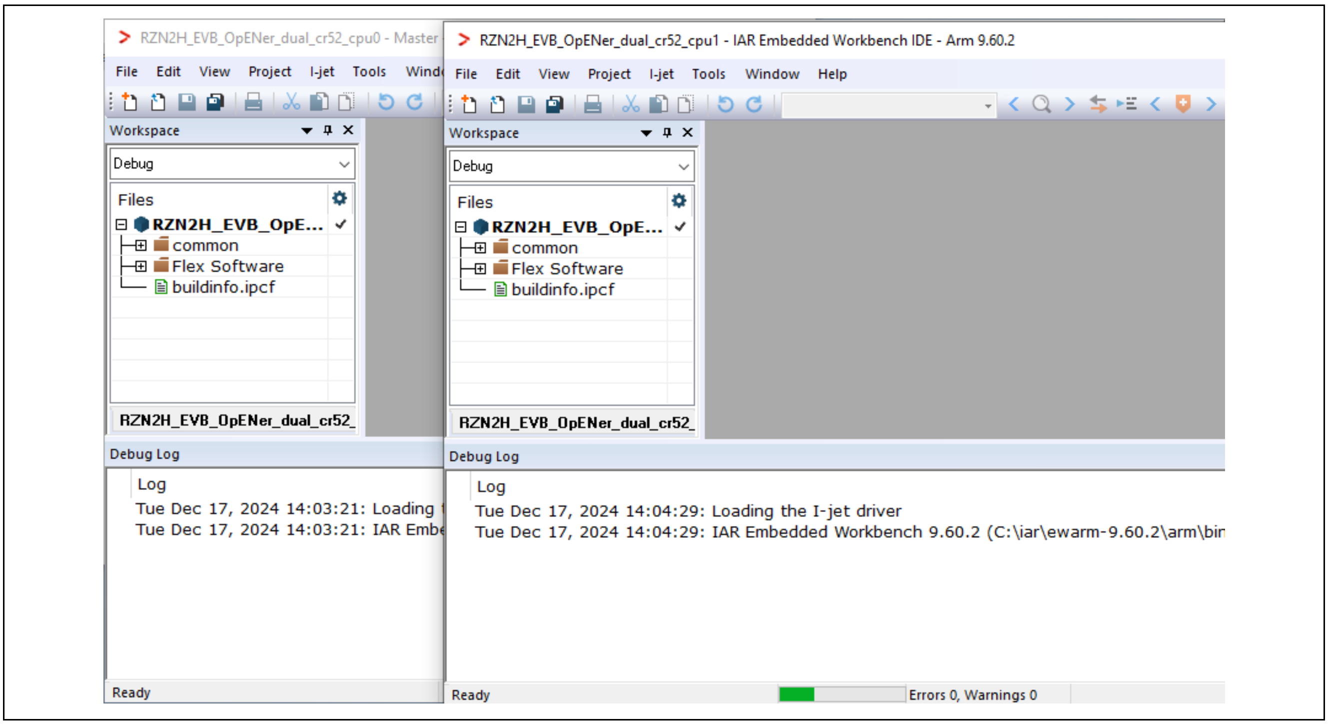


Figure 6-227 EWARM windows

6.8.2.2 RAM execution / Flash boot mode

(1) How to generate source code and how to build

A) Click the “Tools” -> “FSP Smart Configurator” on the CPU1 Workspace.

Note: If you have not set up FSP Smart Configurator yet on EWARM, refer to r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf in which section 5.4 describes how to set up it.

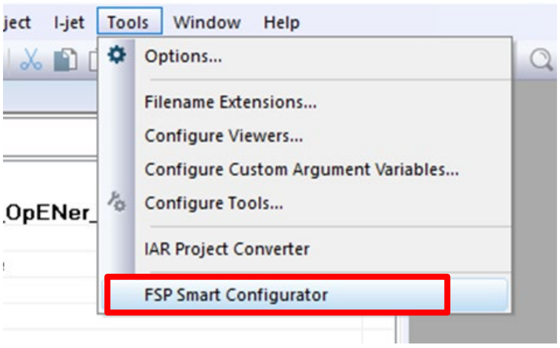


Figure 6-228 Tools tab on the CPU1 Workspace

B) Click “Generate Project Content” button then generate rzn, rzn_gen, rzn_cfg folder.

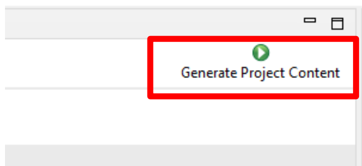


Figure 6-229 Generate Project Content

C) Change the device name in the buildinfo.ipcf file.

R9A09G087M44_CR52_0 -> R9A09G087M44_R52_1

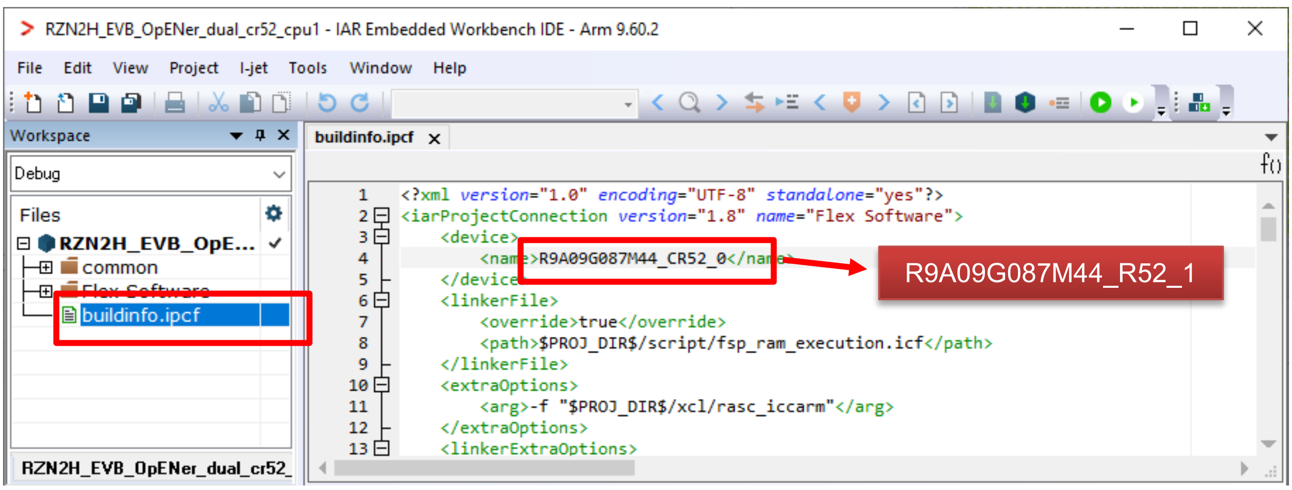


Figure 6-230 Change the device name in the ipcf file

D) (**Flash boot mode only**): - If necessary to enable Type-0 Reset and Type-1 Reset, modify the macro definition "`SAMPLE_EXECUTE_SYSTEM_RESET_PROCESS`" (opener_port_main.c) to "1" enable the Reset process.

E) Click the Make button to build the CPU1 project. Once the build is completed, the build message is displayed in the Build Console window.

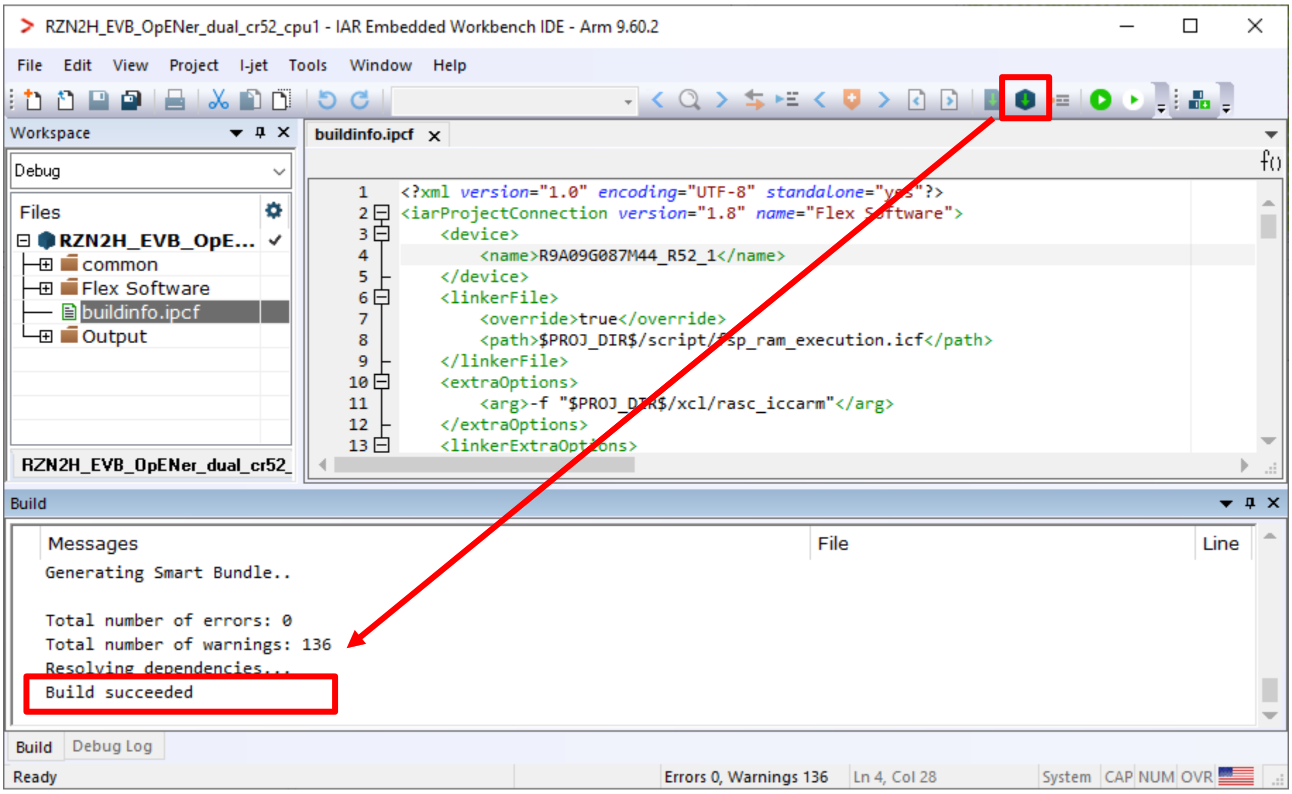


Figure 6-231 Make button on the CPU1 Workspace

F) Click the "Tools" -> "FSP Smart Configurator" on the CPU0 Workspace.

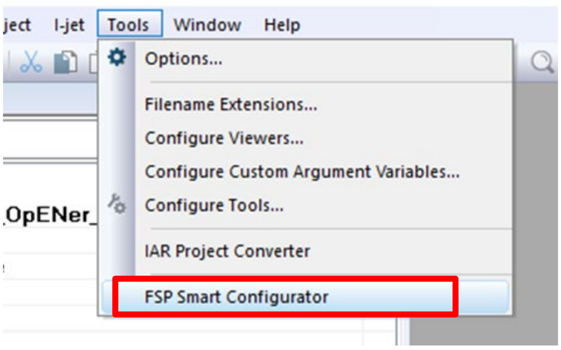


Figure 6-232 Tools tab on the CPU0 Workspace

G) Click “Generate Project Content” button then generate rzn, rzn_gen, rzn_cfg folder.

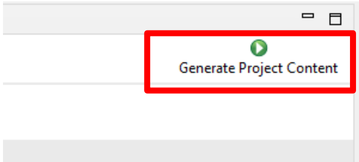


Figure 6-233 Generate Project Content

H) Change the device name in the buildinfo.ipcf file.

R9A09G087M44_CR52_0 -> R9A09G087M44_R52_0

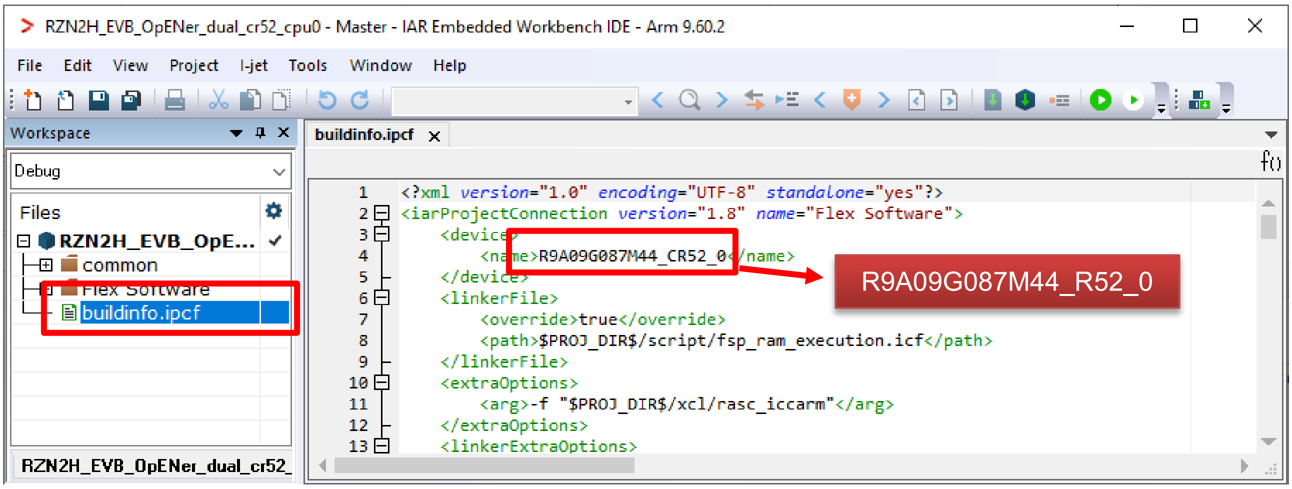


Figure 6-234 Change the device name in the ipcf file

I) Click the Make button to build the CPU0 project. Once the build is completed, the build message is displayed in the Build Console window that displays compilation target files and the number of error / warnings.

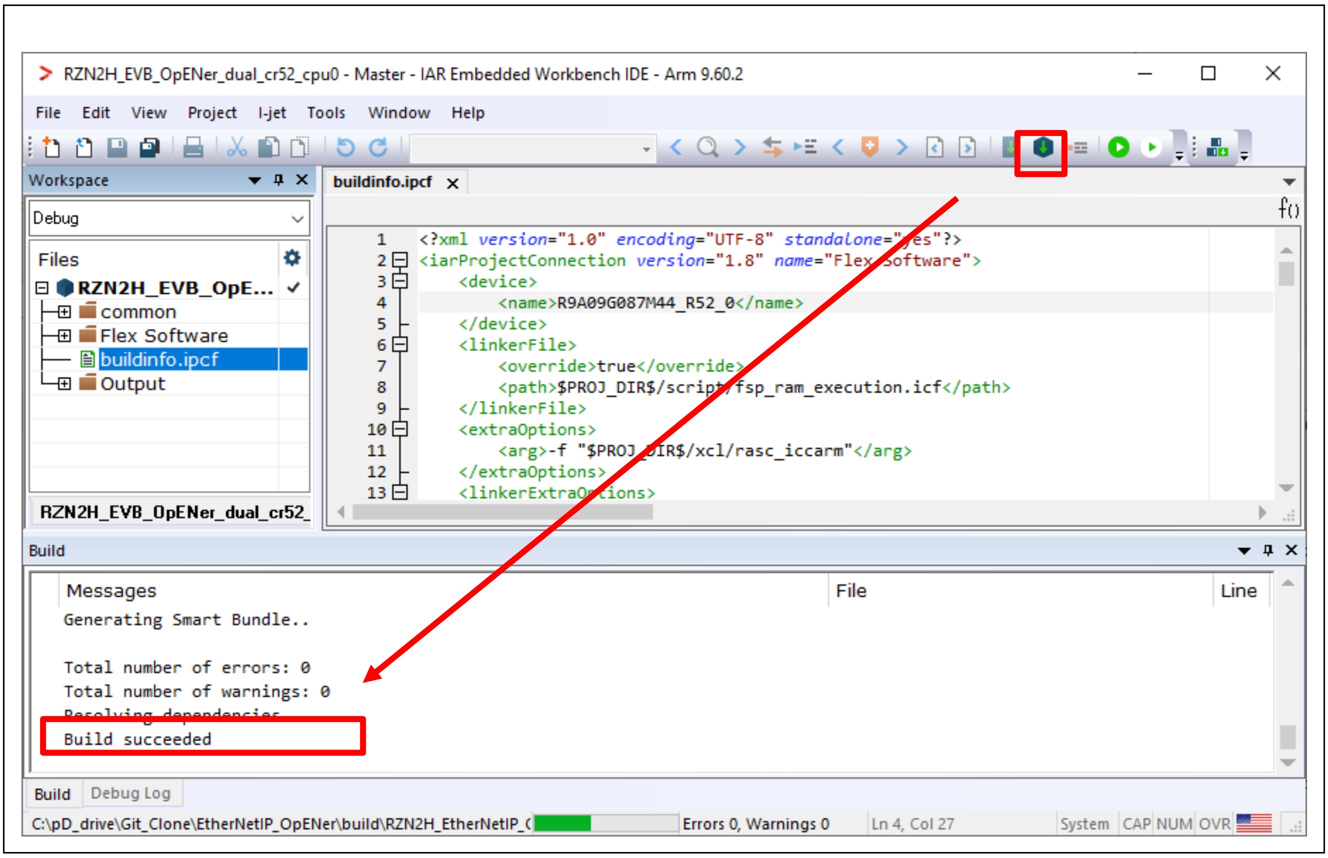


Figure 6-235 Make button on the CPU1 Workspace

J) Close the CPU1 Workspace.

(2) (For RAM-execution): Download application and run debugger

A) Click the Debug button to download the CPU0 application program and launch the debugger. Then automatically the CPU1 Workspace will open.

Figure 6-236 Debug button

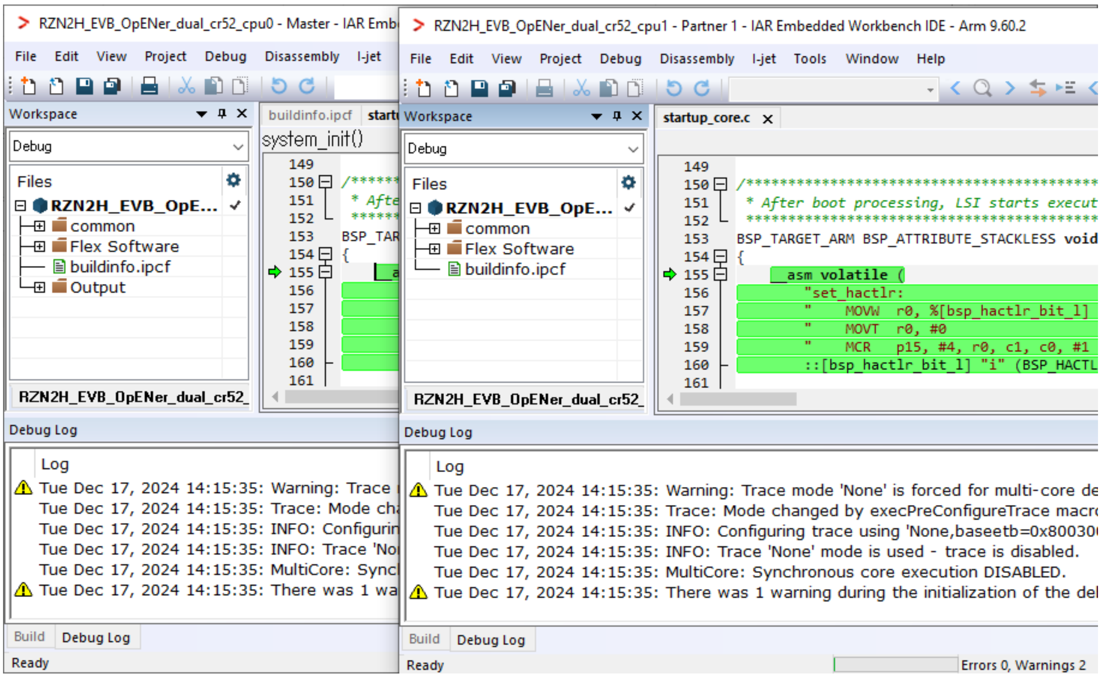


Figure 6-237 The CPU1 Workspace automatically opens

B) Click the Go button on the CPU0 Workspace.

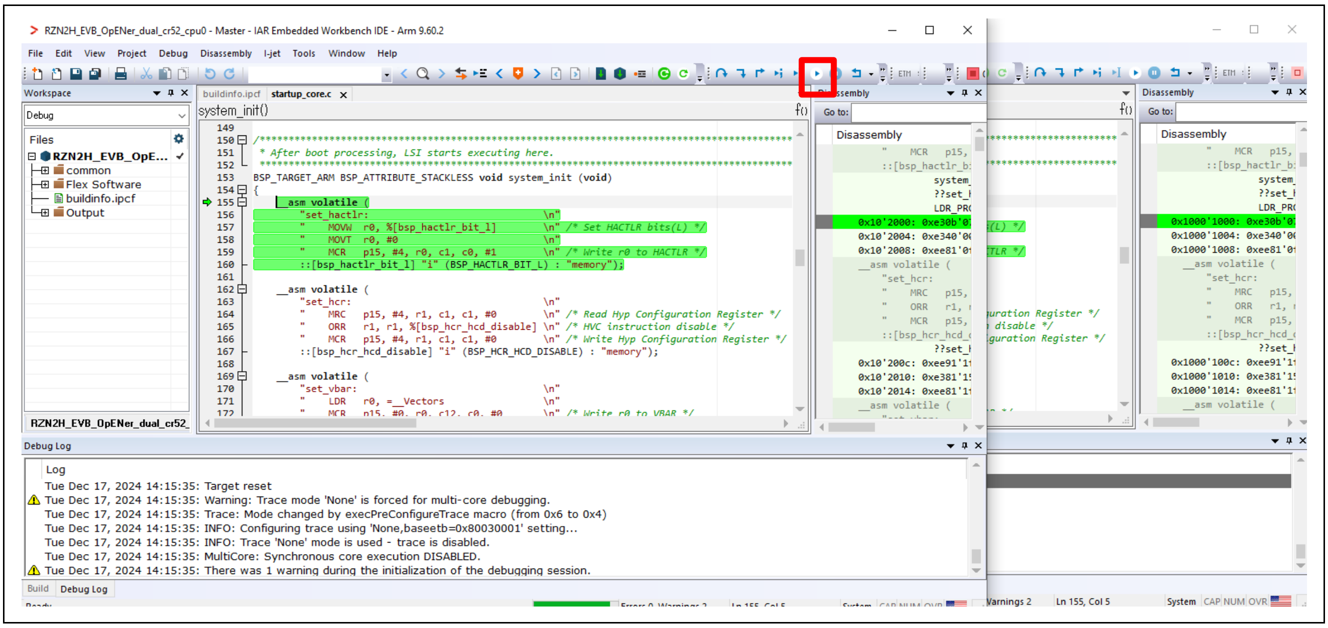


Figure 6-238 Go button on the CPU0 Workspace

C) Click the Go button on the CPU1 Workspace.

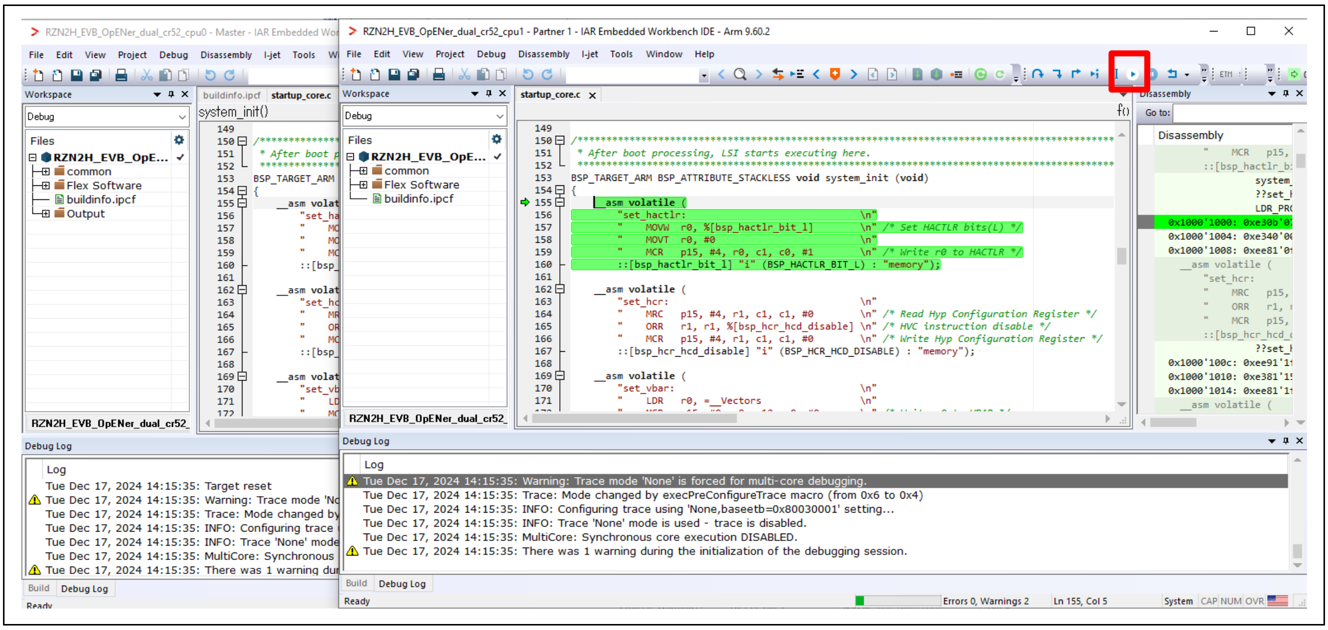


Figure 6-239 Go button on the CPU1 Workspace

Note : Please try to execute "Rebuild All" command on the CPU1 and CPU0 project before downloading if the project doesn't operate.

(3) (For Flash boot mode): Download application

A) Set the Asymmetric multicore setting to “Disabled” in the property of the CPU0 Workspace.

a-1. Click Project > Options….

a-2. Click Debugger > Multicore.

a-3. Select Disable in Asymmetric multicore.

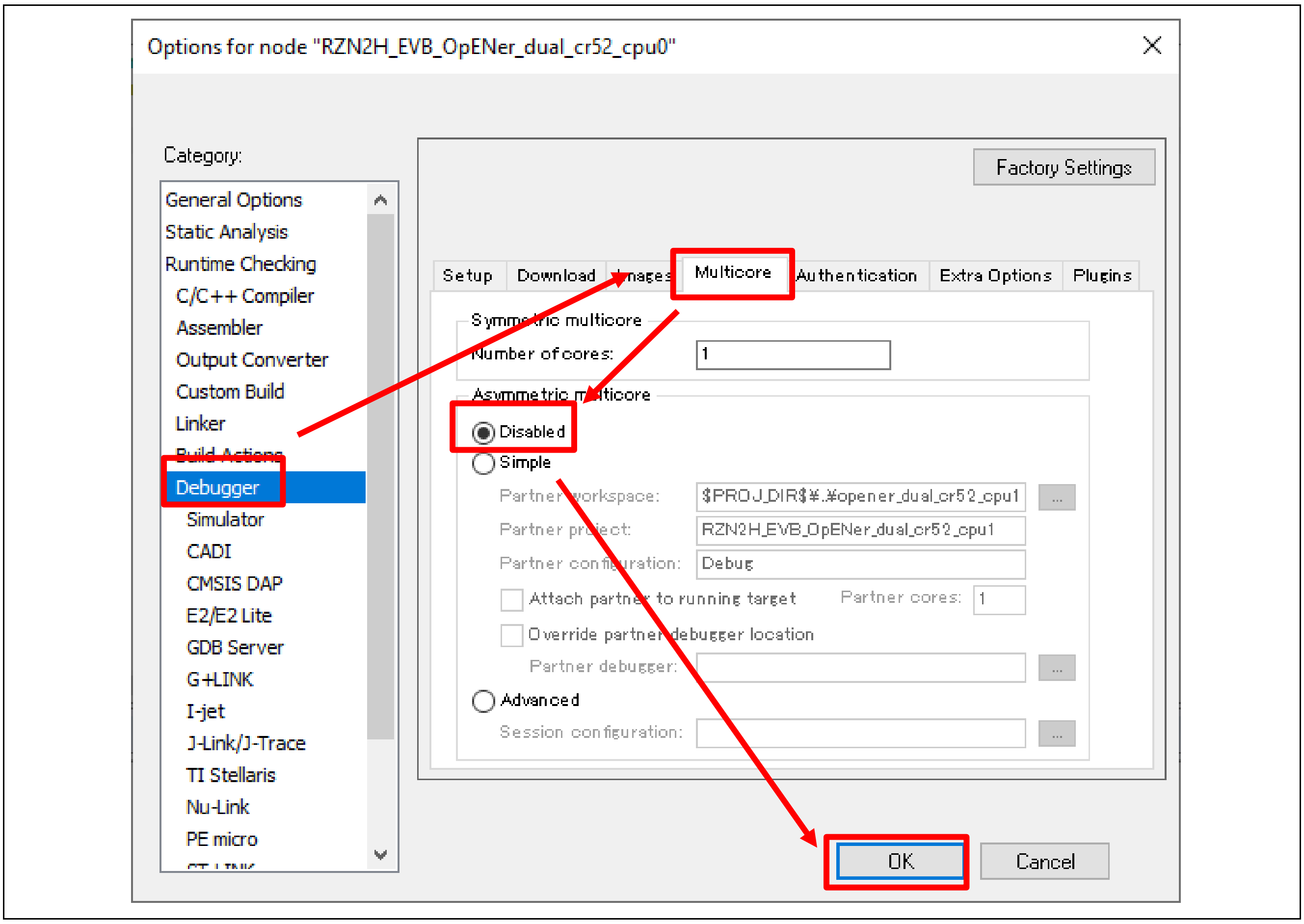


Figure 6-240 Disable multicore debug mode in CPU0 project property

B) In the CPU0 project, click Project > Download > Download file… and select out file of the primary project.

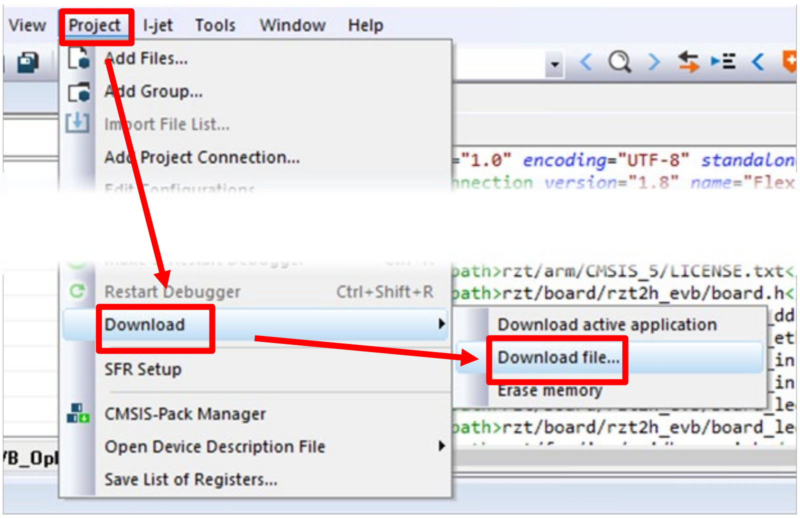
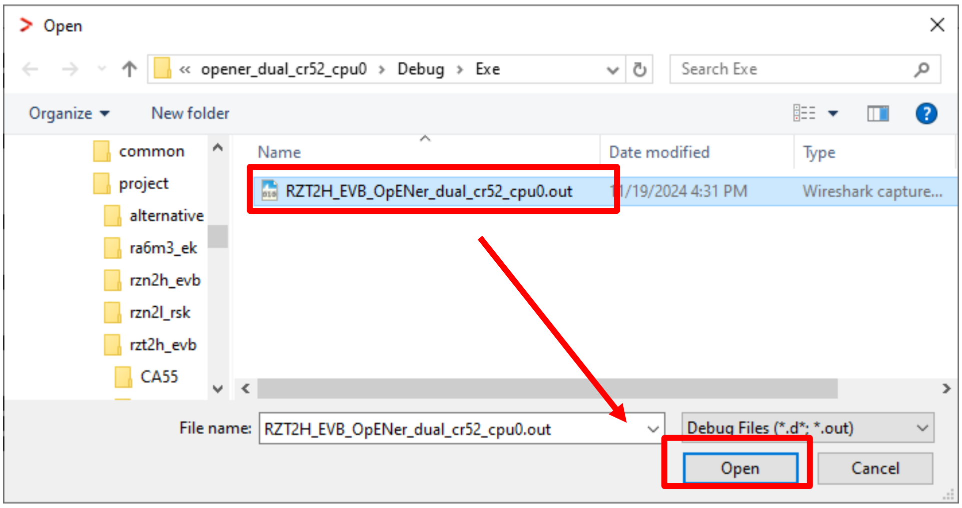


Figure 6-241 IAR EWARM Download File



C) Re-power up the board.

Note : Please try to execute "Rebuild All" command on the CPU1 and CPU0 project before downloading if the project doesn't operate.

7. Demonstration of the application with the CODESYS

The demonstration uses 4 LEDs and 4 Switches. These are named differently on each board. In the explanation in this chapter, they are uniformly referred to as LED 1 ~ LED 4 and SW 1 ~ SW 4.

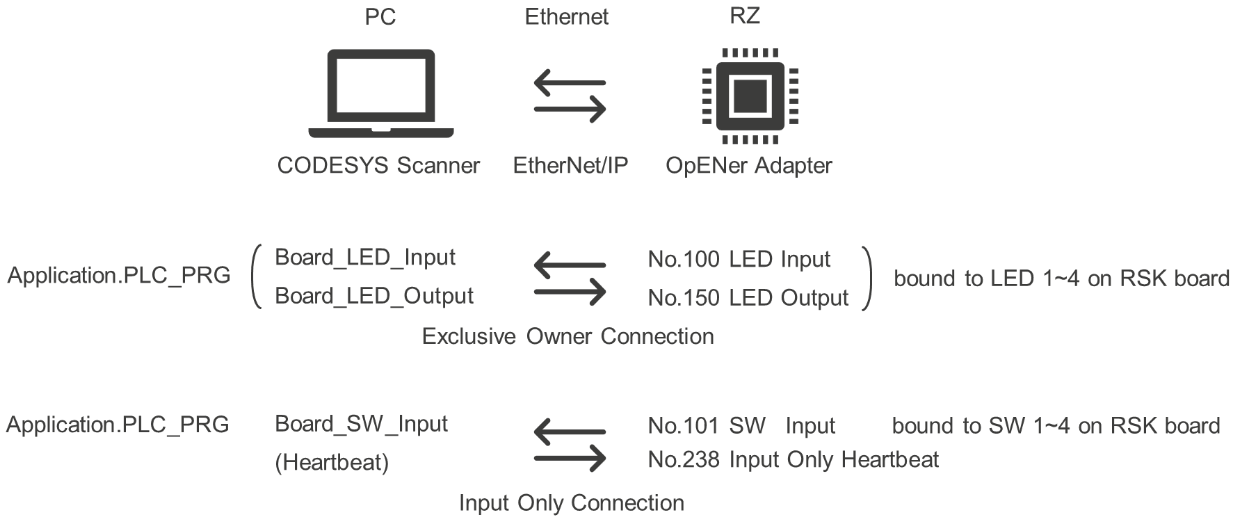
The following table shows the corresponding LEDs and Switches used on each board. Assignments are defined in the opener_port_main.c file.

Table 7-1 LEDs and Switches used for the demonstration on each board

Device Name	Board Name	Used LEDs	Used Switches
RZ/T2M RZ/T2ME	RZ/T2M RSK Board RZ/T2ME RSK Board	LED0 LED1 LED2 LED3	SW3-0 SW3-1 SW3-2 SW3-3
RZ/T2H	RZ/T2H Evaluation Board	LED0 LED1 LED7 LED8	SW12-1 SW12-2 SW12-3 SW12-4
RZ/N2L	RZ/N2L RSK Board	LED0 LED1 LED2 LED3	SW3-0 SW3-1 SW3-2 SW3-3
RZ/N2H	RZ/N2H Evaluation Board	LED5 LED6 LED7 LED8	DSW1-1 DSW1-2 DSW1-3 DSW1-4
Expression in this chapter ->		LED 1 LED 2 LED 3 LED 4	SW 1 SW 2 SW 3 SW 4

7.1 Application Behavior

For demonstration, the application of OpENer RZ/T2 and RZ/N2 port has an Exclusive Connection and an Input Only Connection. The connections between the RZ sample application and the CODESYS project application are shown below.



The PLC program of CODESYS project is shown below. The PLC program is described by ST (Structured Text) language.

```
CycleCounter := CycleCounter+Board_SW_Input;
IF CycleCounter > CyclePerTick THEN
    CycleCounter := CycleCounter-CyclePerTick;
    IF 0 < Board_LED_Input AND Board_LED_Input < 8 THEN
        Board_LED_Output := Board_LED_Input * 2;
    ELSE
        Board_LED_Output := 1;
    END_IF
END_IF
```

- The Exclusive Owner Connection is bound to LED on RSK board.
  - The PLC program makes the LEDs light by shifting every CyclePerTick.
- The Input Only Connection is bound to SW 1 ~ SW 4 on RSK board.
  - The PLC program read the SW values and change the incrementation value of CycleCounter for controlling the frequency of LED light shifting.

The CODESYS application is shown in the figure below.

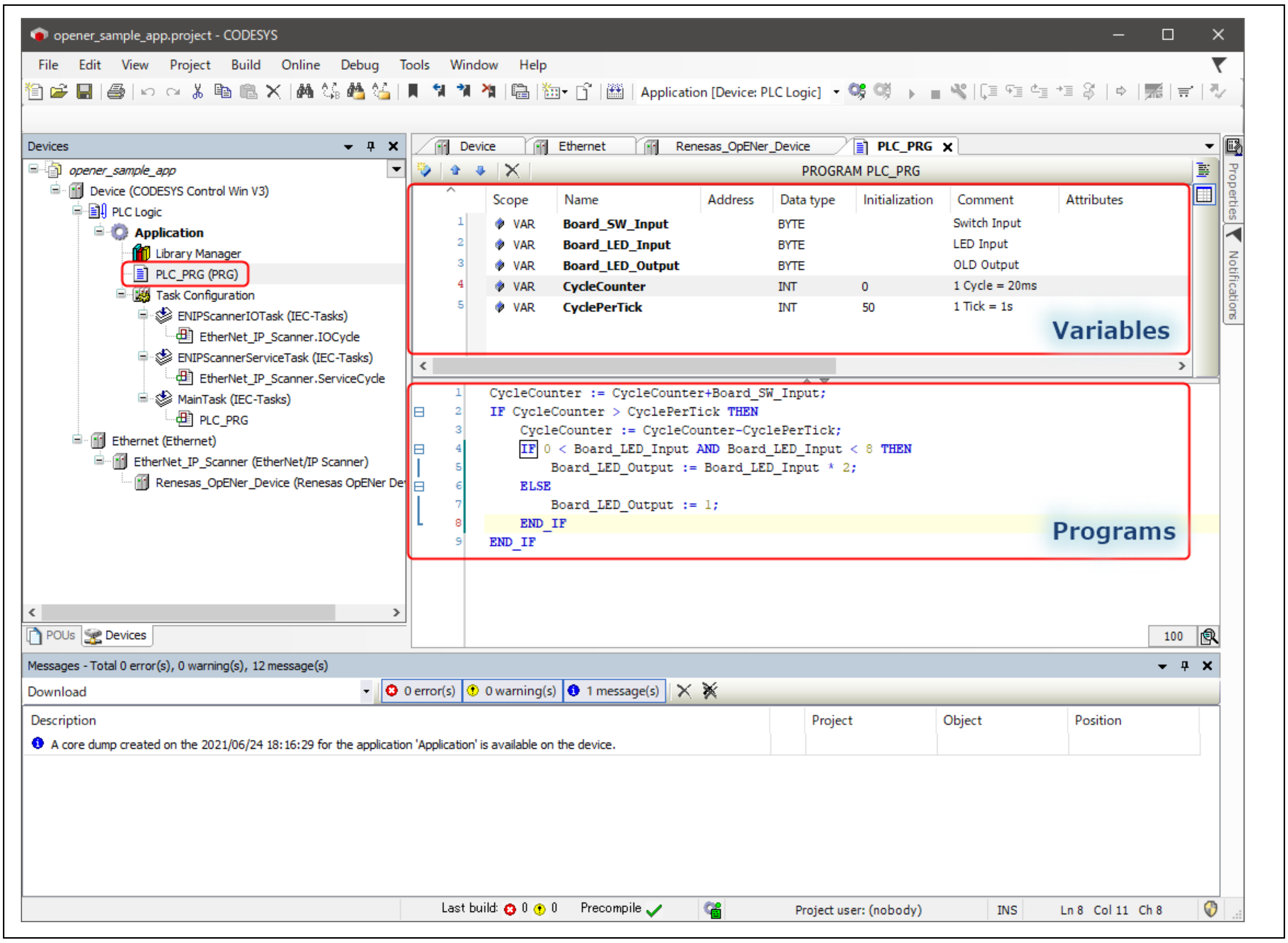


Figure 7-1 CODESYS Project included in this sample program

7.2 IP and MAC Address Configuration

7.2.1 IP Configuration

This program sets its own IP and related parameters to the following values defined statically by “g_lwip_port0_netif_cfg” variable in “common\renesas\application\opener_port_instance.c”.

If you change the IP configuration, please check the following parameters.

Table 7-2 IP Configuration

Item	Value	Variable
IP Address	192.168.1.170- (or 192.168.1.10)	g_lwip_port0_netif_cfg.ip_address
Net Mask	255.255.255.0	g_lwip_port0_netif_cfg.subnet_mask
Gateway Address	192.168.1.2	g_lwip_port0_netif_cfg.gateway_address
Host Name	OPENER_NETIF0	g_lwip_port0_netif_cfg.p_host_name
DHCP	LWIP_PORT_DHCP_DISABLE	g_lwip_port0_netif_cfg.dhcp

If using DHCP, please set “g_lwip_port0_netif_cfg.dhcp” to “LWIP_PORT_DHCP_ENABLE”. After the program starts, DHCP process is executed to get an IP address dynamically. If the program gets an IP address, EtherNet/IP communication starts.

7.2.2 MAC Address Configuration

The MAC address is set to the following values defined statically by FSP Configurator.

Table 7-3 MAC Address Configuration

Item	Value (Decimal)
MAC Address	00:11:22:33:44:55

Table 7-4 MAC Address Configuration (Only RZ/N2L)

Item	Value (Decimal)
MAC Address	74:90:50:10:05:9A

If you change the MAC address, please change the value on FSP configuration.

Regarding FSP configuration, please see the "RZ/T2, RZ/N2 Getting Started with Flexible Software Package" (r01an6434ejxxxx-rzt2-rzn2-fsp-getting-started.pdf) document.

7.3 Startup Software PLC

7.3.1 Open CODESYS project

Please select “File” > “Open project...” in CODESYS tool bar and open “opener_sample_app.project” in “scanner/codesys”.

You may be prompted to update the versions of some libraries.

- Do not update the version of three libraries
  - Standard: 3.5.15.0
  - StringUtils: 3.5.15.0
  - SysSocket Implementation: 3.5.15.0
- If other libraries are displayed, please update them to the latest version.

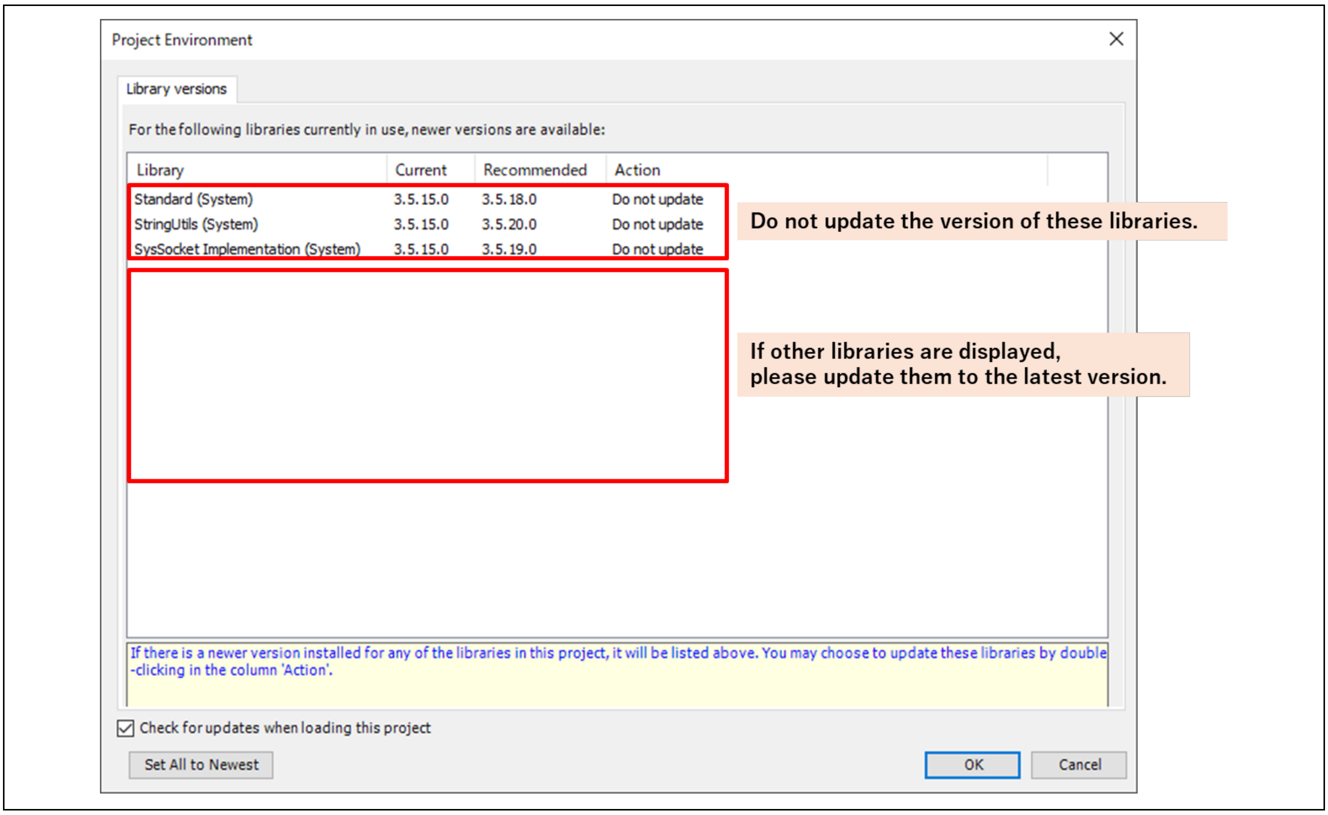


Figure 7-2 Update Codesys libraries

If the project is opened properly, the opened project is shown in “Device” section located at left in the following window.

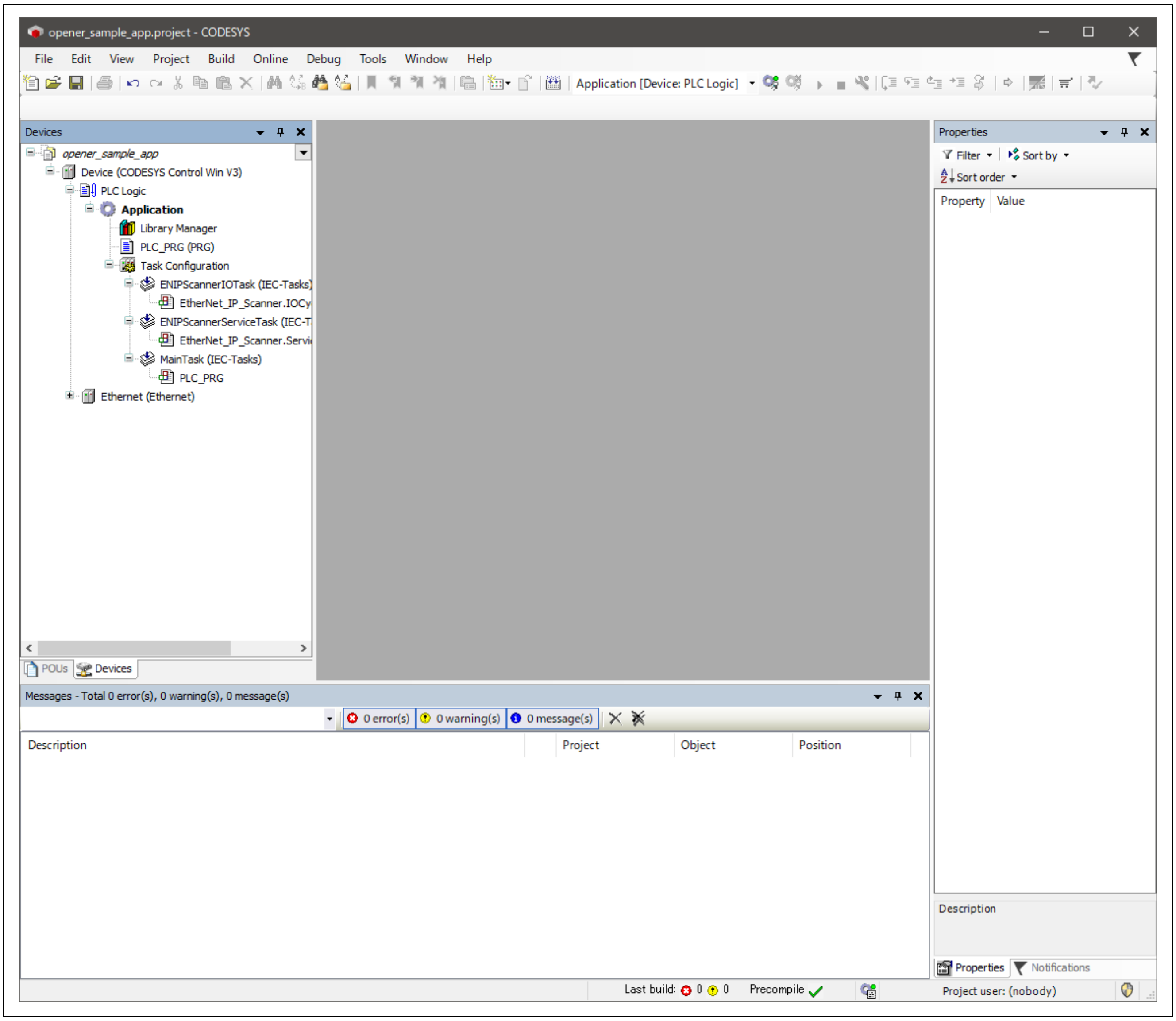


Figure 7-3 Open a CODESYS Project

7.3.2 Network Configuration

Please double-click “Device (CODESYS Control Win V3)” to open “Communication Settings” at center section, and please click “Scan Network...” to open “Select Device” window.

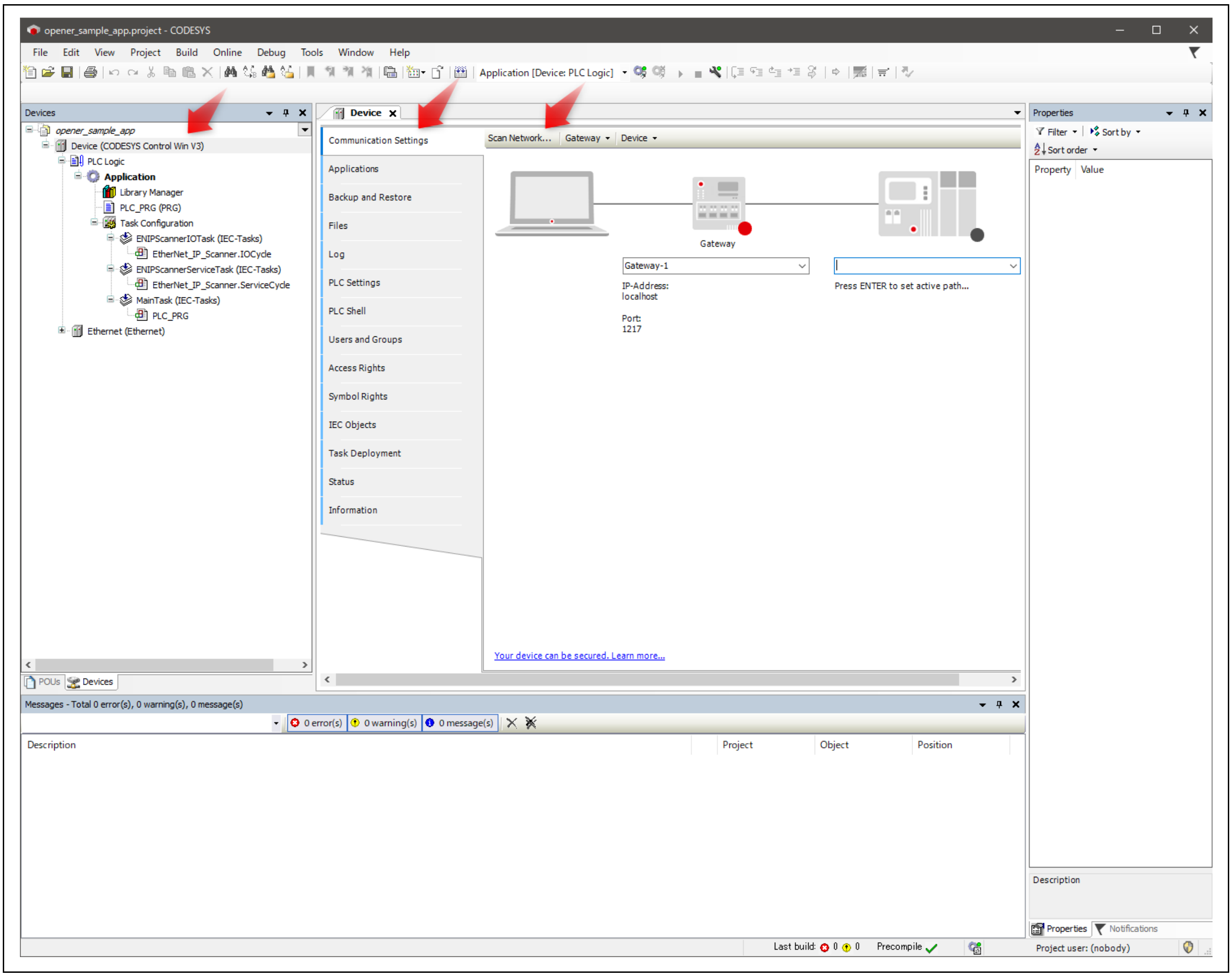


Figure 7-4 Communication Settings

If the software PLC is found after scanning the network, the device name (here, PC name) is shown under Gateway tree. Please double-click this device name (in blue portion).

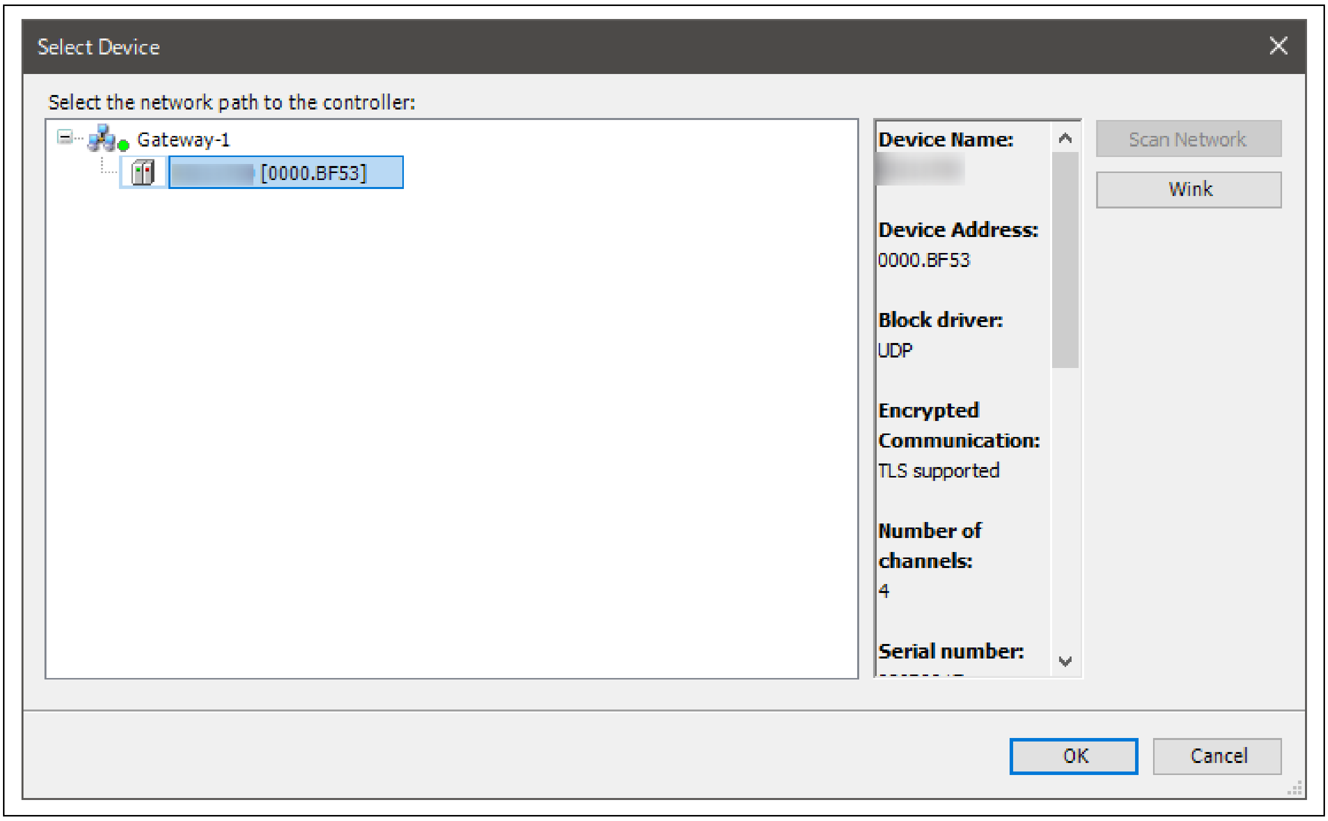


Figure 7-5 Select Device

Click the pull-down button next to the PLC name and select the device name (in this case, the PC name) displayed in Scan Network. If the network is configured properly, its configuration is shown in "Communication Settings" tab, and there are the green marks at gateway and device portions.

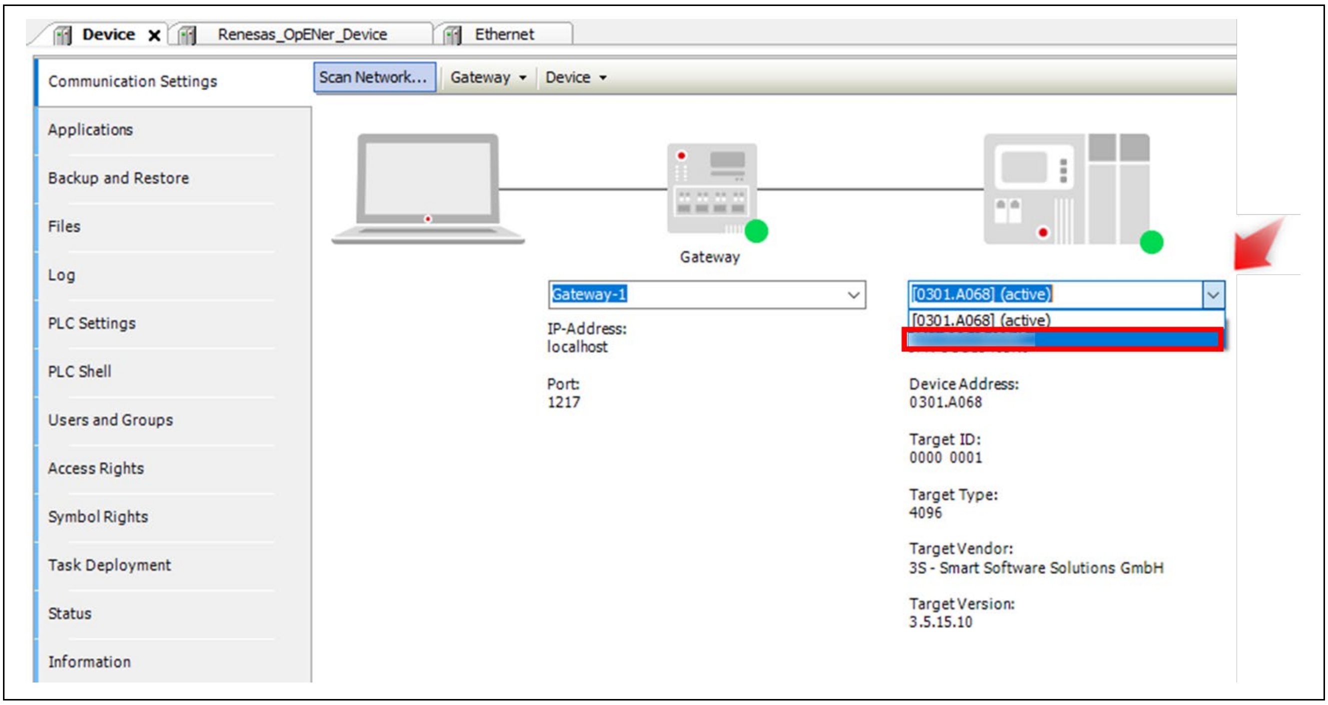


Figure 7-6 Green Marks at Gateway and Device Portions

7.3.3 Interface and IP address configuration

Please click “Ethernet (Ethernet)” in the left section to open “Ethernet” tab in center section.

After that, please select “...” button to select network interface ethernet which relates to RSK board, and please configure the IP address and related address values of the ethernet network interface.

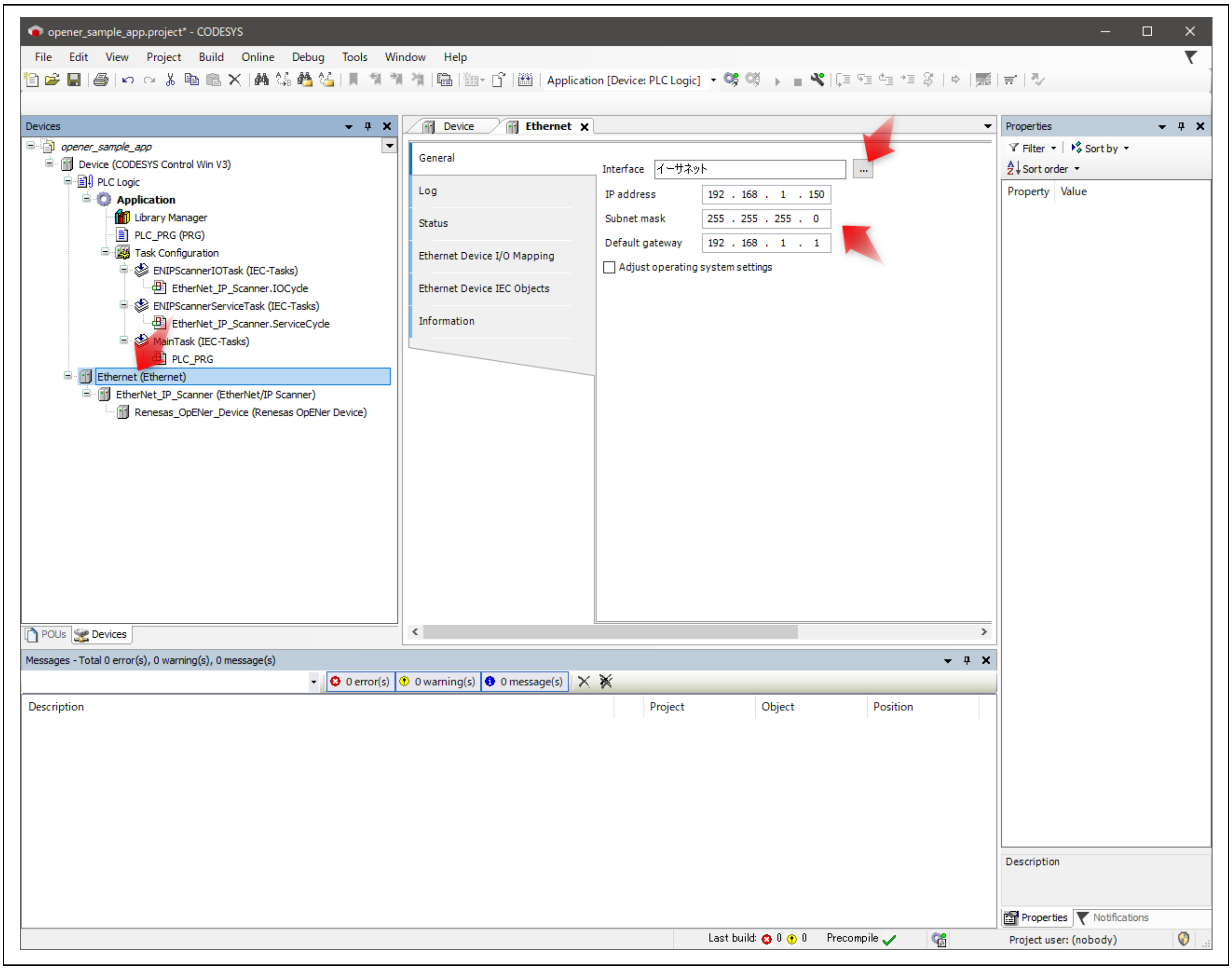


Figure 7-7 Open Ethernet tab

Next, please double-click “Renesas_OpENer_Device (Renesas OpENer Device)” in the left section to open “Renesas_OpENer_Device” tab in center section.

Enter the IP address of the sample code in the IP Address form. (Default setting is 192.168.1.170 or 192.168.1.10)

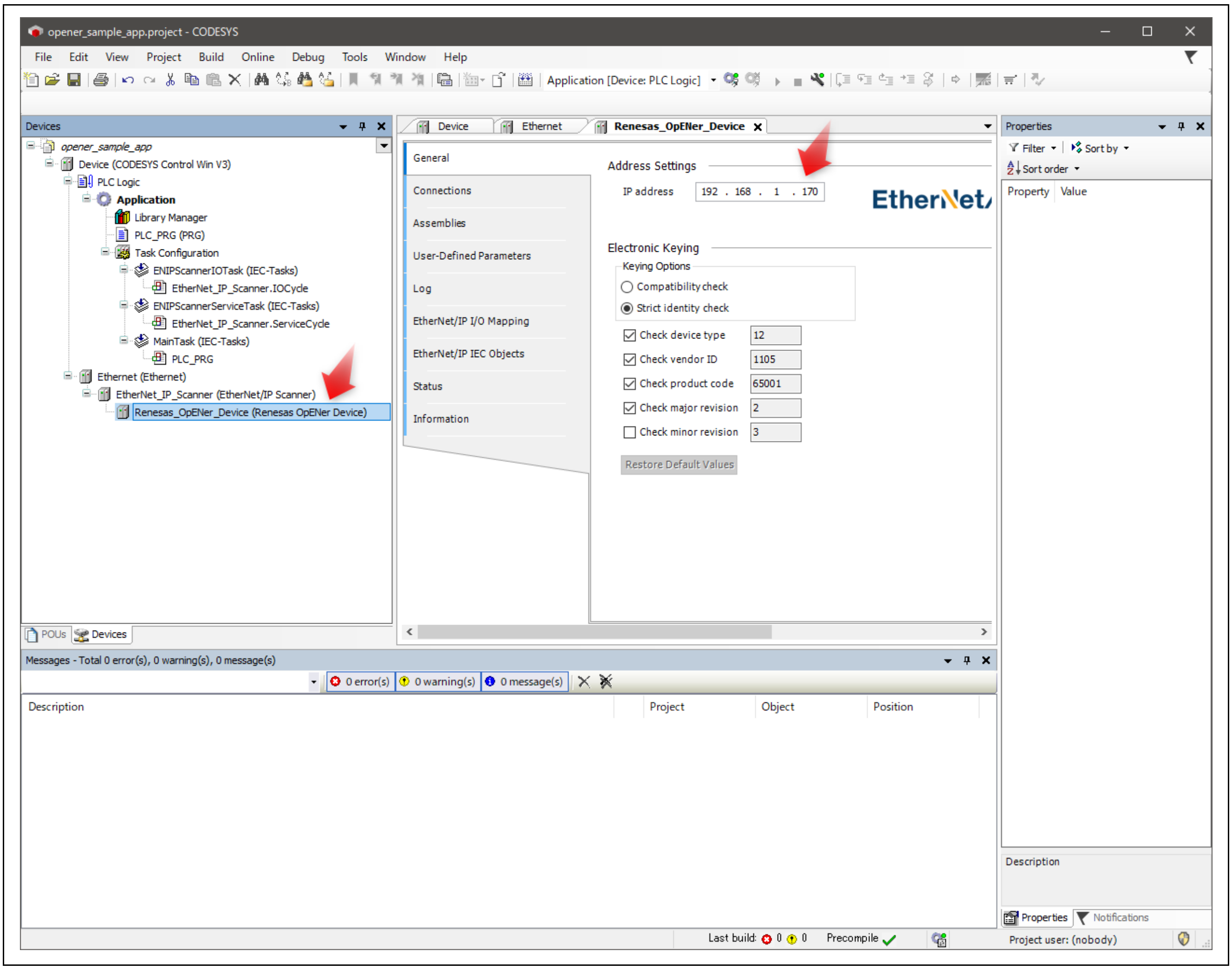


Figure 7-8 Renesas_OpENer_Device (Renesas OpENer Device)

7.4 Operation Check

7.4.1 Build Project and Start Application

Follow the following steps and figure to build the project and start the application.

1. Click “Build” button in the tool bar to build the CODESYS project.
2. Click “Login” button in the tool bar to login to the network.
3. Click “Start” button in the tool bar to run network and application.

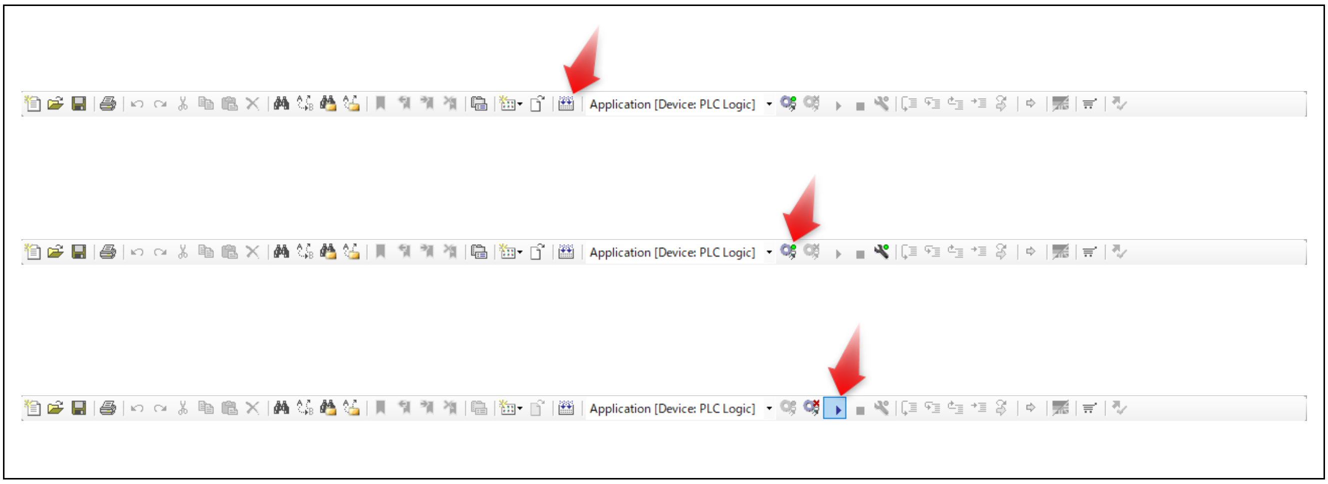


Figure 7-9 Build Project and Start Application

7.4.2 Check Network Connection

If the CODESYS application on PC connects with OpENer application properly, “Device”, “Ethernet”, “EtherNet_IP_Scanner”, and “Renesas_OpENer_Device” in left section are marked with green cycle mark.

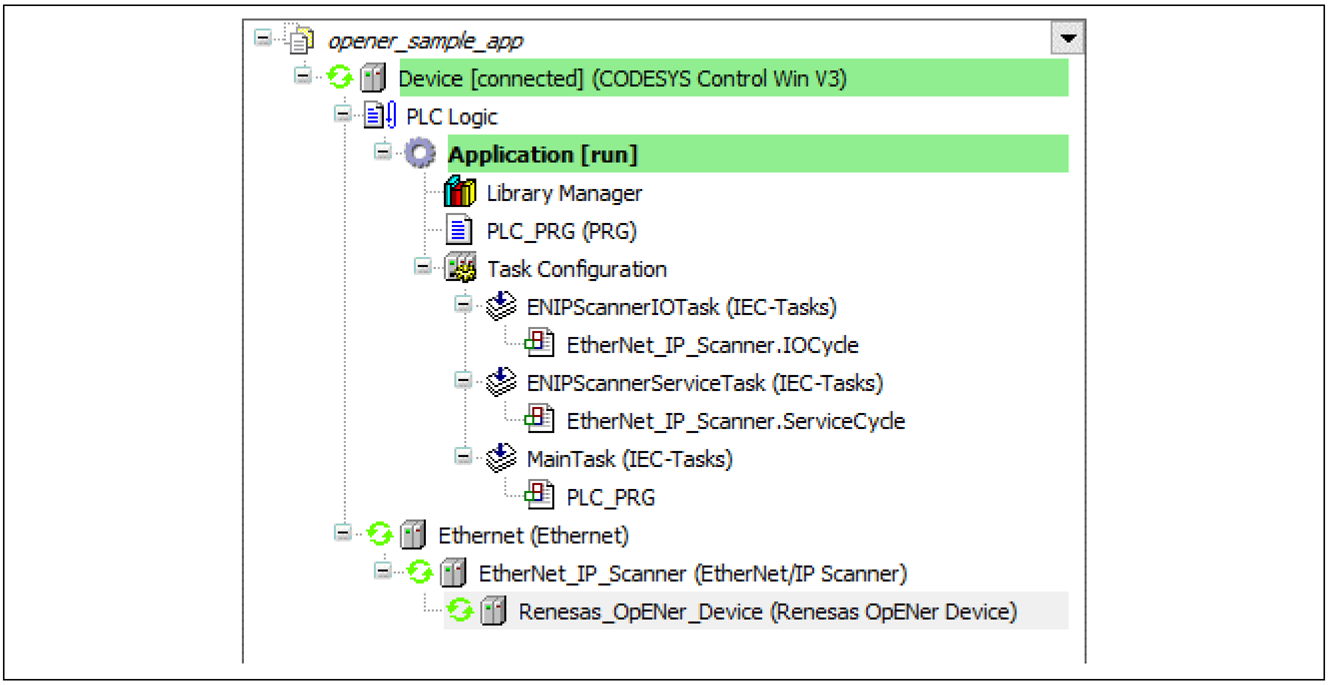


Figure 7-10 Check Network Connection

7.4.3 Check Application Behavior

Please open “EtherNet/IP I/O Mapping” page in “Renesas_OpENer_Device” tab.

This page shows the Exclusive Owner Connection and Input Only Connection mapping which indicates the connections between the CODESYS PLC application and the Assembly objects on OpENer application. The “Current Value” column indicates the LED and SW values.

- The Exclusive Owner Connection is bound to LED on RSK board.
  - The LEDs lights by shifting with [SW Value] Hz.
- The Input Only Connection is bound to SW 1 ~ SW 4 on RSK board.
  - The SW values indicate the frequency of LED light shifting.

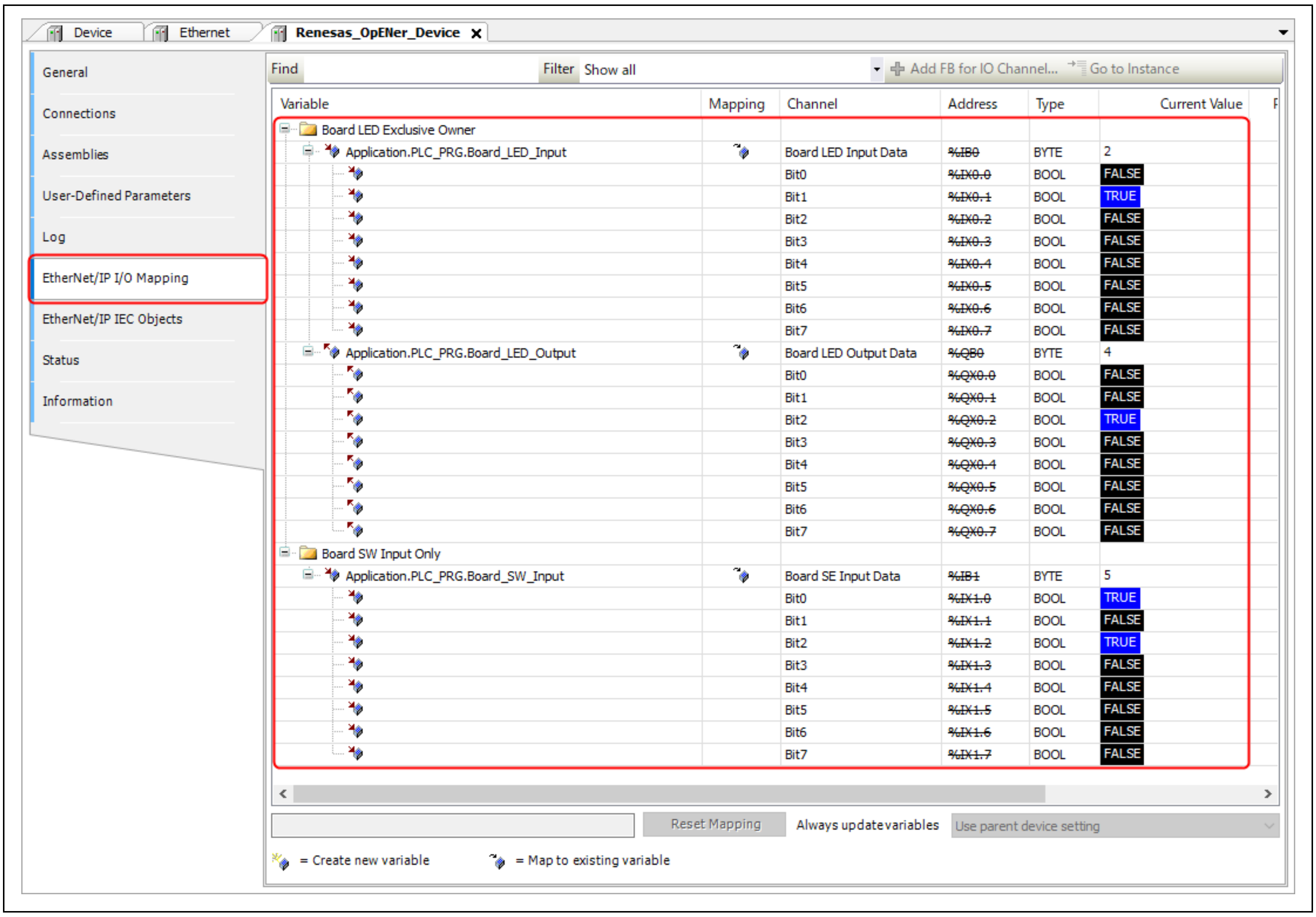


Figure 7-11 Check Application Behavior

8. Software Specification

This chapter shows the specifications of the sample code and other information.

8.1 Stack Diagram

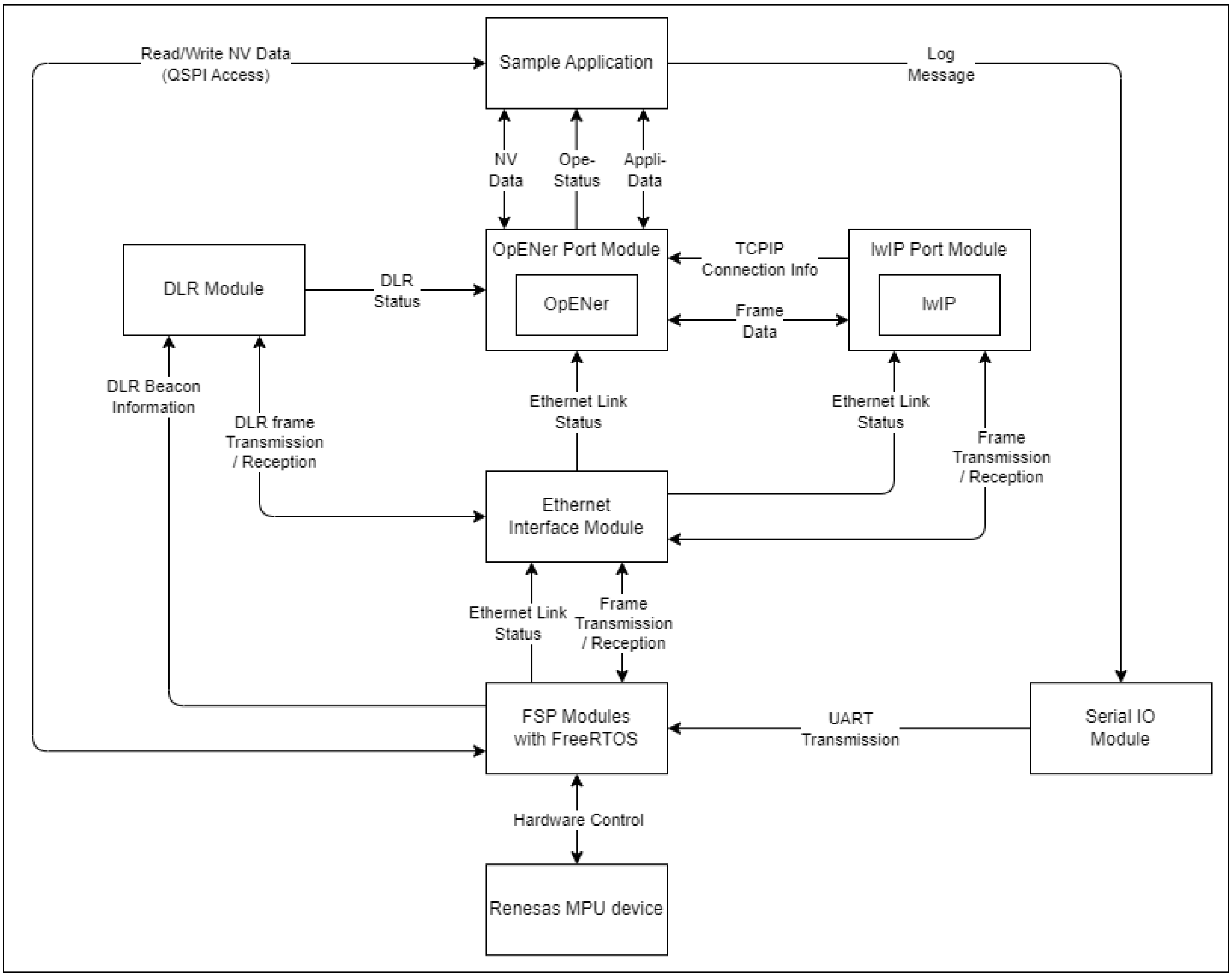


Figure 8-1 Software Stack Diagram

8.2 System Configuration Diagram

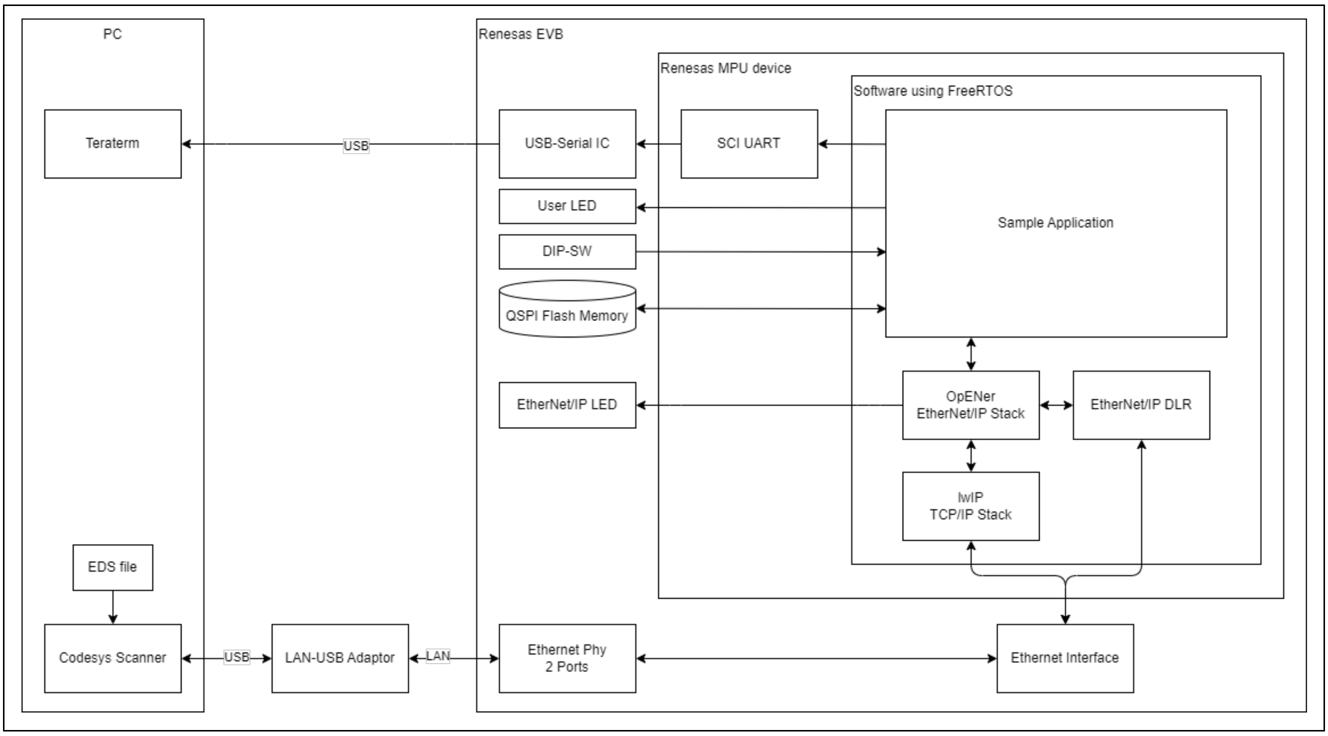


Figure 8-2 System Configuration Diagram

8.3 Support Object Classes, Services and Attributes

Object classes, Services and Attributes supported in this sample code are shown below.

Table 8-1 CIP Object Class on OpENer port

Object Class #	Object Class Name
0x01	Identity
0x02	Massage Router
0x04	Assembly
0x06	Connection Manager
0x47	Device Level Ring
0xF5	TCP/IP Interface
0xF6	Ethernet Link
0x48	QoS
0x109	LLDP Management

Table 8-2 0x01: Identity

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	○	-
2	Max Instance	○	-
3	Number of Instance	○	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	○	-
7	Max Number Instance Attributes	○	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
1	Vendor ID	○	-
2	Device Type	○	-
3	Product Code	○	-
4	Revision	○	-
5	Status	○	-
6	Serial Number	○	-
7	Product Name	○	-
Instance Service			
#	**Service Name**		
0x01	Get_Attributes_All		
0x05	Reset		
0x0E	Get_Attribute_Single		

*: The values of instance attributes #1~#4, #6, and #7 are configured by macros in “devicedata.h" in “common\renesas\oss_deps\opener” directory.

Table 8-3 0x02: Massage Router

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	○	-
2	Max Instance	○	-
3	Number of Instance	○	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	○	-
7	Max Number Instance Attributes	○	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
No instance attributes are implemented.			
Instance Service			
#	**Service Name**		
0x0E	Get_Attribute_Single		

Table 8-4 0x04: Assembly

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	○	-
2	Max Instance	○	-
3	Number of Instance	○	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	○	-
7	Max Number Instance Attributes	○	-
Class Services			
#	**Service Name**		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
3	Data	○	○
4	Size	○	-
Instance Service			
#	**Service Name**		
0x0E	Get_Attribute_Single		
0x10	Set_Attribute_Single		

Table 8-5 0x06: Connection Manager

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	◯	-
2	Max Instance	◯	-
3	Number of Instance	◯	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	◯	-
7	Max Number Instance Attributes	◯	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
No instance attributes are implemented.			
Instance Service			
#	**Service Name**		
0x0E	Get_Attribute_Single		
0x4E	Forward_Close		
0x54	Forward_Open		
0x5a	Get_Connection_Owner		
0x5b	Large_Forward_Open		

Table 8-6 0x47: Device Level Ring

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	◯	-
2	Max Instance	◯	-
3	Number of Instance	◯	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	◯	-
7	Max Number Instance Attributes	◯	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
1	Network Topology	◯	-
2	Network Status	◯	-
10	Active Supervisor Address	◯	-
12	Capability Flags	◯	-
Instance Service			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		

Table 8-7 0x48: QoS

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	○	-
2	Max Instance	○	-
3	Number of Instance	○	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	○	-
7	Max Number Instance Attributes	○	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
1	802.1Q Tag Enable	○	-
2	DSCP PTP Event	○	-
3	DSCP PTP General	○	-
4	DSCP Urgent	○	○
5	DSCP Scheduled	○	○
6	DSCP High	○	○
7	DSCP Low	○	○
8	DSCP Explicit	○	○
Instance Service			
#	**Service Name**		
0x0E	Get_Attribute_Single		
0x10	Set_Attribute_Single		

Table 8-8 0xF5: TCP/IP Interface

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	〇	-
2	Max Instance	〇	-
3	Number of Instance	〇	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	〇	-
7	Max Number Instance Attributes	〇	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
1	Status	〇	-
2	Configuration Capacity	〇	-
3	Configuration Control	〇	〇
4	Physical Link Object	〇	-
5	Interface Configuration	〇	-
6	Host name	〇	-
8	TTL Value	〇	-
9	Mcast Config	〇	-
10	Select ACD	〇	〇
11	LastConflictDetected	〇	〇
13	Encapsulation Inactivity Timeout	〇	〇
Instance Service			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
0x10	Set_Attribute_Single		

Table 8-9 0xF6: Ethernet Link

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	○	-
2	Max Instance	○	-
3	Number of Instance	○	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	○	-
7	Max Number Instance Attributes	○	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
1	Interface Speed	○	-
2	Interface Flag	○	-
3	Physical Address	○	-
7	Interface Type	○	-
11	Interface Capability	○	-
Instance Service			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
0x4C	Get_and_Clear		

Table 8-10 0x109 LLDP Management

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	○	-
2	Max Instance	○	-
3	Number of Instance	○	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	○	-
7	Max Number Instance Attributes	○	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
1	LLDP Enable	○	○
2	msgTxInterval	○	○
3	msgTxHold	○	○
4	LLDP Database	○	-
5	Last Change	○	-
Instance Service			
#	**Service Name**		
0x0E	Get_Attribute_Single		
0x10	Set_Attribute_Single		

8.4 Device Data Configuration

Configurations that are often set for individual devices are listed in this section.

Table 8-11 Device data configuration List

Directory and file	Line	Item	Macro definition	Initial setting
common\renesas\application\opener_port_instance.c	L71	IP address	SAMPLE_STATIC_IP_ADDRESS	192.168.1.170 (or 192.168.1.10 for RZ/N2L)
	L72	Subnet mask	SAMPLE_STATIC_SUBNET_MASK	255,255,255,0
	L73	Gateway address	SAMPLE_STATIC_GATEWAY_ADDRESS	192,168,1,2
	L74	Host name	SAMPLE_HOSTNAME	"OPENER_NETIF0"
	L54	Device serial number	SAMPLE_CIP_DEVICE_SERIAL_NUMBER	0x12345678
common\renesas\oss_deps\opener\devicedata.h	L15	Device vendor ID	OPENER_DEVICE_VENDOR_ID	1105
	L16	Device type	OPENER_DEVICE_TYPE	12
	L17	Device name	OPENER_DEVICE_NAME	"Renesas OpENer Device"
	L20	Product code	OPENER_DEVICE_PRODUCT_CODE	65001
	L21	Device major revision	OPENER_DEVICE_MAJOR_REVISION	2
	L22	Device minor revision	OPENER_DEVICE_MINOR_REVISION	3
common\renesas\oss_deps\lwip\lwipopts.h	L103	MIB2 system description	SNMP_LWIP_MIB2_SYSDESC	"Renesas Electronics Corporation EtherNet/IP Sample"
	L104	MIB2 system contact	SNMP_LWIP_MIB2_SYSCONTACT	"sysContact not set"
	L105	MIB2 system name	SNMP_LWIP_MIB2_SYSNAME	"sysName not set"
	L106	MIB2 system location	SNMP_LWIP_MIB2_SYSLOCATION	"sysLocation not set"

9. Appendix

Appendix A: OSS implemented in the sample code

The following three OSS are used in the sample code.

OpENer

The "OpENer" is an open-source software for I/O communication adapters of EtherNet/IP. Please refer to the following link for the details.

https://github.com/EIPStackGroup/OpENer

- Git comment ID used in this sample software: 05cdd03

This package is the RZ/T2 and RZ/N2 port of OpENer and includes OpENer source codes. Regarding the open-source license of OpENer, please see the following file.

```
common\oss\OpENer\license.txt
```

FreeRTOS

FreeRTOS is an open-source software for real-time operating system (RTOS) for microcontrollers. Please refer to the following link for the details.

https://aws.amazon.com/freertos/

https://github.com/aws/amazon-freertos

- Git comment ID used in this sample software: a038063

The RZ/T2 and RZ/N2 port of OpENer includes FreeRTOS source codes as RTOS kernel. Regarding the open-source license of OpENer, please see the following file.

```
common\oss\amazon-freertos\LICENSE
```

lwIP

The lwIP is an open-source software for a small independent implementation of the TCP/IP protocol suite. Please refer to the following link for the details.

https://savannah.nongnu.org/projects/lwip/

https://github.com/lwip-tcpip/lwip

- Git comment ID used in this sample software: 79cd89f

The RZ/T2 and RZ/N2 port of OpENer includes lwIP source codes as a TCP/IP stack. Regarding the open-source license of lwIP, please see the following file.

```
common\oss\lwip\COPYING
```

Appendix B: Assembly Objects and I/O Connections

This sample application has 7 instances of Assembly object. 5 of these instances are bound to the following array variables.

Table 9-1 Assembly Objects and I/O Connections

Instance No.	Type	Description	Bound variables
100 (0x64)	Static Input	Input of Exclusive Owner I/O Connection bound to LED	`g_assembly_data_led_input[1]`
101 (0x65)	Static Input	Input of Input Only I/O Connection bound to SW	`g_assembly_data_sw_input[1]`
150 (0x96)	Static Output	Input of Exclusive Owner I/O Connection bound to LED	`g_assembly_data_led_output[1]`
151 (0x97)	Static Configuration	Configuration of I/O Connections	`g_assembly_data_config[4]`
154 (0x9A)	Static I/O	Accessing by explicit message connection only	`g_assembly_data_explicit[4]`
238 (0xEE)	Static Output	Heartbeat output of Input only I/O Connection bound to SW	-
237 (0xED)	Static Output	Heartbeat output of listen only I/O Connection.	-

Appendix C: CIP Safety feature

Overview

This section describes how to enable the CIP Safety feature, as well as the software structure and behavior when it is enabled.

Note: The enabled CIP Safety feature has only been tested for basic operation in the connection configurations described later. A self-test based on the official ODVA guidelines has not been conducted.

How to enable CIP Safety feature

To enable the CIP Safety feature, remove “DISABLE_OPENER_CIP_SAFETY” and add “OPENER_CIP_SAFETY” in the project's Defined Symbols.

Place of the project's Defined Symbols

- e² studio case
  - ➢ Project Properties -> C/C++ Build -> Settings -> Tool Settings -> Cross ARM C Compiler -> Preprocessor -> Defined Symbols
- EWARM case
  - ➢ Project’s Options -> C/C++ Compiler -> Preprocessor -> Defined symbols

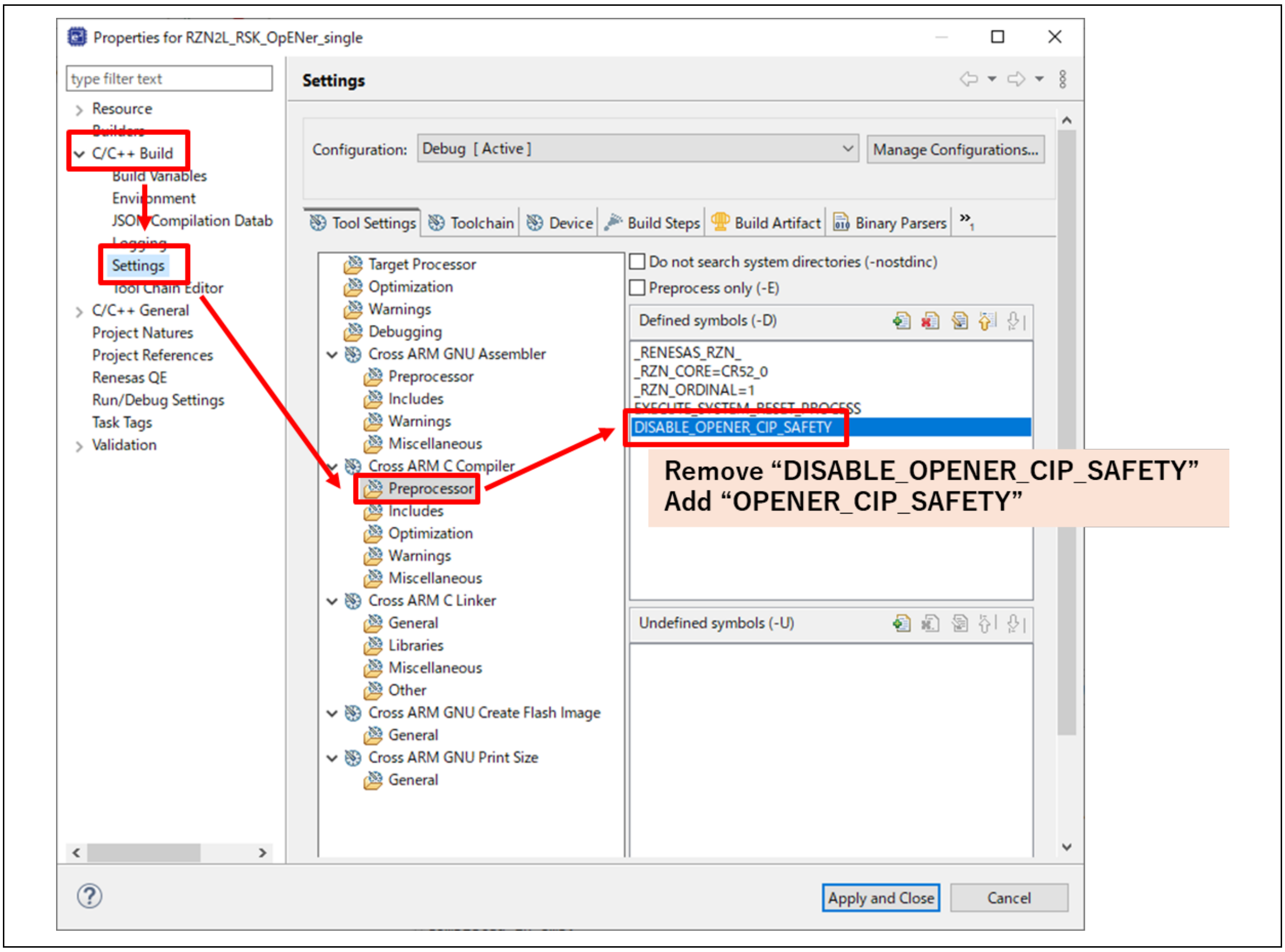


Figure 9-1 How to enable CIP Safety feature, e² studio case

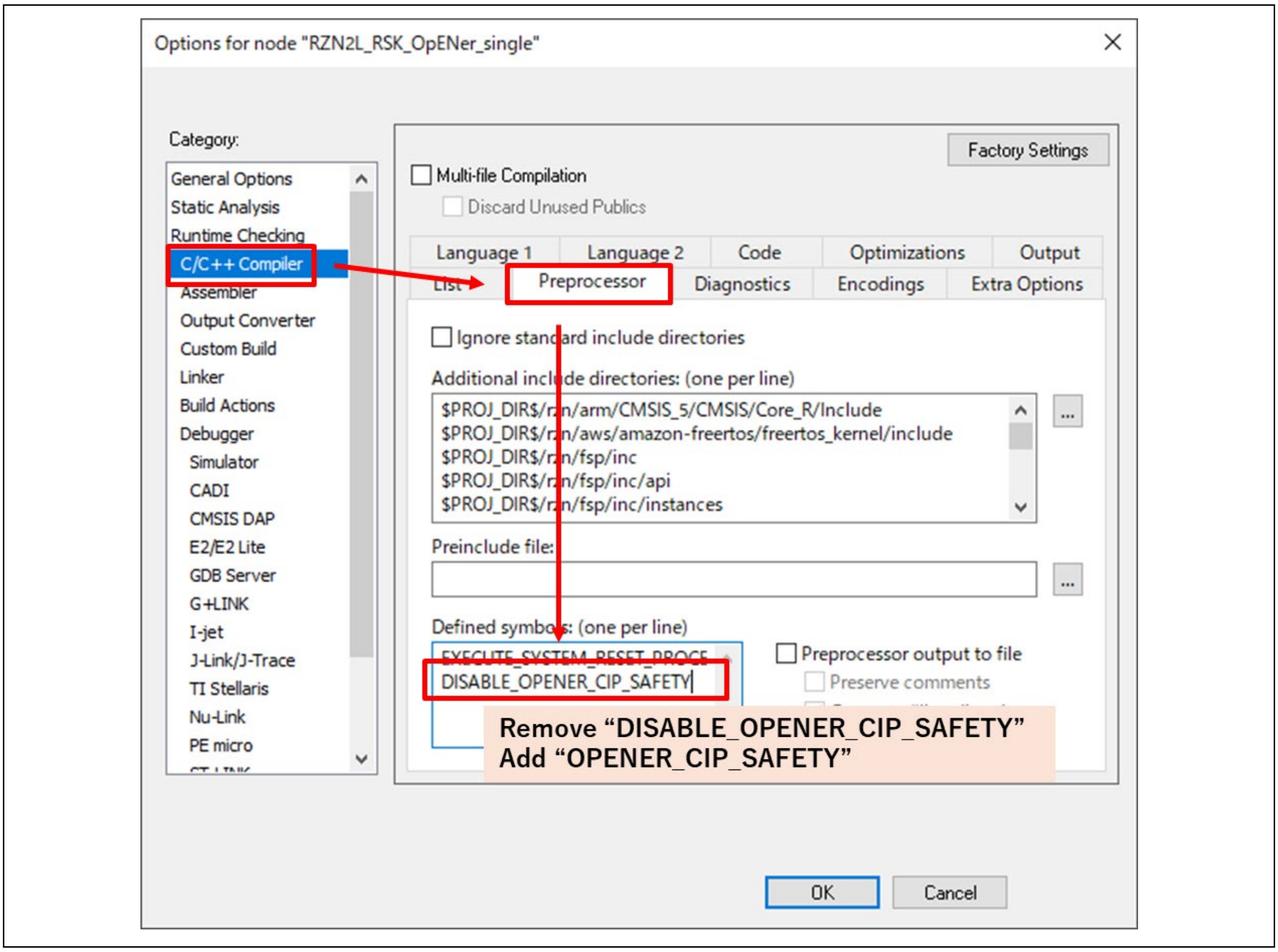


Figure 9-2 How to enable CIP Safety feature, EWARM case

Hardware Setup and Start

Please follow the steps below to complete the setup.

The RSK-RZ/N2L board with this sample code downloaded functions as a black channel for functional safety with respect to the RZ/T2L Functional Safety Reference Board on which the CIP Safety Reference Software is downloaded.

1. Please prepare the contents of Table 9-2.
    - For Renesas product and software, please refer to the Renesas website or contact a distributor.
    - For products from other manufacturers, please contact the respective manufacturer's support.

Table 9-2 Required devices and software

#	Category	Product name/ Model name	Manufacturer
1	Safety PLC (Originator)	Compact GuardLogix SIL3 0.6.0,3M Motion : 5069-L306ERMS3	Rockwell Automation
2	RZ/T2L Functional Safety Reference Board	RTK0EF0179D01001BJ	Renesas Electronics
3	RZ/T2L CIP Safety Reference Software	RTK0EF0200F01001SJ_CI	Renesas Electronics

2. Download the RZ/T2L CIP Safety Reference Software to the RZ/T2L Functional Safety Reference Board.
    - Refer to the software documentation for the download procedure.

3. Download the software to the RSK-RZ/N2L.
    - Enable CIP Safety feature by following the former section, and Download this sample program by following the instructions in Chapter 6.

4. Connect the devices as shown in Figure 9-3 and Table 9-3.

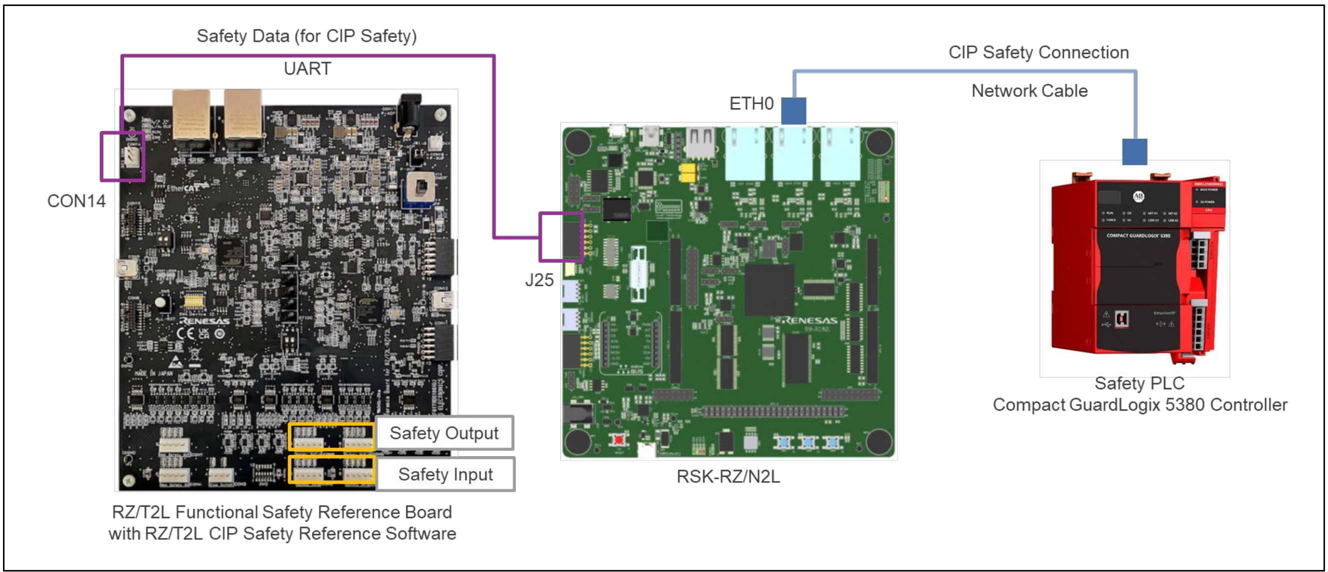


Figure 9-3 Topology when CIP Safety feature is enabled

Table 9-3 Connection between RSK and FuSa Board

RSK-RZ/N2L		RZ/T2L Functional Safety Reference Board	
J25		CON14	
#	**Connector Pin Name**	**#**	**Connector Pin Name**
3	SCI_RXD	3	CON_TXD_A
2	SCI_TXD	2	CON_RXD_A
5	GROUND	1	DGND

5. Power on all devices.

The RSK-RZ/N2L will begin EtherNet/IP operation upon receiving a wakeup request from the Functional Safety Reference Board. For the overall operation as a functional safety system, please refer to the documentation of the CIP Safety Reference Software.

Operation

The RSK-RZ/N2L board with this sample code downloaded functions as a black channel for functional safety with respect to the RZ/T2L Functional Safety Reference Board on which the CIP Safety Reference Software is downloaded.

Stack Diagram

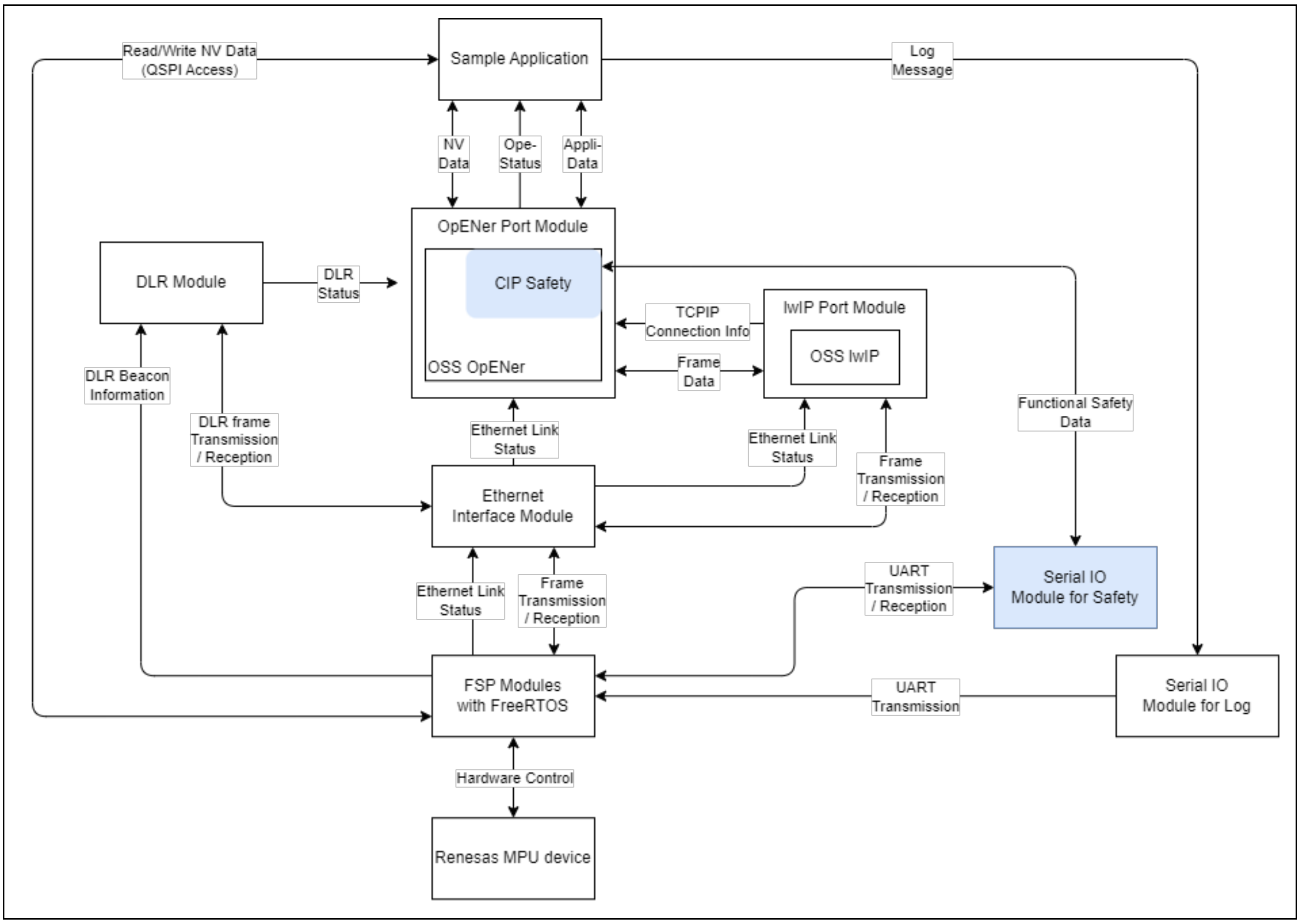


Figure 9-4 Software Stack Diagram when CIP Safety feature is enabled

System Configuration Diagram

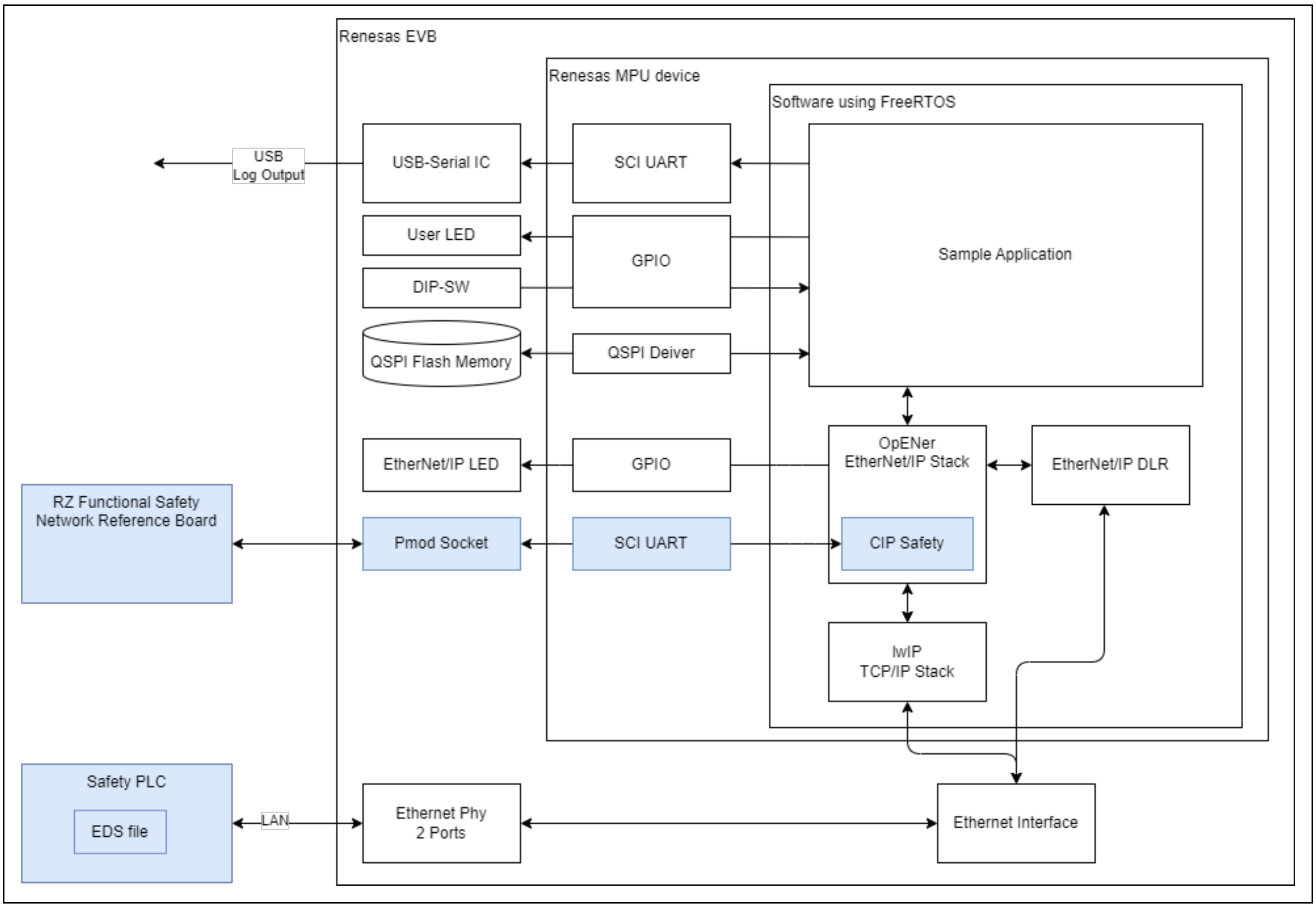


Figure 9-5 System Configuration Diagram when CIP Safety feature is enabled

Support Objects, Services and Attributes

Object classes, Services and Attributes supported in this sample code are shown below. The sections indicated in red are the ones that are modified when CIP Safety is enabled.

Table 9-4 Object Classes supported when CIP Safety feature is enabled.

Object Class #	Object Class Name
0x01	Identity
0x02	Massage Router
0x04	Assembly
0x06	Connection Manager
0x39	Safety Supervisor
0x3A	Safety Validator
0x47	Device Level Ring
0x48	QoS
0x49	Safety Analog Input Point
0x64	Safety Analog Output Point
0xF5	TCP/IP Interface
0xF6	Ethernet Link
0x109	LLDP Management

Table 9-5 0x01: Identity

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	○	-
2	Max Instance	○	-
3	Number of Instance	○	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	○	-
7	Max Number Instance Attributes	○	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
1	Vendor ID	○	-
2	Device Type	○	-
3	Product Code	○	-
4	Revision	○	-
5	Status	○	-
6	Serial Number	○	-
7	Product Name	○	-
Instance Service			
#	**Service Name**		
0x01	Get_Attributes_All		
0x05	Reset **		
0x0E	Get_Attribute_Single		

*: The values of instance attributes #1~#4, #6, and #7 are configured by macros in “devicedata.h" in “common\renesas\oss_deps\opener” directory.

**: In accordance with the SRS61 specification, the Identity Reset service will respond with error code 0x10 if even a single I/O connection is established.

Table 9-6 0x02: Massage Router

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	〇	-
2	Max Instance	〇	-
3	Number of Instance	〇	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	〇	-
7	Max Number Instance Attributes	〇	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
No instance attributes are implemented.			
Instance Service			
#	**Service Name**		
0x0E	Get_Attribute_Single		

Table 9-7 0x04: Assembly

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	〇	-
2	Max Instance	〇	-
3	Number of Instance	〇	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	〇	-
7	Max Number Instance Attributes	〇	-
Class Services			
#	**Service Name**		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
3	Data	〇	〇
4	Size	〇	-
Instance Service			
#	**Service Name**		
0x0E	Get_Attribute_Single		
0x10	Set_Attribute_Single		

Table 9-8 0x06: Connection Manager

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	〇	-
2	Max Instance	〇	-
3	Number of Instance	〇	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	〇	-
7	Max Number Instance Attributes	〇	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
No instance attributes are implemented.			
Instance Service			
#	**Service Name**		
0x0E	Get_Attribute_Single		
0x4E	Forward_Close		
0x54	Forward_Open		
0x5a	Get_Connection_Owner		
0x5b	Large_Forward_Open		

Table 9-9 0x39: Safety Supervisor

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	〇	-
2	Max Instance	〇	-
3	Number of Instance	〇	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	〇	-
7	Max Number Instance Attributes	〇	-
Class Services			
#	**Service Name**		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
11	Device Status	〇	-
12	Exception Status	〇	-
13	Exception Detail Alarm	〇	-
14	Exception Detail Warning	〇	-
15	Alarm Enable	〇	〇
16	Warning Enable	〇	〇
24	Configuration Lock	〇	-
25	Configuration UNID	〇	-
26	Safety Configuration Identifier	〇	-
27	Target UNID	〇	-
28	Output Connection Point Owners	〇	-
29	Proposed UNID	〇	-
Instance Service			
#	**Service Name**		
0x0E	Get_Attribute_Single		
0x10	Set_Attribute_Single		
0x0D	Apply_Attributes		
0x4F	Configure_Request		
0x50	Validate_Configuration		
0x51	Set_Password		
0x52	Configuration_Lock/Unlock		
0x54	Safety_Reset (Type0 Power Cycle, Type 1 Factory Reset)		
0x55	Reset_Password		
0x56	Propose_TUNID		
0x57	Apply_TUNID		

Table 9-10 0x3A: Safety Validator

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	〇	-
2	Max Instance	〇	-
3	Number of Instance	〇	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	〇	-
7	Max Number Instance Attributes	〇	-
8	Safety Connection Fault Count	〇	-
Class Services			
#	**Service Name**		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
1	Safety Validator Status	〇	-
2	Safety Validator Type	〇	-
12	Max Data Age	〇	〇
13	Application Data Path	〇	-
15	Producer/Consumer Fault Counters	〇	-
Instance Service			
#	**Service Name**		
0x0E	Get_Attribute_Single		

Table 9-11 0x47: Device Level Ring

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	○	-
2	Max Instance	○	-
3	Number of Instance	○	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	○	-
7	Max Number Instance Attributes	○	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
1	Network Topology	○	-
2	Network Status	○	-
10	Active Supervisor Address	○	-
12	Capability Flags	○	-
Instance Service			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		

Table 9-12 0x48: QoS

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	○	-
2	Max Instance	○	-
3	Number of Instance	○	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	○	-
7	Max Number Instance Attributes	○	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
1	802.1Q Tag Enable	○	-
2	DSCP PTP Event	○	-
3	DSCP PTP General	○	-
4	DSCP Urgent	○	○
5	DSCP Scheduled	○	○
6	DSCP High	○	○
7	DSCP Low	○	○
8	DSCP Explicit	○	○
Instance Service			
#	**Service Name**		
0x0E	Get_Attribute_Single		
0x10	Set_Attribute_Single		

Table 9-13 0x49: Safety Analog Input Point

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	○	-
2	Max Instance	○	-
3	Number of Instance	○	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	○	-
7	Max Number Instance Attributes	○	-
Class Services			
#	**Service Name**		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
2	Safety Analog Input Value	○	-
3	Input Range	○	○
4	Input Channel Mode	○	○
5	Input Point Status	○	-
6	Point Fault Reason	○	-
Instance Service			
#	**Service Name**		
0x0E	Get_Attribute_Single		
0x10	Set_Attribute_Single		

Table 9-14 0x64: Safety Analog Output Point

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	○	-
2	Max Instance	○	-
3	Number of Instance	○	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	○	-
7	Max Number Instance Attributes	○	-
Class Services			
#	**Service Name**		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
2	Safety Analog Output Value	○	-
3	Output Range	○	○
4	Output Channel Mode	○	○
5	Output Point Status	○	-
6	Point Fault Reason	○	-
Instance Service			
#	**Service Name**		
0x0E	Get_Attribute_Single		
0x10	Set_Attribute_Single		

Table 9-15 0xF5: TCP/IP Interface

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	〇	-
2	Max Instance	〇	-
3	Number of Instance	〇	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	〇	-
7	Max Number Instance Attributes	〇	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
1	Status	〇	-
2	Configuration Capacity	〇	-
3	Configuration Control	〇	〇
4	Physical Link Object	〇	-
5	Interface Configuration	〇	-
6	Host name	〇	-
7	Safety Network Number	〇	-
8	TTL Value	〇	-
9	Mcast Config	〇	-
10	Select ACD	〇	〇
11	LastConflictDetected	〇	〇
13	Encapsulation Inactivity Timeout	〇	〇
Instance Service			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
0x10	Set_Attribute_Single		

Table 9-16 0xF6: Ethernet Link

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	○	-
2	Max Instance	○	-
3	Number of Instance	○	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	○	-
7	Max Number Instance Attributes	○	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
1	Interface Speed	○	-
2	Interface Flag	○	-
3	Physical Address	○	-
7	Interface Type	○	-
11	Interface Capability	○	-
Instance Service			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
0x4C	Get_and_Clear		

Table 9-17 0x109 LLDP Management

Class Attributes			
#	**Attribute Name**	**Get**	**Set**
1	Revision	○	-
2	Max Instance	○	-
3	Number of Instance	○	-
4	Optional Attribute List	-	-
5	Optional Service List	-	-
6	Max Number Class Attributes	○	-
7	Max Number Instance Attributes	○	-
Class Services			
#	**Service Name**		
0x01	Get_Attributes_All		
0x0E	Get_Attribute_Single		
Instance Attributes*			
#	**Attribute Name**	**Get**	**Set**
1	LLDP Enable	○	○
2	msgTxInterval	○	○
3	msgTxHold	○	○
4	LLDP Database	○	-
5	Last Change	○	-
Instance Service			
#	**Service Name**		
0x0E	Get_Attribute_Single		
0x10	Set_Attribute_Single		

Device Data Configuration

Configurations that are often set for individual devices are listed in this section when CIP Safety feature is enabled.

Table 9-18 Device data configuration List

Directory and file	Line	Item	Macro definition	Initial setting
common\renesas\application\opener_port_instance.c	L71	IP address	SAMPLE_STATIC_IP_ADDRESS	192.168.1.10
	L72	Subnet mask	SAMPLE_STATIC_SUBNET_MASK	255,255,255,0
	L73	Gateway address	SAMPLE_STATIC_GATEWAY_ADDRESS	192,168,1,2
	L74	Host name	SAMPLE_HOSTNAME	"OPENER_NETIF0"
	L54	Device serial number	SAMPLE_CIP_DEVICE_SERIAL_NUMBER	0x12345678
common\renesas\oss_deps\opener\devicedata.h	L15	Device vendor ID	OPENER_DEVICE_VENDOR_ID	1105 (0x0451, Renesas Electronics)
	L16	Device type	OPENER_DEVICE_TYPE	42 (0x002A, Safety Analog I/O Device)
	L17	Device name	OPENER_DEVICE_NAME	“RZ Adaptor Safety”
	L20	Product code	OPENER_DEVICE_PRODUCT_CODE	21258 (0x530A)
	L21	Device major revision	OPENER_DEVICE_MAJOR_REVISION	3
	L22	Device minor revision	OPENER_DEVICE_MINOR_REVISION	7
common\renesas\oss_deps\lwip\lwipopts.h	L103	MIB2 system description	SNMP_LWIP_MIB2_SYSDESC	"Renesas Electronics Corporation EtherNet/IP Sample"
	L104	MIB2 system contact	SNMP_LWIP_MIB2_SYSCONTACT	"sysContact not set"
	L105	MIB2 system name	SNMP_LWIP_MIB2_SYSNAME	"sysName not set"
	L106	MIB2 system location	SNMP_LWIP_MIB2_SYSLOCATION	"sysLocation not set"

Note: The sections indicated in red are the ones that are modified when CIP Safety is enabled.

Appendix D: DLR Operation Check

This section describes how to check DLR operation.

The sample code implements the function as a DLR node only (not Supervisor or Redundant Gateway). Refer to Section 8.3 for supported Attributes and Services.

Check Procedure

1. Prepare Table 9-19 devices and Table 9-20 software tools.
2. Establish the Figure 9-6 topology and power up.
3. Download the program to the RSK board and run it on e^2 studio or EWARM. (Please refer to Chapter 6.6)
4. Monitor DLR Instance Attribute values and DLR operation with software tools.

Table 9-19 Used Devices

Product Name	Catalog #:	Vendor	Comment
Allen Bradley Compact Logic 384KB DI/O Controller	1769-L16ER-BB1B	Rockwell Automation	Used as a DLR Supervisor
Allen Bradley 3 port Ethernet Tap	1783-ETAP	Rockwell Automation	Used as a DLR node (Sub-supervisor)

Table 9-20 Used Software Tools

Software Name	Released by
EIP Tools	Molex
RSLinx	Rockwell Automation
DLR Tool	Rockwell Automation

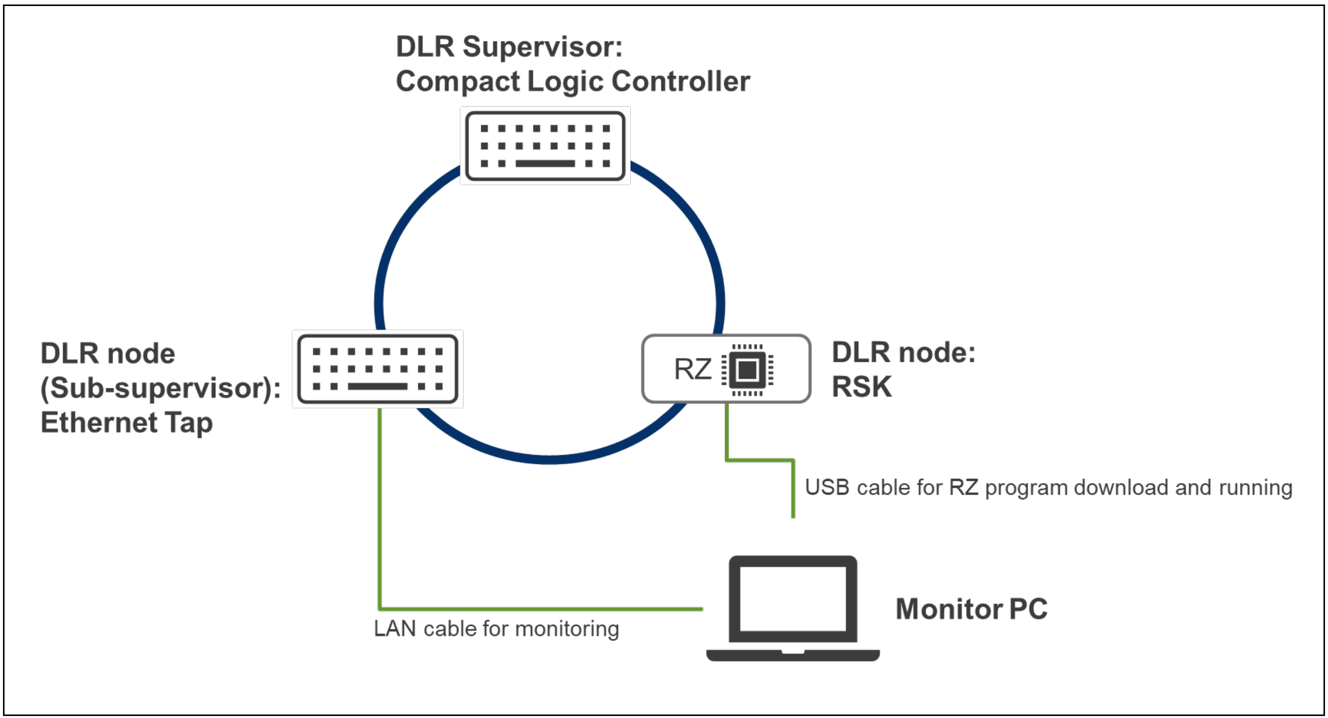


Figure 9-6 Topology for check

Check DLR Instance Attribute values by EIP Tools

The Instance Attribute value can be checked using the EIP Tools. The procedure is described below.

Open EIP Tools and enter the IP address of the target device. The default setting for the IP address written in the sample code is 192.168.1.170 (or 192.168.1.10).

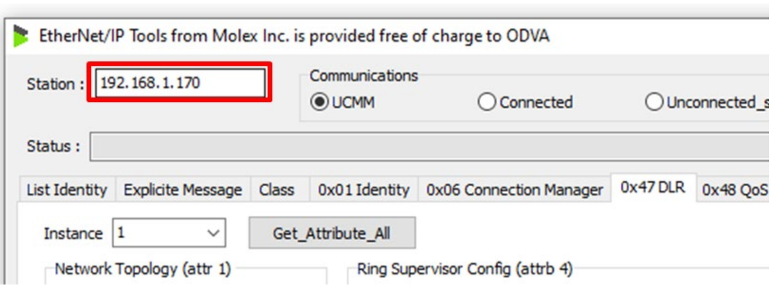
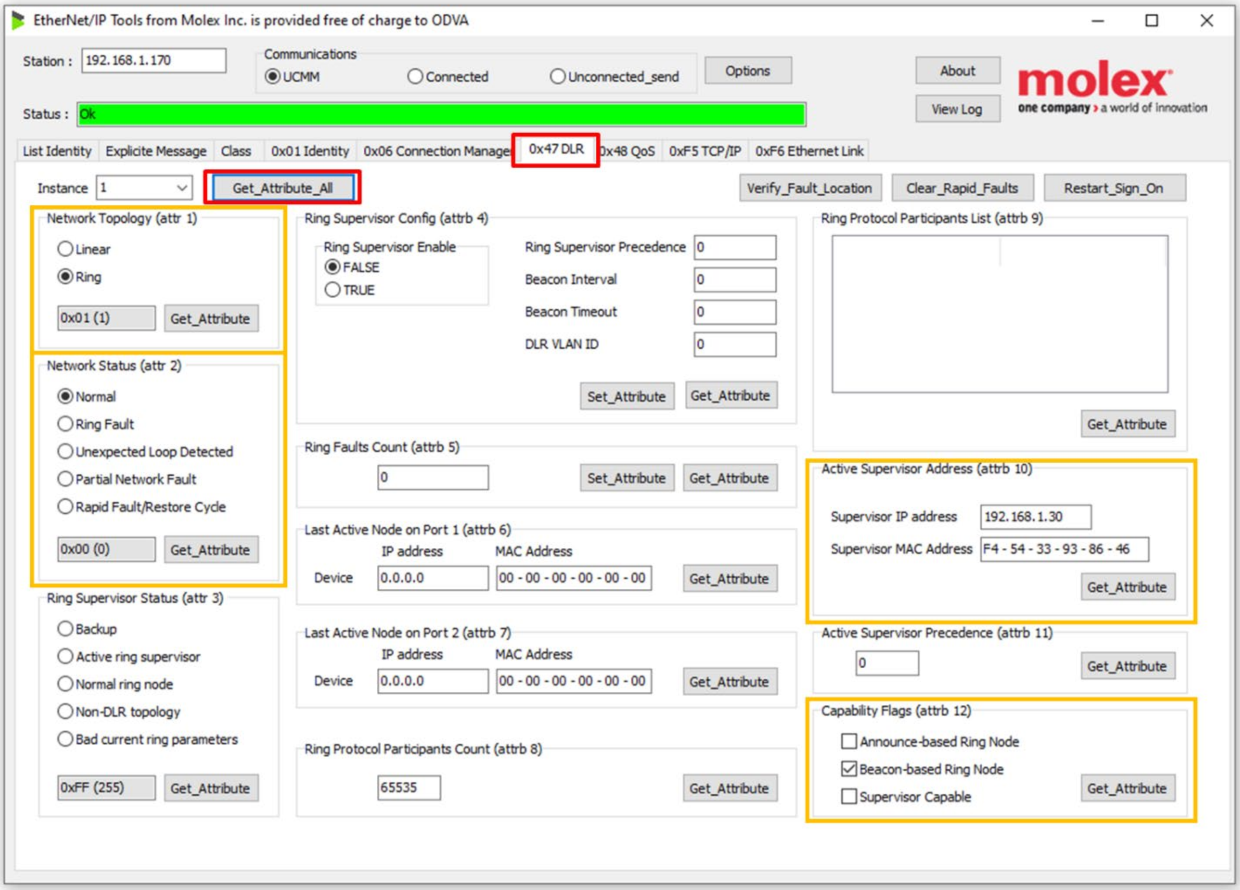


Figure 9-7 Enter the IP address of the target device in EIP Tools

Select the 0x47 DLR tab and click Get_Attributes_All or Get_Attribute_Single. You can see the value of each.



Note: The sample code supports Instance Attribute #1, 2, 10, and 12. (The yellow circled area)

Figure 9-8 Execute Get_Attributes_All in EIP Tools

Check DLR operation by RSLinx and DLR Tool

DLR operation can be checked with the DLR Tool. RSLinx driver settings are required as a preliminary step.

■RSLinx driver settings

Launch RSLinx. If the RSWho window is not open, click RSWho from the Communication tab.

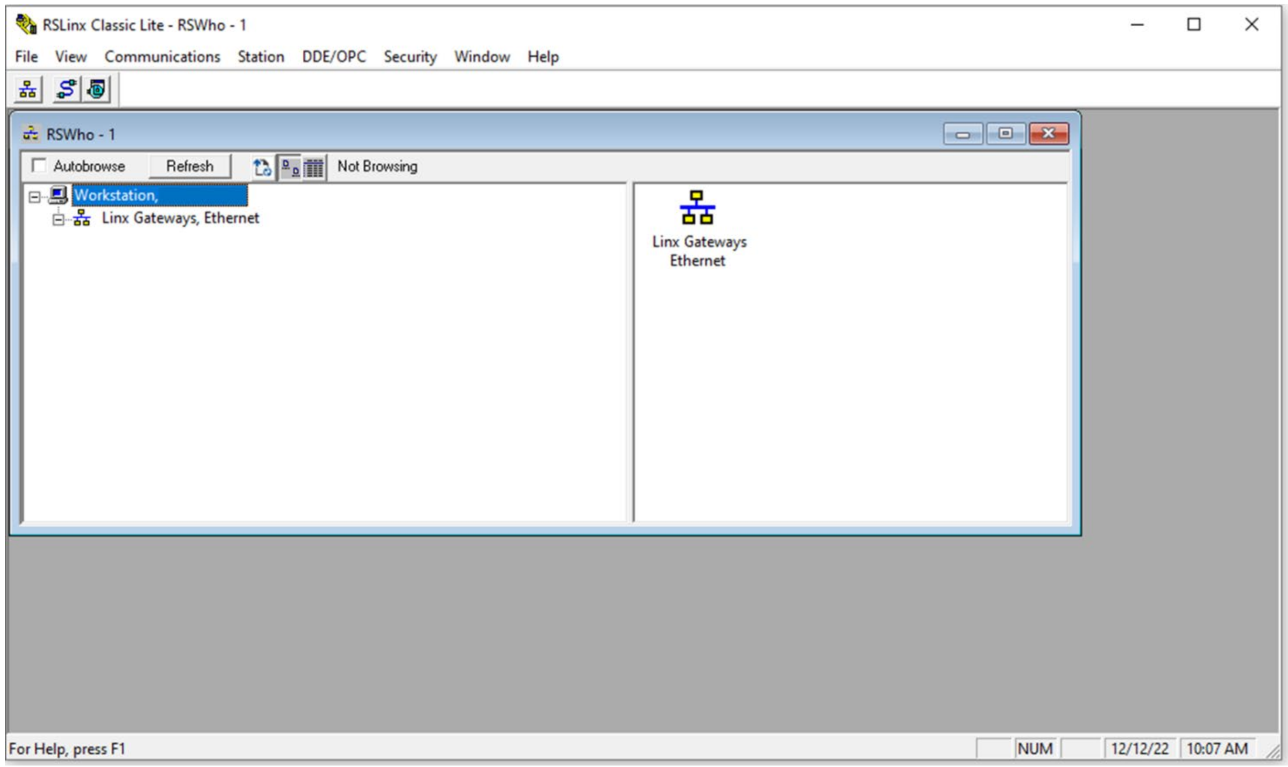


Figure 9-9 Start RSLinx

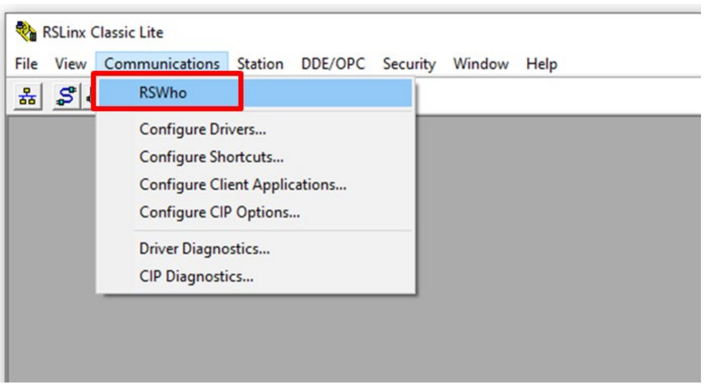


Figure 9-10 Open RSWho window

Create a new driver to monitor the DLR status. Click Configure Driver from the Communication tab.

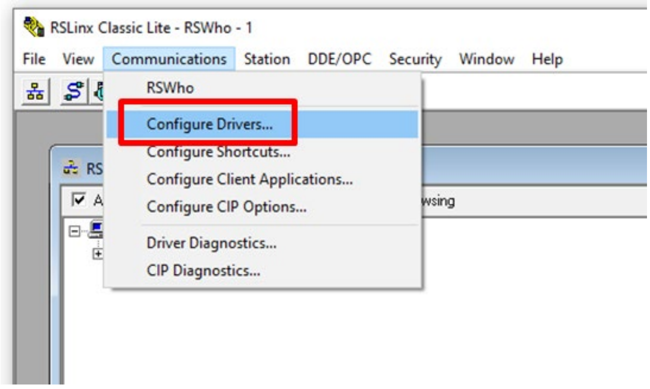


Figure 9-11 Open Configure Drivers window

Select EtherNet/IP Driver from “Available Driver Types” and click “Add New”.

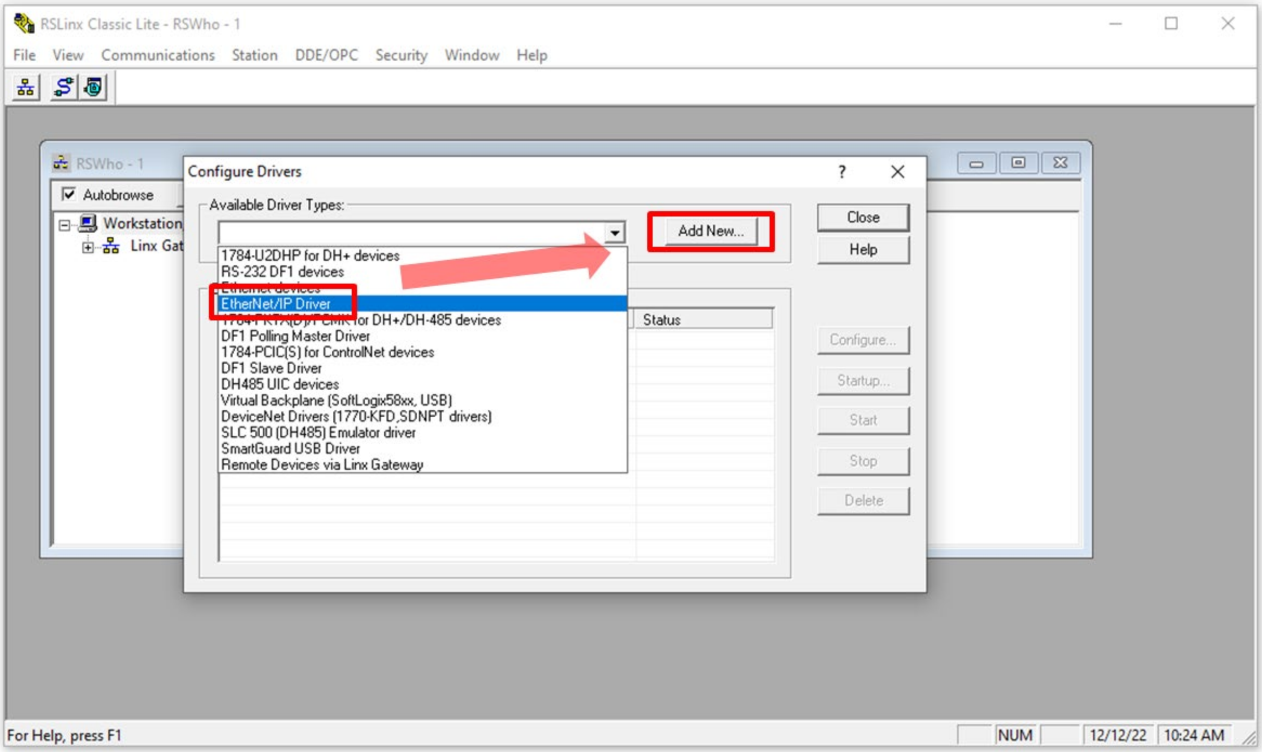


Figure 9-12 Create new EtherNet/IP Driver

Enter any driver name.

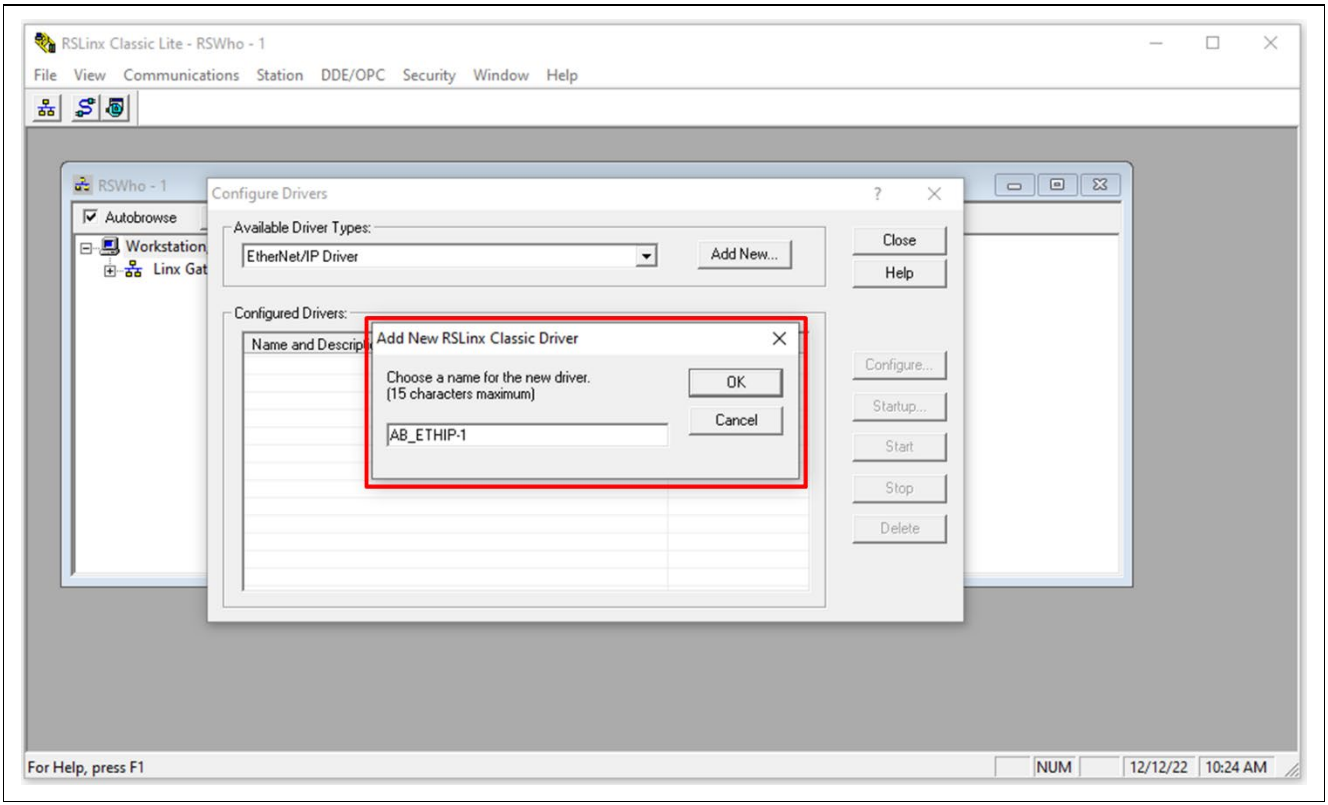


Figure 9-13 Enter new driver name

Select the Ethernet adapter connected to the DLR device, click Apply, and then click OK.

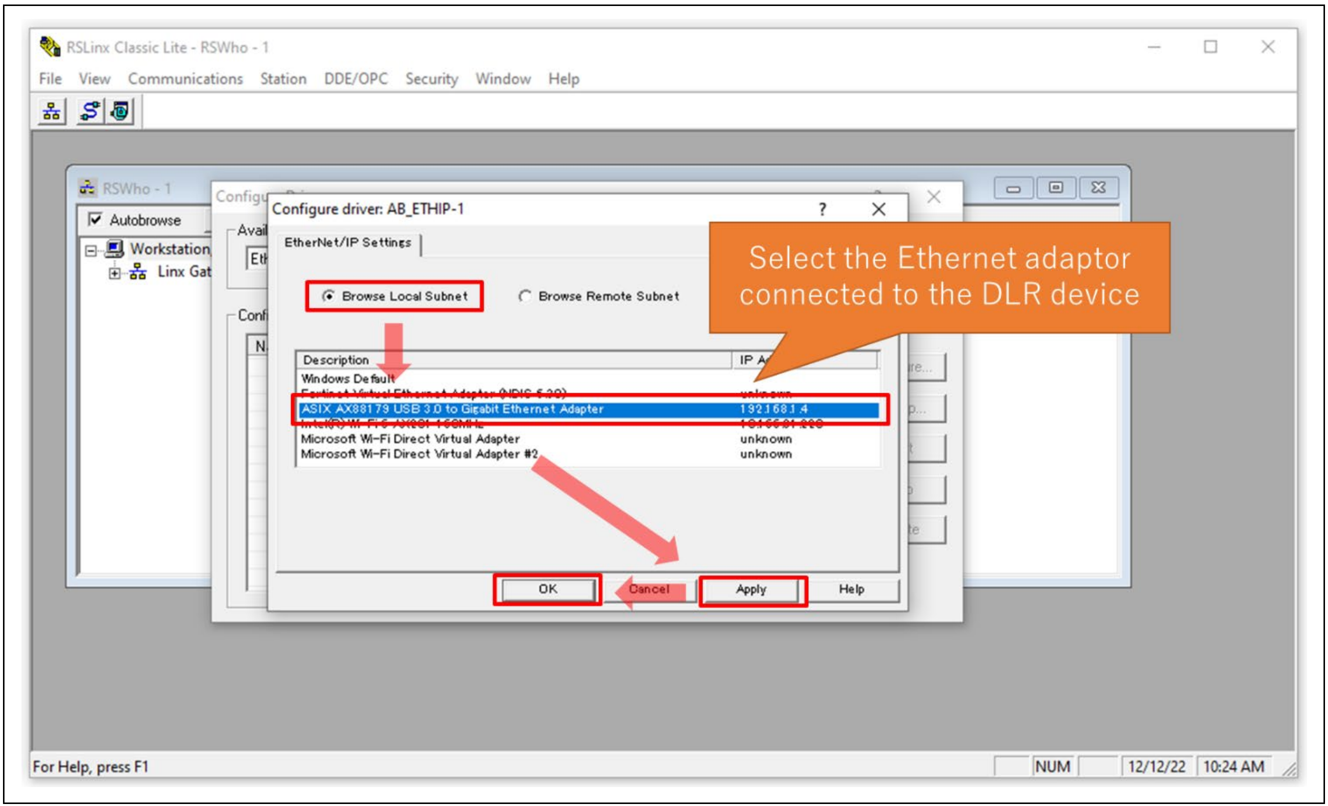


Figure 9-14 Select the Ethernet adaptor connected to the DLR device

Confirm that the driver has been added and Close the Configure Drivers window.

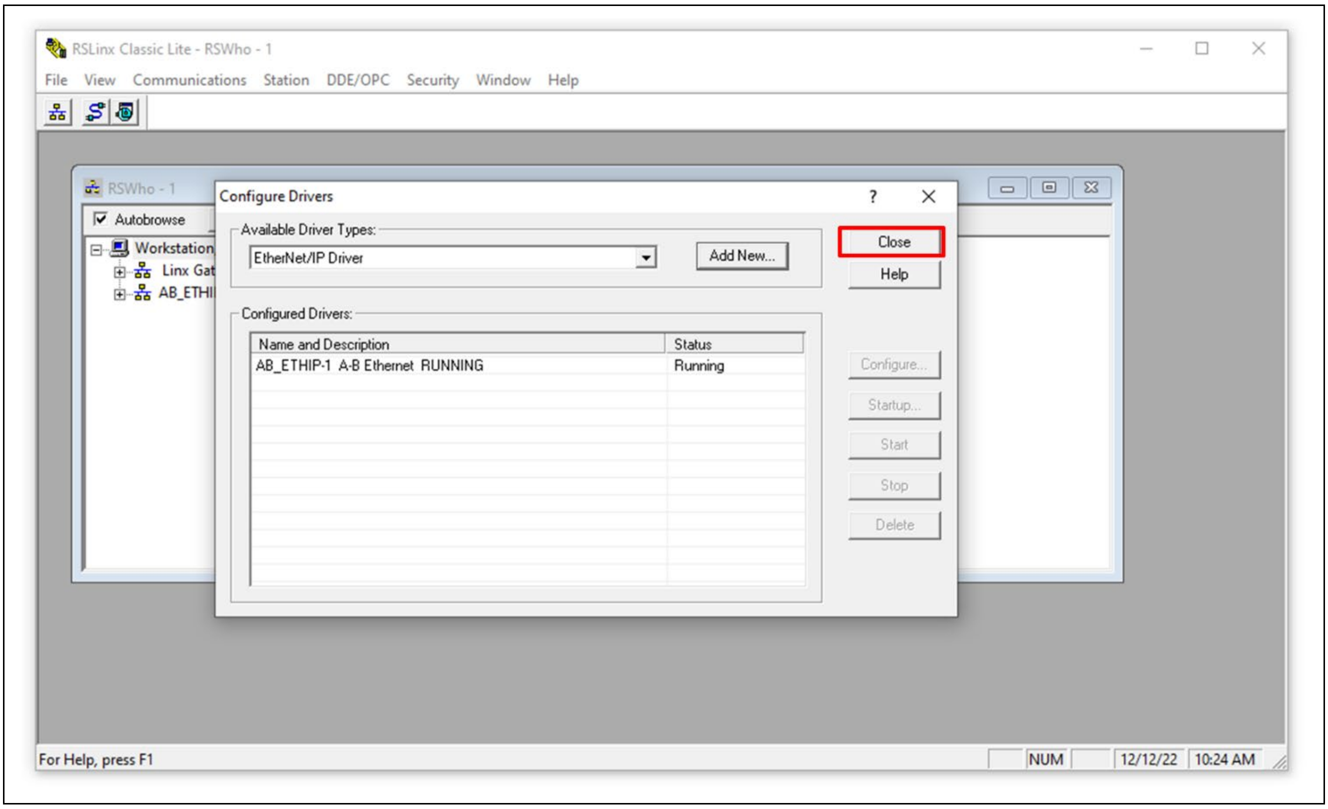


Figure 9-15 Close Configure Drives window

Make sure Autobrowse is enabled and expand the driver you have created. Confirm that the device is recognized as shown in Figure 9-16.

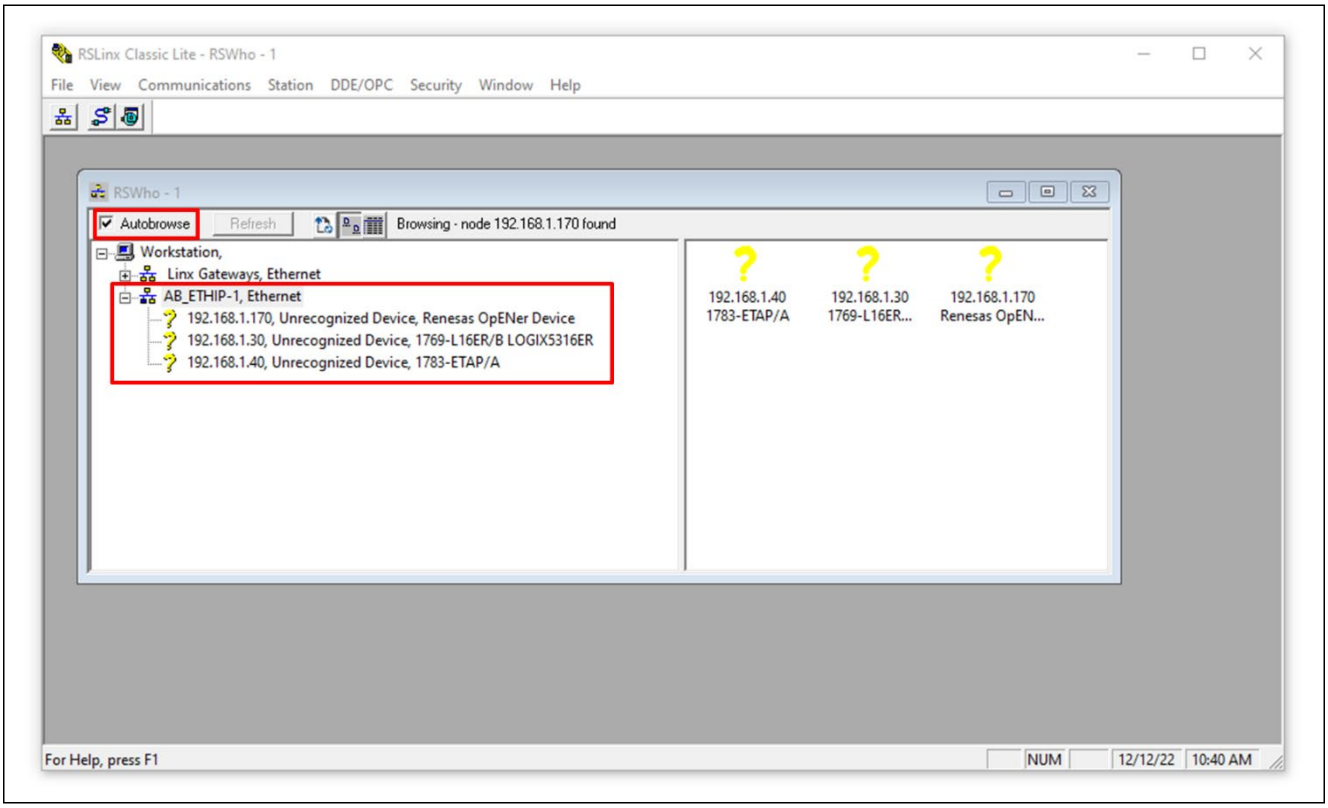


Figure 9-16 Check Autobrowse setting and devices

Note:

> Yellow question marks are resolved by registering the EDS files of each device. See "Register EDS files of devices" in RSLinx Classic Help for details.
> The EDS file for the sample code is stored in the scanner folder (top level of the sample code zip file directory).

It is now ready to use the DLR Tool.

■Monitor DLR operation with the DLR Tool

Launch the DLR Tool; click the Connect button.

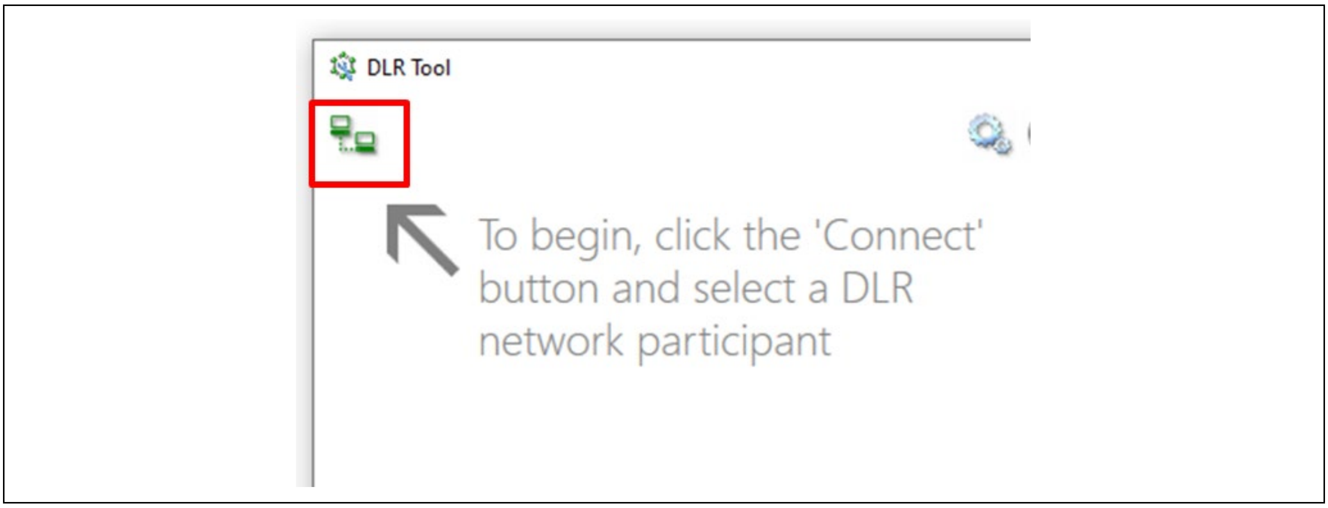


Figure 9-17 Launch DLR Tool and Click Connect button

Expand the added drivers, select one of the devices, and click the Select button.

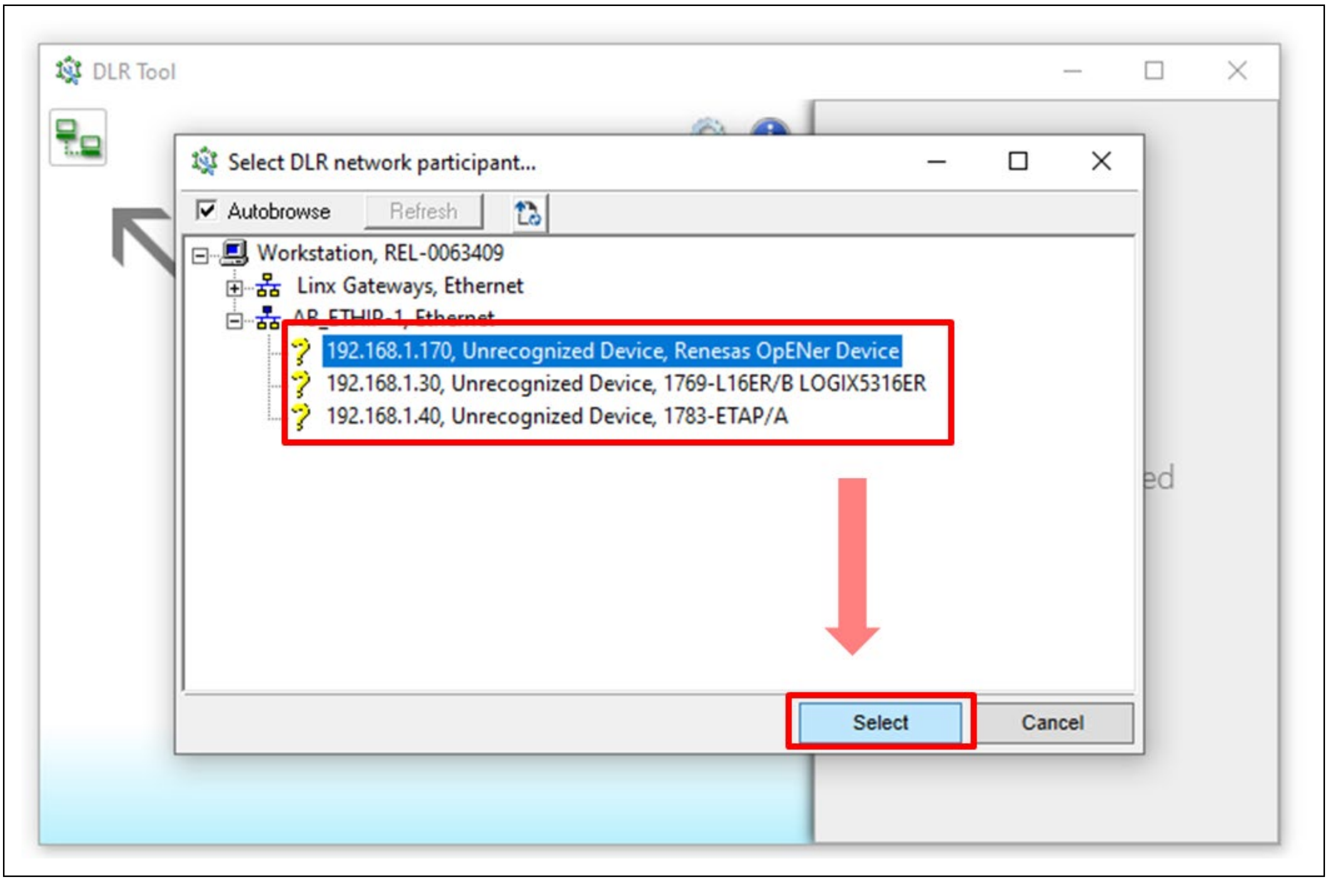


Figure 9-18 Click one of DLR devices and Select button

The status of the DLR operation is graphically displayed and can be monitored.

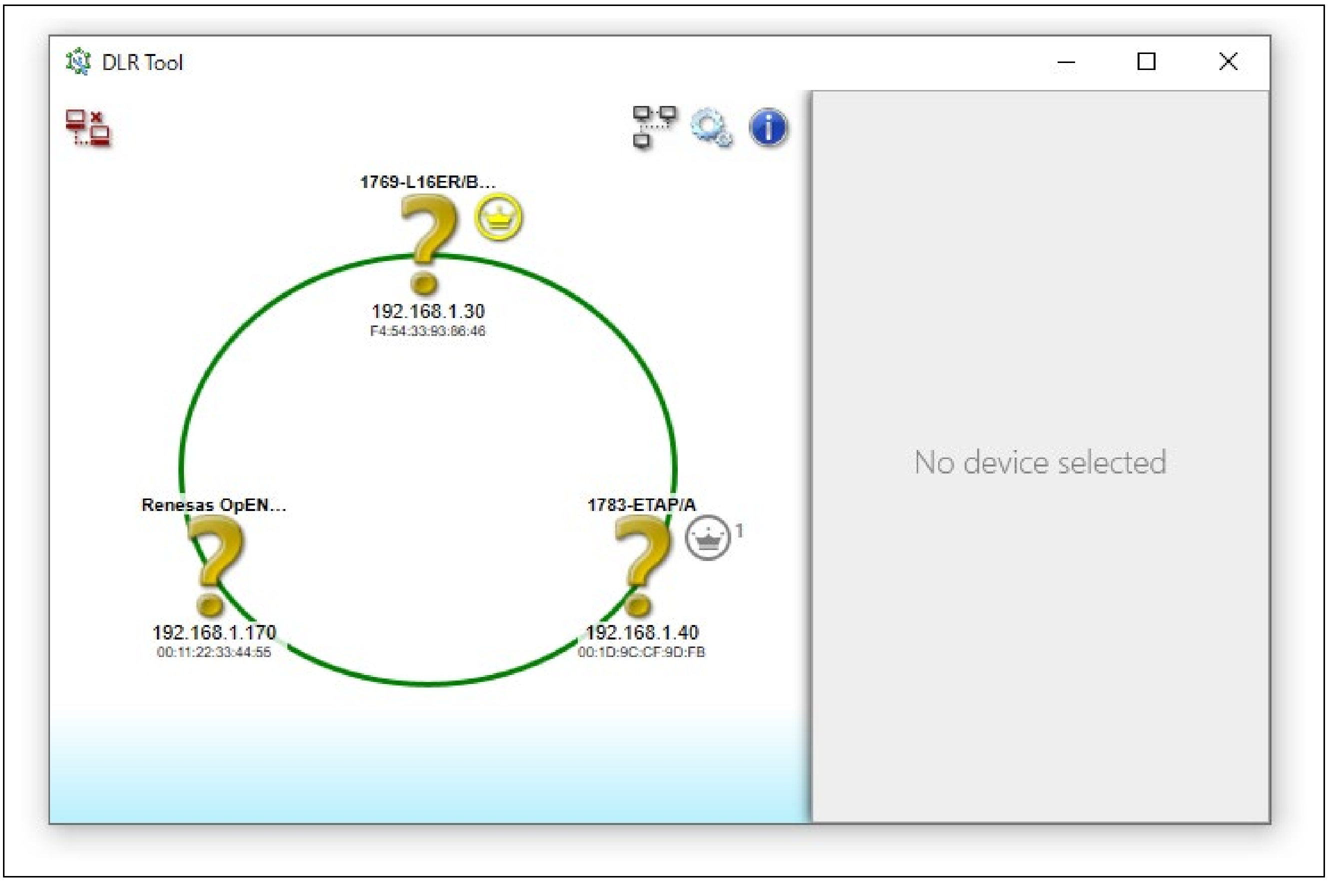


Figure 9-19 DLR operation is graphically displayed on DLR Tool

If any of the topology route is disconnected, the area is highlighted in red. It will be restored when reconnecting.

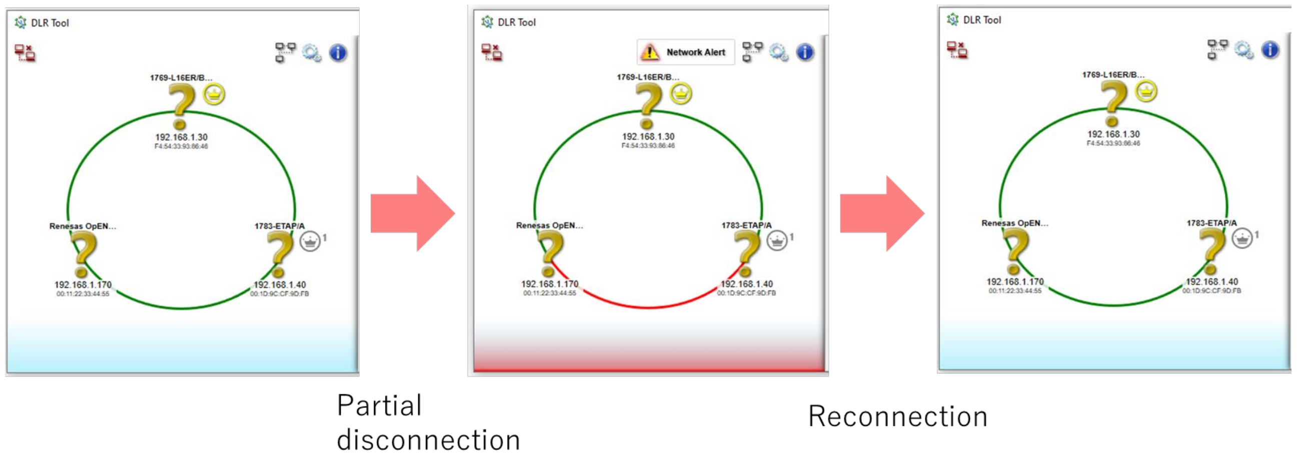


Figure 9-20 Partial disconnection indication and Reconnection

By clicking on a device, information about the device is displayed.

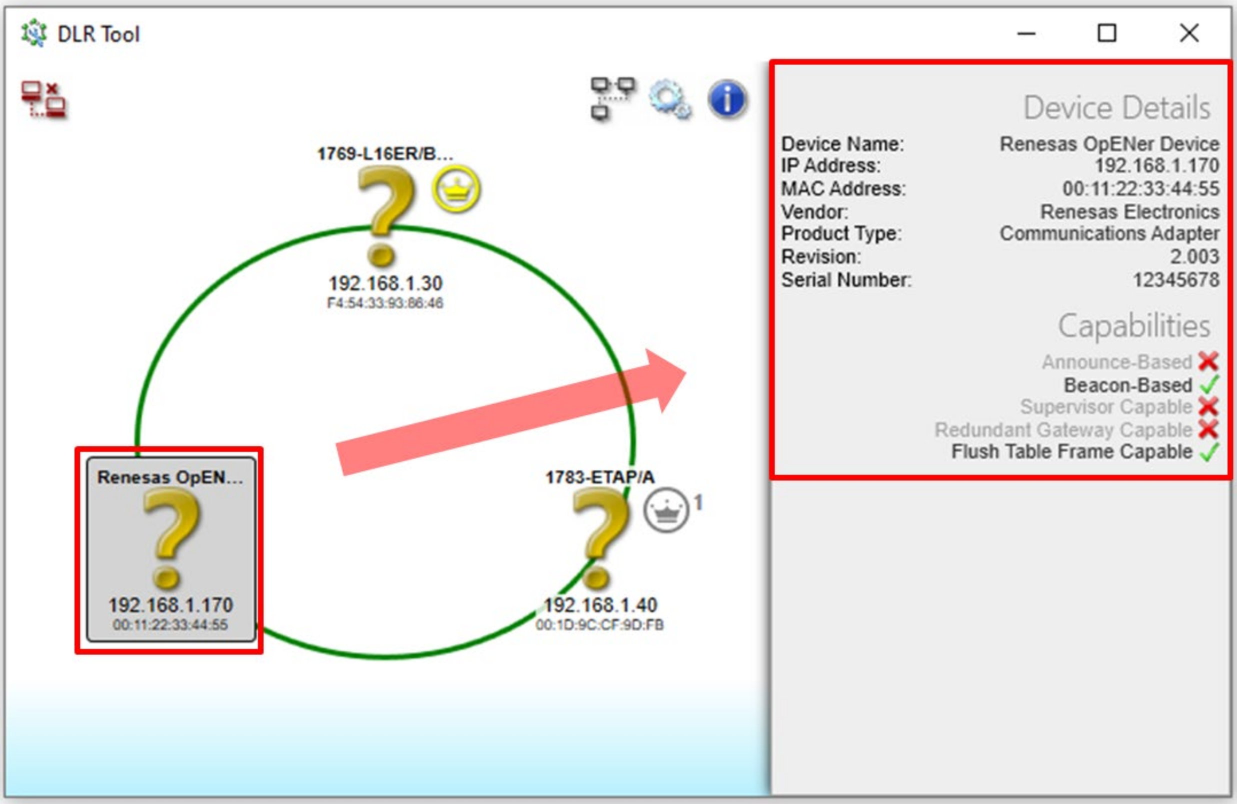


Figure 9-21 Device Details field

Appendix E: How to set fixed Interface Speed 100Mbps Full-Duplex

In testing EtherNet/IP devices that support DLR, Auto-Negotiation may have to be disabled, and the interface speed fixed to 100 Mbps Full-Duplex.

This section describes how to set fixed Interface Speed.

1. Open FSP Configuration Window

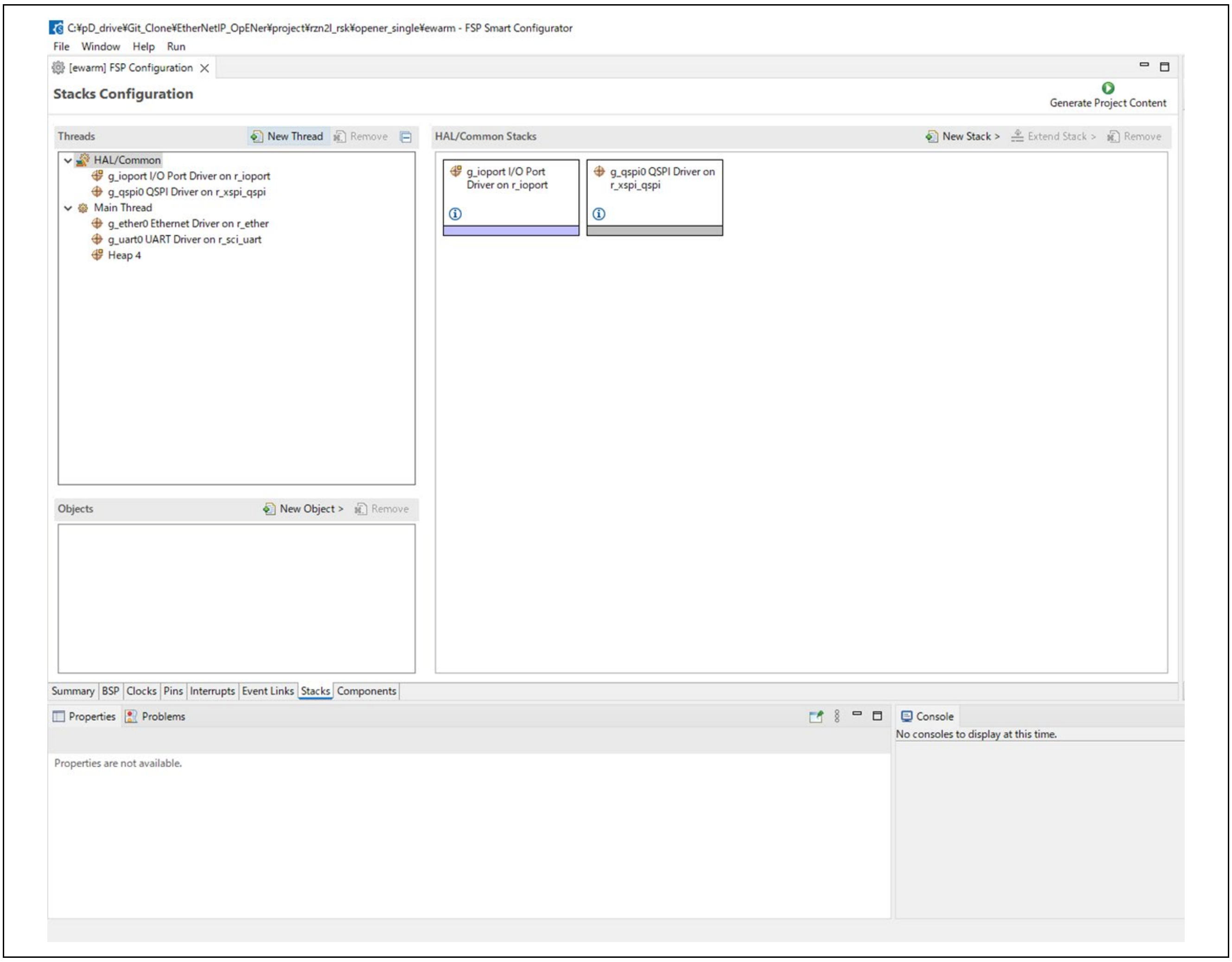


Figure 9-22 FSP Configuration Window, in the case of Smart Configurator

2. Please select “g_ether0 Ethernet Driver” -> “g_ether_phy0 Ethernet Driver on r_ether_phy” (ETH 0 case), and check Properties tab.

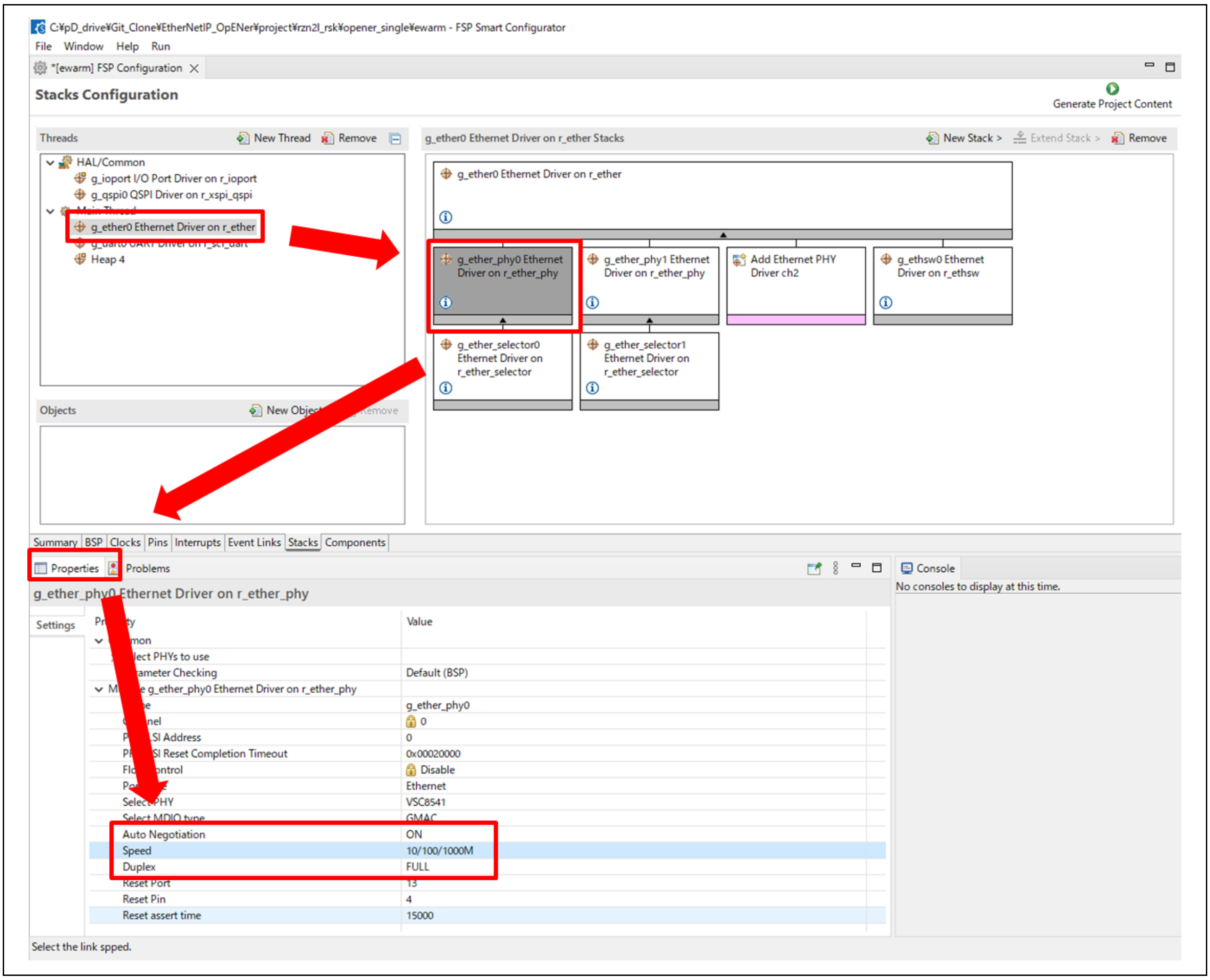


Figure 9-23 Property tab in r_ether_phy

3. Change Interface settings.
   - Auto Negotiation: ON -> OFF
   - Speed: 10/100/1000M -> 100M
   - Duplex: Full (No need to change)

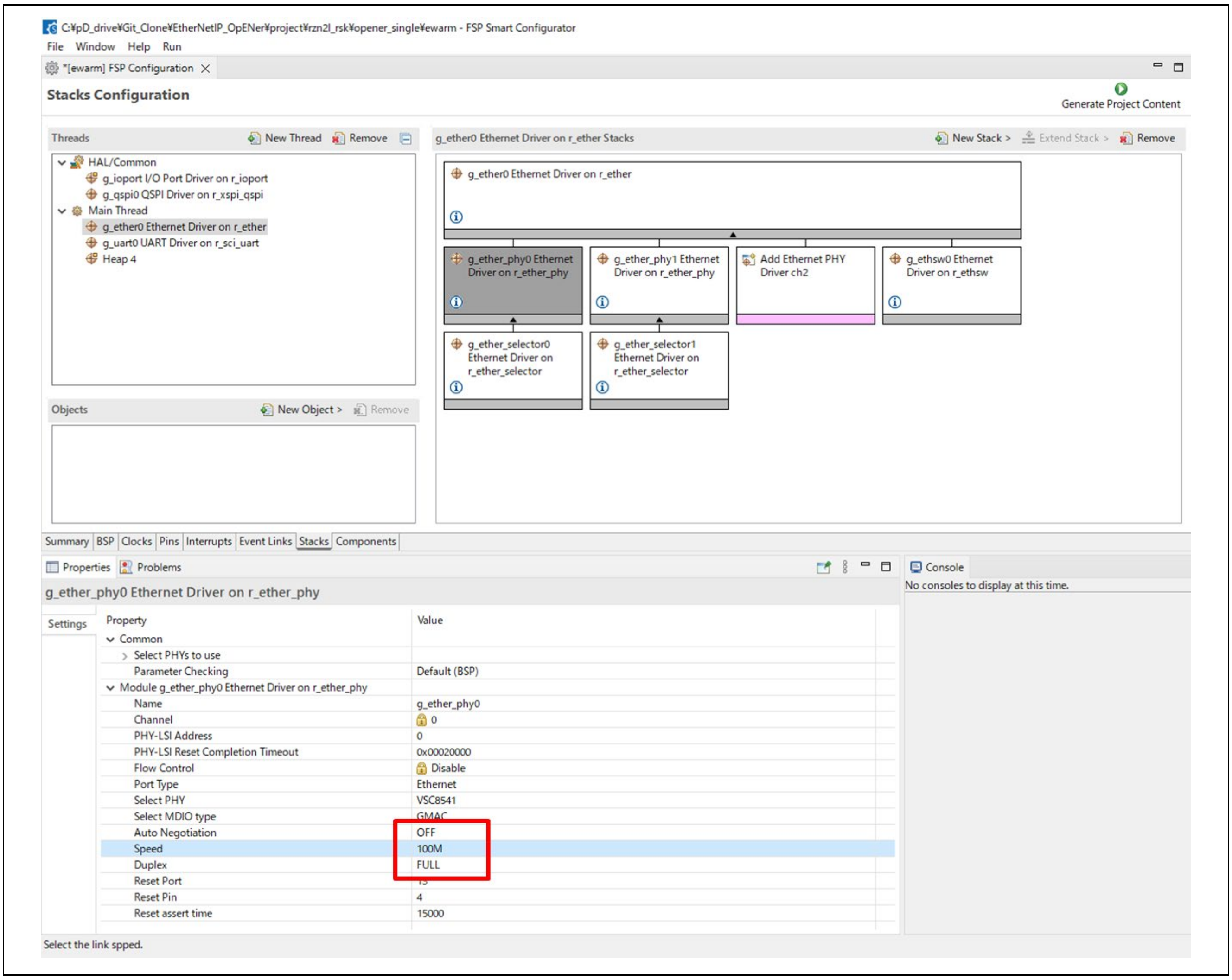


Figure 9-24 Change values in Interface Speed settings

4. Re-generate FSP files and build.

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- OpENer

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- FreeRTOS

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`common\oss\lwip\COPYING`

Revision History

Rev.	Date	Description	
		Page	**Summary**
1.00	Sep. 30, 2022	-	First issued
1.01	Feb. 10, 2023	All	RZ/T2M supported
		4	Add Limitations and Restrictions
		4	Add 0x47 DLR to Appendix C support CIP Object Class
		50	Create new section Appendix D about DLR.
1.02	Jan. 26, 2024	4	Limitation updated - Eliminate the need for extended r_ethsw driver (r_ethsw_extend) - Add a note about reset process - Add restriction about download in ROM operation project Remove Conformance Test errors
		5	Feature updated
		6	Requirement (Development environment) updated Update FSP version (RZT v1.3.0, RZN v1.3.0) etc.
		9	Configuration to enable QSPI is added
		5, 44, 50	LLDP Management object supported
		61-63	Add Appendix-E How to set fixed Interface Speed
		64	Add Appendix F Device data configuration
1.03	July 5, 2024	4	Add Limitations and Restrictions
		6	Requirement (Development environment) updated Update FSP version (FSP for Renesas RZ/N series v2.0.0) etc.
		8,10	Add xSPI0 boot mode SW settings
		21-25, 34-36	Added instructions on how to switch between RAM boot and flash boot within the FSP settings.
		26	Remove CPSR register setting procedure
		28-31, 39	Added instructions on how to write and run applications to flash
1.04	July 31, 2024	All	RZ/T2ME supported
		5	Add a limitation for EWARM 9.60.1 with FSP SC v2.1.0
		7	Requirement (Development environment) updated Update FSP version (FSP for Renesas RZ/T series v2.0.0). Add a table for RZ/T2ME
		10, 13	Add explanation about RSK+RZ/T2ME to Chapter 4
		21-103	Divide Chapter 6 into RZ/T2M and RZ/N2L.And add Dual-core project (for RZ/T2M) guidance. Add a section for RZ/T2ME.
1.05	Nov. 22, 2024	All	RZ/T2H supported
		5	Update the Limitations and Restrictions - Remove the limitation of HW reset on xSPI boot mode - Remove the limitation of program download on xSPI boot mode
		87-115	Add the new procedures to the Running the Sample Application Chapter. That's for - RZT2H CR52 Dual-core projects and RZT2H CA55 Single-core projects.
Later	-	-	Please find it in the next page.

Rev.	Date	Description	
		Page	Summary
1.06	Dec. 20, 2024	All	Correction of typographical errors and omissions
		All	RZ/N2H supported
		All	Remove the indication of the number of lines, as it changes with each code update. (“SAMPLE_EXECUTE_SYSTEM_RESET_PROCESS” in opener_port_main.c)
		6	Add 2 items as known issues. For RZ/T2H and RZ/N2H, some files generated by FSP need to be kept. It's countermeasure against Known issue No.39 “The CA55 project with non-cache sections aborts when debugging with flash boot mode on IAR EWARM.” described in the FSP quick start guide
		145-174	Add the new procedures to the Running the Sample Application Chapter. That's for - RZN2H CR52 Dual-core projects and RZN2H CA55 Single-core projects.
		175	Add a correspondence table of LEDs and switches to each board.
1.07	Feb. 7, 2025	6	Change the description of Chapter 1.3 Limitations from bullet to table. Add limitations regarding the linker files.
		133	Add guidance regarding replacing the e2 studio linker file as a Note.
		140	Add guidance regarding replacing the ewarm linker file as a Note.
		-	Correct other typos
2.00	Jun. 13, 2025	-	Implement the CIP Safety feature and ACD for RZ/N2L
		6	Delete the restriction regarding Flash writing because it is resolved. Delete the sentence regarding the icf file replacing.
		7	Add explanation about CIP Safety feature to Chapter 2 Features
		8-10	Remove annotations related to the CODESYS project
		11-23	Exchange the section order in Chapter 4 for easy to read.
		24	Remove the instructions for downloading a specific version, as well as any references to non-existent appendices.
		133	Delete guidance regarding replacing the e2 studio linker file as a Note.
		137	Add guidance regarding replacing the ewarm linker file as a Note.
		170	Add a guide regarding library versions.
		178-187	Create a new chapter which describes software specifications. Move some appendixes to this chapter.
		184	Update support Attributes and Servies for the implementation of ACD
		190-210	Create a new appendix which describes CIP Safety feature

General Precautions in the Handling of Microprocessing Unit and Microcontroller Unit Products

The following usage notes are applicable to all Microprocessing unit and Microcontroller unit products from Renesas. For detailed usage notes on the products covered by this document, refer to the relevant sections of the document as well as any technical updates that have been issued for the products.

1. Precaution against Electrostatic Discharge (ESD)
   A strong electrical field, when exposed to a CMOS device, can cause destruction of the gate oxide and ultimately degrade the device operation. Steps must be taken to stop the generation of static electricity as much as possible, and quickly dissipate it when it occurs. Environmental control must be adequate. When it is dry, a humidifier should be used. This is recommended to avoid using insulators that can easily build up static electricity. Semiconductor devices must be stored and transported in an anti-static container, static shielding bag or conductive material. All test and measurement tools including work benches and floors must be grounded. The operator must also be grounded using a wrist strap. Semiconductor devices must not be touched with bare hands. Similar precautions must be taken for printed circuit boards with mounted semiconductor devices.
2. Processing at power-on
   The state of the product is undefined at the time when power is supplied. The states of internal circuits in the LSI are indeterminate and the states of register settings and pins are undefined at the time when power is supplied. In a finished product where the reset signal is applied to the external reset pin, the states of pins are not guaranteed from the time when power is supplied until the reset process is completed. In a similar way, the states of pins in a product that is reset by an on-chip power-on reset function are not guaranteed from the time when power is supplied until the power reaches the level at which resetting is specified.
3. Input of signal during power-off state
   Do not input signals or an I/O pull-up power supply while the device is powered off. The current injection that results from input of such a signal or I/O pull-up power supply may cause malfunction and the abnormal current that passes in the device at this time may cause degradation of internal elements. Follow the guideline for input signal during power-off state as described in your product documentation.
4. Handling of unused pins
   Handle unused pins in accordance with the directions given under handling of unused pins in the manual. The input pins of CMOS products are generally in the high-impedance state. In operation with an unused pin in the open-circuit state, extra electromagnetic noise is induced in the vicinity of the LSI, an associated shoot-through current flows internally, and malfunctions occur due to the false recognition of the pin state as an input signal become possible.
5. Clock signals
   After applying a reset, only release the reset line after the operating clock signal becomes stable. When switching the clock signal during program execution, wait until the target clock signal is stabilized. When the clock signal is generated with an external resonator or from an external oscillator during a reset, ensure that the reset line is only released after full stabilization of the clock signal. Additionally, when switching to a clock signal produced with an external resonator or by an external oscillator while program execution is in progress, wait until the target clock signal is stable.
6. Voltage application waveform at input pin
   Waveform distortion due to input noise or a reflected wave may cause malfunction. If the input of the CMOS device stays in the area between V_{IL} (Max.) and V_{IH} (Min.) due to noise, for example, the device may malfunction. Take care to prevent chattering noise from entering the device when the input level is fixed, and also in the transition period when the input level passes through the area between V_{IL} (Max.) and V_{IH} (Min.).
7. Prohibition of access to reserved addresses
   Access to reserved addresses is prohibited. The reserved addresses are provided for possible future expansion of functions. Do not access these addresses as the correct operation of the LSI is not guaranteed.
8. Differences between products
   Before changing from one product to another, for example to a product with a different part number, confirm that the change will not lead to problems. The characteristics of a microprocessing unit or microcontroller unit products in the same group but having a different part number might differ in terms of internal memory capacity, layout pattern, and other factors, which can affect the ranges of electrical characteristics, such as characteristic values, operating margins, immunity to noise, and amount of radiated noise. When changing to a product with a different part number, implement a system-evaluation test for the given product.

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